Ambersong

Hardware Bible

Hardware revision A: what is in the radio and how it is joined

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Hardware revision	A
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Author	Zeroshin

Ambersong is a 1960s LLOYDS TM-838N AM/SW/FM tube radio rebuilt in a wooden cabinet with two ESP32 boards. The tube set still receives; the boards add a four-digit LED-filament clock, a stepper-driven dial needle that calibrates itself, Bluetooth audio, an AUX input and a web portal.

Colophon

Item	Value
Release	Version 1, declared 2026-11-10
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Hardware revision	A
Firmware version	Main board (ESP32-S3) `v.1.5.20260927T141322Z`. Audio board (ESP32) `v.1.4.20260926T153301Z`.
Author	Zeroshin
Licences	This document: `CC-BY-SA-4.0`. Hardware design (KiCad, STL and SketchUp files): `CERN-OHL-S-2.0`. Firmware: `GPL-3.0-or-later`. Photos and renders: `CC-BY-NC-ND-4.0`. The running hare on the dial: all rights reserved.
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Callouts in this document:

> ⚠ DANGER Hazard that will kill or seriously injure, its consequence, and how to avoid it.

> ⚠ WARNING Hazard that could kill or seriously injure, its consequence, and how to avoid it.

> NOTICE Risk of damage to the radio or to what it is connected to, and how to avoid it. No injury.

> Note A fact not to miss. It is not a hazard.

The other book, the Markdown editions of both books and every source file are in the Ambersong repository: start with its `README.md`.

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About this book

This book describes the hardware of Ambersong: a vintage tube radio, rebuilt in a wooden cabinet with two small computer boards, a digital clock, a motor-driven dial needle and a modern amplifier. It says what every part is, where it sits when that is known, and what it is joined to. With it you should be able to repair, debug, change or rebuild a machine like this one without any other document.

What this book covers

- The machine: the two microcontroller boards, the power and the grounds, the audio chain, the clock display, the dial needle and its sensors, the front-panel controls and lamps, and every cable.
- The tube radio inside it, a LLOYDS TM-838N, stage by stage. Its parts are given as fitted, with the original factory and service-data values beside them for reference.
- The FM antenna as built and installed, the amplifier and its speakers, and the physical build: the cabinet, the front panel, the tuning and needle mechanics, and the Faraday cages.
- Datasheet limits for the parts used here (Appendix A (p. 139)). They describe the catalogue part, not a measurement of the one fitted.

What it does not cover

- The firmware. What the two boards run, what you see on the radio, and how to update them are in the companion book, the Ambersong Firmware Gospel. This book states the wire and the part; the Gospel states what the code does with them.
- The amplifier's own circuit. Chapter 12 says what the amplifier is and how it connects, but not what is on its boards.
- Build details: wire colours (except the amplifier supply's four wires, chapter 5 (p. 21)), which side of a connector is pin 1, crimp and connector housings, the length of each cable (the only length facts: every cable inside is under 14 in, and the DAC-to-amplifier cable is under 12 in), cable routing, the screws of the printed plates, and the order of the contacts inside a connector.

Appendix C (p. 149) lists every open point and every gap.

How to use it

Read chapter 2 (p. 7) before you open the cabinet. Mains is present inside even with the front switch off, and the tube radio's metal chassis is wired to mains neutral.

Then find your job in the table below; it names the chapter to start at.

Table 1. How to use it

If you want to...	Start at
know what the machine is and how the parts fit together	chapter 1 (p. 3)
open the cabinet safely	chapters 2 (p. 7) and 3
find what a GPIO pin does	chapter 4 (p. 12)
trace a power or ground problem	chapter 5 (p. 21)
trace a sound problem	chapters 6 (p. 28) and 12
fix the clock display	chapter 7 (p. 38)
fix the dial needle or tuning sensors	chapter 8 (p. 46)
fix a front-panel control or lamp	chapter 9 (p. 52)
repair or remake a cable	chapter 10 (p. 60)
service the tube radio	chapter 13 (p. 98)

Most chapters end with a When something is wrong table. It holds only what this book knows: the wiring, and what has been stated about the fitted parts.

Much of this book describes one machine: this cabinet, this radio, this amplifier and these boards. If you build your own, adapt it.

The drawings. The machine was drawn in KiCad to show a person what was built, not to make a circuit board. Some drawings place a part where it reads best rather than where it sits. You do not need them to use this book. If you do read them, this book is right where the two differ, and Appendix B (p. 146) lists the places where a drawing can mislead.

The firmware's own text. Where a comment or a name in the firmware disagrees with this book, the hardware stated here is right.

Notation

A few conventions run through the whole book. Learn them once here.

The two boards. The book calls them by what they do:

Table 2. Notation

In this book	The board	Also written
main board	the ESP32-S3 development board: clock display, needle, tuning sensors, panel lamps	S3
audio board	the ESP32 development board: sound, Bluetooth, front controls, real-time clock	A32

Signal names are written in `code font`, exactly as on the drawings and the cable sheet: `GPIO5`, `A_IO22`, `S3_to_A32`. On the main board a signal is usually named after its pin (`GPIO38`); on the audio board it carries an `A_` prefix (`A_IO19` is the audio board's GPIO19). In the text, an audio-board pin is written with its board named, as in "the audio board's GPIO25".

The four grounds. The machine has four ground names, kept different on purpose. Three of them, Earth, the DC ground and the amplifier supply return, are all joined. The fourth, the radio chassis, is joined to none of them:

it sits on mains neutral. Chapter 5, section 5.3 (p. 23), explains each one and how they are joined.

Table 3. Notation

Name in this book	Written as
Earth	`Earth`
DC ground	`GND`
Amplifier supply return	`20V_GND`
Radio chassis	`CHASSIS`

Rails with two names. Some conductors carry two names on the drawings. The two names are the same wire. The last row is the one rail that is easy to confuse with them: it has one name and is a separate wire.

Table 4. Notation

Name	Same as	What it is
`5VDC_S`	`5VDC`	the 5 V supply's output; `5VDC_S` is its name before the bus
`PSU_GND`	`GND`	the 5 V supply's ground; `PSU_GND` is its name before the bus
`A_VDC`	`A_3V3`	the 3.3 V bus, fed from the audio board's 3.3 V output
`S3_3V3`	—	the main board's own 3.3 V output, a separate rail

CABLE S/FTP on a connector pin means the cable's shield and foil end there. Unless a cable's page says otherwise, they are joined to `GND` at the board end and left unconnected at the far end (chapter 10 (p. 60)).

Pin numbers on cables are the cable sheet's, counted along the drawn connector. They are not the numbers moulded on a real connector.

Parts are named by what they are. Where you need a reference to find a part on a drawing, it follows in brackets: "the safety capacitor between the antenna shield and the chassis (C51)". The tube radio's references clash with the main drawing's, so chapter 13 (p. 98) always gives a name with them.

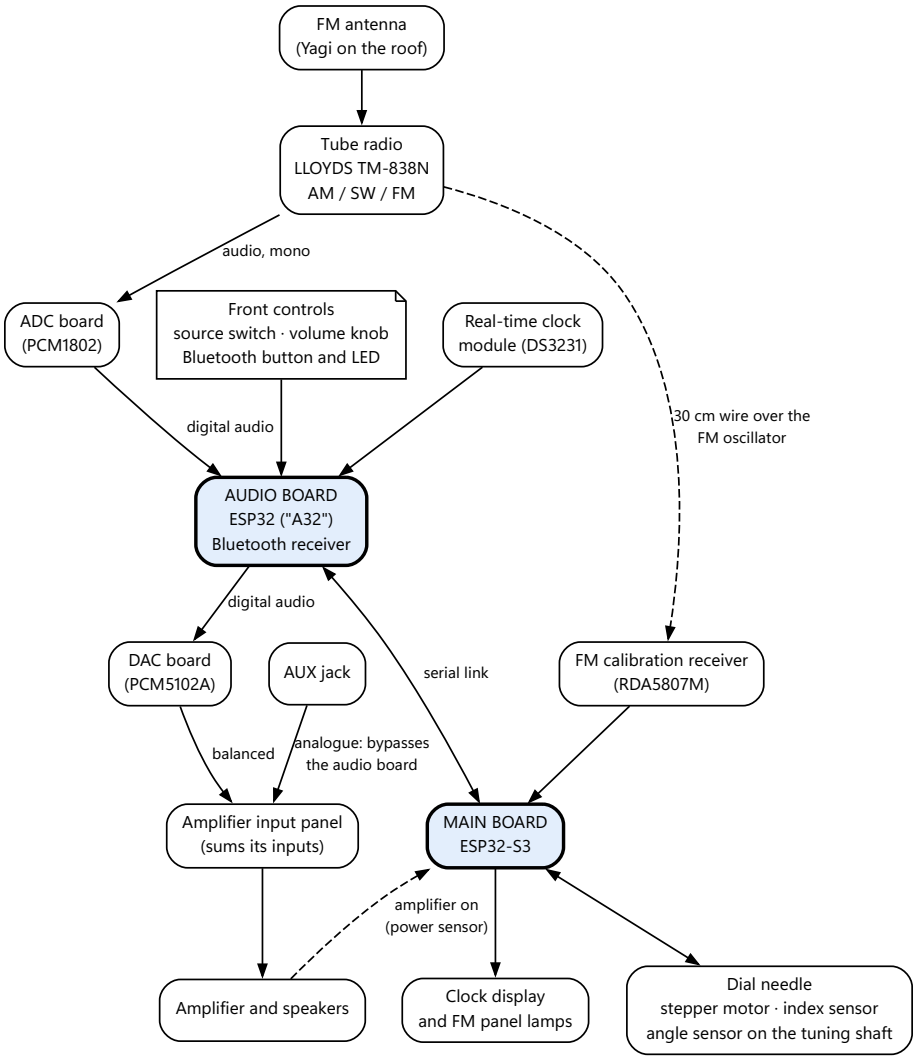
Values are as fitted. Capacitors drawn as polarised are fitted the right way round as drawn; the others are film or ceramic.

1. What Ambersong is

Ambersong is a vintage vacuum-tube radio, a LLOYDS TM-838N AM, shortwave (SW) and FM set, joined to modern audio and control electronics. The tube radio still makes the sound when you listen to the radio. Two small ESP32 boards beside its chassis turn that sound into digital audio, run a clock, and move the dial needle with a motor. They also add Bluetooth and an auxiliary input. This chapter shows how the pieces fit together before the rest of the book takes them one at a time.

1.1 The machine at a glance

Figure 1. The whole machine: where the sound goes, and which board runs what



Everything sits in one wooden cabinet: the tube radio's original chassis, the two boards, the converters, the amplifier with its speakers, and the power supplies. Chapter 3 describes the cabinet.

The table names each part, says what it does, and gives the chapters that cover it.

Table 5. The machine at a glance

Part	What it does	Chapter
Tube radio (LLOYDS TM-838N)	Receives AM, SW and FM and makes the radio's audio. Its output transformer, loaded by a 7.5 Ω resistor, feeds the ADC board (below).	13
Main board (ESP32-S3)	Drives the clock display, the dial-needle motor and its index sensor, reads the angle sensor on the tuning shaft and the FM calibration receiver, dims the FM panel lamps, and senses when the amplifier is on.	4, 7, 8, 9
Audio board (ESP32, "A32")	Takes the radio's digital audio from the ADC, receives Bluetooth, and sends the chosen source to the DAC. Reads the source switch, the volume knob, the Bluetooth button, and the real-time clock; lights the Bluetooth LED.	4, 6, 9
ADC board (analogue-to-digital converter, PCM1802)	Turns the tube radio's audio into digital audio for the audio board.	6
DAC board (digital-to-analogue converter, PCM5102A)	Turns the audio board's digital audio back into an analogue signal for the amplifier.	6
Amplifier and speakers	A powered speaker pair's amplifier, with its two speakers. Its input panel adds up everything plugged into it.	12
Clock display	Four seven-segment digits and a colon, built for this front panel.	7
Dial needle	The radio's original brass needle, moved by a stepper motor, not by the dial cord. A magnet on its mount passes a Hall-effect index sensor, the needle's position reference.	8
Angle sensor (AS5600)	Reads the angle of the tuning capacitor's shaft through a magnet on the shaft.	8
FM calibration receiver (RDA5807M)	A small FM receiver chip, the main board's instrument for calibrating the dial. Its antenna is a 30 cm wire inside the radio's cage, bent over the tube radio's FM oscillator.	8
Real-time clock module (DS3231)	Keeps the time for the audio board.	4
FM antenna	A three-element Yagi on the roof, built for this machine, on the radio's antenna input.	11

The two boards talk to each other over one serial link: two signal wires and a ground, in a shielded cable (chapter 10 (p. 60)).

1.2 The sound path

The front source switch picks one of three sources: RADIO, BT (Bluetooth) or AUX. Each one reaches the speakers by its own path.

Table 6. The sound path

Source	Path to the speakers
RADIO	tube radio → its output transformer, loaded by a 7.5 Ω 5 W resistor → shielded cable → ADC → audio board → DAC → amplifier input panel → amplifier → speakers
BT	a paired Bluetooth device → audio board → DAC → amplifier input panel → amplifier → speakers
AUX	AUX jack → amplifier input panel → amplifier → speakers

AUX is analogue and bypasses the audio board entirely. It never goes through the DAC. On AUX the audio board stays silent, and nothing it does can change AUX loudness or mute it.

The radio feed is mono. One side of the radio's output reaches the ADC; the ADC's left and right inputs are fed from the same point (chapter 6 (p. 28)).

The amplifier's input panel sums its inputs; it does not choose between them. The DAC reaches it through its balanced input. The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn. If two of them carry sound at once, you hear both.

Three volume controls are in the chain. Chapter 6, section 6.8 (p. 37), sets out all three.

- The tube radio's own, kept at about one eighth of its travel.
- The front volume knob. The audio board reads it, so it acts on RADIO and BT only.
- The amplifier's own, reached only from the back.

1.3 The control side

The main board runs everything you see move or light: the clock digits and colon, the dial needle, and the four FM panel lamps. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor; nothing mechanical joins it to the needle. The angle sensor reads the capacitor's shaft, and the main board moves the needle to match. The FM calibration receiver is the main board's instrument for calibrating the dial. How the firmware uses these sensors is in the Firmware Gospel.

The audio board reads the front controls (source switch, volume knob, Bluetooth button), lights the Bluetooth LED, and keeps the real-time clock.

A small amplifier power sensor, an optocoupler lit by the amplifier's supply, tells the main board when the front power switch has turned the amplifier on (chapter 9 (p. 52)).

1.4 The power at a glance

One mains inlet feeds the whole machine. A 5 V supply runs whenever the cord is plugged in. It powers both boards and everything digital. The 3.3 V parts on the audio side run from the audio board's own 3.3 V output. The front power switch turns on only the amplifier's ±20 V supply and the tube radio. So the clock keeps running with the switch off, and mains is still present inside the cabinet.

Chapter 2 lists the dangers. Chapter 5 follows the power from the inlet to every board and explains the four grounds.

1.5 The front panel at a glance

From the front you reach the tuning knob, the volume knob, the source switch, the power switch and the Bluetooth LED, which is also the Bluetooth button. You also see the clock display and the four FM panel lamps. On the

FM panel the needle shows the rough FM position against a blackened brass plate. Chapter 3, section 3.2 (p. 10), lists each one.

2. Safety before you open it

Ambersong runs on mains. It holds a tube radio built in the old “AC/DC” way: no mains transformer, and its metal chassis wired straight to the mains. This short chapter lists every point where mains or high voltage is present, and when. Chapter 5 explains the same wiring in detail.

> ⚠ DANGER the front power switch does not remove mains from the cabinet. Mains can still shock or burn you with the switch off. The only way to remove mains is to pull the cord out of the mains inlet. Do it before you open the back panel or the front panel.

2.1 Where the mains is, and when

The table lists each point that carries mains or high voltage, and what must be on for it to be live.

Table 7. Where the mains is, and when

Point	Live when	Notes
Mains inlet (IEC C14) and its wiring	the cord is plugged in	Live, neutral and earth.
5 V power supply, mains side	the cord is plugged in	It is wired ahead of the front power switch, so it never switches off.
Front power switch, input side	the cord is plugged in	The switch breaks the live wire only.
Amplifier ±20 V supply, mains side	the front switch is on	Its neutral wire stays connected with the switch off.
Mains line filter, in its own cage	the front switch is on	Its neutral wire stays connected with the switch off.
Tube radio: its mains terminals, its own power switch	the front switch is on	
Tube radio: its 0.75 A fuse, its heater string, and everything after its own power switch	the front switch and the radio’s own power switch are on	The seven tube heaters are wired in series from the fused live to the chassis, which is on neutral.
Tube radio’s high-voltage supply (B+)	the radio is on	Made from the mains by a half-wave rectifier inside the radio (chapter 13 (p. 98)).
Tube radio’s chassis (`CHASSIS`)	always wired to mains neutral	See section 2.3 (p. 8).

B+ is the radio’s high-voltage supply for its tubes. The radio makes it by rectifying the mains directly. The chain is a 28 Ω surge resistor and a silicon rectifier, then three 47 µF 315 V capacitors with two dropping resistors between them. No bleeder resistor is drawn across these capacitors. The numbers in the names of its points (`B130`, `B115_A`) are the original service data’s nominal figures, not measured voltages.

2.2 The front power switch does not make it safe

The front power switch is a single-pole toggle. It breaks only the live wire, and only to two things: the amplifier's ±20 V supply and the tube radio (through the line filter). Three things follow:

- The 5 V power supply stays on. Both microcontroller boards, the clock and everything digital keep running.
- The neutral wire is never switched. It stays connected to the amplifier supply, the line filter and the tube radio's chassis.
- Only unplugging the cord removes mains from the cabinet.

2.3 The radio chassis is on mains neutral

The tube radio has no isolation transformer. Its circuit common, `CHASSIS` — the metal chassis and every part connected to it — is wired to mains neutral through the line filter. It sits near 0 V only while the wall socket's live and neutral are the right way round, and nothing in the machine checks that. If they are reversed, the chassis is on the live wire, and it stays there with the front switch off, because the neutral wire is never switched.

`CHASSIS` is never joined to Earth, to the Faraday cages, to the DC ground, or to the amplifier supply return. The machine uses four ground names. Earth, the DC ground and the amplifier supply return are all joined; the radio chassis is joined to none of them:

Table 8. The radio chassis is on mains neutral

Name	What it is
Earth (`Earth`)	the mains protective earth; the Faraday cages and the antenna connector's shell are on it
DC ground (`GND`)	the return of both boards and everything on 5 V and 3.3 V; reaches the amplifier's ground and so on Earth at almost 0 Ω
Amplifier supply return (`20V_GND`)	the 0 V output of the amplifier's ±20 V supply; its two middle black wires go to the amplifier's ground, which sits on the earth star point of the amplifier's back plate, so it too is on Earth
Radio chassis (`CHASSIS`)	the tube radio's circuit common, on mains neutral

Chapter 5, section 5.3 (p. 23), says what each one is joined to.

The DC ground is on Earth, at almost 0 Ω, through the amplifier's ground (a washer on its Bass pot, chapter 5 (p. 21), section 5.3 (p. 23)). Anything you connect to the DC ground is connected to Earth.

2.4 Safety-rated parts

A few capacitors in the tube radio, on the mains side or on the antenna input, are safety-rated types. The table lists them, with the safety class of each.

Table 9. Safety-rated parts

Part	Where	Fitted
The antenna coupling capacitor (C3)	between the antenna's centre conductor and the radio's FM input	100 pF, class X1Y1
The antenna shield capacitor (C51)	between the antenna coax shield, which is on Earth, and the chassis: the shield's only path to the chassis	0.001 µF, class X1Y2
The mains bypass capacitor (C44)	between the fused live and the chassis	0.01 µF, class X1Y2
Three of the four heater-string bypass capacitors (C47, C49, C52)	from points on the heater string to the chassis	0.0022 µF each, class X1Y2. The fourth (C48) is the original 0.002 µF ceramic, not a safety type.

Fuses. The machine's mains side has no fuse besides the tube radio's own 0.75 A fuse, which sits in the radio's switched live, between its own power switch and the rest of the set.

2.5 The low-voltage side

The rest of the machine runs at low voltage. This is what each part of it is referenced to.

- The amplifier supply gives +20 V and −20 V (±19.33 V measured with no load) around its own return, `20V_GND`. It is isolated from the mains and held off the amplifier's back plate by plastic screws and standoffs.
- The 5 V and 3.3 V buses, and both boards, are on the DC ground, which is on Earth (section 2.3 (p. 8)).

NOTICE each board has a USB-C port on the back panel, carried straight through to the board, power wire included. The boards are also powered from the 5 V bus on their 5V pin whenever the cord is plugged in. Espressif's guide for the ESP32-S3-DevKitC-1, the board the main board is compatible with, calls the USB port, the 5V pin and the 3V3 pin "three mutually exclusive ways to provide power to the board". Espressif's guide for its original ESP32 DevKitC V4 board says the same, and adds that power must come from one and only one of them, "otherwise the board and/or the power supply source can be damaged". Plugging a USB cable into a back-panel port while the cord is in gives the board two of these inputs at once. Chapter 4, section 4.6 (p. 20), describes the ports.

3. The physical build

This chapter says where things are and how they are held, so that you can open the cabinet, find a part, and repair or rebuild it. It describes this cabinet; yours will differ. Screw details for the printed plates, cable routing and the length of each cable are not covered.

⚠ DANGER the front power switch does not remove mains from the cabinet; unplug the cord first. Read chapter 2 (p. 7) before you open it.

3.1 The cabinet and how to open it

A wooden cabinet holds everything. The tube radio's original case has been stripped; its original chassis sits in the cabinet. The table says how each main piece is held, so you know what to undo.

Table 10. The cabinet and how to open it

What	How it is held
Back panel	An inverted-T panel of ½ in plywood behind the radio, held by ten countersunk 3 mm screws.
Front panel	Held from the inside by five 3 mm screws.
Tube radio	Two #8 ½ in wood screws through its tabs, one on each side.
Everything else	Almost every board and mechanical part is on a small perfboard or module in its own printed PLA case, or on a printed PLA carrying plate, screwed to the cabinet with #8 screws.

Every cable inside is under 14 in long, long enough to service the machine with the panels open and nothing unplugged.

3.2 The front panel

The table lists the controls and displays on the front panel, with the chapter that covers each one.

Table 11. The front panel

Control or display	What it is	Chapter
Tuning knob	A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor. Clockwise raises the frequency.	8
Volume knob	The audio board's volume control, read digitally. It is not the amplifier's volume control, which is reached only from the back.	9
Source switch	A three-position switch (double-pole, on–off–on): RADIO and BT at the two ends, AUX in the centre.	9
Power switch	The front power switch. It turns on the amplifier and the tube radio, but does not remove mains from the cabinet.	2, 5
Bluetooth LED	A single part that is both the Bluetooth LED and the Bluetooth button.	9
Clock display	Four seven-segment digits and a colon, made for this panel.	7
FM panel lamps	Four LEDs, dimmed together by the main board.	9
FM panel	The needle shows the rough FM position against a blackened brass plate.	8

The source switch and the power switch are fixed in the wooden front panel itself, each by its own nut and then plenty of hot glue.

The tube radio's own power and tone switch (S2) and its own volume control (R1) are not on the front panel. The radio's volume control sits at about one eighth of its travel (chapter 13 (p. 98), section 13.11 (p. 138)).

The AUX jack is not in this list either (chapter 6 (p. 28), section 6.7 (p. 37)).

3.3 Tuning, needle and index

The printed front knob, on a printed shaft extension, turns the radio's original tuning knob, which drives the dial cord and the tuning capacitor. A magnetic angle sensor (AS5600) reads the capacitor's shaft. The needle sits on a printed mount on a small stepper motor, which alone moves it. A magnet on that mount passes a Hall-effect switch, the needle's only position reference. Chapter 8, section 8.1 (p. 47), describes the mechanics and how each part is mounted.

3.4 Cooling

Two fans move air (chapter 5 (p. 21), section 5.6 (p. 27)):

- The box fan sits at the top of the back panel and blows air out of the cabinet. Fresh air comes in through two small louvred printed vents at the other two ends of the back panel's inverted T.
- The cage fan is screwed to the tube radio's Faraday cage and blows air into it.

3.5 The Faraday cages

Three earthed cages shield the tube radio, the angle sensor and the mains line filter. The cages over the tube radio and the angle sensor are printed PLA boxes lined with aluminium tape. The tape is covered with Kapton tape. The cage over the tube radio stands well clear of it. The line filter's cage is layers of tape over the circuit. Chapter 5, section 5.4 (p. 25), describes each cage and how it is earthed.

3.6 Connectors and labels

This section lists the main connectors, and how the cables are marked.

- Mains inlet: a standard IEC C14 inlet, the one that came on the amplifier's aluminium back plate. That plate is at the back of the cabinet. The amplifier's own volume control is reached only from the back.
- Connectors that fit only one place. Some connector types are used once only, so that a cable cannot be plugged where it would break something. The mains feed to the front power switch is one of them.
- Labels. Each cable is marked at both ends, with an evident name or with its name on the cable sheet. This book uses the cable sheet's names (chapter 10 (p. 60)).
- Power sockets. The 5 V and 3.3 V buses are rows of small 2-pin JST XH sockets (chapter 5 (p. 21), section 5.5 (p. 26)).
- Back-panel USB-C ports. One for each microcontroller board (chapter 4 (p. 12), section 4.6 (p. 20)).
- Antenna connector: a BNC (J10) on the back panel, where the FM antenna's coax arrives (chapter 11 (p. 92)).

3.7 Printed parts

Every printed part of this build is listed in Appendix D (p. 152). All are PLA except the membrane of the passive radiator, which is TPU.

4. The two microcontroller boards

Two ESP32 development boards run the machine. The main board, an ESP32-S3, runs everything you see move or light: the clock display, the dial needle and its sensors, the FM calibration receiver and the panel lamps. The audio board, an ESP32 called “A32”, handles the sound, Bluetooth, the front controls and the real-time clock. This chapter gives each board’s identity, every pin it uses, the small parts around those pins, the back-panel USB-C ports and the real-time clock module. What the firmware does with each pin is in the Firmware Gospel.

4.1 The boards

Both are development boards: a module on a small board with pin headers and USB. The table gives what each one is, so you can identify it.

Table 12. The boards

	Main board	Audio board
Also written	S3	A32
Board marking	ESP32-S3 N16R8 YD-ESP32-23 2022-v1.3	FCC IDL 2BB77-ESP32-32X (exactly as marked)
Module	marked ESP32-S3-N16R8, FCC ID 2AD6X-ESP32S3N16R8	meant to carry an ESP32-WROOM-32E; whether it matches that module is not established
Chip	ESP32-S3 (QFN56), revision v0.2	ESP32-D0WD-V3, revision v3.1; dual core, 240 MHz
Radio	Wi-Fi, Bluetooth LE	Wi-Fi, Bluetooth
Flash	16 MB, quad; flash voltage 3.3 V, set by eFuse (a one-time setting burned into the chip)	4 MB; flash voltage 3.3 V, set by a strapping pin (a pin the chip reads at power-up)
PSRAM (extra memory)	8 MB, inside the chip	—
Crystal	40 MHz	40 MHz
USB	the chip’s own USB port (USB-Serial/JTAG); USB ID `303A:1001`	a Silicon Labs CP210x USB-to-serial bridge; USB ID `10C4:EA60`
On-board LED	an RGB LED on GPIO48	—

The main board is compatible with Espressif’s ESP32-S3-DevKitC-1 in firmware, and pin-compatible with it for every pin this machine uses. The pins it does not use have not been checked. Neither module’s maker has been identified (Appendix C (p. 149)).

4.2 Main board pin map

Figure 2. Schematic: the main board (ESP32-S3), as drawn on the main sheet

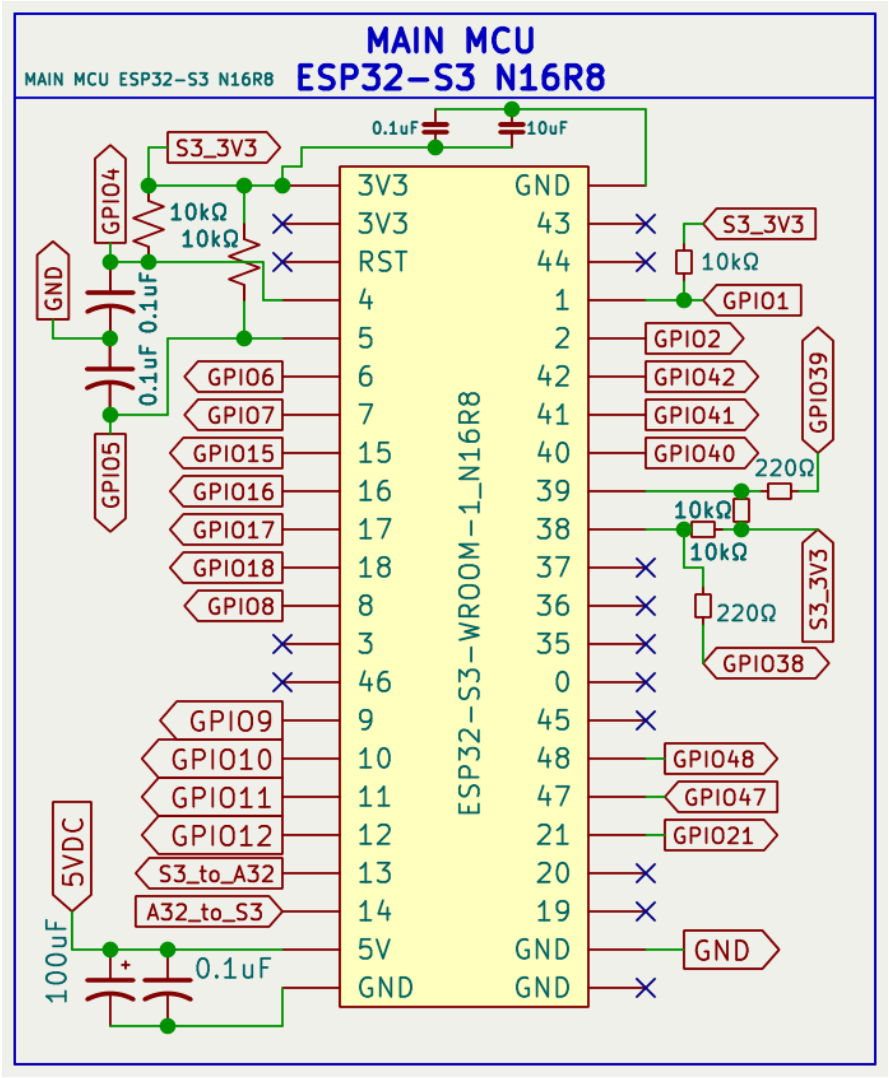
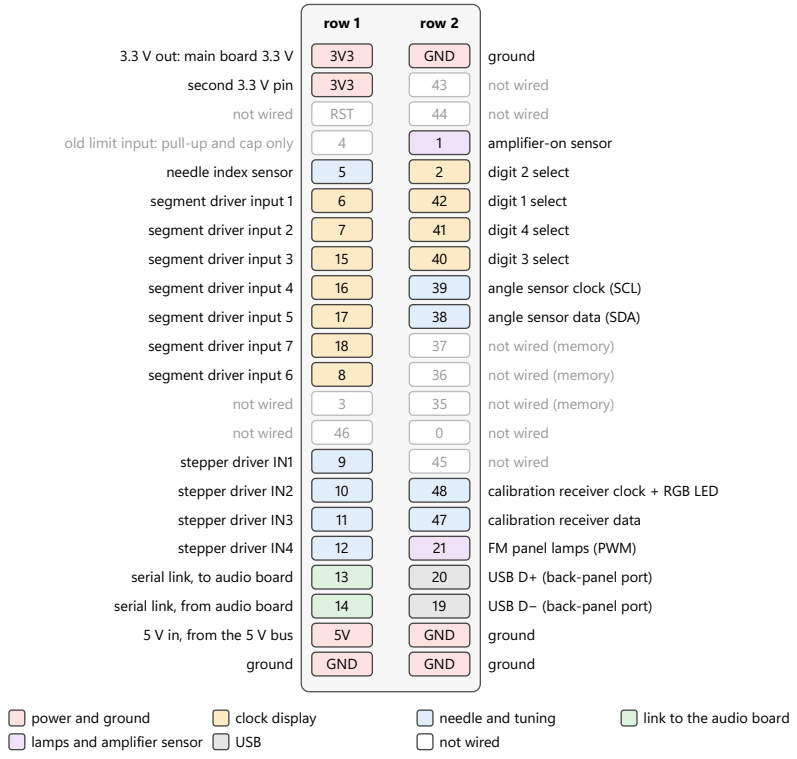


Figure 3. The main board's two pin rows, with the job of each pin



Numbers are GPIO numbers. Row order as on Espressif's ESP32-S3-DevKitC-1; positions of unused pins not checked on this board.

The table lists every main-board pin this machine uses. Signal is the name on the drawings. Cable names the cable's connectors as on the cable sheet (chapter 10 (p. 60)); an arrow joins a cable's two ends.

Table 13. Main board pin map

Pin	Signal	What it does	Connects to	Cable
GPIO1	`GPIO1`	Amplifier-on input	The output transistor (collector) of the optocoupler in the amplifier power sensor. 10 kΩ pull-up to the main board's 3.3 V. Low when the amplifier is on.	`S3_PWR_DET` → `POWER_DET`
GPIO2	`GPIO2`	Digit 2 select	Input 2 of the digit driver, then the P-channel MOSFET that switches digit 2	`S3_DIGIT` → `DIGIT_IN`
GPIO4	`GPIO4`	Nothing (old limit-switch input)	10 kΩ pull-up and 0.1 µF to ground. Wired to the main board's `S3_LIMITS` socket, but its contact is not carried in the cable.	—
GPIO5	`GPIO5`	Needle index	The output of the Hall-effect index switch (A3144). 10 kΩ pull-up to the main board's 3.3 V and 0.1 µF to ground at the pin.	`S3_LIMITS` → `LIMIT`
GPIO6	`GPIO6`	Segment drive 1	Segment driver input 1	`S3_DISPLAY` → `DISPLAY_IN`
GPIO7	`GPIO7`	Segment drive 2	Segment driver input 2	same
GPIO15	`GPIO15`	Segment drive 3	Segment driver input 3	same
GPIO16	`GPIO16`	Segment drive 4	Segment driver input 4	same
GPIO17	`GPIO17`	Segment drive 5	Segment driver input 5	same
GPIO8	`GPIO8`	Segment drive 6	Segment driver input 6	same

Pin	Signal	What it does	Connects to	Cable
GPIO18	GPIO18	Segment drive 7	Segment driver input 7	same
GPIO9	GPIO9	Stepper coil drive	Stepper driver input IN1	S3_STEPPER → STEPPER_IN
GPIO10	GPIO10	Stepper coil drive	Stepper driver input IN2	same
GPIO11	GPIO11	Stepper coil drive	Stepper driver input IN3	same
GPIO12	GPIO12	Stepper coil drive	Stepper driver input IN4	same
GPIO13	S3_to_A32	Serial link, transmit	The audio board's GPIO26	S3_UART → A32_UART
GPIO14	A32_to_S3	Serial link, receive	The audio board's GPIO27	same
GPIO19	—	USB D−	The board's own USB port, which reaches a back-panel USB-C port (section 4.6 (p. 20))	—
GPIO20	—	USB D+	The same	—
GPIO21	GPIO21	FM panel lamps, dimming	Through 100 Ω to the gate of the lamps' MOSFET (chapter 9 (p. 52))	S3_FM_LED → FM_LED_IN
GPIO38	GPIO38	Angle sensor data (SDA)	10 kΩ pull-up to the main board's 3.3 V at the pin, then 220 Ω in series, then the AS5600's SDA	S3_TUNER
GPIO39	GPIO39	Angle sensor clock (SCL)	10 kΩ pull-up to the main board's 3.3 V at the pin, then 220 Ω in series, then the AS5600's SCL	S3_TUNER
GPIO40	GPIO40	Digit 3 select	Input 3 of the digit driver	S3_DIGIT → DIGIT_IN
GPIO41	GPIO41	Digit 4 select	Input 4 of the digit driver	same
GPIO42	GPIO42	Digit 1 select	Input 1 of the digit driver	same
GPIO47	GPIO47	Calibration receiver data (SDIO)	The RDA5807M's SDIO. No pull-up resistors.	S3_RDA → RDA_I2C
GPIO48	GPIO48	Calibration receiver clock (SCLK)	The RDA5807M's SCLK. No pull-up resistors. Also drives the board's own RGB LED.	S3_RDA → RDA_I2C
3V3	S3_3V3	The main board's 3.3 V output	The pull-ups above; the angle sensor and the FM calibration receiver, through their cables. Not joined to the 3.3 V bus.	S3_TUNER, S3_3V3 → RDA_3V3
3V3 (the second one)	—	Nothing	Nothing is wired to it. Both 3V3 pins are the same on-board 3.3 V rail.	—
5V	5VDC	Power in	The 5 V bus	5 V bus → S3_5VIN
GND	GND	Ground	The DC ground	with the 5 V cable

Not used: GPIO0, GPIO3, GPIO35, GPIO36, GPIO37, GPIO43, GPIO44, GPIO45, GPIO46, and the reset pin (RST). GPIO35 to GPIO37 are taken by the 8 MB PSRAM and cannot be used, according to Espressif's datasheet.

Digit numbering. Digit 1 is the rightmost digit, the units of minutes; digits 2, 3 and 4 run to its left. The digit pins are therefore not in number order: GPIO42, GPIO2, GPIO40 and GPIO41 select digits 1 to 4. Which segment each segment drive lights is in chapter 7 (p. 38).

The amplifier-on input. The optocoupler's output transistor pulls GPIO1 to ground while the amplifier's supply lights its LED; the pull-up holds GPIO1 high otherwise. Chapter 9 describes the sensor.

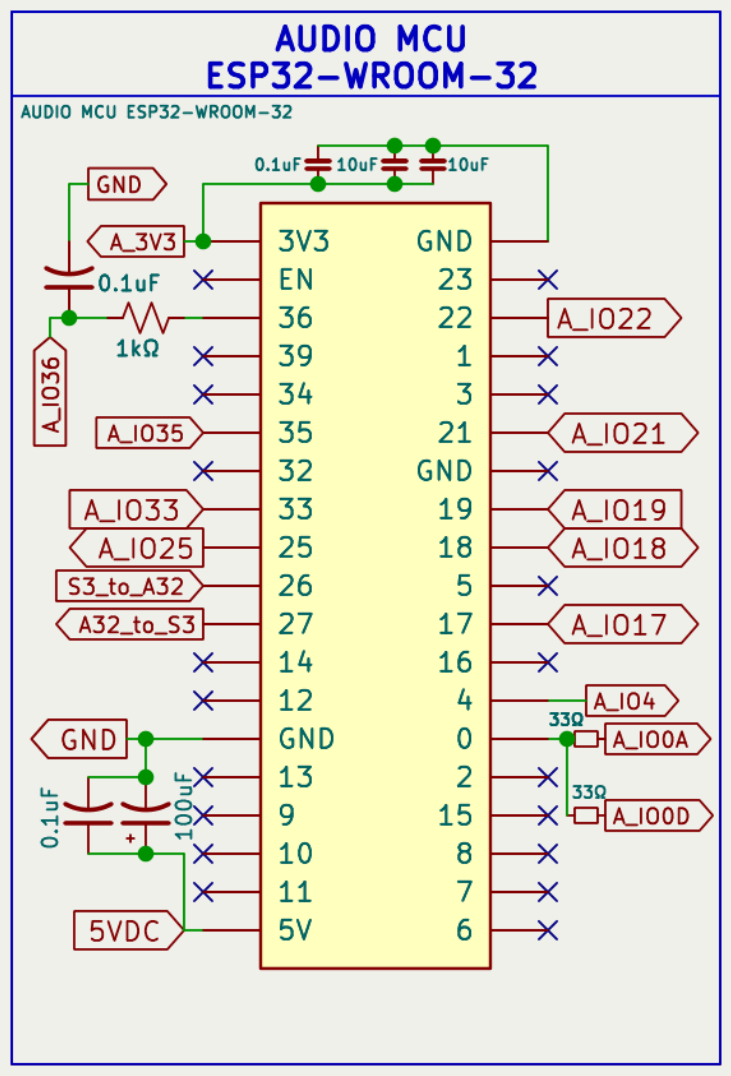
Pins with a second job on the chip. Some used pins have other duties on the ESP32-S3; Appendix A (p. 139) lists the chip's limits.

Table 14. Main board pin map

Pins	Other duty on the chip	Here
GPIO19, GPIO20	USB D− and D+ of the chip's own USB port	used for USB only
GPIO39 to GPIO42	the chip's JTAG debug interface	used for the angle sensor clock and three digit selects
GPIO0, GPIO3, GPIO45, GPIO46	strapping pins, read at power-up	not used

4.3 Audio board pin map

Figure 4. Schematic: the audio board (ESP32), as drawn on the main sheet



The table lists every audio-board pin this machine uses, with the same columns as section 4.2 (p. 13). The two converters are the ADC and the DAC (chapter 6 (p. 28)).

Table 15. Audio board pin map

Pin	Signal	What it does	Connects to	Cable
GPIO0	A_IO0A, A_IO0D	Master clock (SCK) for both converters	Two 33 Ω resistors at the pin, one per converter: A_IO0A to the ADC's SCK, A_IO0D to the DAC's SCK	A32_I2S (chapter 10 (p. 60), section 10.15 (p. 74))
GPIO4	A_IO4	Audio data to the DAC (DIN)	The DAC's DIN, with a 100 kΩ pull-down to ground	A32_I2S
GPIO17	A_IO17	Word clock (LRCK)	The LRCK of both the ADC and the DAC	A32_I2S
GPIO18	A_IO18	Bit clock (BCK)	The BCK of both the ADC and the DAC	A32_I2S
GPIO19	A_IO19	Audio data from the ADC (DOUT)	The ADC's DOUT	A32_I2S
GPIO21	A_IO21	Real-time clock, clock line (SCL)	The clock module's SCL	A32_RTC → RTC_I2C
GPIO22	A_IO22	Real-time clock, data line (SDA)	The clock module's SDA	A32_RTC → RTC_I2C
GPIO25	A_IO25	Bluetooth button	One side of the button; the other side is on ground. Active low: pressing it pulls the pin to ground. No pull-up resistor is drawn; the firmware uses the chip's internal one.	A32_BT_LED → BT_LED
GPIO26	S3_to_A32	Serial link, receive	The main board's GPIO13	A32_UART → S3_UART
GPIO27	A32_to_S3	Serial link, transmit	The main board's GPIO14	same
GPIO33	A_IO33	Bluetooth LED	The LED's cathode; its anode is on 3.3 V (A_3V3). Active low: the pin lights the LED by pulling low. No series resistor is drawn.	A32_BT_LED → BT_LED
GPIO35	A_IO35	Volume knob	The wiper of the front volume knob, a potentiometer with its ends on 3.3 V (A_VDC) and ground	A32_VOL → VOLUME
GPIO36	A_IO36	Source switch	Through 1 kΩ to the source switch's resistor ladder; 0.1 µF to ground at the pin. Reads 3.3 V on RADIO, 0.81 V on BT, 0 V on AUX.	A32_SWITCH → LISTENER
3V3	A_3V3	The audio board's 3.3 V output	Feeds the 3.3 V bus (A_VDC)	A32_3V3
5V	5VDC	Power in	The 5 V bus	5 V bus → A32_5VIN
GND	GND	Ground	The DC ground	with the power cables

Not used: GPIO1, GPIO2, GPIO3, GPIO5, GPIO6 to GPIO11, GPIO12, GPIO13, GPIO14, GPIO15, GPIO16, GPIO23, GPIO32, GPIO34, GPIO39, and the enable pin (EN). GPIO6 to GPIO11 belong to the module's flash memory. GPIO34 and GPIO39 are not joined to GPIO36 or to each other on this board.

One clock, two names. A_IO0A and A_IO0D are both the audio board's GPIO0. Each converter gets its own 33 Ω series resistor, fitted at the audio board's pin; the two names only make each converter's wiring read clearly. Chapter 6 explains the audio bus.

GPIO36 is the selector signal only. On AUX the audio itself never reaches the audio board. The source switch, its ladder and the firmware's thresholds for the three levels are in chapter 9 (p. 52), section 9.3 (p. 55).

GPIO34, GPIO35, GPIO36 and GPIO39 are input-only on this chip. The volume knob (GPIO35) and the source switch (GPIO36) use two of them (Appendix A (p. 139)).

4.4 Main board support parts

These are the small parts fitted around the main board's pins, and why each is there.

Table 16. Main board support parts

Pin	Parts	Why
3V3	0.1 µF and 10 µF to ground	decoupling of the main board's 3.3 V
5V	100 µF and 0.1 µF to ground	decoupling of the 5 V input
GPIO1	10 kΩ pull-up to the main board's 3.3 V	the optocoupler only pulls the line low
GPIO4	10 kΩ pull-up to the main board's 3.3 V, 0.1 µF to ground	left from the old limit switch; the pin goes nowhere else
GPIO5	10 kΩ pull-up to the main board's 3.3 V, 0.1 µF to ground	the Hall switch's output only pulls low; the pull-up sets its high level at 3.3 V
GPIO38, GPIO39	10 kΩ pull-up to the main board's 3.3 V at the pin, then 220 Ω in series toward the sensor	the angle sensor's two-wire bus
GPIO47, GPIO48	none	the calibration receiver's bus has no external pull-ups

The shields of the `S3_LIMITS` and `S3_TUNER` cables are on `GND` at the main board. The GPIO4 and GPIO5 capacitors return to the `S3_LIMITS` cable's ground and shield contact.

4.5 Audio board support parts

These are the small parts fitted around the audio board's pins, and why each is there.

Table 17. Audio board support parts

Pin	Parts	Why
3V3	0.1 µF, 10 µF and 10 µF to ground	decoupling of the 3.3 V that feeds the 3.3 V bus
5V	0.1 µF and 100 µF to ground	decoupling of the 5 V input
GPIO0	two 33 Ω in series, at the pin, one per converter	one clock line for each converter
GPIO4	100 kΩ to ground	pull-down on the DAC's data input
GPIO36	1 kΩ in series from the ladder; 0.1 µF to ground on the pin side	filters the source switch's level
GPIO36 ladder	22 kΩ from the ladder node to ground; 68 kΩ in the BT leg	sets the three levels (chapter 9 (p. 52))

The 22 kΩ and 68 kΩ ladder resistors are on the audio board, not at the switch.

4.6 The back-panel USB-C ports

The back panel has two USB-C sockets, one for each board. Each passes all four USB wires straight through to its board's USB-C connector: the 5 V power wire (VBUS), D−, D+ and ground. Each socket's other three ground pins are joined together.

Table 18. The back-panel USB-C ports

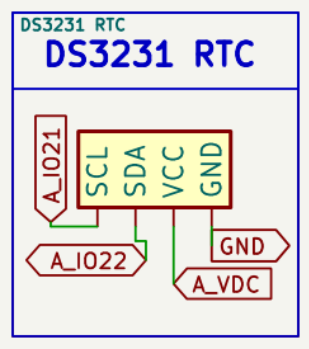
Back-panel port	Goes to
main (P2)	the main board's own USB port: the chip's USB-Serial/JTAG, USB ID `303A:1001`
audio (P1)	the audio board's USB-C: the CP210x USB-to-serial bridge, USB ID `10C4:EA60`

How the firmware uses the ports is in the Firmware Gospel.

> NOTICE VBUS reaches each board from the back panel while the 5 V bus also feeds the board's 5V pin. Espressif's guides treat these as mutually exclusive ways to power the board (chapter 2 (p. 7), section 2.5 (p. 9)); powering a board from both at once risks damaging it.

4.7 The real-time clock module

Figure 5. Schematic: the real-time clock module, as drawn on the main sheet



The real-time clock is a DS3231 module of the common ZS-042 style, read by the audio board. The same module carries an AT24C32 EEPROM at I2C address 0x57, and a CR2032 backup cell. Most modules of this kind carry such a cell, but not all, so a replacement module may not.

The table shows how the module is wired.

Table 19. The real-time clock module

Module connection	Goes to	Cable
VCC	the 3.3 V bus (`A_VDC`)	`A32_VDC` pin 1 → `RTC_3V3` pin 1
GND	DC ground	`A32_VDC` pin 2 → `RTC_3V3` pin 2
SDA	audio board GPIO22 (`A_IO22`)	`A32_RTC` → `RTC_I2C`
SCL	audio board GPIO21 (`A_IO21`)	`A32_RTC` → `RTC_I2C`

The pin numbers are the cable sheet's. The cable sheet uses the name `A32_VDC` for a second jack with the opposite pin order (chapter 5 (p. 21), section 5.5 (p. 26)).

The module's I2C socket (`RTC_I2C`) is a 4-pin JST with only two pins connected, SDA and SCL. The signal cable's shield ends on ground at the audio board only.

The DS3231's datasheet limits are not included in this book.

4.8 When something is wrong

Find what you see in the left column; the right column says what to check.

Table 20. When something is wrong

You see	Check
The main board's RGB LED is lit brightly	Nothing: it is normal. GPIO48 drives both that LED and the calibration receiver's clock.
The needle cannot find its index	The index switch on GPIO5: its cable (`S3_LIMITS` → `LIMIT`), its 5 V supply, and the 10 kΩ pull-up. GPIO4 on the same socket goes nowhere and is not the fault.
The angle sensor does not answer	The `S3_TUNER` cable, the main board's 3.3 V on it, and the two 220 Ω series resistors and 10 kΩ pull-ups at GPIO38 and GPIO39.
The calibration receiver does not answer	The `S3_RDA` signal cable and the `S3_3V3` power cable to the receiver. Its bus has no pull-up resistors; that is as built.
The two boards do not talk	The `S3_UART` cable: main board GPIO13 to audio board GPIO26, audio board GPIO27 to main board GPIO14, and its ground.
The clock forgets the time when the cord is pulled	The module's CR2032 backup cell.
A board is dead while the other runs	Its own 5 V cable from the bus (`S3_5VIN` or `A32_5VIN`), and its polarity at both ends.

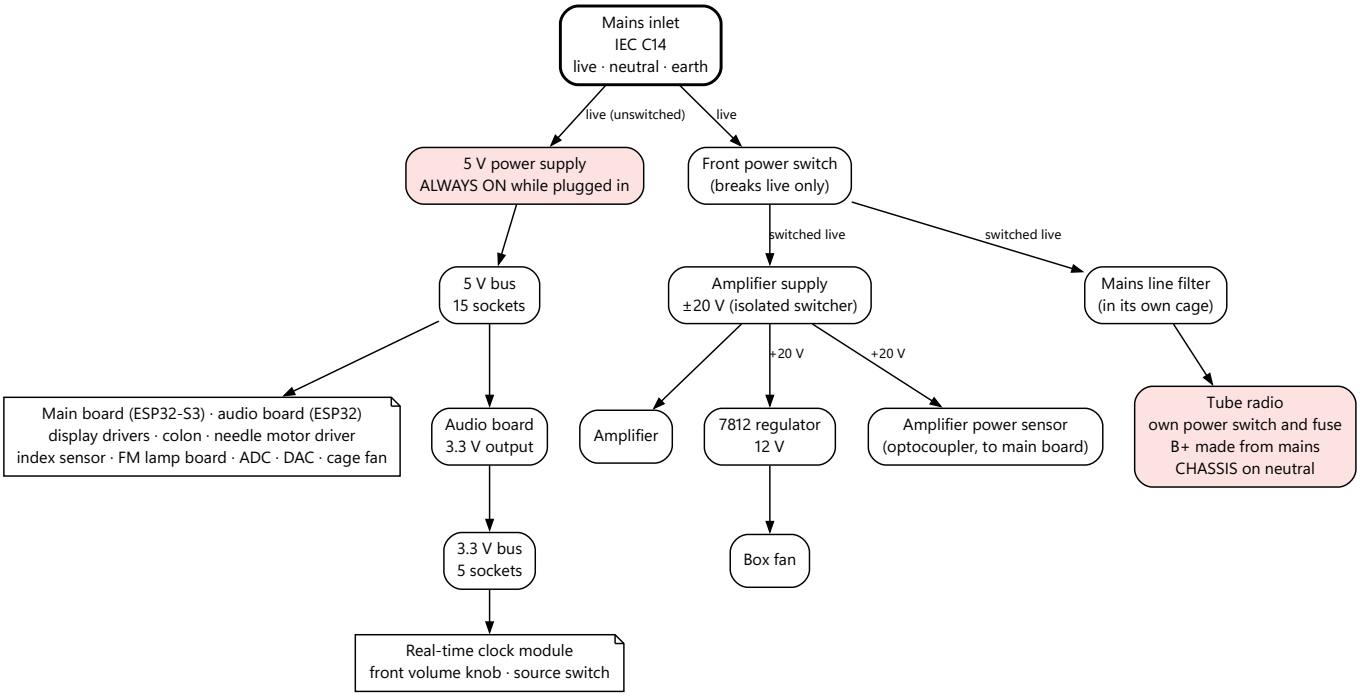
5. Power, earth and grounds

This chapter follows the power from the wall socket to every board. It also explains the four different "grounds" in the machine. One of them, the tube radio's chassis, sits on mains neutral, so read section 5.3 (p. 23) before you touch anything.

> ⚠ DANGER the front power switch does not remove mains from the cabinet. The 5 V power supply is always live while the cord is plugged in. The tube radio's metal chassis sits on mains neutral, and it is never joined to Earth. Unplug the cord before you open the cabinet. Chapter 2 lists every live point.

5.1 The power at a glance

Figure 6. The power tree: from the mains inlet to every board



Mains enters through one socket: a standard IEC C14 inlet, the same kind as a computer's. It is the inlet that came on the amplifier's aluminium back plate, at the back of the cabinet. From there the power splits in two:

- Before the front power switch: the 5 V power supply. It runs whenever the cord is plugged in. It feeds the 5 V bus, which powers both microcontroller boards and everything digital.
- After the front power switch: the amplifier's ±20 V power supply, and the tube radio (through the mains line filter). Switching the front switch off silences the radio and the amplifier, but not the clock or the computers.

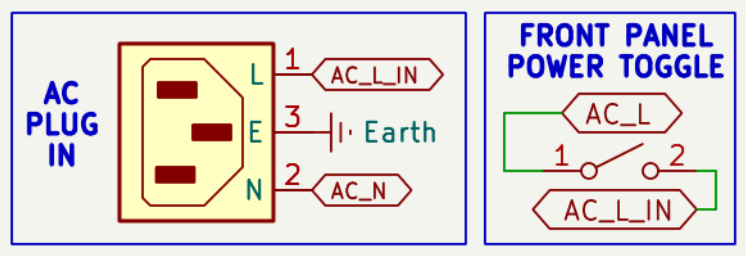
The table shows where each part takes its power from, and when it is on.

Table 21. The power at a glance

What	Where its power comes from	On when
5 V power supply	mains live, before the front switch	the cord is plugged in
Both microcontroller boards, display, needle motor, sensors, DAC, ADC, cage fan	the 5 V bus	the cord is plugged in
3.3 V parts (real-time clock module, front volume knob, source switch)	the audio board's own 3.3 V output	the cord is plugged in
Amplifier's ±20 V supply	mains live, after the front switch	the front switch is on
Amplifier, box fan, amplifier power sensor	the ±20 V supply	the front switch is on
Tube radio	mains live after the front switch, through the line filter	the front switch is on and the radio's own power switch is on

5.2 The mains side

Figure 7. Schematic: the mains inlet and the front power switch, as drawn on the main sheet



The table follows the mains wiring point by point, from the inlet to the tube radio.

Table 22. The mains side

Point	What it is
Mains inlet	IEC C14, three pins: live, neutral, earth.
Front power switch	A single-pole on/off toggle on the front panel. It breaks only the live wire, and only to the amplifier supply and the tube radio. The mains to it uses a connector type found nowhere else in the cabinet, so no other cable can be plugged into it by mistake.
Live, always on	Inlet live → 5 V power supply, and → the input side of the front switch.
Live, switched	Front switch output → amplifier supply, and → line filter → tube radio's live terminal.
Neutral	Inlet neutral → 5 V power supply, amplifier supply, and line filter → tube radio's neutral terminal.
Mains line filter	An external EMI filter between the switched mains and the tube radio. It took the place of two of the radio's original coils, which were removed by accident. It sits under its own small Faraday cage (section 5.4 (p. 25)).
Fuses	The tube radio has its own 0.75 A fuse; the mains side carries no other fuse (chapter 2 (p. 7), section 2.4 (p. 8)).

The tube radio has no isolation transformer. Its circuit common — the metal chassis and every part connected to it — is wired straight to mains neutral. It sits near 0 V only while the wall socket's live and neutral are the right way round, and nothing in the machine checks that. The radio also makes its high-voltage supply (B+) by rectifying the mains directly (chapter 13 (p. 98)), and that supply is present whenever the radio is on.

5.3 The four grounds

The machine has four ground names, kept different on purpose. They are not four separate grounds. Three of them, Earth, the DC ground and the amplifier supply return, are all joined. The fourth, the radio chassis, is joined to none of them. The table says what each one is joined to.

Table 23. The four grounds

Name	Also written in this book	What it is	Joined to
Earth	`Earth`	The mains protective earth, from the inlet's earth pin.	The amplifier supply's earth pin, all three Faraday cages, the antenna connector's metal shell, and the amplifier's aluminium back plate, whose earth star point carries the amplifier's ground (below). The DC ground and the amplifier supply return are on it too.
DC ground	`GND`	The common return of everything on the 5 V and 3.3 V side: both microcontroller boards, the display, the motor, the sensors, the ADC and DAC.	All its wires meet at one star point: the ground pins of the 5 V and 3.3 V buses, and from there reach the amplifier's ground (below), so it is on Earth at almost 0 Ω.
Amplifier supply return	`20V_GND`	The middle (0 V) output of the amplifier's ±20 V supply.	The amplifier's ground, through the supply's two middle black wires (section 5.6 (p. 27)), and so Earth. Also the amplifier power sensor's LED cathode, the box-fan regulator's return and the box fan's −.
Radio chassis	`CHASSIS`	The tube radio's circuit common. On mains neutral.	Mains neutral only. It is not joined to Earth, to the Faraday cages, to the DC ground, or to the amplifier supply return. The antenna's shield reaches it only through one safety-rated capacitor (section 5.4 (p. 25)).

How the three are joined. Earth, `GND` and `20V_GND` are all joined. The amplifier's ground sits on the earth star point of its aluminium back plate, through a washer on the Bass pot (the amplifier's bass control). This joint is part of the amplifier's original design; it is not on any drawing. The amplifier supply's ground wires, the two middle black ones (`20V_GND` on the drawings), go to that amplifier ground. The DC side's `GND`, both boards and the rest, reaches that same amplifier ground and so Earth, at almost 0 Ω. The amplifier supply itself is insulated from the plate by plastic screws and standoffs, and the amplifier's volume and treble controls do not touch the plate. `CHASSIS` is joined to none of them.

The amplifier power sensor. The sensor is an optocoupler. Its LED is lit from the amplifier supply (the `20V_GND` side) and its transistor is read by the main board (the `GND` side). Inside the sensor no wire joins the two sides; the "amplifier is on" signal crosses by light. The two returns are still joined outside it, as described above. The sensor itself is in chapter 9 (p. 52).

5.4 Earth, cages and the antenna

Figure 8. Schematic: the Faraday cage over the radio, as drawn on the main sheet

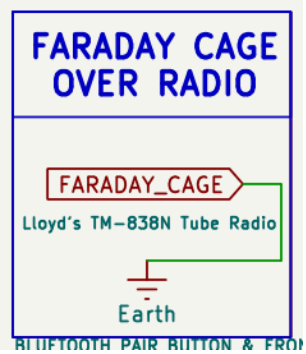
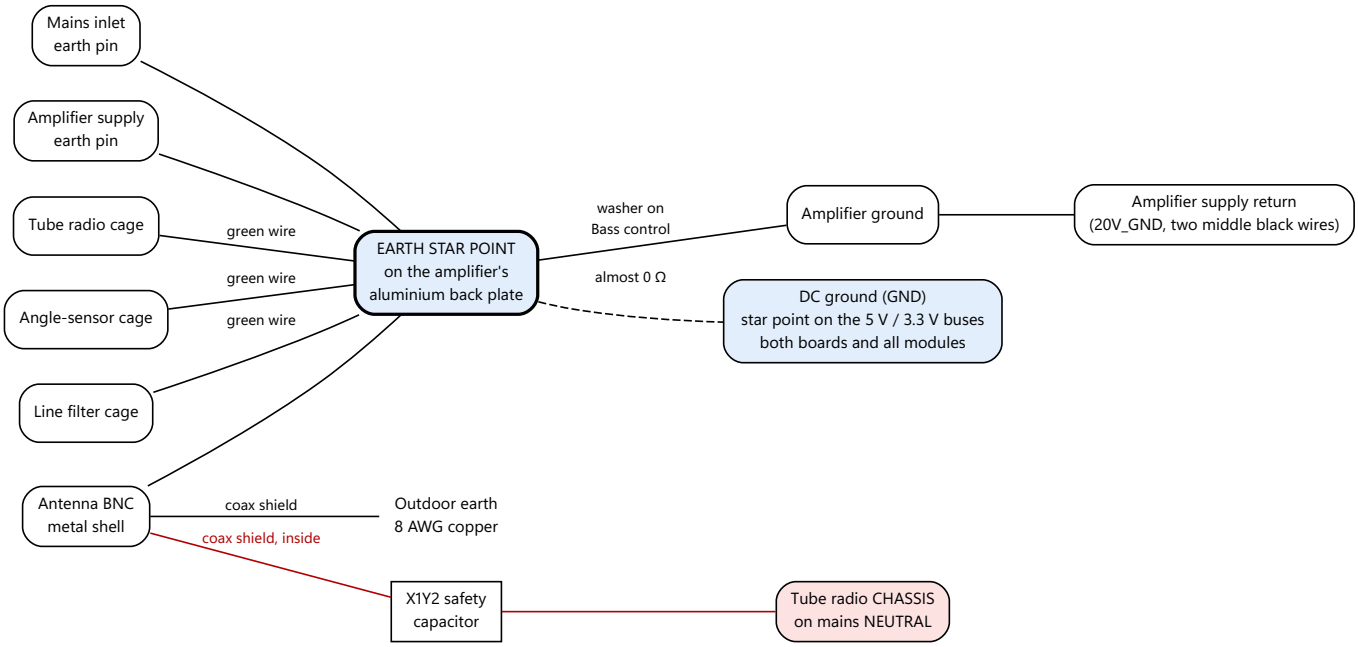


Figure 9. Where Earth goes, and the one path from the antenna to the radio's chassis



The three Faraday cages shield the noisiest or most sensitive parts:

Table 24. Earth, cages and the antenna

Cage	What it covers	How it is built
Tube radio cage	the whole tube radio, well clear of it	A PLA box lined with aluminium tape, covered with Kapton tape. A cage fan on the 5 V bus blows air into it.
Sensor cage	the magnetic angle sensor on the tuning shaft (AS5600)	Built the same way. The sensor is mounted on the cage's base, clear of the aluminium and copper tape.
Line filter cage	the mains line filter	Kapton tape over the circuit, then aluminium tape, then Kapton again.

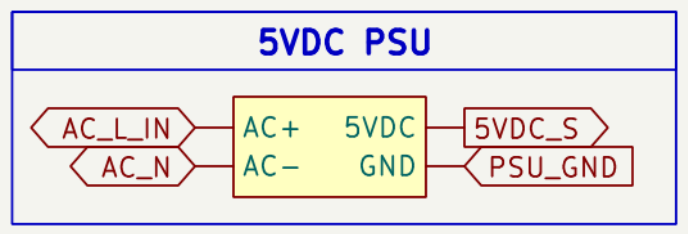
Each cage is earthed the same way: a stranded green wire, held against the aluminium by copper tape with conductive adhesive.

The antenna. The FM antenna's coax arrives at the antenna connector, a BNC on the back panel. The connector's shell, and so the coax shield, is on Earth. Outside, an 8 AWG stranded copper run goes from the antenna's grounding

block to a lug on the meter base (chapter 11 (p. 92)). Inside, the shield reaches the radio's chassis only through one capacitor: a safety-rated X1Y2 type inside the tube radio.

5.5 The 5 V and 3.3 V buses

Figure 10. Schematic: the 5 V supply, as drawn on the main sheet



All the low-voltage power is shared out from two star-point buses: rows of small 2-pin JST XH sockets, held in their own small PLA cases. Devices plug their power cables into these sockets. Each socket carries power on one pin and DC ground on the other. Every ground pin is the same wire: this is the "single star point" of the DC ground.

Table 25. The 5 V and 3.3 V buses

Bus	Positions	Fed from	What runs on it
5 V bus	15 sockets	the 5 V power supply	main board (ESP32-S3), audio board (ESP32), clock display colon, display driver board, needle motor driver, needle index sensor, FM panel-lamp board, ADC board, DAC board, cage fan
3.3 V bus	5 sockets, plus the two jacks named for the audio board (`A32_VDC`, `A32_3V3`) on the same wire	the audio board's own 3.3 V output	the same 3.3 V wire also feeds the real-time clock module, the front volume knob and the source switch

The 3.3 V bus is fed from the audio board's 3.3 V pin, which is decoupled there by one 0.1 µF and two 10 µF capacitors. The main board's own 3.3 V output does not go to the bus. It feeds only the two parts on its own cables: the FM calibration receiver (RDA5807M) and the angle sensor (AS5600).

Two jacks named **`A32_VDC`**. The cable sheet gives the name `A32_VDC` to two different jacks, and draws them with opposite pin orders:

Table 26. The 5 V and 3.3 V buses

Jack named **`A32_VDC`**	Pin 1	Pin 2	Its cable goes to
the real-time clock's supply	`A_VDC` (3.3 V)	`GND`	the clock module's `RTC_3V3`
the audio board's 3.3 V cable	`GND`	`A_3V3` (3.3 V)	the audio board's `A32_3V3`

The pin numbers are the cable sheet's and settle nothing physical.

NOTICE as drawn, the two jacks named A32_VDC carry 3.3 V and ground on opposite pins. The name alone does not tell which of the two jacks a cable belongs to; check before you plug one in.

NOTICE the 5 V cable to the clock display's colon has its polarity reversed at the COLON_IN side, compared with the other 5 V cables. Check polarity before you plug it in.

A JST XH socket is rated 3 A, for wire from 30 to 22 AWG.

5.6 The amplifier supply and the box fan

Figure 11. Schematic: the amplifier supply, as drawn on the main sheet

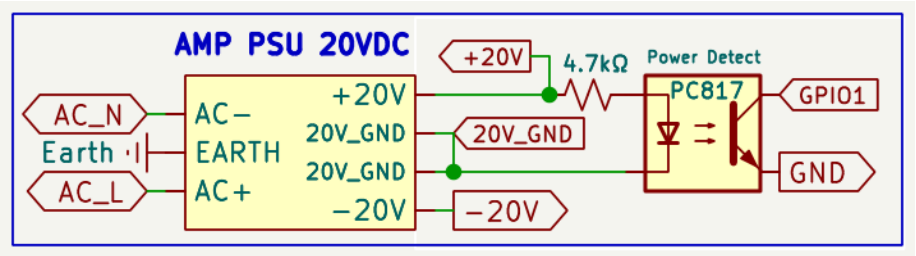
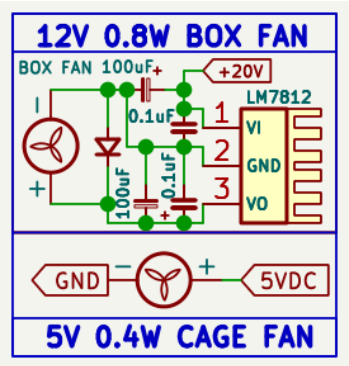


Figure 12. Schematic: the box fan, as drawn on the main sheet



The amplifier supply is the one that came with the amplifier: an isolated mains switching supply, measured at ±19.33 V with no load. It reaches the amplifier on four wires:

Table 27. The amplifier supply and the box fan

Wire colour	Carries
red	+20 V
black (the two middle wires)	0 V, the amplifier supply return
yellow	−20 V

These colours are not verified inside the supply. The two middle black wires go to the amplifier's ground (section 5.3 (p. 23)). The same supply also feeds the box fan and the amplifier power sensor, from its +20 V output.

The box fan sits at the top of the back panel and blows air out of the cabinet. Fresh air comes in through two small louvred PLA vents at the other

two ends of the back panel's inverted T. The fan runs on 12 V, made from the +20 V output by an LM7812 regulator. It therefore turns only while the front switch is on. The regulator's parts:

Table 28. The amplifier supply and the box fan

Part	Value
Regulator	LM7812, input from +20 V, output 12 V
Input capacitors	0.1 µF and 100 µF, to the supply return
Output capacitors	0.1 µF and 100 µF, to the supply return
Protection diode	one diode across the 12 V output

The fan harness has two 2-pin plugs: FAN_IN brings +20 V and its return, and FAN_OUT takes 12 V and its return to the fan.

The cage fan is a small 5 V fan screwed to the tube radio's Faraday cage, blowing air into it. It plugs into the 5 V bus, so it runs whenever the cord is plugged in.

5.7 When something is wrong

Find what you see in the left column; the right column says what to check.

Table 29. When something is wrong

You see	Check
Nothing at all works: no clock, no sound	The cord and the wall socket. The 5 V supply has no switch; if it is dead, everything digital is dead.
Clock and needle work, but no sound from radio or amplifier	The front power switch. Then the amplifier supply: +20 V on the red wire, −20 V on the yellow.
Radio silent but Bluetooth plays	The tube radio's own power switch, and its fuse (chapter 13 (p. 98)).
One board or module dead, the others fine	Its own power cable to the bus, and the polarity at both ends.
Box fan stopped while the radio plays	The FAN_IN and FAN_OUT plugs, then 12 V at the regulator's output.

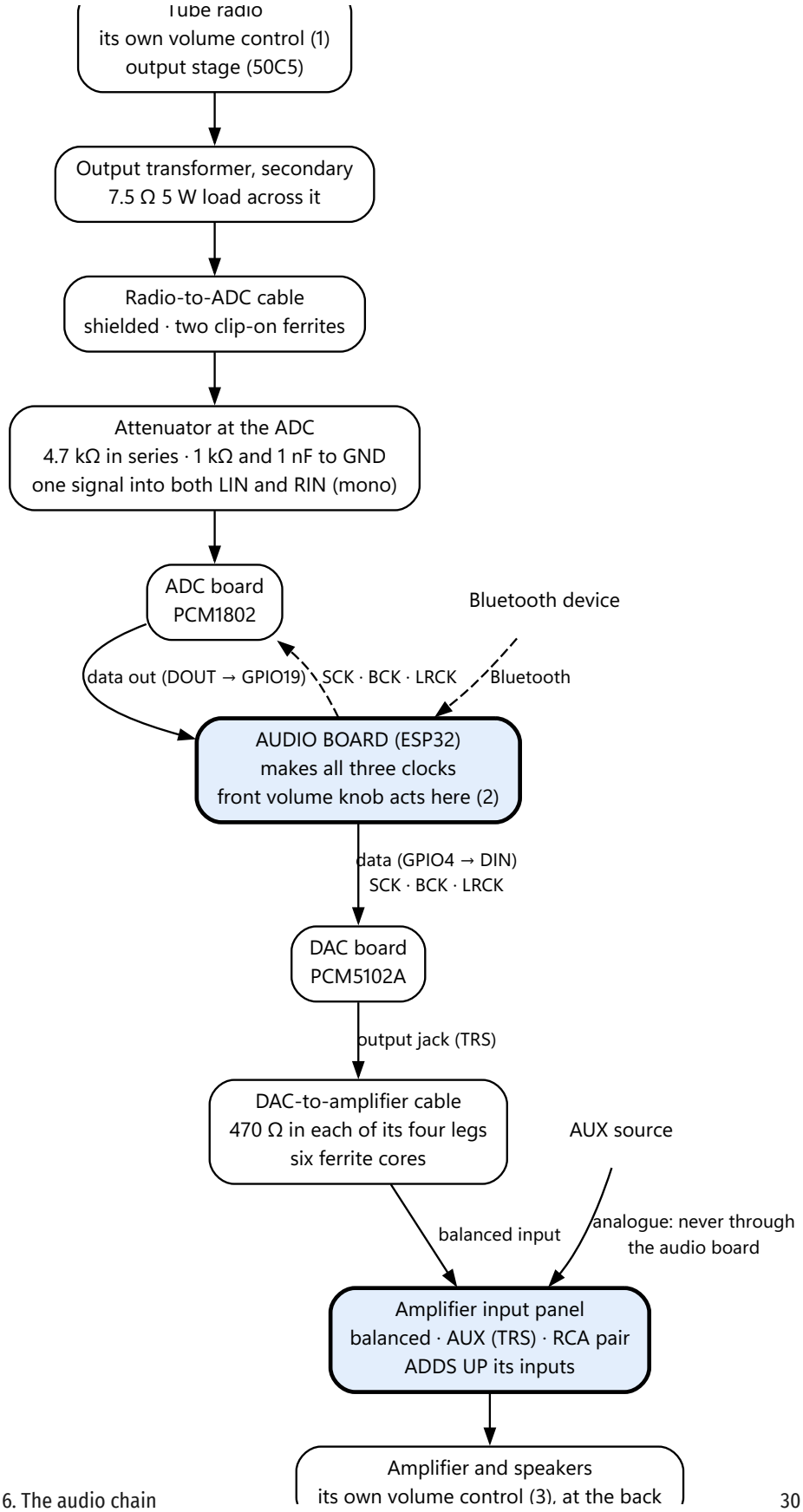
6. The audio chain

This chapter follows the sound from each source to the amplifier. The tube radio's audio is turned into digital audio by an ADC (analogue-to-digital converter) and passed through the audio board. A DAC (digital-to-analogue converter) then turns it back into an analogue signal. Bluetooth joins at the audio board. AUX, the analogue input you plug a cable into, goes straight to the amplifier. When the sound is missing, distorted or noisy, this chapter tells you which piece to look at.

What the audio board's firmware does with the sound (sample rate, volume, muting, source changes) is in the Firmware Gospel, chapter 10.

6.1 The path at a glance

Figure 13. The audio chain: from each source to the speakers, with the three volume controls numbered



The machine has three sources, and each takes its own road:

Table 30. The path at a glance

Source	What carries it	Converted by
RADIO	the tube radio's output transformer → the radio-to-ADC cable → the audio board	the ADC, then the DAC
BT	Bluetooth → the audio board	the DAC
AUX	an analogue run to the amplifier's input panel	nothing: it never reaches the audio board

Every source ends at the amplifier's input panel, which adds up whatever reaches it (section 6.7 (p. 37)). The amplifier and its speakers are in chapter 12 (p. 96).

6.2 The digital audio bus (I2S)

I2S is a simple way to send digital audio between chips over a few wires: a bit clock (BCK) that ticks once per data bit, a word clock (LRCK) that marks left and right samples, one data line per direction, and a master clock (SCK) that the converters run their insides from.

The audio board makes all three clocks. The ADC runs in slave mode: it takes its clocks from outside. The DAC is given its master clock. So neither converter makes its own timing. Both converters share the same bit clock and word clock, so they always run at the same rate.

The table gives each clock and data line, from the audio board's pin to each converter's pin:

Table 31. The digital audio bus (I2S)

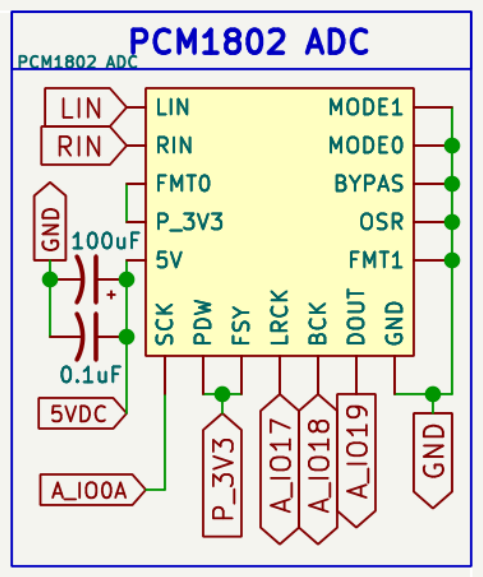
Signal	Audio board pin	ADC (PCM1802) pin	DAC (PCM5102A) pin	Notes
Master clock	GPIO0	SCKI, as `A_IO0A`	SCK, as `A_IO0D`	One pin, two branches, each through its own 33 Ω at the audio board's pin
Bit clock	GPIO18 (`A_IO18`)	BCK	BCK	shared
Word clock	GPIO17 (`A_IO17`)	LRCK	LRCK	shared
Data from the ADC	GPIO19 (`A_IO19`)	DOUT	—	
Data to the DAC	GPIO4 (`A_IO4`)	—	DIN	100 kΩ pull-down to ground

`A_IO0A` and `A_IO0D` are the same GPIO0. They carry two names only so that each converter's wiring reads clearly (chapter 4 (p. 12), section 4.3 (p. 17)). GPIO0 is also one of the chip's strapping pins, read at power-up (Appendix A (p. 139)).

All of these signals travel in one shielded harness from the audio board. The harness splits into an ADC branch and a DAC branch. The shield is on ground at the audio board end only. Chapter 10, section 10.15, gives it wire by wire.

6.3 The ADC (PCM1802)

Figure 14. Schematic: the ADC (PCM1802), as drawn on the main sheet



The ADC (analogue-to-digital converter) is a PCM1802 on a small module board. It turns the tube radio's audio into digital samples for the audio board.

Power. The module's 5V pin is on the 5 V bus, through its own cable (chapter 10 (p. 60), section 10.25 (p. 83)), with 100 µF and 0.1 µF to ground. The module has its own 3.3 V pin, named `P_3V3` on the drawing. Only one thing is wired to that pin: the FMT0 setting pin (below).

> Note the module's own 3.3 V pin is named `P_3V3`, and only FMT0 is wired to it. The drawing also uses the same name, `P_3V3`, for a separate label that carries two other setting pins (PDWN and FSYNC, below); that label appears on no other drawing and is not joined to the module's pin. The name alone does not tell you which node you are looking at, so confirm which `P_3V3` you are on before you probe or disconnect it.

Setting pins. The PCM1802 is configured by tying pins high or low. The table gives each pin, what it is tied to, and what that means by the datasheet (Appendix A (p. 139)).

Table 32. The ADC (PCM1802)

Pin	Tied to	Meaning
MODE0, MODE1	ground	Slave mode: the clocks come from the audio board
FMT1	ground	with FMT0 high: I2S, 24-bit
FMT0	the module's own 3.3 V pin (`P_3V3`), and nothing else	high (see FMT1)
OSR	ground	oversampling ×64
BYPAS	ground	high-pass filter on: DC is cut from the input
PDWN (power-down, active low) and FSYNC (frame sync)	joined together, as drawn, on the separate `P_3V3` label. That label goes nowhere else, and it is not joined to the module's own 3.3 V pin.	see below

By the chip's datasheet, PDWN, BYPAS, OSR, FMT0, FMT1, MODE0 and MODE1 have internal pull-downs (about 50 kΩ). In slave mode the PCM1802 sends data only while FSYNC is high, and it runs only while PDWN is high. These are facts about the chip. On the fitted module the converter runs, so PDWN is not held low.

Data and clocks reach the module on a 6-pin header: data out, bit clock, word clock and master clock; two of its pins are unused. Its pin order is in chapter 10 (p. 60), section 10.15 (p. 74).

6.4 The tube radio into the ADC

Figure 15. Schematic: the tube radio's output into the ADC, as drawn on the main sheet

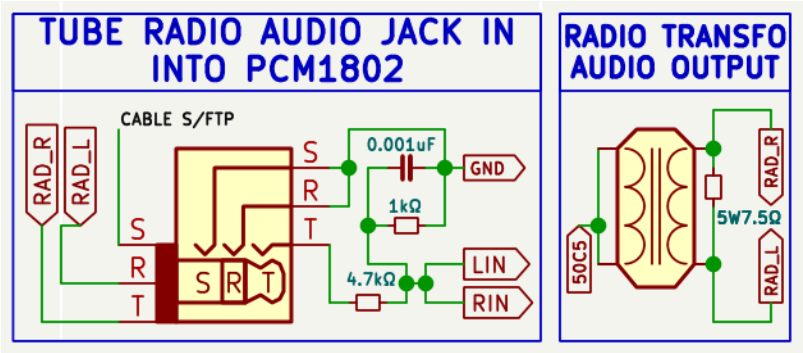
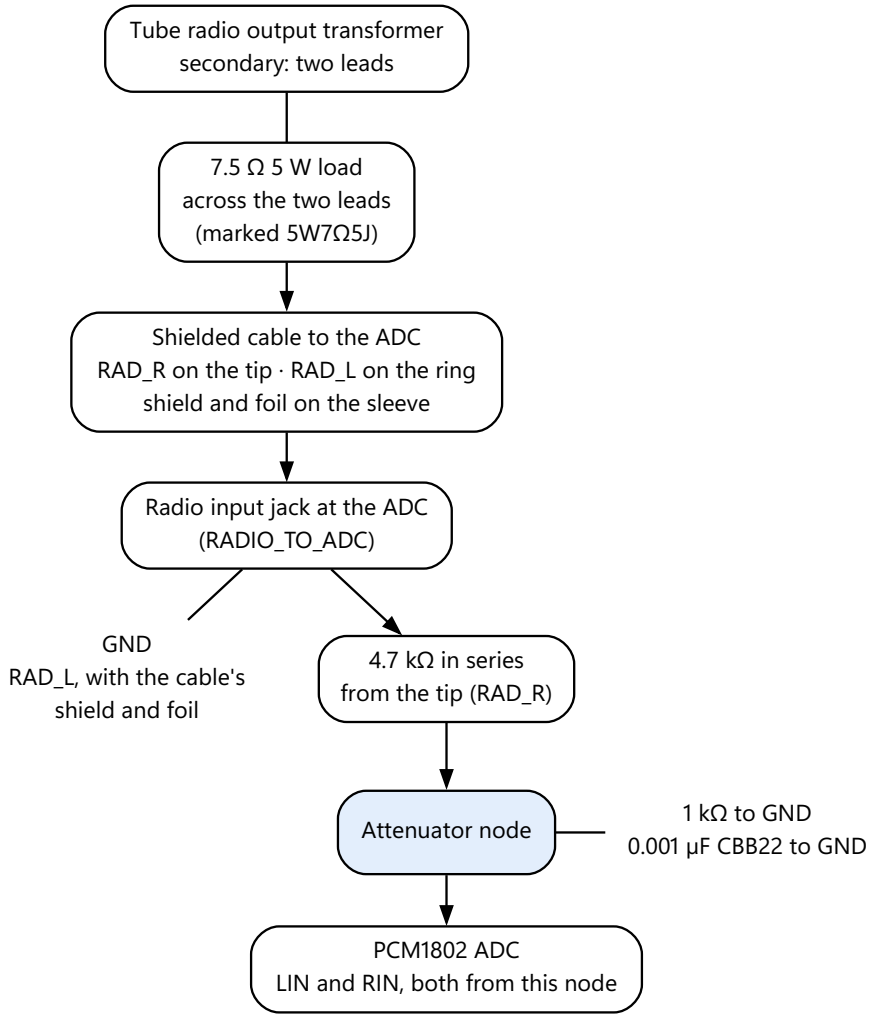


Figure 16. The radio's output, from the transformer to the ADC's inputs



The load. The tube radio's output transformer has a two-lead secondary. In the original set it fed the radio's speakers; here it feeds the ADC. A 7.5 Ω 5 W resistor sits across the two leads as the transformer's load (it is marked 5W7Ω5J). The transformer's primary belongs to the radio's output stage, a 50C5 tube (chapter 13 (p. 98)).

The cable. The two secondary leads, `RAD_L` and `RAD_R`, run in a shielded cable to the radio input jack at the ADC (J51, `RADIO_TO_ADC`), on a three-pole (TRS: tip, ring, sleeve) plug:

Table 33. The tube radio into the ADC

Plug contact	Carries
tip	`RAD_R`, one transformer lead
ring	`RAD_L`, the other transformer lead
sleeve	the cable's shield and foil

Tip and ring are interchangeable: which transformer lead goes to which does not matter. Two clip-on ferrite cores sit on this cable, one near the

transformer and one near the ADC. Chapter 10, section 10.22, has the cable and the cores.

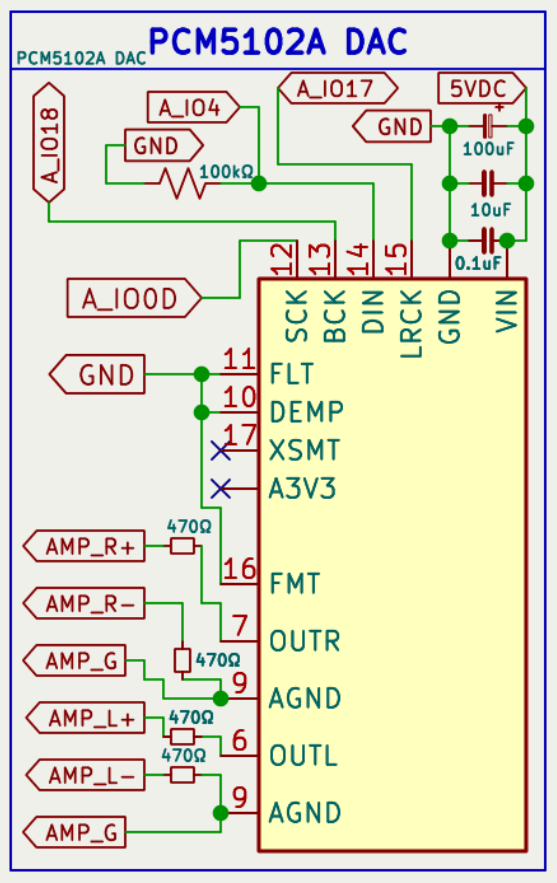
The jack's output side. Toward the ADC, the jack's sleeve and ring are on ground and its tip goes into a 4.7 kΩ resistor. So RAD_L goes to ground, with the cable's shield and foil, and RAD_R goes through the 4.7 kΩ.

The attenuator. The 4.7 kΩ leads to the attenuator node. From that node a 1 kΩ resistor and a 0.001 µF CBB22 capacitor (C59) go to ground. The node feeds both the ADC's left input (LIN) and its right input (RIN).

The radio feed is mono. One signal reaches both ADC inputs.

6.5 The DAC (PCM5102A)

Figure 17. Schematic: the DAC (PCM5102A), as drawn on the main sheet



The DAC (digital-to-analogue converter) is a PCM5102A on a small module board. It turns the audio board's digital audio, from the radio or from Bluetooth, back into an analogue signal.

Power. The module's VIN is on the 5 V bus. Its 5 V lead joins the audio harness's DAC branch (chapter 10 (p. 60), section 10.15 (p. 74)). At VIN there are three capacitors to ground, all fitted: 100 µF, 10 µF ceramic and 0.1 µF.

Data input. The DAC's DIN is the audio board's GPIO4, with a 100 kΩ pull-down to ground (chapter 4 (p. 12), section 4.5 (p. 19)).

Setting pins. Like the ADC, the DAC is configured by tying pins high or low:

Table 34. The DAC (PCM5102A)

Pin	Tied to	Meaning
FLT	ground	normal-latency filter
DEMP	ground	de-emphasis off
FMT	ground	I2S format
XSMT (soft mute)	pulled up to 3.3 V on the module itself	un-muted. Nothing outside the module reaches it.

The module's own 3.3 V pin (A3V3) is not connected to anything. The audio board's GPIO16 is unused: no wire runs from it to the DAC.

The DAC mutes itself on silence. By its datasheet, after 1024 word-clock periods of zero data on both channels (21 ms at 48 kHz) the PCM5102A puts its output in full analogue mute (Appendix A (p. 139)).

Output. The module has a three-pole (TRS: tip, ring, sleeve) output jack:

Table 35. The DAC (PCM5102A)

Jack contact	Carries
tip	left output (OUTL)
ring	right output (OUTR)
sleeve	the DAC's analogue ground (AGND)

6.6 From the DAC to the amplifier

A plug in the DAC's output jack feeds a short cable to the balanced input on the amplifier's input panel. A balanced input takes each channel on two wires, a plus and a minus, plus a ground. The cable puts a 470 Ω resistor in each of its four signal legs. The two on the plus legs are at the DAC end. Each amplifier input is fed like this:

Table 36. From the DAC to the amplifier

Amplifier input	Comes from
`AMP_L+`	the plug's tip (left output), through 470 Ω
`AMP_R+`	the plug's ring (right output), through 470 Ω
`AMP_G`	the plug's sleeve (DAC analogue ground), directly
`AMP_L−`	the plug's sleeve, through its own 470 Ω
`AMP_R−`	the plug's sleeve, through its own 470 Ω

So each minus leg is the DAC's ground seen through 470 Ω. The cable is under 12 inches long and carries six ferrite cores, three at each end. Each end has one HDMI-type sleeve, one small toroid and one large toroid, and the cable passes once through each. Chapter 10, section 10.21 (p. 80), gives the cable and the cores; chapter 12 (p. 96) gives what was measured at the amplifier's input.

What was observed. A frying noise was heard at the amplifier. It went away when this cable was unplugged at the amplifier, and almost went away when it was unplugged at the DAC.

Separately, once the clip-on ferrites were on the radio-to-ADC cable, no pops were audible. That was before the 470 Ω resistors on the plus legs and this cable's HDMI-type ferrite sleeves were added.

6.7 The AUX path and the amplifier's input panel

AUX is analogue and bypasses the audio board entirely. It never goes through the ADC or the DAC. The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn. On AUX the audio board sends silence to the DAC. Nothing the audio board does can change the AUX loudness or mute it.

The front source switch only tells the audio board which source you chose (chapter 9 (p. 52), section 9.3 (p. 55)). It does not switch any audio.

The input panel adds up its inputs; it does not choose between them. It has three inputs:

Table 37. The AUX path and the amplifier's input panel

Input	What reaches it	With nothing plugged in
Balanced (L+, L–, ground, R–, R+)	the DAC, by the cable of section 6.6 (p. 36): RADIO and BT	not grounded
AUX (TRS jack)	see below	grounded
RCA pair	—	grounded

Because the run from the AUX jack is not drawn, this book does not say which of the panel's inputs it lands on.

If two inputs carry sound at the same time, you hear both. On RADIO or BT, anything playing into AUX is still heard.

6.8 The three volume controls

Three volume controls sit in the chain, one at each stage. Their numbers match the figure in section 6.1 (p. 29):

Table 38. The three volume controls

No.	Control	Where	Acts on
1	The tube radio's own volume control	on the tube radio	RADIO only, before the ADC. In normal use it sits at about one eighth of its travel; above that, the sound is too loud further down the chain.
2	The front volume knob	front panel; read by the audio board (chapter 9 (p. 52), section 9.4 (p. 58))	RADIO and BT, inside the audio board. Not AUX.
3	The amplifier's own volume control	on the amplifier's board, reached only from the back	everything. It does not reach silence at its minimum.

The front knob is not the amplifier's volume control: it is read digitally by the audio board and does nothing to the analogue audio. The amplifier's own control is a B10k dual-gang potentiometer (chapter 12 (p. 96)).

6.9 When something is wrong

Table 39. When something is wrong

You see	Check
No radio sound, but Bluetooth plays	The radio side only. The tube radio itself: its power switch and its own volume control (chapter 13 (p. 98)). The radio-to-ADC cable and its plug in the radio input jack. The ADC branch of the audio harness and the ADC's 5 V cable (chapter 10 (p. 60), sections 10.15 (p. 74) and 10.25). By the chip's datasheet, PDWN and FSYNC must both be high for the ADC to send data (section 6.3 (p. 32)).
No sound from RADIO or BT, but AUX plays	The DAC side. The DAC branch of the audio harness and its 5 V lead (chapter 10 (p. 60), section 10.15 (p. 74)), the DAC-to-amplifier cable, and the front volume knob.
No sound from any source	The amplifier: the front power switch and the amplifier supply (chapter 5 (p. 21)), and the amplifier's own volume control.
Two sources heard at once	Normal: the input panel adds its inputs. Unplug whatever is playing into AUX.
Radio sound too loud further down the chain	The tube radio's own volume control: keep it near one eighth of its travel.
A frying noise at the amplifier	The DAC-to-amplifier cable: its four 470 Ω resistors and its ferrites. Unplugging it at the amplifier tells you whether it is the path.
The amplifier is never fully silent	Normal: its volume control does not reach silence at its minimum.

7. The clock display

The clock display shows the time on the front panel: four seven-segment digits and a colon, made for this front panel from single LEDs. The display is called the Leditron. The main board drives it through a small driver board. This chapter gives the digit numbering, which LED is which segment, how each digit and segment is switched, and the colon's own simple circuit.

What the main board shows on the display, and how fast it switches the digits, is in the Firmware Gospel, chapter 4 (p. 12).

7.1 The display at a glance

Figure 18. Schematic: the clock display, as drawn on the main sheet

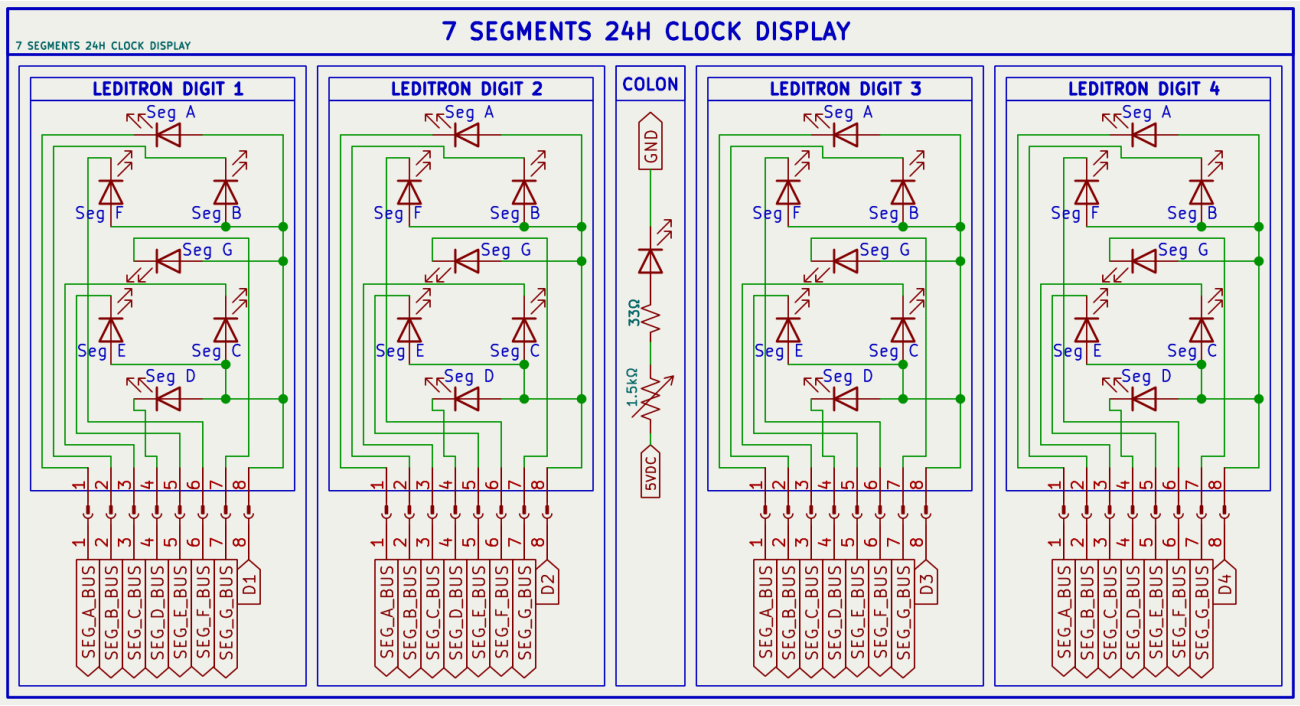
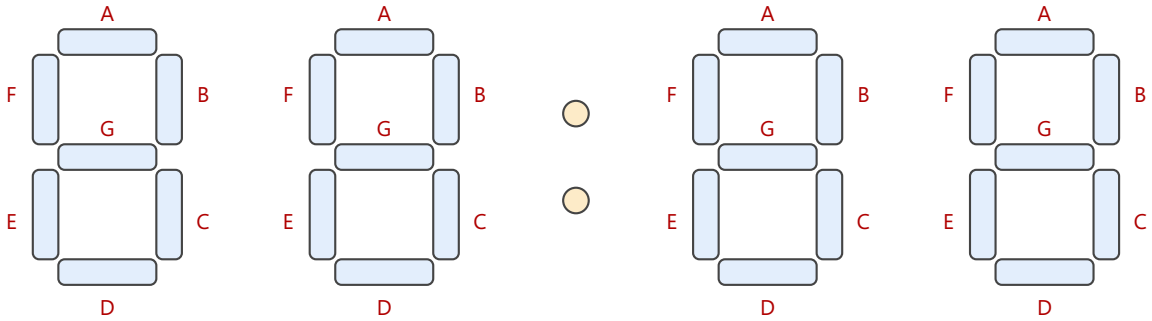


Figure 19. The clock display from the front: digit numbers, the pin that selects each digit, and the segment letters



Seen from the front. Each segment is one LED; red letters name the segments.

Digit numbering. Digit 1 is the rightmost digit, the units of minutes. Digits 2, 3 and 4 run to its left. The main board selects them with GPIO42, GPIO2, GPIO40 and GPIO41, in that order. So the select pins are not in number order. This order is exact: it is the order the firmware drives, and the order the drawings show (section 7.4 (p. 43)).

What it is made of. Each segment is one LED: 28 segment LEDs in all, seven per digit. In each digit the seven LEDs share one common anode (their plus

side) and each has its own cathode. The colon is one more LED with its own supply (section 7.6 (p. 45)).

The display is built from four parts:

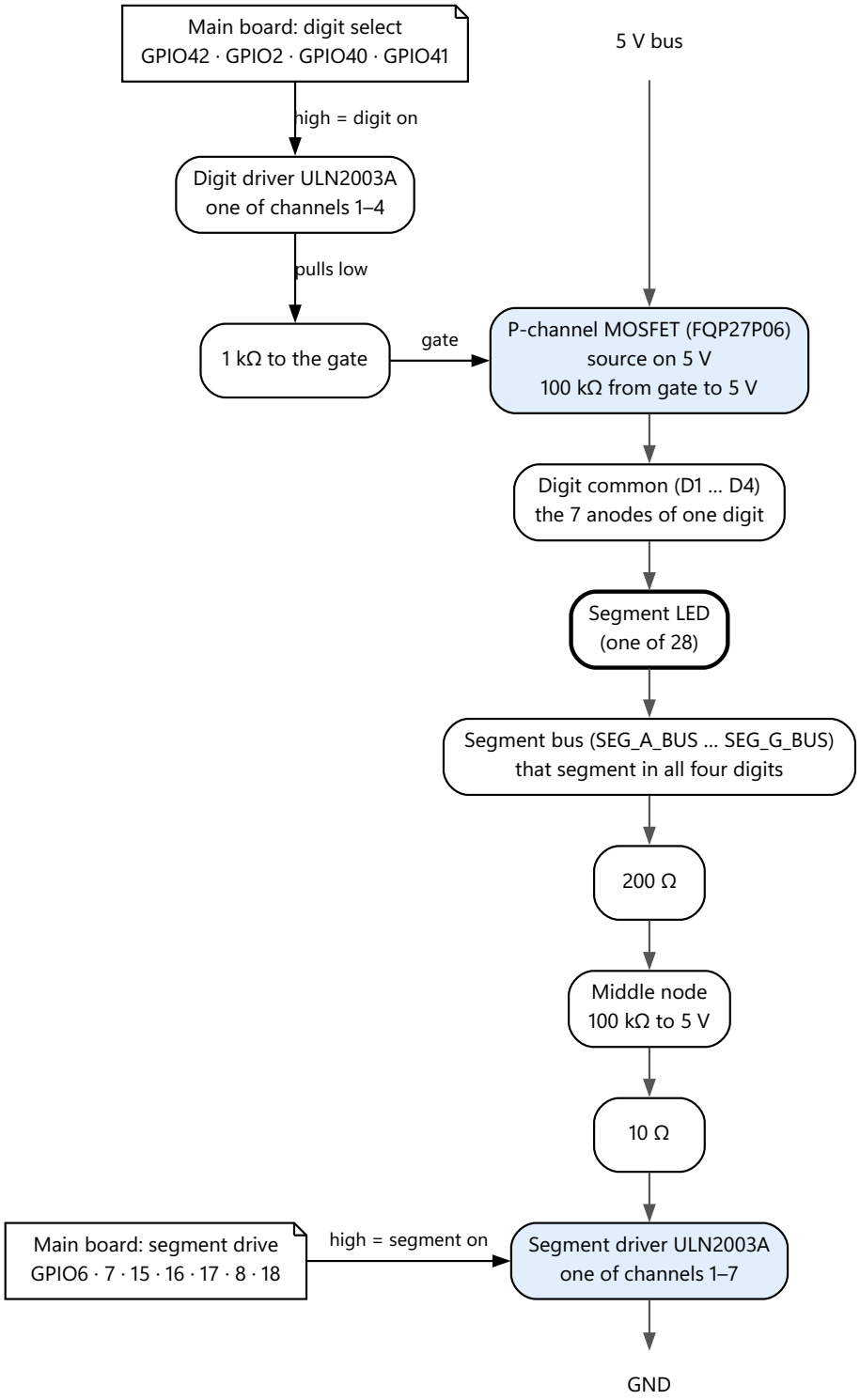
Table 40. The display at a glance

Part	What it does
Four digits	Seven segment LEDs each, on an 8-pin header: pins 1 to 7 are the cathodes of segments A to G, pin 8 is the common anode.
Display bus board	It takes the seven segment lines and the four digit commons from the driver board, and shares them out on four 8-pin digit outputs. It also carries the colon's supply. It sits in its own printed case.
Display driver board	A perfboard that switches the digits and the segments: one ULN2003A for the four digit selects, four P-channel MOSFETs, and one ULN2003A for the seven segments.
Colon	One LED with a 33 Ω resistor and a variable resistor, fed straight from the 5 V bus.

A ULN2003A is a chip of seven switches to ground (Darlington transistor pairs). When an input is driven high, its output pulls down toward ground. It accepts the main board's 3.3 V logic directly.

7.2 How it is driven

Figure 20. One digit channel and one segment channel



The four digits share the seven segment lines: segment A of every digit is on one wire, segment B on another, and so on. Each digit has its own common,

switched to 5 V by its own MOSFET. A segment LED lights only when its digit's common is on 5 V and its segment line is pulled low.

So, to show four different numbers, the main board lights one digit at a time and changes the segment lines between digits, fast enough that all four look lit. In short:

Table 41. How it is driven

To light	The main board drives high
a digit	that digit's select pin (section 7.4 (p. 43))
a segment of that digit	that segment's drive pin (section 7.5 (p. 44))

7.3 The segment map

This table gives, for each segment: the main board pin that drives it, its pin on every digit's 8-pin header, and which LED is that segment in each digit.

Table 42. The segment map

Segment	Drive pin	Header pin on each digit	LED in digit 1	LED in digit 2	LED in digit 3	LED in digit 4
A	GPIO6	1	D1	D24	D17	D10
B	GPIO7	2	D3	D27	D20	D13
C	GPIO15	3	D5	D28	D21	D14
D	GPIO16	4	D6	D26	D19	D12
E	GPIO17	5	D4	D23	D16	D9
F	GPIO8	6	D2	D22	D15	D8
G	GPIO18	7	D7	D25	D18	D11
common anode	—	8	all seven	all seven	all seven	all seven

The LED references (D1 to D28) are only there to find a LED on a drawing. Do not confuse them with the four digit commons, whose signal names are `D1` to `D4` (section 7.4 (p. 43)).

Each digit's header plugs into its own 8-pin socket, pin for pin: pin 1 of all four sockets is the segment A line, pin 2 the segment B line, and so on to pin 7; pin 8 of each socket is that digit's own common. The display bus board's four 8-pin digit outputs carry the same signals on the same pin numbers (chapter 10 (p. 60), section 10.34 (p. 90)).

7.4 The digit drivers

Figure 21. Schematic: the digit driver, as drawn on the main sheet

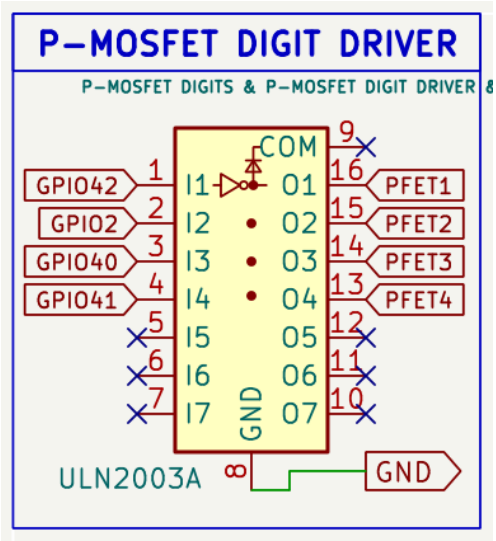
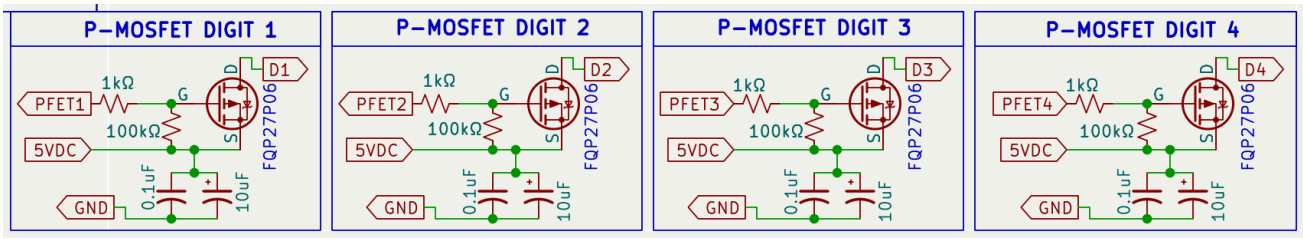


Figure 22. Schematic: the four digit switches, as drawn on the main sheet



Each digit's common anode is switched to 5 V by a P-channel MOSFET (FQP27P06). One ULN2003A, the digit driver, sits between the main board's select pins and the four MOSFET gates. Each digit has its own channel. The table is the chain as drawn:

Table 43. The digit drivers

Digit	Select pin	Digit driver input	Drives	Digit common
1 (rightmost)	GPIO42	1	the first MOSFET	D1
2	GPIO2	2	the second MOSFET	D2
3	GPIO40	3	the third MOSFET	D3
4 (leftmost)	GPIO41	4	the fourth MOSFET	D4

One channel, step by step:

1. The select pin drives the digit driver's input.
2. The driver's output (PFET1 to PFET4) goes through a 1 kΩ resistor to the MOSFET's gate.
3. A 100 kΩ resistor pulls the gate up to 5 V, the MOSFET's source. With the select pin low, this holds the MOSFET off.
4. With the select pin high, the driver pulls the gate low, the MOSFET turns on, and its drain puts 5 V on the digit's common.

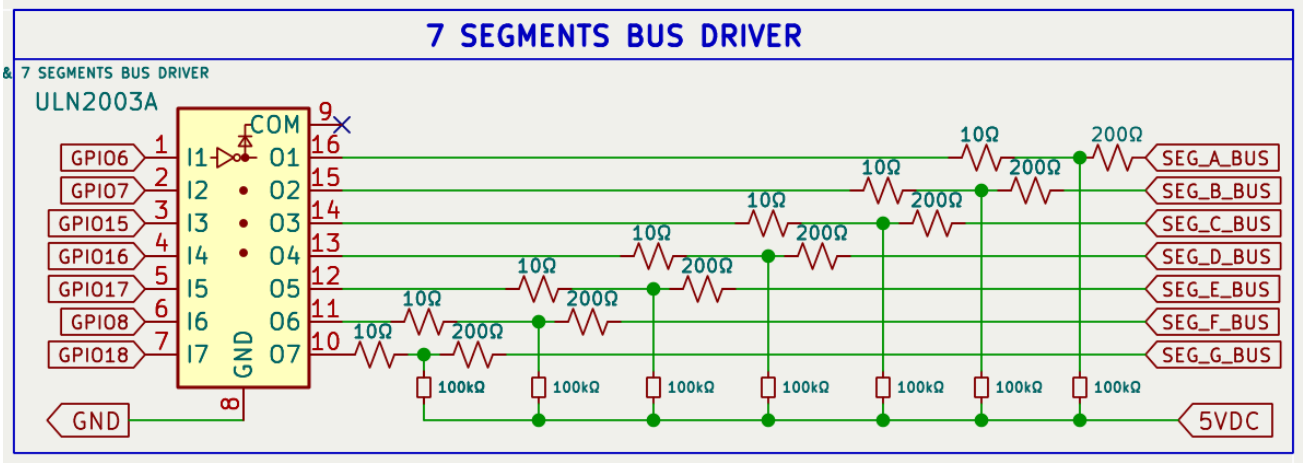
Per digit, a 0.1 µF and a 10 µF capacitor go from the 5 V at the MOSFET's source to ground.

The digit driver's ground pin is on the DC ground. Its COM pin (the common of its built-in protection diodes) is not connected, and its channels 5 to 7 are unused.

The digit order itself is certain (section 7.1 (p. 39)). The drawing does not settle where along the physical chain, from select pin to digit, the wiring sets that order. It is believed to be at the digit driver board.

7.5 The segment drivers

Figure 23. Schematic: the segment drivers, as drawn on the main sheet



The second ULN2003A (the segment driver) pulls each segment line low. Each segment has its own channel:

Table 44. The segment drivers

Segment	Drive pin	Segment driver input	Segment line
A	GPIO6	1	SEG_A_BUS
B	GPIO7	2	SEG_B_BUS
C	GPIO15	3	SEG_C_BUS
D	GPIO16	4	SEG_D_BUS
E	GPIO17	5	SEG_E_BUS
F	GPIO8	6	SEG_F_BUS
G	GPIO18	7	SEG_G_BUS

Each channel runs: driver output → 10 Ω → a middle node → 200 Ω → the segment line. The middle node has its own 100 kΩ to 5 V, which holds the line high (segment off) while the driver is off. The seven channels are fully separate: seven lines, one 100 kΩ each.

The segment driver's ground pin is on the DC ground. Its COM pin is not connected.

7.6 The colon

The colon is not switched by either board. It is lit whenever the 5 V bus is on.

Its circuit, in order: 5 V → 33 Ω → the colon LED → a variable resistor → ground. The variable resistor sits at the colon module. It is glued at 1.5 kΩ.

The colon has its own 2-pin power cable from the 5 V bus, `COLON_IN`. It is the one 5 V cable whose polarity is reversed at the `COLON_IN` side (chapter 5 (p. 21), section 5.5 (p. 26)).

7.7 Boards and cables

The display hangs on seven cables. Chapter 10 draws each one wire by wire.

Table 45. Boards and cables

Cable	From → to	Carries	Cable page
`S3_DISPLAY` → `DISPLAY_IN`	main board → driver board	the seven segment drive pins	section 10.5 (p. 65)
`S3_DIGIT` → `DIGIT_IN`	main board → driver board	the four digit select pins	section 10.8 (p. 68)
5 V bus → `DRIVERS_5VIN`	5 V bus → driver board	the driver board's 5 V and ground	section 10.26 (p. 84)
`DIGIT_OUT` → `DRIVER_IN`	driver board → display bus board	the four digit commons `D1` to `D4` one shielded cable over two plugs	section 10.27 (p. 85)
`DISPLAY_OUT` → `LEDITRON_IN`	driver board → display bus board	the seven segment lines	section 10.28 (p. 86)
5 V bus → `COLON_IN`	5 V bus → display bus board	the colon's 5 V and ground, reversed at **`COLON_IN`**	section 10.23 (p. 82)
colon output	display bus board → colon	the two colon LED leads	section 10.34 (p. 90)

7.8 When something is wrong

Table 46. When something is wrong

You see	Check
One segment dark in all four digits	That segment's channel: its drive pin, its segment driver channel, the 10 Ω and 200 Ω, its 100 kΩ, and its wire in the segment-line cable (`DISPLAY_OUT` → `LEDITRON_IN`).
One segment dark in one digit only	That segment's LED (section 7.3 (p. 42)), and its pin on that digit's 8-pin header.
One whole digit dark	That digit's select chain: its select pin, its digit driver channel, the 1 kΩ, the MOSFET, its wire in the digit-rail cable, and pin 8 of the digit's header.
All digits dark, colon lit	The driver board's 5 V cable (`DRIVERS_5VIN`), then the main board itself (chapter 4 (p. 12)). The digits also go dark while the main board takes a firmware update (Firmware Gospel).
All digits and the colon dark	The 5 V bus and the 5 V supply (chapter 5 (p. 21)).
Colon dark, digits fine	The colon's cable, and its polarity at `COLON_IN` the 33 Ω, the variable resistor and the colon LED.
Digits in the wrong places after a rebuild	The digit order is GPIO42, GPIO2, GPIO40, GPIO41 for digits 1 to 4, rightmost first. Check each select pin's chain against section 7.4 (p. 43).

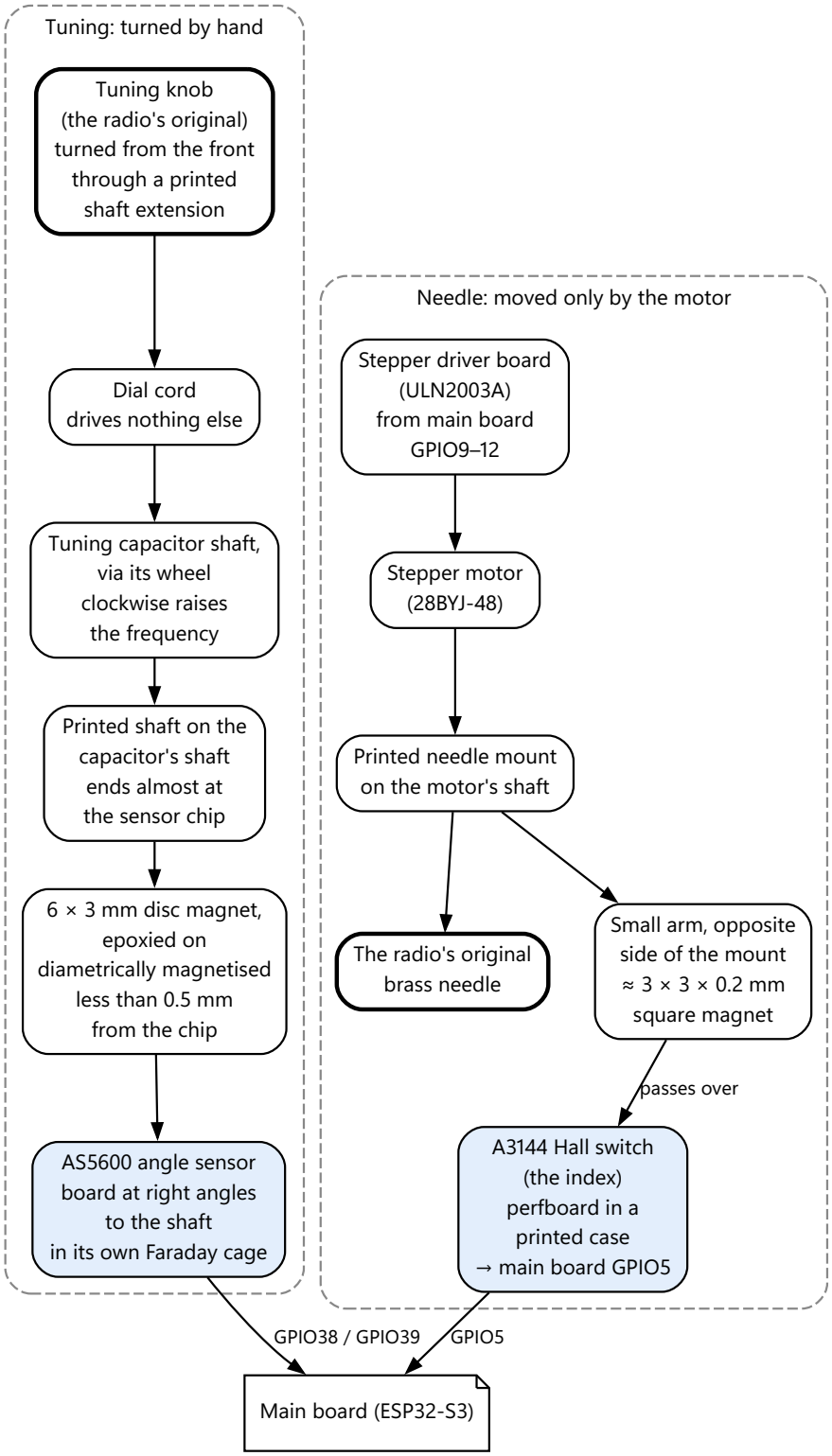
8. Dial, needle and tuning sensors

The tuning knob and the dial needle are not joined. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor as it always did. A magnetic angle sensor reads that capacitor's shaft, and a small stepper motor moves the needle to match. A magnetic index switch gives the needle its one fixed reference point. A small FM receiver chip serves as the instrument that calibrates the dial. This chapter describes that hardware: the parts, how they are mounted, and how they are wired.

What the main board does with them (homing, following the knob, the calibration) is in the Firmware Gospel: chapter 5 (p. 21) for the needle, chapter 6 (p. 28) for tuning and calibration.

8.1 The mechanics

Figure 24. The two halves of the dial: the tuning path, turned by hand, and the needle path, moved only by the motor



Tuning. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob. The original knob drives the dial cord, and the dial cord turns the tuning capacitor's shaft through its wheel. The cord drives nothing else. Turning the knob clockwise raises the frequency. The printed shaft extension is for tuning only.

The angle sensor's magnet. Another printed shaft, the angle sensor's own, extends the tuning capacitor's shaft almost to the angle sensor's chip. A disc magnet is epoxied to its end. The sensor's board stands at right angles to the shaft, with the shaft's axis through the centre of the chip (section 8.4 (p. 50)).

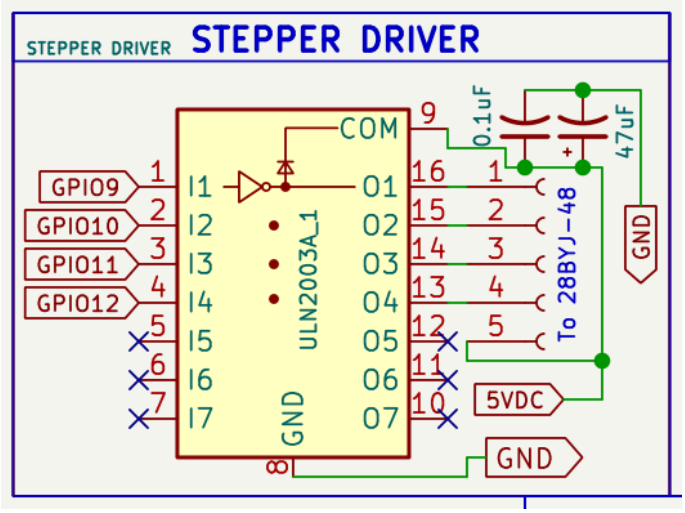
The needle. The tube radio's original brass needle sits on a printed mount on the stepper motor's shaft. The motor alone moves it; it has never been on the dial cord.

The index. A small arm on the opposite side of the same mount carries a tiny magnet. As the needle moves, the magnet passes over the index Hall switch (section 8.3 (p. 49)).

The angle sensor sits under its own earthed Faraday cage (chapter 5 (p. 21), section 5.4 (p. 25)).

8.2 The stepper motor and its driver

Figure 25. Schematic: the stepper motor driver, as drawn on the main sheet



The needle motor is a 28BYJ-48 stepper motor. It is driven by a ULN2003A on a driver board; the motor's own lead plugs into the 5-pin socket that comes on the common ULN2003A boards. A ULN2003A is a chip of seven switches to ground (chapter 7 (p. 38), section 7.1 (p. 39)). The four coil lines run like this:

Table 47. The stepper motor and its driver

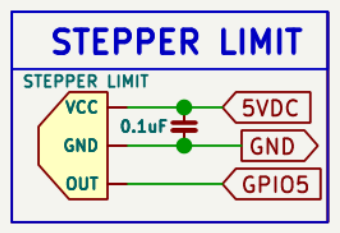
Main board pin	Driver input	Driver output	Motor socket pin	Coil, as named on the drawing
GPIO9	IN1	OUT1	1	coil 4
GPIO10	IN2	OUT2	2	coil 3
GPIO11	IN3	OUT3	3	coil 2
GPIO12	IN4	OUT4	4	coil 1
—	COM	—	5	motor common, on 5 V

Power. The driver board takes 5 V and ground from the 5 V bus on its own 2-pin cable (`STEPPER_5VIN`). The ULN2003A's COM pin, the common of its built-in protection diodes, and the motor's common (socket pin 5) are both on 5 V. A 0.1 µF and a 47 µF capacitor go from that 5 V to ground. The ULN2003A's ground pin is on the DC ground. Its channels 5 to 7 are unused.

The step sequence, speeds and microstepping are the firmware's (Firmware Gospel, chapter 5).

8.3 The index sensor (A3144) and its magnet

Figure 26. Schematic: the index sensor, as drawn on the main sheet



The index is the needle's only position reference. It is an A3144 Hall-effect switch: a sensor whose output switches on when a magnet comes close enough.

Table 48. The index sensor (A3144) and its magnet

What	Detail
Where	On a small perfboard in its own printed case, under the path of the needle mount's small arm.
Magnet	A square neodymium magnet, about 3 × 3 mm and 0.2 mm thick, on the arm.
Supply	5 V from the 5 V bus, with a 0.1 µF capacitor across the sensor's supply.
Output	Main board GPIO5. A 10 kΩ pull-up to the main board's 3.3 V and a 0.1 µF capacitor to ground sit at the main board's pin.

How it switches. The A3144′s output is open-collector: it can only pull the line low. By its datasheet, it pulls low when the magnetic field rises above its operate point, and lets go when the field falls below its release point. The pull-up sets the high level at 3.3 V, the main board's own level, even though the sensor runs on 5 V.

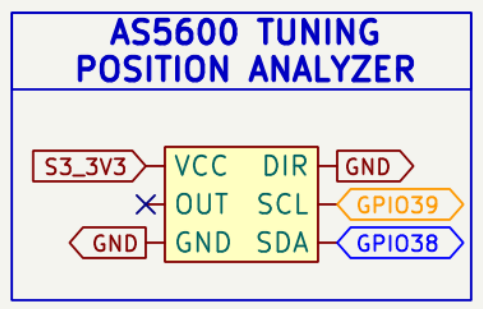
Pins (datasheet, viewed from the branded side): 1 supply, 2 ground, 3 output. On the sensor's cable (`LIMIT`): wire 1 is 5 V to the supply pin, wire 2 is ground, wire 3 is the output to GPIO5.

Replacing it. The A3144 is discontinued; its maker names the A1104 as the replacement (Appendix A (p. 139)).

The main board's GPIO4, an old limit-switch input, is on the same main board socket but goes nowhere (chapter 4 (p. 12), section 4.2 (p. 13)).

8.4 The angle sensor (AS5600)

Figure 27. Schematic: the angle sensor (AS5600), as drawn on the main sheet



The AS5600 is a magnetic angle sensor: a chip that reads the direction of a magnet spinning above it. It reads the tuning capacitor's shaft, through the magnet on the printed shaft that extends that shaft (section 8.1 (p. 47)).

Table 49. The angle sensor (AS5600)

What	Detail
Magnet	A 6 × 3 mm disc neodymium magnet, epoxied to the end of the printed shaft that extends the capacitor's shaft. It must be diametrically magnetised (its north and south poles on opposite sides of the disc, not on its faces). Less than 0.5 mm from the chip.
Mounting	Its board at right angles to the tuning capacitor's shaft, with the shaft's axis through the centre of the chip; the printed shaft reaches almost to the chip. Inside its own Faraday cage, on the cage's base, clear of the aluminium and copper tape.
Supply	The main board's own 3.3 V (`S3_3V3`) and ground. Not the 3.3 V bus. By the datasheet, the chip's VDD5V and VDD3V3 pins must be tied together for 3.3 V operation (Appendix A (p. 139)).
Direction pin (DIR)	On ground. By the datasheet, the reading then increases clockwise.
Analogue output (OUT)	Not connected.
Bus	I2C, a two-wire bus: data (SDA) and clock (SCL).

Each bus line has a series resistor and a pull-up:

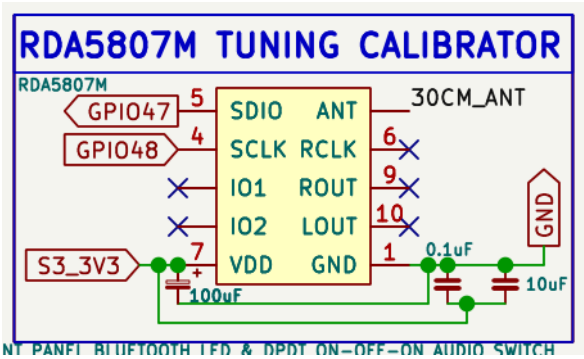
Table 50. The angle sensor (AS5600)

Sensor pin	Main board pin	Parts
SDA	GPIO38	220 Ω in series; 10 kΩ pull-up to the main board's 3.3 V at the main board's pin
SCL	GPIO39	the same: 220 Ω in series, 10 kΩ pull-up at the pin

The AS5600 answers at I2C address 0x36 (Appendix A (p. 139)). Its cable (S3_TUNER) carries ground with the shield, 3.3 V, SDA and SCL; the shield is on ground at the main board end.

8.5 The FM calibration receiver (RDA5807M)

Figure 28. Schematic: the FM calibration receiver (RDA5807M), as drawn on the main sheet



The RDA5807M is a single-chip FM receiver on a small module board. Here it plays no audio: it is the main board's instrument for calibrating the dial. Its antenna wire listens to the tube radio's own FM oscillator. That oscillator runs one IF (the radio's intermediate frequency, 10.6 MHz) below the station.

Table 51. The FM calibration receiver (RDA5807M)

What	Detail
Bus	Data (SDIO) on main board GPIO47, clock (SCLK) on GPIO48. No pull-up resistors: that is as built. GPIO48 also drives the main board's own RGB LED (chapter 4 (p. 12), section 4.2 (p. 13)).
Address	The fitted chip answers at I2C address 0x11. Its datasheet documents only 0x10.
Supply	The main board's own 3.3 V (S3_3V3) and ground. Across the supply, at the module: 100 µF electrolytic, 10 µF ceramic and 0.1 µF ceramic.
Unused pins	The clock input (RCLK), both audio outputs (LOUT, ROUT), and IO1 and IO2 are not connected.
Antenna	A 30 cm length of wire inside the tube radio's Faraday cage, bent in a U over the radio's FM oscillator section.
Antenna plug	The module reaches the wire through a 2 × 3 pin grid, so the module can be unplugged from its antenna there during a repair.

On the receiver's signal cable the shield ends at the receiver end, on ground, and is left open at the main board: the reverse of most cables (chapter 10 (p. 60), section 10.3 (p. 64)).

8.6 Cables

The dial hardware hangs on seven cables. Chapter 10 draws each one wire by wire.

Table 52. Cables

Cable	Carries	Cable page
S3_STEPPER → STEPPER_IN	GPIO9 to GPIO12 to the driver's IN1 to IN4. The pin order reverses end to end; follow the signal names.	section 10.6 (p. 66)
5 V bus → STEPPER_5VIN	the motor driver's 5 V and ground	section 10.24 (p. 83)
motor lead → motor socket	the motor's four coils and its common	section 10.32 (p. 89)
S3_LIMITS + 5 V bus → LIMIT	the index sensor's output, ground and 5 V; one harness that joins a bus jack	section 10.9 (p. 69)
S3_TUNER	the angle sensor's ground, 3.3 V, SDA and SCL	section 10.10 (p. 70)
S3_3V3 → RDA_3V3	the calibration receiver's 3.3 V and ground	section 10.2 (p. 63)
S3_RDA → RDA_I2C	the calibration receiver's SDIO and SCLK	section 10.3 (p. 64)

8.7 When something is wrong

Table 53. When something is wrong

You see	Check
The needle never moves	The motor driver board's 5 V cable, the stepper input cable, and the motor's lead in its socket.
The needle moves erratically or the wrong way	That each of GPIO9 to GPIO12 reaches IN1 to IN4 in order, by signal name, and that the motor lead is in its socket the right way.
The needle cannot find its index (the firmware reports NEEDLE FAULT; Firmware Gospel, chapter 5)	The index sensor: its 5 V from the bus jack on its harness, its cable to GPIO5, the 10 kΩ pull-up, and the magnet on the needle mount's arm.
The needle does not follow the knob	The angle sensor: its cable, the main board's 3.3 V on it, the two 220 Ω resistors and the two 10 kΩ pull-ups. Then the magnet: diametrically magnetised, and within 0.5 mm of the chip.
The calibration receiver does not answer	Its two cables. It answers at 0x11, not the 0x10 of its datasheet. Its bus has no pull-ups; that is as built.
The receiver answers but finds no signal	Its antenna wire, the 2 × 3 pin grid it plugs through, and the wire's place over the radio's FM oscillator. The tube radio must be on.

9. Front-panel controls and lamps

The front panel carries the controls you touch and the lamps you see. The main board dims the four FM panel lamps and senses when the amplifier is switched on. The audio board reads the source switch, the volume knob and the Bluetooth button, and lights the Bluetooth LED. This chapter gives the circuit behind each one. Chapter 3, section 3.2 (p. 10), lists them. The tuning knob is in chapter 8 (p. 46).

What the firmware does with these controls (dimming, pairing, the volume law) is in the Firmware Gospel: chapter 4 (p. 12) for the lamps, chapter 10 (p. 60) for the audio board's controls. The source switch's thresholds are given here, in section 9.3 (p. 55).

9.1 The FM panel lamps

Figure 29. Schematic: the FM panel lamps' driver, as drawn on the main sheet

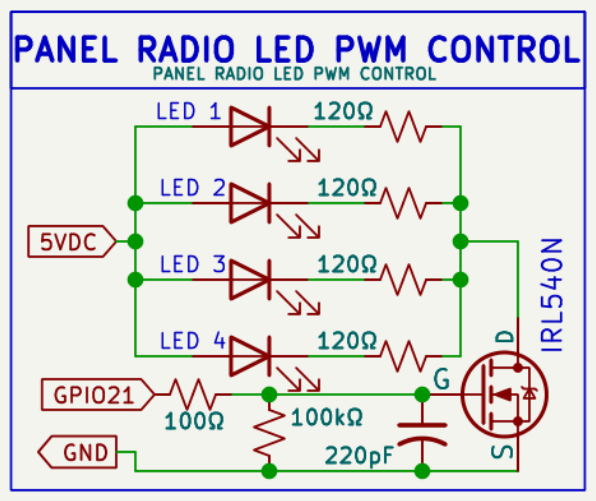
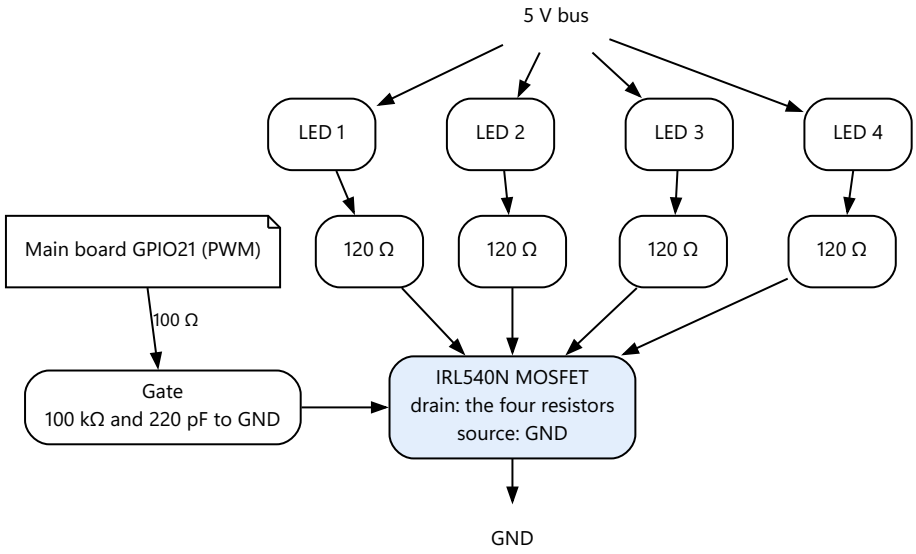


Figure 30. The FM panel lamps: four LEDs, each with its own resistor, switched together by one MOSFET



The four FM panel lamps are LEDs, dimmed together by the main board. The main board dims them with PWM (pulse-width modulation): GPIO21 switches the lamps on and off very fast, and the share of "on" time sets the brightness. The lamp circuit is:

Table 54. The FM panel lamps

Part	Value	What it does
LED anodes	on 5 V	all four, from the 5 V bus
LED resistors	120 Ω, one per LED	each LED's cathode returns through its own resistor. The value was chosen for maximum brightness control.
MOSFET	IRL540N	its drain takes the four resistors, its source is on ground. A logic-level MOSFET: its gate threshold is 1 to 2 V (Appendix A (p. 139)).
Gate resistor	100 Ω	from GPIO21 to the gate
Gate pull-down	100 kΩ to ground	holds the lamps off while GPIO21 is not driven
Gate capacitor	220 pF to ground	across the pull-down

The lamps light and dim with the PWM.

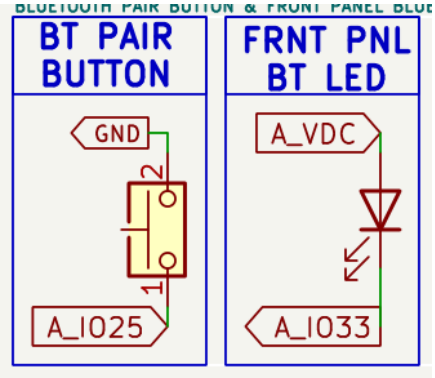
The panel-lamp board. The MOSFET and its parts sit on a small board of their own. It has three connections:

Table 55. The FM panel lamps

Connector	Carries	Cable page
`FM_LED_IN`, 3 pins	GPIO21 from the main board (shielded, shield on ground at the main board end), and 5 V and ground from a 5 V bus jack	section 10.11 (p. 71)
5 V out (`5VOUT`), 2 pins	pin 1 the board's 5 V, pin 2 the four LED anodes	section 10.33 (p. 89)
lamp returns, 4 pins	the four LED cathodes, each to its own 120 Ω on the board	section 10.33 (p. 89)

9.2 The Bluetooth button and LED

Figure 31. Schematic: the Bluetooth button and LED, as drawn on the main sheet



On the front panel, one part is both the Bluetooth LED and the Bluetooth button. Both sides are active low: pressing the button pulls its pin to ground, and the audio board lights the LED by pulling its pin low. Each side has its own audio board pin:

Table 56. The Bluetooth button and LED

Side	Audio board pin	Wiring
Button	GPIO25 (`A_IO25`)	between the pin and ground. No pull-up resistor is drawn; the firmware turns on the chip's internal one.
LED	GPIO33 (`A_IO33`)	anode on the audio board's 3.3 V (`A_3V3`), cathode to the pin. No series resistor is drawn.

One 4-wire cable, with no shield, carries 3.3 V, GPIO33, ground and GPIO25 to the front panel. Its pin order crosses from one end to the other (chapter 10 (p. 60), section 10.14 (p. 73)).

9.3 The source switch

Figure 32. Schematic: the source switch, as drawn on the main sheet

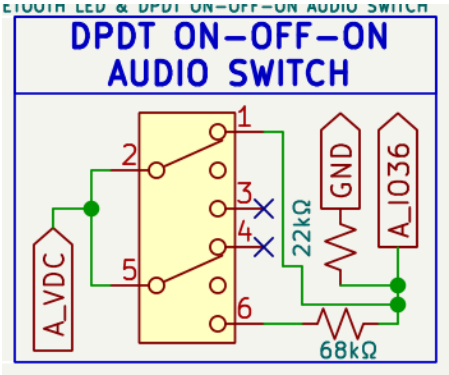
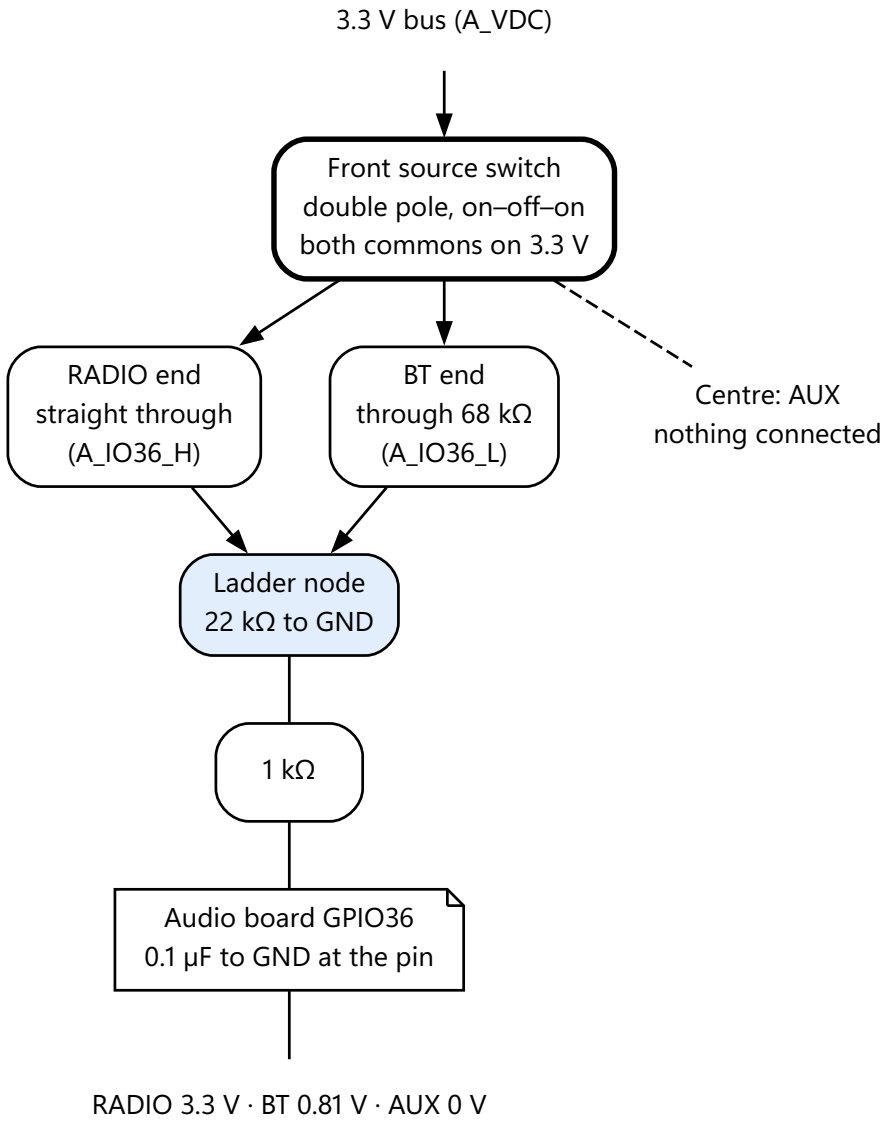


Figure 33. The source switch's resistor ladder: one pin reads three voltages



The front source switch chooses RADIO, BT or AUX. It is a double-pole, on–off–on switch: RADIO and BT at the two ends, AUX in the centre. It switches no audio. It only puts one of three voltages on the audio board's GPIO36, and the audio board picks its source from that. AUX audio never passes through the audio board at all (chapter 6 (p. 28), section 6.7 (p. 37)).

The ladder. A resistor ladder turns the switch position into one of three voltages on one pin:

- Both of the switch's commons are on the 3.3 V bus (`A_VDC`).
- At one end, one pole connects 3.3 V straight to the ladder node (the leg named `A_IO36_H`).

- At the other end, the other pole connects 3.3 V through 68 kΩ to the ladder node (`A_IO36_L`).
- The two remaining end contacts are not connected.
- A 22 kΩ resistor holds the ladder node to ground.
- The node reaches GPIO36 through 1 kΩ, with 0.1 µF to ground on the pin side.

Each position puts its own voltage on the pin:

Table 57. The source switch

Position	What reaches the node	Voltage on GPIO36	Unit
RADIO (one end)	3.3 V directly	3.3	V
BT (other end)	3.3 V through 68 kΩ, against the 22 kΩ to ground	0.81	V
AUX (centre, off)	nothing; the 22 kΩ pulls the node down	0	V

The 22 kΩ and 68 kΩ ladder resistors are on the audio board, not at the switch, and the 0.1 µF is at the audio board's pin.

The thresholds. The audio board's firmware reads GPIO36 in counts of its analogue input and sorts the reading like this. These numbers are one of the few facts this book takes from the firmware.

Table 58. The source switch

Reading on GPIO36	Source chosen	Measured in that position
below 424 counts	AUX	0 counts (0 V)
from 424 up to 2470 counts	BT	848 counts (0.81 V)
2471 counts and up	RADIO	4095 counts (3.3 V)

The wiring. Two cables meet on a small source-switch board, which is wired on to the switch. One comes from the audio board, and one from a 3.3 V jack.

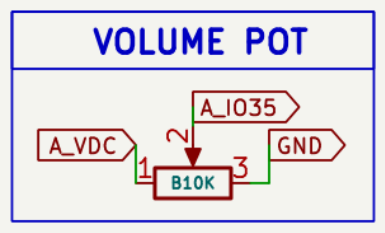
Table 59. The source switch

Cable	Carries	Cable page
`A32_SWITCH`	the two legs, `A_IO36_L` and `A_IO36_H`, in a shielded cable (shield on ground at the audio board end only)	section 10.20 (p. 79)
`A32_VDC_LISTENER`	3.3 V (`A_VDC`) to the switch's commons	section 10.20 (p. 79)

No ground wire goes to the switch.

9.4 The front volume knob

Figure 34. Schematic: the front volume knob, as drawn on the main sheet



The front volume knob turns a potentiometer read by the audio board. It is not the amplifier's volume control, which is reached only from the back (chapter 6 (p. 28), section 6.8 (p. 37)). The knob does nothing to the analogue audio: the audio board reads its position and sets the volume digitally, so it acts on RADIO and BT only, never on AUX. The potentiometer (RV4) is drawn as B10K. Its three pins go to:

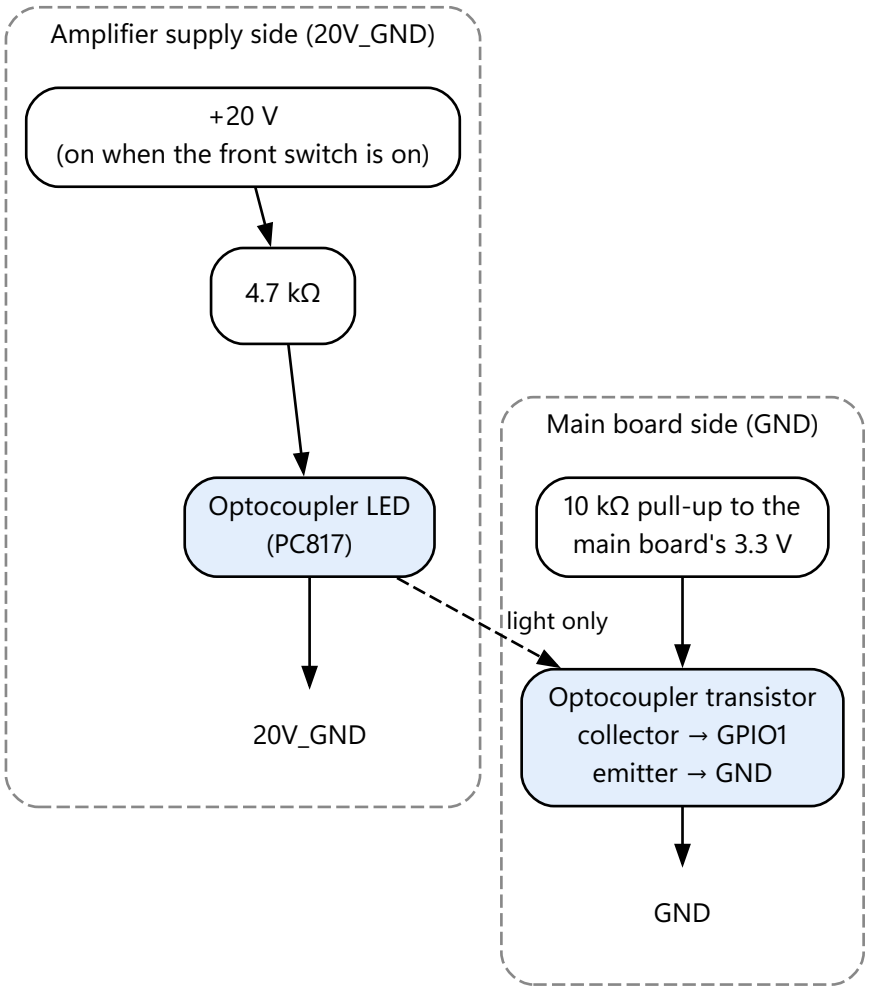
Table 60. The front volume knob

Potentiometer pin	Goes to
one end	the 3.3 V bus (A_VDC)
wiper	the audio board's GPIO35 (A_IO35), an input-only pin
other end	ground

Two cables meet in the potentiometer's 3-pin connector (VOLUME). One carries the wiper, shielded, with the shield on ground at the audio board end only. The other carries 3.3 V and ground (chapter 10 (p. 60), section 10.13 (p. 72)). The knob itself is a printed part (Appendix D (p. 152)).

9.5 The amplifier power sensor

Figure 35. The amplifier power sensor: the signal crosses by light; no wire inside the sensor joins its two sides



The main board needs to know when the front power switch has turned the amplifier on. A small board tells it, through an optocoupler (PC817): an LED and a light-sensitive transistor in one package, with no electrical path between them. Its two sides are wired like this:

Table 61. The amplifier power sensor

Side	Wiring
LED	+20 V from the amplifier supply → 4.7 kΩ → the LED's anode; the LED's cathode → the amplifier supply return (`20V_GND`)
Transistor	collector → main board GPIO1, with a 10 kΩ pull-up to the main board's 3.3 V; emitter → DC ground (`GND`)

How it reads. The +20 V exists only while the front power switch is on (chapter 5 (p. 21)). Then the LED lights, the transistor conducts, and GPIO1 reads low. With the amplifier off, the pull-up holds GPIO1 high. Inside the

sensor no wire joins its two returns, 20V_GND and GND outside it they are joined, as Earth, GND and 20V_GND all are (chapter 5 (p. 21), section 5.3 (p. 23)).

Pins (datasheet): 1 anode, 2 cathode, 3 emitter, 4 collector.

Its cable to the main board carries GPIO1 and ground, shielded, with the shield on ground at the main board end (chapter 10 (p. 60), section 10.7 (p. 67)). What the firmware does when the amplifier goes on or off is in the Firmware Gospel.

9.6 When something is wrong

Table 62. When something is wrong

You see	Check
The FM panel lamps never light	The panel-lamp board's cable: 5 V and ground from its bus jack, and GPIO21. Then the MOSFET, and the 5 V lead to the LEDs.
One FM panel lamp dark	That LED, its 120 Ω, and its return lead to the board.
The Bluetooth button does nothing	Its wires to GPIO25 and to ground in the Bluetooth cable.
The Bluetooth LED never lights	3.3 V (A_3V3) and GPIO33 in the Bluetooth cable, and the LED's polarity.
The radio picks the wrong source	The voltage on GPIO36 in each position: 3.3 V, 0.81 V, 0 V. Then 3.3 V on A32_VDC_LISTENER, the two legs in A32_SWITCH, and the 68 kΩ and 22 kΩ on the audio board.
The volume knob does nothing	The wiper cable to GPIO35, and 3.3 V and ground on the knob's other cable. On AUX this is normal: the knob never acts on AUX.
The main board does not see the amplifier come on	GPIO1 should read low with the amplifier on and 3.3 V with it off. Check +20 V at the sensor board, the 4.7 kΩ, and the sensor's cable.

10. Cables and harnesses

Every board, sensor and control in the machine is joined to the others by a cable that plugs in at both ends. This chapter draws each of those cables, wire by wire, so that you can test one, trace a fault to it, or make a replacement. It is the only chapter that draws the cables themselves, wire by wire; other chapters draw circuits.

10.1 How to read the cable pages

Most of this chapter follows the cable sheet: the drawing that defines the important and less evident cables of the machine. Sections 10.2 to 10.36 each describe one cable, or a few wires that belong together. Every page uses the same headings in the same order, so you always know where to look. A heading is left out when there is nothing under it: a cable that uses all its pins has no Not used, for example.

Table 63. How to read the cable pages

Heading	What it tells you
Carries	What the cable is for: the signals or the power on it, and as much of the circuit at each end as you need to understand them.
Shape	How the cable is built: how many wires, whether it is shielded, the connector at each end with its name on the cable sheet, and whether the pin numbers cross from one end to the other.
(drawing)	The cable, drawn by signal name with the cable sheet's pin numbers. How to read it is at the end of this section.
Wire by wire	A table with one row per wire: the pin at one end, the signal, what it is, and the pin at the other end. A dash (—) means that the wire has no pin at that end. Where two cables meet in one connector, each cable has its own table.
Shield	Where the shield and foil end, or "none". A missing ground wire is noted here too, with where the board gets its ground instead.
Not used	Pins that carry no wire.
(notes and cautions)	Anything else you must know before you work on the cable: a warning box, parts built into the cable, ferrite cores.
If it fails	What you see when a wire in this cable breaks. Section 10.37 turns this round: it goes from what you see to the cable to check.

The cables in sections 10.29 (p. 86) to 10.36 are not on the cable sheet. Their pages are short, and they draw only what is recorded.

Pin numbers are the cable sheet's. The cable sheet numbers each connector's pins from the top of the connector's symbol, as that symbol is defined in the drawing's parts library: pin 1, pin 2, and so on. That number is not a number moulded on the connector. A pin number says nothing about where a wire sits in the real connector: each wire is identified by its signal name, not by its position. Where this chapter says that pins cross or run in opposite order, it speaks of that numbering only, not of the real connector.

Signals are named as on the cable sheet. The names you will meet:

Table 64. How to read the cable pages

Name	What it is
`GPIOn`	In a signal name, pin n of the main board (ESP32-S3, "S3"). In the text, "the audio board's GPIOn" is pin n of the audio board, whose signal names are `A_IOn`.
`A_IOn`	Pin n of the audio board (ESP32, "A32")
`5VDC`	The 5 V bus (chapter 5 (p. 21))
`S3_3V3`	The main board's own 3.3 V output
`A_VDC`, `A_3V3`	The audio side's 3.3 V: the audio board's 3.3 V output, shared out by the 3.3 V bus. The two names are one rail.
`GND`	DC ground
`CABLE S/FTP`	The cable's shield and foil (below)

The shield rule. Where it is possible and useful, the cables are twisted pairs, shielded, foiled, or both. A shielded cable here is S/FTP: a braided shield (S) and a foil (F) around twisted pairs (TP). `CABLE S/FTP` on a pin means that the cable's shield and foil end on that pin, joined to ground on that board, and are left unconnected at the cable's other end, unless the page says

otherwise. Where a pin reads `GND + CABLE S/FTP`, both a ground wire and the shield end there.

Length and labels. Every cable inside the cabinet is under 14 inches: long enough to service the box with its panels open and nothing unplugged. Each cable is marked at both ends, with a name that makes its job evident or with its name on the cable sheet. Wire colours are given only for the amplifier supply's four wires, not verified inside the supply (section 10.30 (p. 87)).

Connectors. The 5 V bus jacks are JST XH; their rating is in chapter 5 (p. 21), section 5.5 (p. 26). Some connector types are used for one cable only, so that no cable can be plugged where it would break something. The mains feed to the front power switch is one of them.

In the drawings, each connector box gives its name, its pins and, at the bottom, what it belongs to. The middle box is the cable. Each of its lines follows one wire, left to right: the connector, pin and signal at one end, written `connector:pin:signal` the wire's number and its signal; then the connector, pin and signal at the other end, written the same way. It repeats what the end boxes show, so that you can follow one wire along one line. `+ S` means the cable has a shield. A pin marked `(not used)` carries no wire. A wire that ends in the cable box with no connector after it goes to an end this chapter does not draw.

All the cables

This table lists every page in the chapter. In the Wires column, "+ shield" means that the cable is shielded. A second "+" adds the wires of another cable or connector described on the same page.

Table 65. All the cables

Section	Cable	From → to	Wires	Carries
10.2	FM receiver power	main board → FM calibration receiver	2	3.3 V and ground
10.3	FM receiver control	main board → FM calibration receiver	2 + shield	control bus (I2C)
10.4	Main board 5 V	5 V bus → main board	2	5 V and ground
10.5	Segment drive	main board → display driver board	7	which segments light
10.6	Needle motor control	main board → needle motor driver	4	the four coil signals
10.7	Amplifier power sensor	main board → power sensor board	2 + shield	"amplifier is on"
10.8	Digit select	main board → display driver board	4	which digit lights
10.9	Needle index sensor	main board and 5 V bus → index sensor	2 + shield, + 1	sensor output, ground, 5 V
10.10	Tuning angle sensor	main board → angle sensor	4 + shield	3.3 V, ground, control bus (I2C)
10.11	Panel-lamp control	main board and 5 V bus → panel-lamp board	1 + shield, + 2	lamp brightness, the board's 5 V
10.12	Board link	main board ↔ audio board	3 + shield	serial link both ways, ground

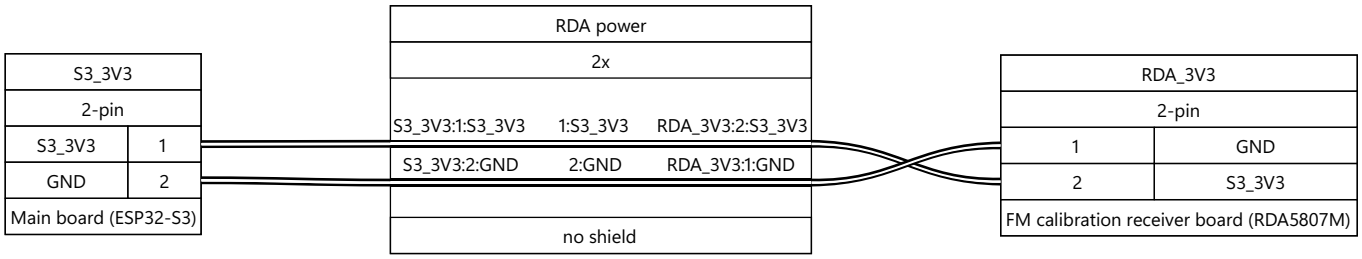
Section	Cable	From → to	Wires	Carries
10.13	Front volume knob	audio board and 3.3 V → volume knob	1 + shield, + 2	knob position, 3.3 V, ground
10.14	Bluetooth button and LED	audio board → front panel	4	button and LED
10.15	Audio harness	audio board → ADC and DAC; 5 V bus → DAC	6 + shield, + 2	digital audio, the DAC's 5 V
10.16	Clock module power	3.3 V jack → real-time clock module	2	3.3 V and ground
10.17	Clock module control	audio board → real-time clock module	2 + shield	control bus (I2C)
10.18	Audio board 5 V	5 V bus → audio board	2	5 V and ground
10.19	Audio board 3.3 V	audio board → 3.3 V jack	2	the audio board's 3.3 V output
10.20	Source switch	audio board and 3.3 V → source-switch board	2 + shield, + 1	the RADIO / BT / AUX position
10.21	DAC to amplifier	DAC board → amplifier input	3 at the plug, 5 at the amplifier	analogue audio, balanced
10.22	Radio to ADC	tube radio's output transformer → ADC	2 + shield	the radio's audio, mono
10.23	Colon 5 V	5 V bus → display bus board, for the colon	2	5 V and ground, reversed at `COLON_IN`
10.24	Motor driver 5 V	5 V bus → needle motor driver	2	5 V and ground
10.25	ADC 5 V	5 V bus → ADC board	2	5 V and ground
10.26	Driver board 5 V	5 V bus → display driver board	2	5 V and ground
10.27	Digit rails	display driver board → display bus board	4 + shield	each digit's switched 5 V
10.28	Segment buses	display driver board → display bus board	7	the seven segment lines
10.29	Box-fan harness	amplifier supply → fan regulator → box fan	2 + 2	+20 V in, 12 V out
10.30	Amplifier supply wires	amplifier supply → amplifier	4	±20 V and ground
10.31	USB	back panel → each board's USB port	4 each	USB, straight through
10.32	Needle motor lead	needle motor driver → needle motor	5	four coils and their common
10.33	Panel-lamp leads	panel-lamp board → the four FM lamps	1 + 4	5 V out, four returns
10.34	Inside the clock display	display bus board → colon and digits	2; 8 per digit	the colon LED, the digits
10.35	FM receiver antenna and the antenna coax	—	—	the calibration receiver's antenna wire; the FM antenna's coax (neither drawn as a cable)
10.36	Mains and earth wiring	—	—	see chapter 5 (p. 21)

10.2 FM receiver power — main board to FM calibration receiver

Carries: the main board's own 3.3 V (`S3_3V3`) and ground to the FM calibration receiver, an RDA5807M FM receiver chip on its own small module.

Shape: one 2-wire cable, with a 2-pin connector at each end: `S3_3V3` at the main board, `RDA_3V3` at the receiver board. The pins cross in the cable sheet's numbering: pin 1 at one end is pin 2 at the other.

Figure 36. The FM receiver power cable, by signal name, as on the cable sheet



Wire by wire.

Table 66. FM receiver power — main board to FM calibration receiver

Main board end, pin	Signal	What it is	Receiver end, pin
1	`S3_3V3`	The main board's 3.3 V output	2
2	`GND`	Ground	1

Shield: none.

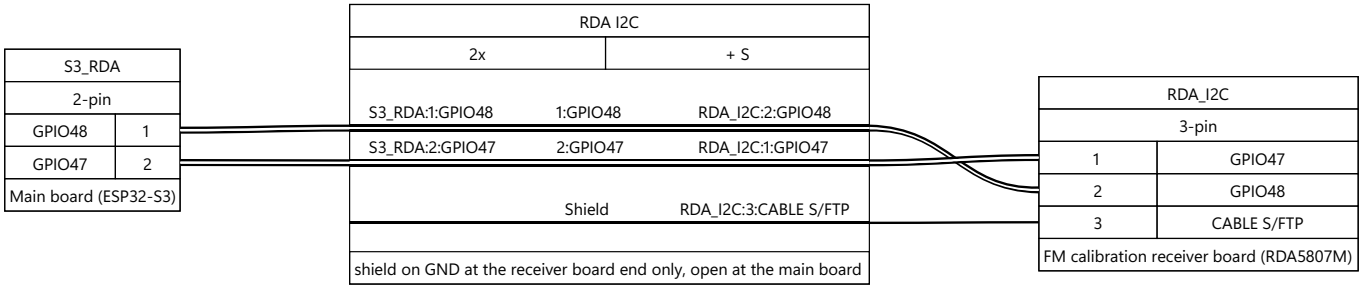
If it fails: the receiver loses its power and stops answering the main board.

10.3 FM receiver control — main board to FM calibration receiver

Carries: the two-wire control bus (I2C, a clock line and a data line) between the main board and the FM calibration receiver: `GPIO47` is the data line, `GPIO48` the clock line. Neither line has an external pull-up resistor.

Shape: one shielded cable with two wires. A 2-pin connector `S3_RDA` at the main board, a 3-pin connector `RDA_I2C` at the receiver board; the third pin takes the shield.

Figure 37. The FM receiver control cable, by signal name, as on the cable sheet



Wire by wire.

Table 67. FM receiver control — main board to FM calibration receiver

Main board end, pin	Signal	What it is	Receiver end, pin
1	`GPIO48`	Clock line (the receiver's SCLK)	2
2	`GPIO47`	Data line (the receiver's SDIO)	1
—	`CABLE S/FTP`	Shield and foil	3

Shield: at the receiver end, unlike most cables here. The shield and foil are joined to ground at the receiver board end, on pin 3. They are left unconnected at the main board end, whose connector has only two pins. There is no ground wire: the receiver's ground comes on its power cable (section 10.2).

GPIO48 also drives the main board's own RGB LED, so that LED is lit while the machine runs.

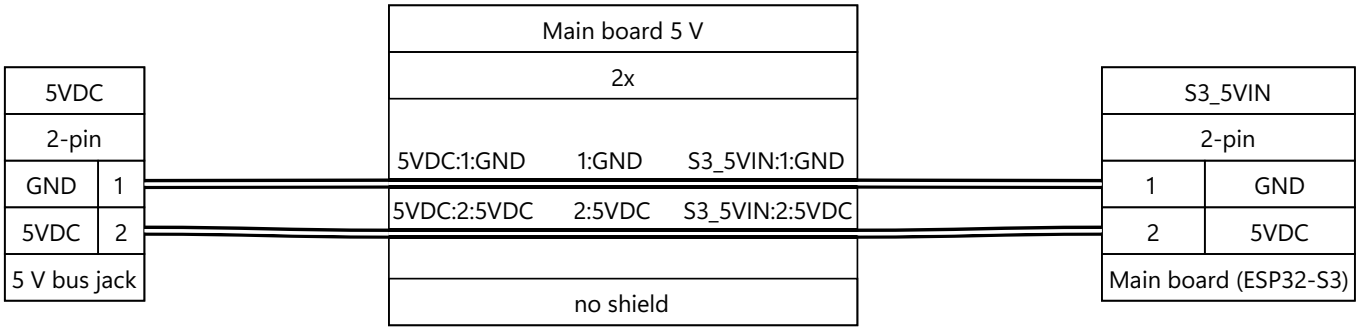
If it fails: the main board can no longer talk to the receiver.

10.4 Main board 5 V — 5 V bus to main board

Carries: 5 V and ground from the 5 V bus to the main board's 5 V pin.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to `S3_5VIN` on the main board.

Figure 38. The main board's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 68. Main board 5 V — 5 V bus to main board

5 V bus jack, pin	Signal	Main board end, pin
1	GND	1
2	5VDC	2

Shield: none.

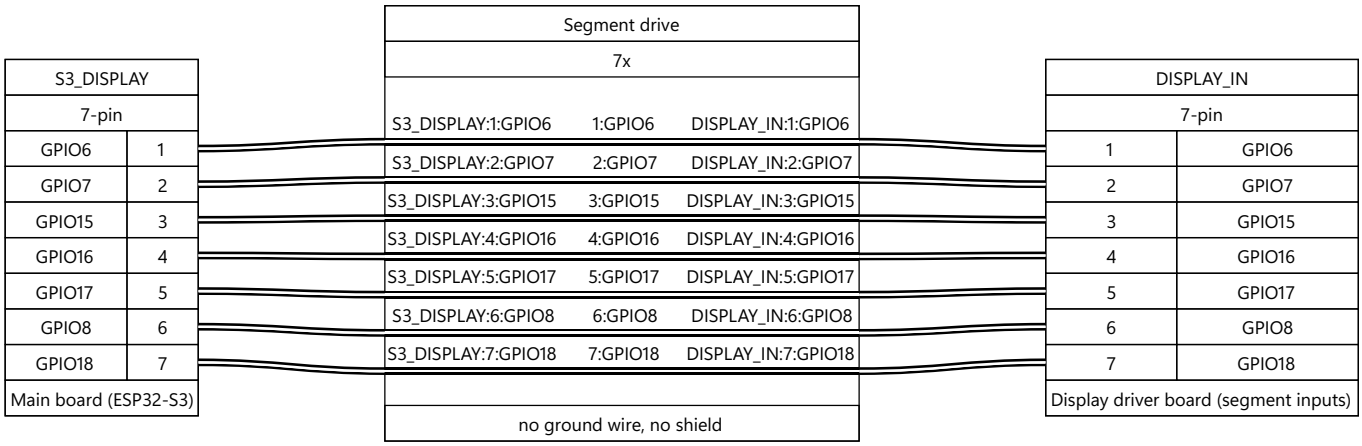
If it fails: the main board goes dead: no clock display, no needle, no panel lamps. The audio board has its own 5 V cable (section 10.18 (p. 77)).

10.5 Segment drive — main board to display driver board

Carries: the seven signals that choose which segments of the clock display light. Each one drives one input of the segment driver on the display driver board, a ULN2003A (a chip of seven transistor switches that sink current to ground). Each segment line is shared by all four digits.

Shape: one 7-wire cable, pin for pin: `S3_DISPLAY` at the main board, `DISPLAY_IN` at the driver board.

Figure 39. The segment drive cable, by signal name, as on the cable sheet



Wire by wire.

Table 69. Segment drive — main board to display driver board

Main board end, pin	Signal	What it is	Driver board end, pin
1	GPIO6	Segment A (driver input 1)	1
2	GPIO7	Segment B (driver input 2)	2
3	GPIO15	Segment C (driver input 3)	3
4	GPIO16	Segment D (driver input 4)	4
5	GPIO17	Segment E (driver input 5)	5
6	GPIO8	Segment F (driver input 6)	6
7	GPIO18	Segment G (driver input 7)	7

Shield: none. There is no ground wire either: the driver board takes its ground from its own power cable (section 10.26 (p. 84)).

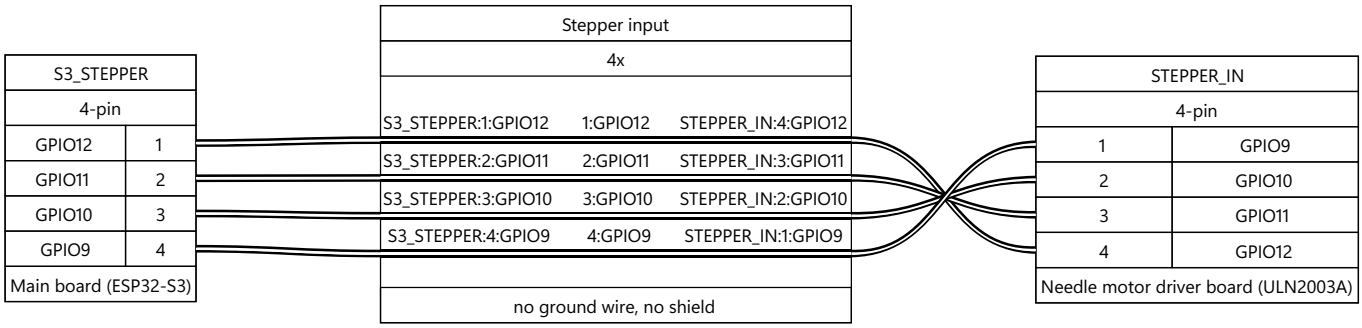
If it fails: a broken wire leaves one segment dark on all four digits.

10.6 Needle motor control — main board to needle motor driver

Carries: the four signals that step the needle motor. They enter the needle motor driver board, a ULN2003A board, on its inputs IN1 to IN4, which drive the motor's four coils (section 10.32 (p. 89)).

Shape: one 4-wire cable, `S3_STEPPER` at the main board, `STEPPER_IN` at the driver board. The pin order runs opposite at the two ends.

Figure 40. The needle motor control cable, by signal name, as on the cable sheet



Wire by wire.

Table 70. Needle motor control — main board to needle motor driver

Main board end, pin	Signal	What it is	Driver board end, pin
1	`GPIO12`	Driver input IN4	4
2	`GPIO11`	Driver input IN3	3
3	`GPIO10`	Driver input IN2	2
4	`GPIO9`	Driver input IN1	1

Shield: none. There is no ground wire either: the driver board takes its ground from its own power cable (section 10.24 (p. 83)).

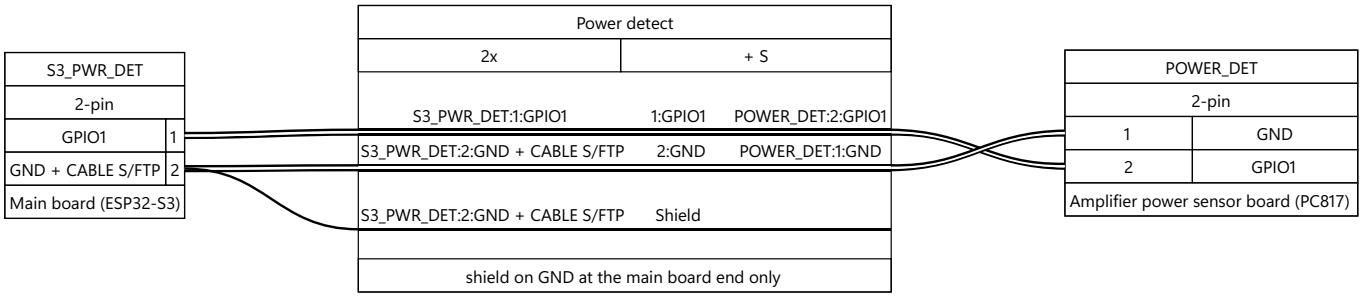
If it fails: one coil is no longer driven, so the needle cannot step properly.

10.7 Amplifier power sensor — main board to power sensor board

Carries: the “amplifier is on” signal. The sensor board holds an optocoupler (PC817): an LED and a light-sensitive transistor in one package, so that a signal crosses by light with no wire between the two sides. The LED is lit from the amplifier’s +20 V supply. The transistor’s collector is on `GPIO1`, which has a 10 kΩ pull-up to `S3_3V3` on the main board; its emitter is on ground. While the amplifier supply is on, the transistor conducts and pulls `GPIO1` low. Chapter 9 has the sensor.

Shape: one shielded 2-wire cable, `S3_PWR_DET` at the main board, `POWER_DET` at the sensor board.

Figure 41. The amplifier power sensor cable, by signal name, as on the cable sheet



Wire by wire.

Table 71. Amplifier power sensor — main board to power sensor board

Main board end, pin	Signal	What it is	Sensor end, pin
1	GPIO1	The optocoupler's collector	2
2	GND + CABLE S/FTP	Ground to the optocoupler's emitter; the shield and foil end here too	1

Shield: joined to ground at the main board end only. The ground wire itself runs end to end.

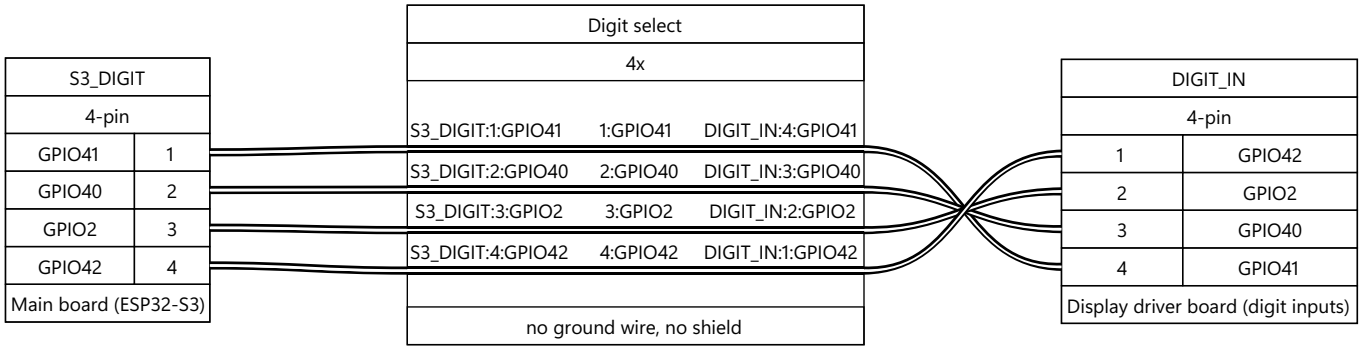
If it fails: with either wire broken, nothing can pull GPIO1 low, so the main board reads "amplifier off" all the time.

10.8 Digit select — main board to display driver board

Carries: the four signals that choose which digit of the clock display lights. Digit 1 is the rightmost digit (units of minutes); digits 2, 3 and 4 run leftwards from it. On the driver board each signal goes through a ULN2003A channel and a 1 kΩ resistor to the gate of a P-channel MOSFET (FQP27P06), which switches 5 V onto that digit (section 10.27 (p. 85)). Chapter 7 has the circuit.

Shape: one 4-wire cable, S3_DIGIT at the main board, DIGIT_IN at the driver board. The pin order runs opposite at the two ends.

Figure 42. The digit select cable, by signal name, as on the cable sheet



Wire by wire.

Table 72. Digit select — main board to display driver board

Main board end, pin	Signal	What it is	Driver board end, pin
1	GPIO41	Selects digit 4 (leftmost)	4
2	GPIO40	Selects digit 3	3
3	GPIO2	Selects digit 2	2
4	GPIO42	Selects digit 1 (rightmost)	1

Shield: none, and no ground wire: the cable has four wires only.

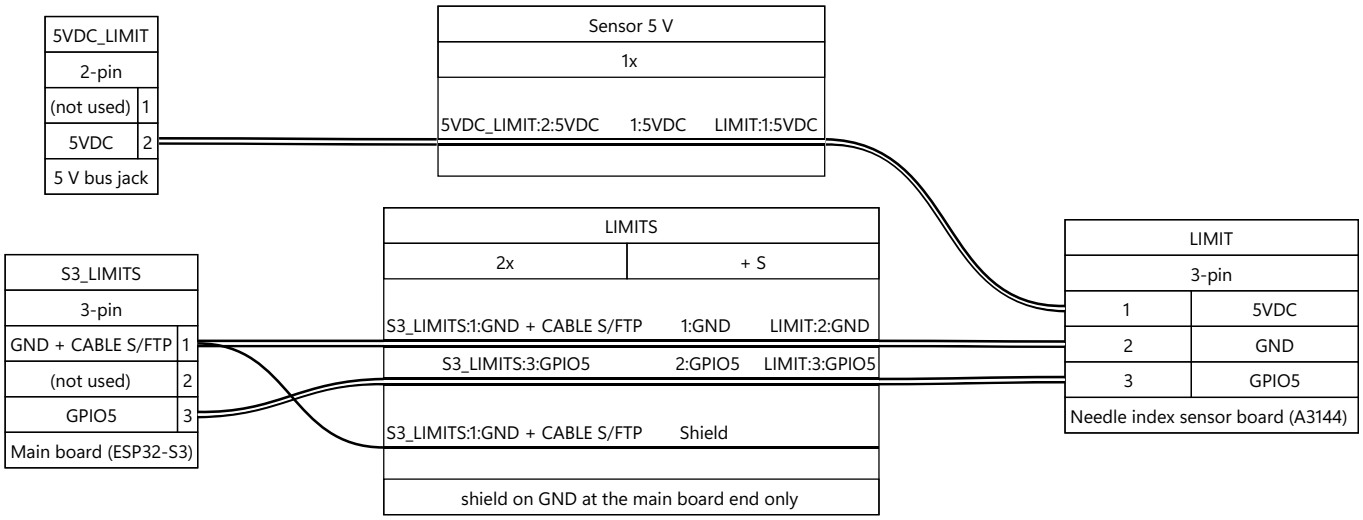
If it fails: a broken wire leaves one digit dark.

10.9 Needle index sensor — main board and 5 V bus to the index sensor

Carries: the index sensor's output, its ground and its 5 V. The sensor is an A3144 Hall switch (it reacts to a magnet) on a small perfboard in its own PLA case. A small magnet on the needle mount passes over it, and its output switches low when the magnet's field is strong enough. That gives the needle a fixed reference point. The output is open-collector: on the main board, `GPIO5` has a 10 kΩ pull-up to `S3_3V3` and 0.1 µF to ground.

Shape: a Y harness. A shielded 2-wire cable from `S3_LIMITS` at the main board, and a single 5 V wire from a bus jack `5VDC_LIMIT`, meet in the sensor's 3-pin connector `LIMIT`.

Figure 43. The needle index sensor harness, by signal name, as on the cable sheet



Wire by wire.

Table 73. Needle index sensor — main board and 5 V bus to the index sensor

Main board end, pin	Signal	What it is	Sensor end, pin
1	`GND + CABLE S/FTP`	Ground for the sensor; the shield and foil end here too	2
2	(not used)	—	—
3	`GPIO5`	The sensor's output	3

Table 74. Needle index sensor — main board and 5 V bus to the index sensor

5 V bus jack, pin	Signal	Sensor end, pin
1	(not used)	—
2	`5VDC`	1

Shield: joined to ground at the main board end only.

Not used: pin 2 at the main board end, and the bus jack's ground pin. The sensor's ground comes from the main board, its 5 V from the bus. At the

main board, GPIO4 is still wired to pin 2, with its own 10 kΩ pull-up and 0.1 µF, but the cable carries nothing on that pin.

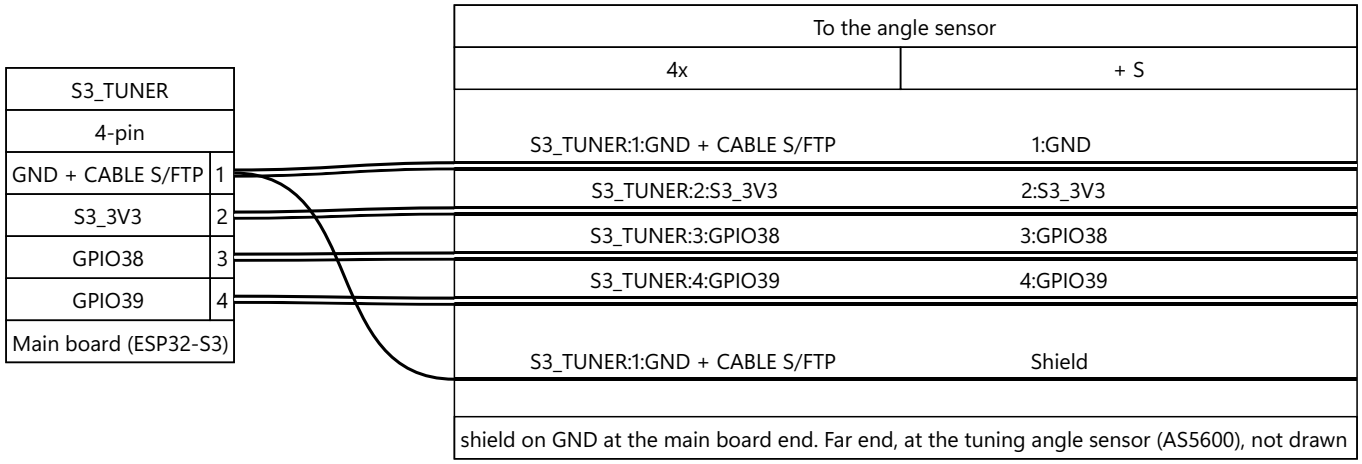
If it fails: if any of the three wires breaks, GPIO5 stays high and the main board never sees the index magnet, so the needle cannot find its reference point.

10.10 Tuning angle sensor — main board to the angle sensor

Carries: power and the control bus (I2C) for the tuning angle sensor, an AS5600 magnetic angle sensor. It sits inside its own Faraday cage, its board perpendicular to the tuning capacitor's shaft, with the shaft's axis through the centre of the chip. It reads the angle of a magnet glued to the end of a PLA extension of that shaft. At the main board, each bus line has a 10 kΩ pull-up to S3_3V3 at the board's pin, then a 220 Ω resistor in series before the connector.

Shape: one shielded 4-wire cable from S3_TUNER at the main board. Only the main board end is drawn. The table gives the sensor pin each wire reaches.

Figure 44. The tuning angle sensor cable



Wire by wire.

Table 75. Tuning angle sensor — main board to the angle sensor

Main board end, pin	Signal	What it is	Sensor pin it reaches
1	GND + CABLE S/FTP	Ground; the shield and foil end here too	GND
2	S3_3V3	The main board's 3.3 V output	VCC
3	GPIO38	Data line (SDA)	SDA
4	GPIO39	Clock line (SCL)	SCL

Shield: joined to ground at the main board end.

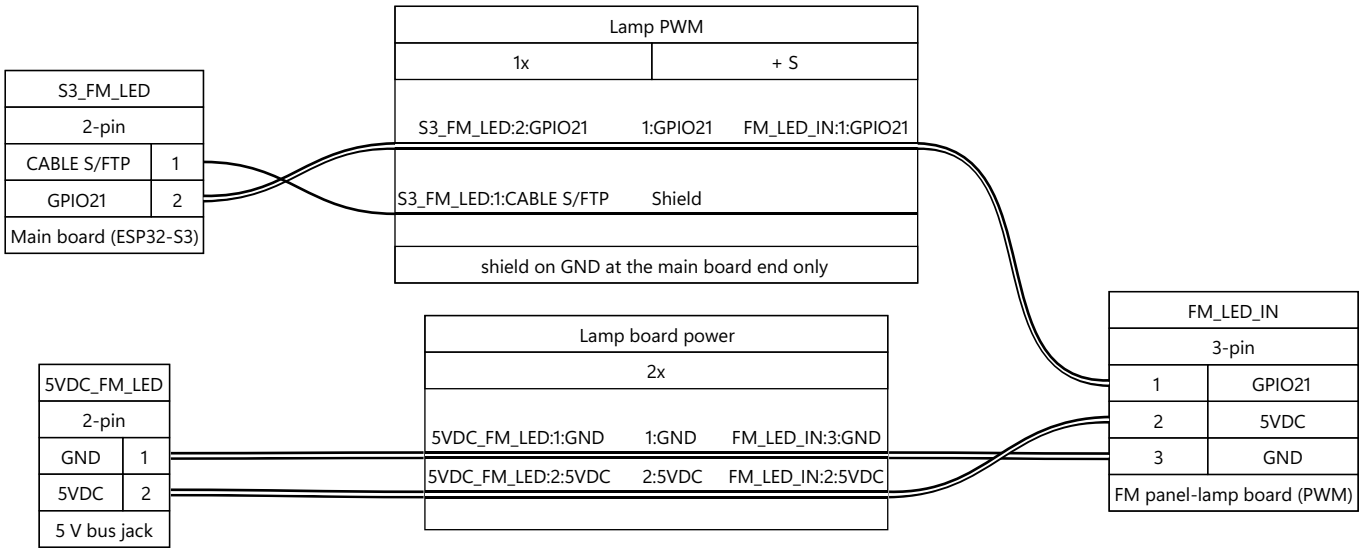
If it fails: the main board can no longer read where the tube radio is tuned.

10.11 Panel-lamp control — main board and 5 V bus to the panel-lamp board

Carries: the brightness signal for the four FM panel lamps, and the lamp board's power. `GPIO21` reaches the gate of a logic-level MOSFET (IRL540N) through 100 Ω; the gate also has 100 kΩ and 220 pF to ground. The MOSFET sinks the four lamps' current. The main board dims the lamps by PWM (pulse-width modulation: switching on and off fast, with a varying on-time). Chapter 9 has the circuit.

Shape: two cables into one 3-pin connector, `FM_LED_IN`, on the panel-lamp board: a shielded 1-wire cable from `S3_FM_LED` on the main board, and a 2-wire power cable from a 5 V bus jack `5VDC_FM_LED`.

Figure 45. The panel-lamp control cables, by signal name, as on the cable sheet



Wire by wire.

Table 76. Panel-lamp control — main board and 5 V bus to the panel-lamp board

Main board end, pin	Signal	What it is	Lamp board end, pin
1	`CABLE S/FTP`	Shield and foil	—
2	`GPIO21`	The lamp brightness signal (PWM)	1

Table 77. Panel-lamp control — main board and 5 V bus to the panel-lamp board

5 V bus jack, pin	Signal	Lamp board end, pin
1	`GND`	3
2	`5VDC`	2

Shield: joined to ground at the main board end, left unconnected at the lamp board. There is no ground wire from the main board: the lamp board takes its ground from the bus jack.

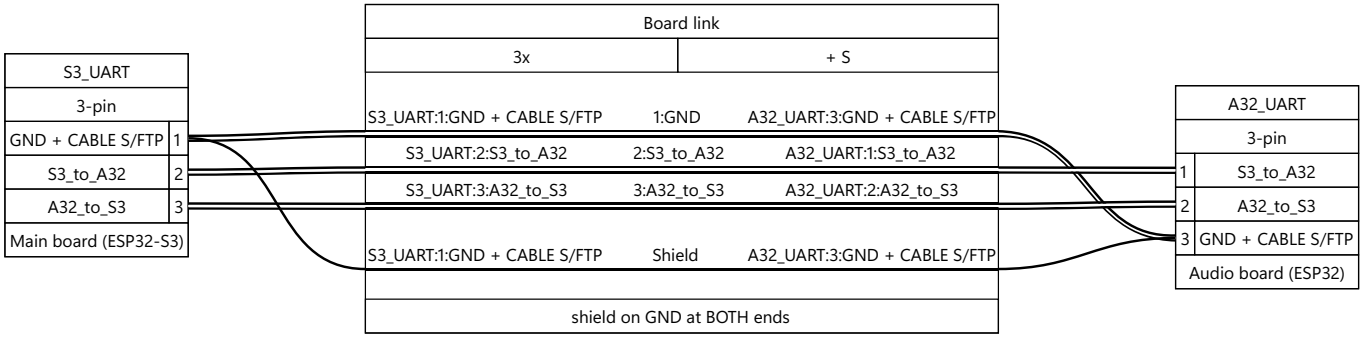
If it fails: if the `GPIO21` wire breaks, the gate's 100 kΩ holds the MOSFET off and the lamps stay dark. A broken power wire also leaves them dark.

10.12 Board link — main board to audio board

Carries: the serial link (UART) between the two boards, one wire each way, and ground. `S3_to_A32` runs from the main board's `GPIO13` to the audio board's `GPIO26`. `A32_to_S3` runs from the audio board's `GPIO27` to the main board's `GPIO14`.

Shape: one shielded 3-wire cable, `S3_UART` at the main board, `A32_UART` at the audio board.

Figure 46. The board link cable, by signal name, as on the cable sheet



Wire by wire.

Table 78. Board link — main board to audio board

Main board end, pin	Signal	What it is	Audio board end, pin
1	`GND + CABLE S/FTP`	Ground; the shield and foil end here too	3
2	`S3_to_A32`	Main board `GPIO13` → audio board `GPIO26`	1
3	`A32_to_S3`	Audio board `GPIO27` → main board `GPIO14`	2

Shield: joined at both ends. Unlike the other shielded cables, this one has its shield and foil joined to ground at both boards.

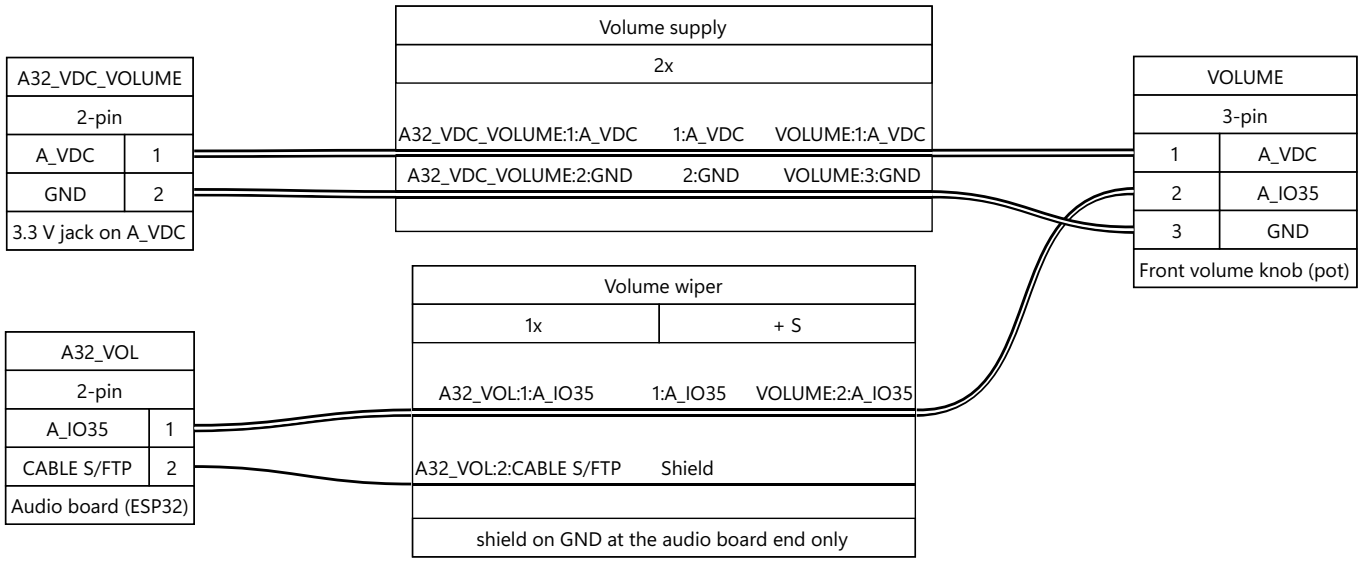
If it fails: the two boards can no longer talk to each other.

10.13 Front volume knob — audio board and 3.3 V to the volume knob

Carries: the front volume knob's position to the audio board. The knob turns a potentiometer (RV4, drawn as B10K) whose two ends sit on 3.3 V and ground; its wiper goes to the audio board's `GPIO35`. This is the audio board's own volume, set digitally. The amplifier's own volume control is a different pot, on the amplifier's board, reached only from the back.

Shape: two cables into one 3-pin connector, `VOLUME`, at the knob. A shielded 1-wire cable from `A32_VOL` on the audio board carries the wiper. A 2-wire cable from `A32_VDC_VOLUME` brings 3.3 V and ground. `A32_VDC_VOLUME` is a jack on the audio side's 3.3 V (`A_VDC`).

Figure 47. The front volume knob cables, by signal name, as on the cable sheet



Wire by wire.

Table 79. Front volume knob — audio board and 3.3 V to the volume knob

Audio board end, pin	Signal	What it is	Knob end, pin
1	A_IO35	The pot's wiper	2
2	CABLE S/FTP	Shield and foil	—

Table 80. Front volume knob — audio board and 3.3 V to the volume knob

3.3 V jack A32_VDC_VOLUME, pin	Signal	What it is	Knob end, pin
1	A_VDC	3.3 V, to the pot's top end	1
2	GND	Ground, to the pot's bottom end	3

Shield: joined to ground at the audio board end, left unconnected at the knob.

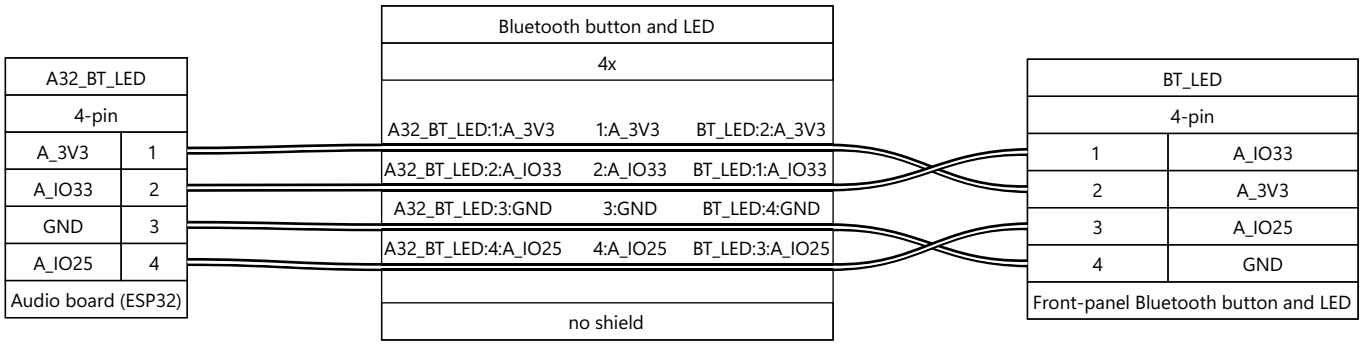
If it fails: the knob no longer changes the volume.

10.14 Bluetooth button and LED — audio board to front panel

Carries: the front-panel Bluetooth LED and the Bluetooth pairing button. On the front panel the LED is also the button. The LED's anode is on 3.3 V and its cathode on GPIO33: the audio board lights it by pulling GPIO33 low. No series resistor is drawn. The button connects GPIO25 to ground when pressed; GPIO25 uses the chip's internal pull-up, so pressed reads low.

Shape: one 4-wire cable, A32_BT_LED at the audio board, BT_LED at the front panel. The pins cross in pairs: 1 and 2 swap, 3 and 4 swap.

Figure 48. The Bluetooth button and LED cable, by signal name, as on the cable sheet



Wire by wire.

Table 81. Bluetooth button and LED — audio board to front panel

Audio board end, pin	Signal	What it is	Front panel end, pin
1	`A_3V3`	3.3 V to the LED's anode	2
2	`A_IO33`	The LED's cathode; low lights it	1
3	`GND`	The button's ground side	4
4	`A_IO25`	The button; pressed pulls it low	3

Shield: none.

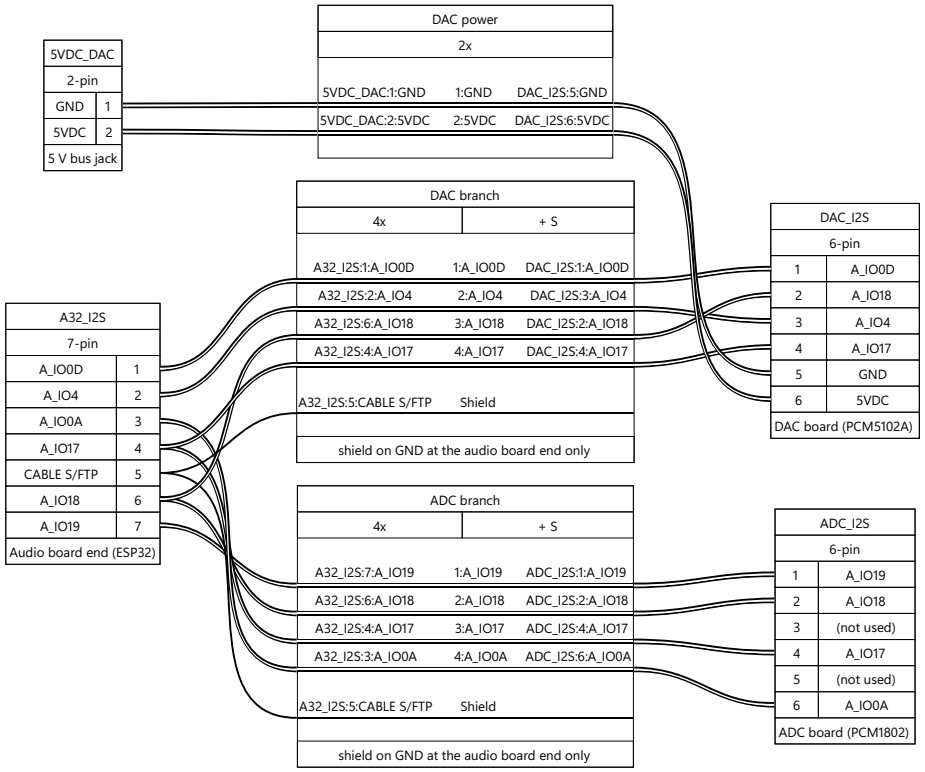
If it fails: a broken pin 1 or 2 wire leaves the LED dark; a broken pin 3 or 4 wire makes the button do nothing.

10.15 The audio harness — audio board to ADC and DAC

Carries: the digital audio (I2S) between the audio board (ESP32, "A32") and its two converters: the ADC that digitises the tube radio, and the DAC that feeds the amplifier. The DAC's 5 V power rides in the same harness.

Shape: one shielded harness that starts at the audio board and splits in two: one branch to the ADC board, one to the DAC board. A short power lead from the 5 V bus joins the DAC branch.

Figure 49. The audio harness, by signal name, as on the cable sheet



Wire by wire. Pin numbers are the cable sheet's.

Table 82. The audio harness — audio board to ADC and DAC

Audio board end, pin	Signal	What it is	ADC end, pin	DAC end, pin
1	A_IO0D	Master clock to the DAC: the audio board's GPIO0, through its own 33 Ω resistor at the board	—	1
2	A_IO4	Audio data to the DAC (GPIO4)	—	3
3	A_IO0A	Master clock to the ADC: the same GPIO0, through its other 33 Ω resistor	6	—
4	A_IO17	Word clock, left/right (LRCK), shared by both converters (GPIO17)	4	4
5	CABLE S/FTP	The harness's shield and foil, on ground at the audio board	—	—
6	A_IO18	Bit clock (BCK), shared by both converters (GPIO18)	2	2
7	A_IO19	Audio data from the ADC (GPIO19)	1	—

Table 83. The audio harness — audio board to ADC and DAC

5 V bus jack, pin	Signal	DAC end, pin
1	GND	5
2	5VDC	6

Shield: the shield and foil are joined to ground at the audio board end only, and left unconnected at both converter ends.

Not used: pins 3 and 5 at the ADC end carry nothing. The ADC takes its power from its own cable on the 5 V bus, not from this harness.

The two clock wires are the same signal. `A_IO0A` and `A_IO0D` both come from the audio board's GPIO0. They have two names only so that each converter's drawing reads clearly. Each has its own 33 Ω series resistor, at the audio board's pin.

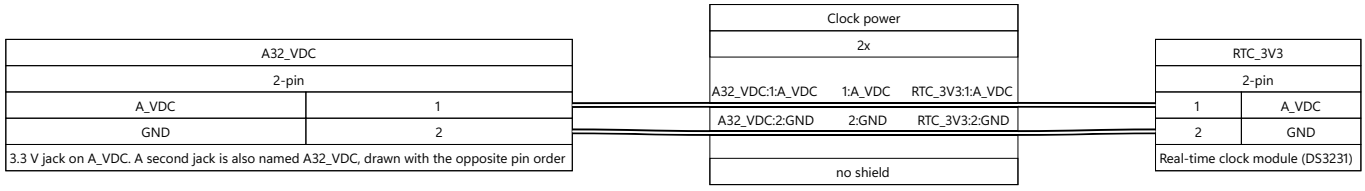
If it fails: no radio sound points to the ADC branch (pins 3, 4, 6, 7); no radio and no Bluetooth sound, while AUX still plays, points to the DAC branch (pins 1, 2, 4, 6) or its power lead. AUX never passes through this harness. Bluetooth plays through the DAC but not the ADC, so it tells the two branches apart.

10.16 Clock module power — 3.3 V jack to the real-time clock module

Carries: 3.3 V and ground to the real-time clock module (a DS3231 module, chapter 4 (p. 12)).

Shape: one 2-wire cable, from a 3.3 V jack the cable sheet names `A32_VDC`, on the audio side's 3.3 V (`A_VDC`), to `RTC_3V3` on the clock module. A second jack has the same name: the one that takes the audio board's 3.3 V cable (section 10.19 (p. 78)).

Figure 50. The clock module power cable, by signal name, as on the cable sheet



Wire by wire.

Table 84. Clock module power — 3.3 V jack to the real-time clock module

3.3 V jack, pin	Signal	Clock module end, pin
1	A_VDC	1
2	GND	2

Shield: none.

> NOTICE two jacks are both named `A32_VDC`, with opposite pin orders as the cable sheet draws them. This jack has pin 1 on 3.3 V (`A_VDC`) and pin 2 on ground. The other `A32_VDC` jack, which takes the audio board's 3.3 V cable (section 10.19 (p. 78)), has pin 1 on ground and pin 2 on 3.3 V (`A_3V3`, the same rail). The name alone does not tell you which jack you are holding, nor which way round its 3.3 V and ground are; check before you plug a cable into either one.

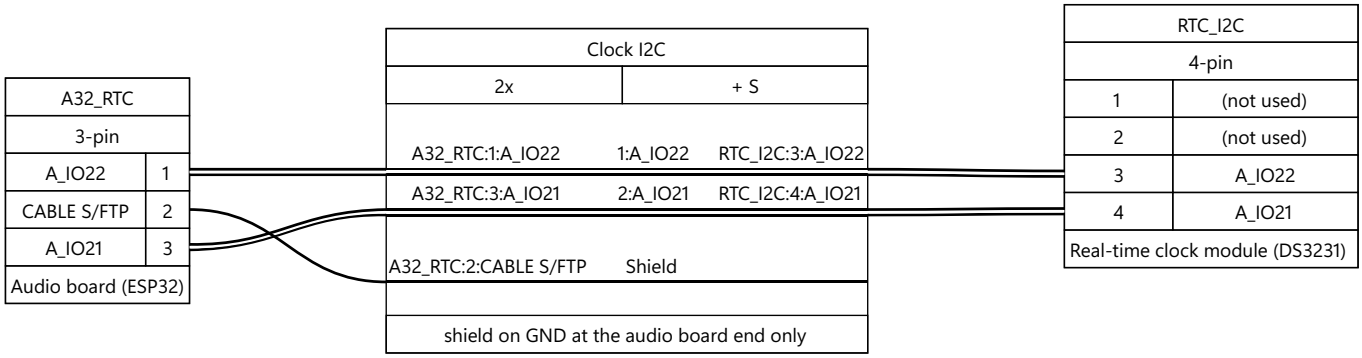
If it fails: the real-time clock module loses its 3.3 V supply.

10.17 Clock module control — audio board to the real-time clock module

Carries: the control bus (I2C) between the audio board and the real-time clock module: `A_IO22` is the data line (SDA), `A_IO21` the clock line (SCL).

Shape: one shielded 2-wire cable, `A32_RTC` at the audio board (3 pins) and `RTC_I2C` at the clock module (4 pins).

Figure 51. The clock module control cable, by signal name, as on the cable sheet



Wire by wire.

Table 85. Clock module control — audio board to the real-time clock module

Audio board end, pin	Signal	What it is	Clock module end, pin
1	`A_IO22`	Data line (SDA)	3
2	`CABLE S/FTP`	Shield and foil	—
3	`A_IO21`	Clock line (SCL)	4

Shield: joined to ground at the audio board end, left unconnected at the module.

Not used: pins 1 and 2 at the clock module end. Only two of that connector's four positions carry wires.

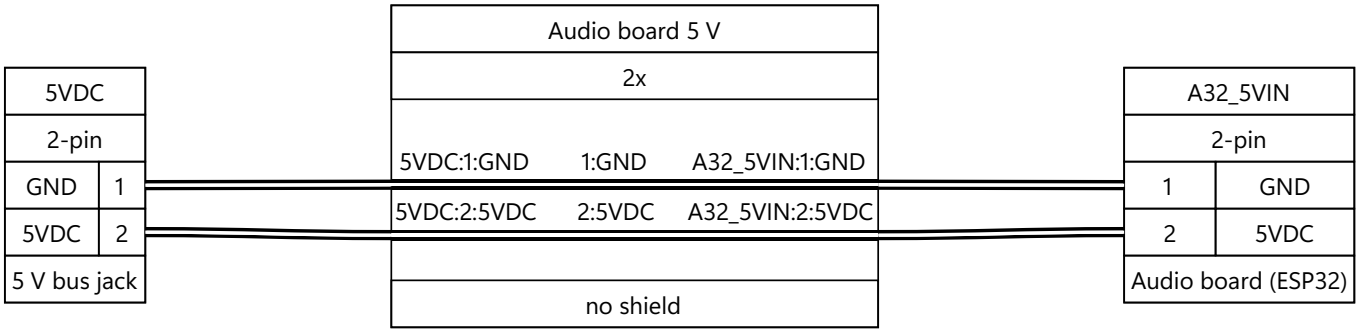
If it fails: the audio board can no longer talk to the real-time clock module.

10.18 Audio board 5 V — 5 V bus to audio board

Carries: 5 V and ground from the 5 V bus to the audio board's 5 V pin.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to `A32_5VIN` on the audio board.

Figure 52. The audio board's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 86. Audio board 5 V — 5 V bus to audio board

5 V bus jack, pin	Signal	Audio board end, pin
1	GND	1
2	5VDC	2

Shield: none.

If it fails: the audio board goes dead, and with it the 3.3 V it makes (section 10.19 (p. 78)). There is no sound from any source but AUX.

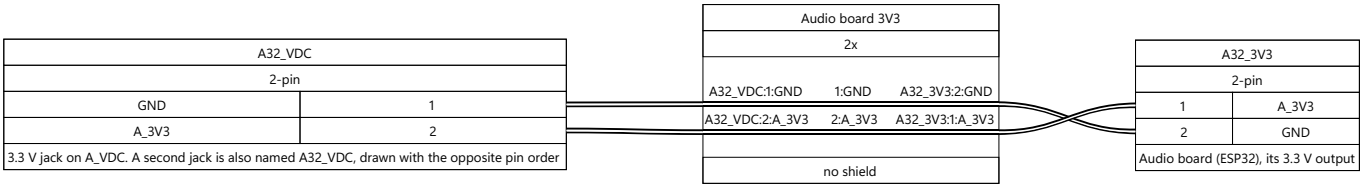
10.19 Audio board 3.3 V — audio board to the 3.3 V supply

Carries: the audio board's own 3.3 V output (`A_3V3`) and ground out to the 3.3 V bus (chapter 5 (p. 21), section 5.5 (p. 26)). The bus's 3.3 V is one wire with the 3.3 V jacks that feed the real-time clock module, the front volume knob and the source switch. The audio board decouples its 3.3 V pin with one 0.1 µF and two 10 µF capacitors.

Shape: one 2-wire cable, from `A32_3V3` on the audio board to a jack the cable sheet names `A32_VDC`, on the 3.3 V supply. The pins cross in the cable sheet's numbering: pin 1 at one end is pin 2 at the other.

Two jacks are named **`A32_VDC`**, with opposite pin orders as drawn. This one has pin 1 on ground and pin 2 on 3.3 V. The other, which feeds the real-time clock module (section 10.16 (p. 76)), has pin 1 on 3.3 V and pin 2 on ground. Both are on the same 3.3 V rail and the same ground. The caution in section 10.16 (p. 76) applies to both.

Figure 53. The audio board's 3.3 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 87. Audio board 3.3 V — audio board to the 3.3 V supply

A32_VDC end, pin	Signal	Audio board end, pin
1	GND	2
2	A_3V3	1

Shield: none.

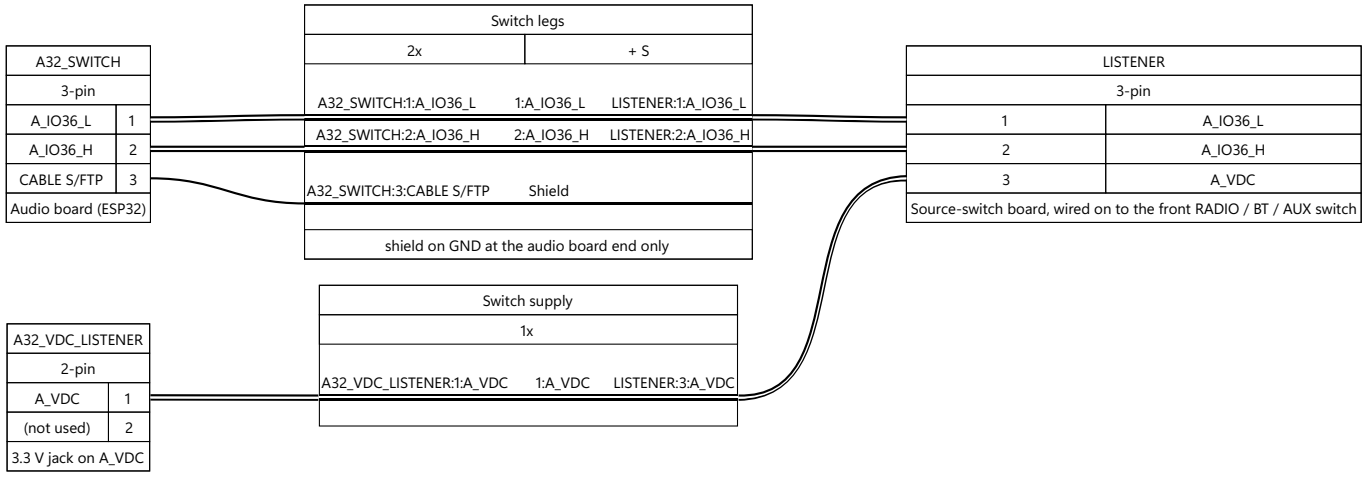
If it fails: the 3.3 V bus loses its supply, and every part fed from a bus position loses its 3.3 V.

10.20 Source switch — audio board and 3.3 V to the source-switch board

Carries: the position of the front RADIO / BT / AUX switch, as a voltage on the audio board's GPIO36. The switch puts 3.3 V on one of two legs: A_IO36_H, the ladder node itself, on RADIO (3.3 V), or A_IO36_L, which reaches the node through 68 kΩ, on BT (about 0.81 V). On AUX it feeds neither, and the node reads 0 V. The ladder's resistors are all on the audio board; chapter 9 (p. 52), section 9.3, has the circuit.

Shape: two cables into one 3-pin connector, LISTENER, on the source-switch board, a small board in its own printed case that the drawings call the IN/LISTENER/OUT board. A shielded 2-wire cable from A32_SWITCH on the audio board carries the two legs. A 1-wire cable from the 3.3 V jack A32_VDC_LISTENER brings 3.3 V. From the source-switch board, wires go on to the front switch. That last leg is not drawn as a cable; it is given by signal at the end of Wire by wire.

Figure 54. The source switch cables, by signal name, as on the cable sheet



Wire by wire.

Table 88. Source switch — audio board and 3.3 V to the source-switch board

Audio board end, pin	Signal	What it is	Switch board end, pin
1	A_IO36_L	The BT leg, through 68 kΩ	1
2	A_IO36_H	The RADIO leg, direct	2
3	CABLE S/FTP	Shield and foil	—

Table 89. Source switch — audio board and 3.3 V to the source-switch board

3.3 V jack **A32_VDC_LISTENER**, pin	Signal	What it is	Switch board end, pin
1	A_VDC	3.3 V for the switch	3
2	(not used)	—	—

The last leg, from the source-switch board to the front switch, by signal.

Table 90. Source switch — audio board and 3.3 V to the source-switch board

Signal	Front switch pin
A_VDC	2 and 5
A_IO36_H	1
A_IO36_L	6

Shield: joined to ground at the audio board end, left unconnected at the switch board.

Not used: the 3.3 V jack's ground pin (pin 2), so no ground goes to the switch; and the front switch's pins 3 and 4.

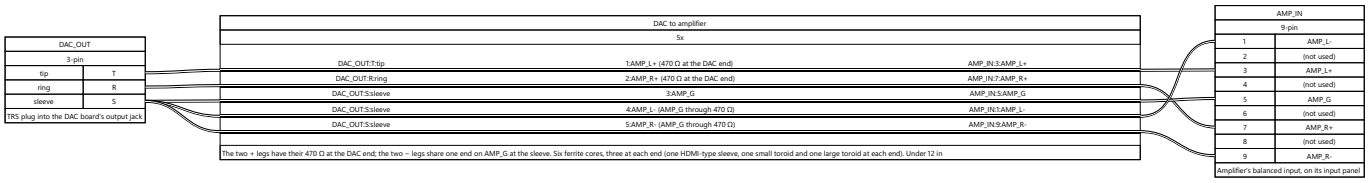
If it fails: with the A_IO36_H wire broken, RADIO reads as AUX. With the A_IO36_L wire broken, BT reads as AUX. With the 3.3 V wire broken, every position reads as AUX.

10.21 DAC to amplifier — DAC board to the amplifier's balanced input

Carries: the analogue audio from the DAC to the amplifier's balanced input (an input that takes each channel as a + and a − wire). The DAC gives left, right and ground; the cable feeds each channel to a + leg and ties each − leg to the DAC's ground through 470 Ω.

Shape: a 3-pole TRS plug, DAC_OUT (tip, ring, sleeve, like a headphone plug), goes into the DAC board's output jack. At the other end, a 9-pin connector AMP_IN plugs into the amplifier's input panel. The plug has three poles, and the amplifier end uses five pins. The cable carries four 470 Ω resistors, one in each signal leg. The two on the + legs sit at the DAC end. The two on the − legs share one end on AMP_G, at the plug's sleeve.

Figure 55. The DAC-to-amplifier cable, by signal name, as on the cable sheet



Wire by wire.

Table 91. DAC to amplifier — DAC board to the amplifier's balanced input

Plug pole	DAC signal	Wire	In the cable	Amplifier end, pin
Tip	left out (OUTL)	`AMP_L+`	470 Ω in series, at the DAC end	3
Ring	right out (OUTR)	`AMP_R+`	470 Ω in series, at the DAC end	7
Sleeve	analogue ground (AGND)	`AMP_G`	direct	5
Sleeve	analogue ground	`AMP_L−`	through its own 470 Ω	1
Sleeve	analogue ground	`AMP_R−`	through its own 470 Ω	9

Shield: none.

Not used: pins 2, 4, 6 and 8 at the amplifier end.

Ferrite cores. Six cores sit on this cable, three at each end: at each end, one HDMI-type sleeve, one small toroid and one large toroid. The cable passes once through each; it is under 12 inches long, so it makes no turns. The table gives the size of each kind of core:

Table 92. DAC to amplifier — DAC board to the amplifier's balanced input

Core	Size
HDMI-type sleeve	17.4 mm outside, 9.7 mm inside, 28.5 mm long (the same type as the antenna's feed-point cores, chapter 11 (p. 92))
Small toroid	16 mm outside, 3 mm wall, 8 mm long
Large toroid	22 mm outside, 4 mm wall, 8 mm long

The toroids are salvaged and unidentified. Their material is not that of the radio cable's clip-on cores (section 10.22 (p. 81)). The four resistors and the six cores are part of this cable.

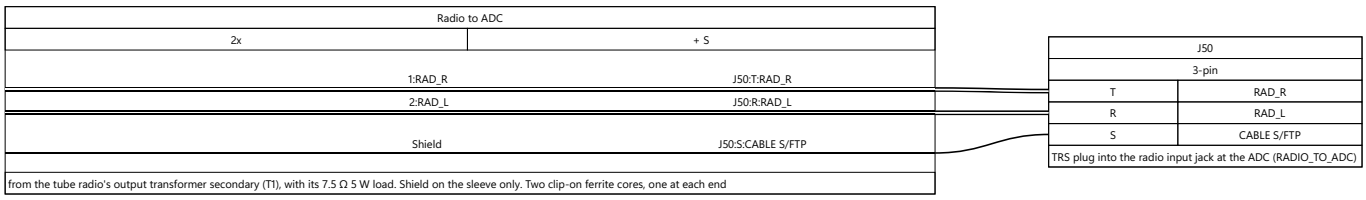
If it fails: radio and Bluetooth go silent at the amplifier while AUX still plays. AUX reaches the amplifier's input panel on its own path and never goes through this cable.

10.22 Radio to ADC — tube radio's output transformer to the ADC

Carries: the tube radio's audio, from the secondary of its output transformer (T1), to the ADC. A 7.5 Ω 5 W resistor (marked 5W7Ω5J) is wired across that secondary as its load. The feed is mono.

Shape: one shielded 2-wire cable. At the ADC end, a 3-pole TRS plug, `J50`, goes into the radio input jack at the ADC (J51, `RADIO_TO_ADC`). At the radio end, the two wires come from the transformer secondary.

Figure 56. The radio-to-ADC cable



Wire by wire.

Table 93. Radio to ADC — tube radio's output transformer to the ADC

Plug pole	Signal	What it is	Radio end
Tip	RAD_R	One side of the transformer secondary	T1 terminal SB
Ring	RAD_L	The other side	T1 terminal SA
Sleeve	CABLE S/FTP	Shield and foil	not connected

The jack at the ADC (J51). Cable side: shield and foil on the sleeve, RAD_L on the ring, RAD_R on the tip. Output side: sleeve and ring on ground, tip into the 4.7 kΩ of the input attenuator (chapter 6 (p. 28), section 6.4 (p. 33)).

Shield: on the plug's sleeve, which is ground at the ADC end. It is left unconnected at the radio end.

Tip and ring can be swapped. The feed is mono: at the ADC one side goes to ground, the other into the input attenuator.

Ferrite cores. Two clip-on cores sit on this cable, one near the transformer end and one near the ADC end. The cable passes once through each. Measured: 10 mm outside, 3.5 mm inside, 3.5 mm walls, 20 mm long. One of these figures is rounded: the hole and the two walls make 10.5 mm, against 10 mm outside. Their material, as supplied: rectangle ratio 20, coercivity 16 A/m, remanence 200 mT, Curie temperature 130 °C, density 4.9 g/cm³, "soft magnetic".

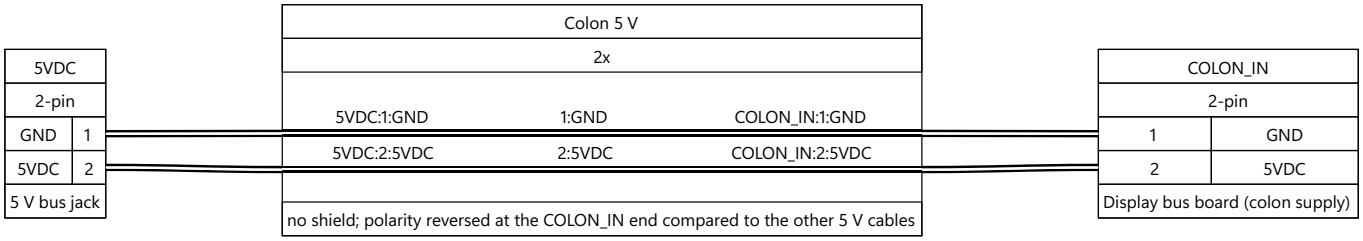
If it fails: the radio goes silent while Bluetooth still plays.

10.23 Colon 5 V — 5 V bus to the clock display's colon

Carries: 5 V and ground to the clock display's colon. At the colon, the 5 V goes through 33 Ω, the colon LED and a variable resistor glued at 1.5 kΩ to ground (section 10.34 (p. 90)).

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to COLON_IN on the display bus board, which feeds the colon.

Figure 57. The colon's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 94. Colon 5 V — 5 V bus to the clock display's colon

5 V bus jack, pin	Signal	**COLON_IN**, pin
1	GND	1
2	5VDC	2

Shield: none.

> NOTICE this cable's polarity is reversed at the COLON_IN side, compared with the other 5 V cables. Check polarity before you plug it in.

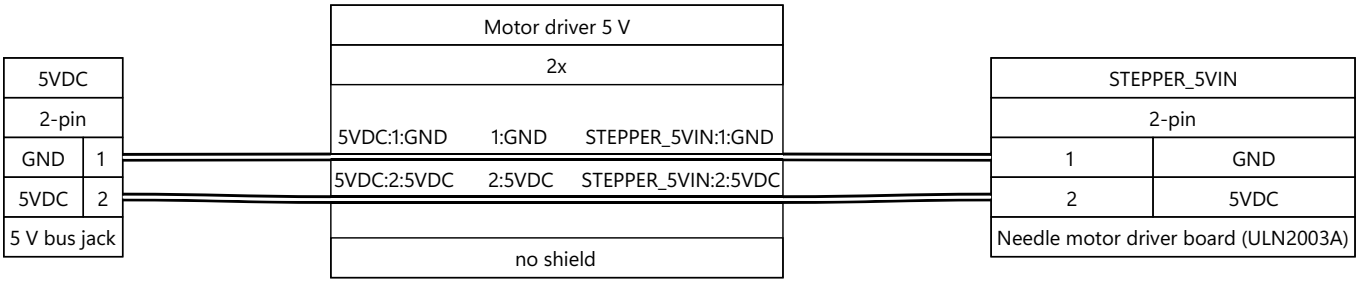
If it fails: the colon stays dark while the digits light.

10.24 Motor driver 5 V — 5 V bus to the needle motor driver

Carries: 5 V and ground to the needle motor driver board. The 5 V feeds the ULN2003A's common pin and the motor's common wire, with 0.1 µF and 47 µF to ground.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to STEPPER_5VIN on the driver board.

Figure 58. The needle motor driver's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 95. Motor driver 5 V — 5 V bus to the needle motor driver

5 V bus jack, pin	Signal	Driver board end, pin
1	GND	1
2	5VDC	2

Shield: none.

If it fails: the needle motor has no power and the needle does not move.

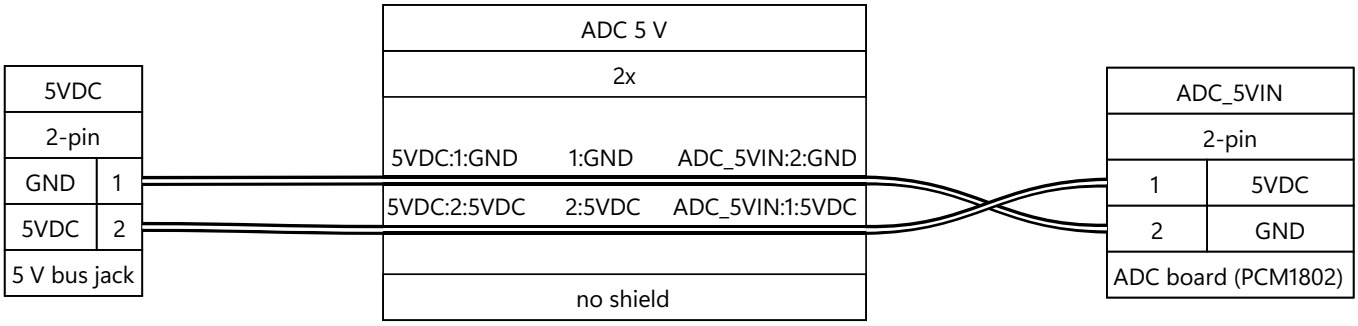
10.25 ADC 5 V — 5 V bus to the ADC board

Carries: 5 V and ground to the ADC board (PCM1802). The ADC module's 5 V pin has 100 µF and 0.1 µF to ground.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to ADC_5VIN on the ADC board. The pin numbers differ at the two ends in the cable sheet's numbering: at the ADC end, pin 1 is 5 V and pin 2 is ground, reversed from the bus-jack end. This cable's pins do swap end to end, but the cable sheet

carries no caution note for it. The Colon 5 V cable is the other way round: its pin numbers match end to end (section 10.23 (p. 82)), yet a note on the sheet calls its polarity reversed.

Figure 59. The ADC's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 96. ADC 5 V — 5 V bus to the ADC board

5 V bus jack, pin	Signal	ADC end, pin
1	GND	2
2	5VDC	1

Shield: none.

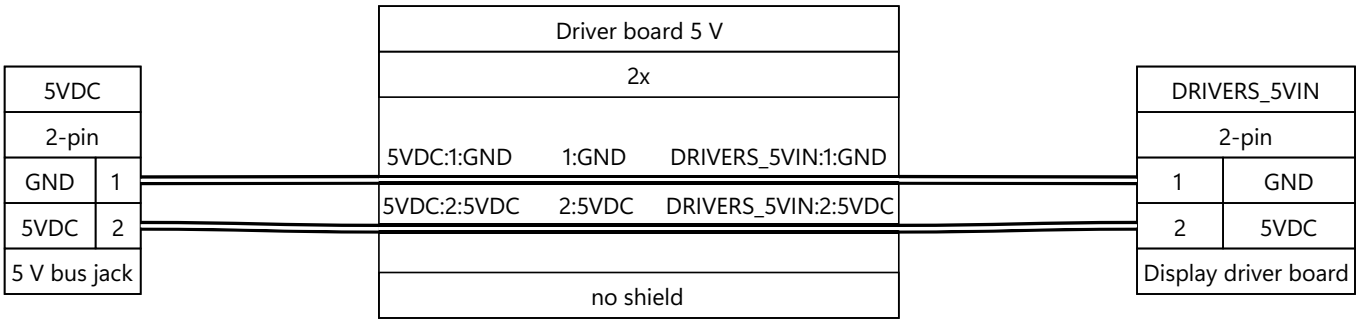
If it fails: the radio goes silent while Bluetooth still plays.

10.26 Driver board 5 V — 5 V bus to the display driver board

Carries: 5 V and ground to the display driver board. The 5 V is what the four digit switches deliver to the digits (section 10.27 (p. 85)), and the ground is the return of both ULN2003A drivers.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to `DRIVERS_5VIN` on the driver board.

Figure 60. The display driver board's 5 V cable, by signal name, as on the cable sheet



Wire by wire.

Table 97. Driver board 5 V — 5 V bus to the display driver board

5 V bus jack, pin	Signal	Driver board end, pin
1	GND	1
2	5VDC	2

Shield: none.

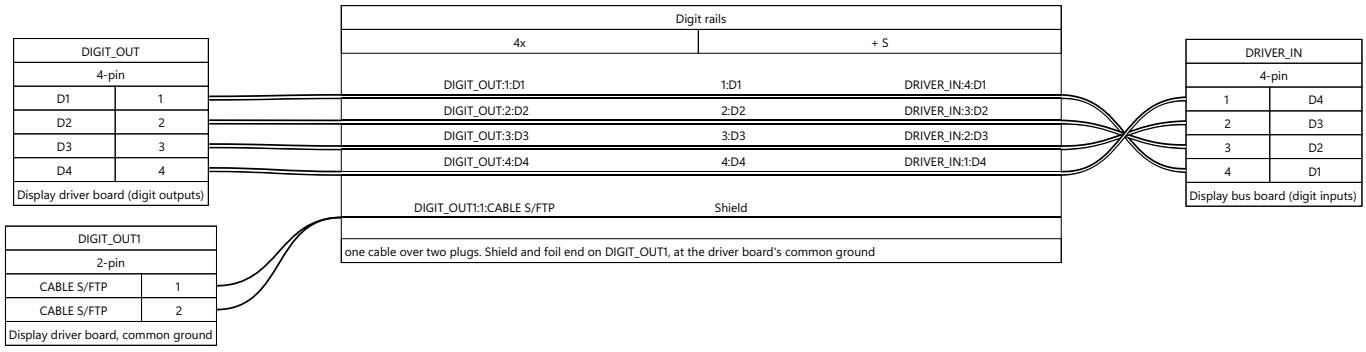
If it fails: the whole clock display goes dark; the colon has its own cable (section 10.23 (p. 82)).

10.27 Digit rails — display driver board to the display bus board

Carries: each digit's switched 5 V. `D1` to `D4` are the drains of the four P-channel MOSFETs on the driver board; each feeds the common anode of one digit, digit 1 being the rightmost.

Shape: one shielded 4-wire cable that ends in two connectors at the driver board: `DIGIT_OUT` (4 pins) for the wires, and `DIGIT_OUT1` (2 pins) for the shield and the foil. At the display bus board it ends in `DRIVER_IN`. The pin order runs opposite at the two ends.

Figure 61. The digit rail cable, by signal name, as on the cable sheet



Wire by wire.

Table 98. Digit rails — display driver board to the display bus board

Driver board end, pin	Signal	What it is	Display bus board end, pin
`DIGIT_OUT` 1	`D1`	5 V for digit 1 (rightmost)	4
`DIGIT_OUT` 2	`D2`	5 V for digit 2	3
`DIGIT_OUT` 3	`D3`	5 V for digit 3	2
`DIGIT_OUT` 4	`D4`	5 V for digit 4 (leftmost)	1
`DIGIT_OUT1` 1 and 2	`CABLE S/FTP`	Shield and foil	—

Shield: the shield and the foil end on the two pins of `DIGIT_OUT1`, at the driver board's common ground.

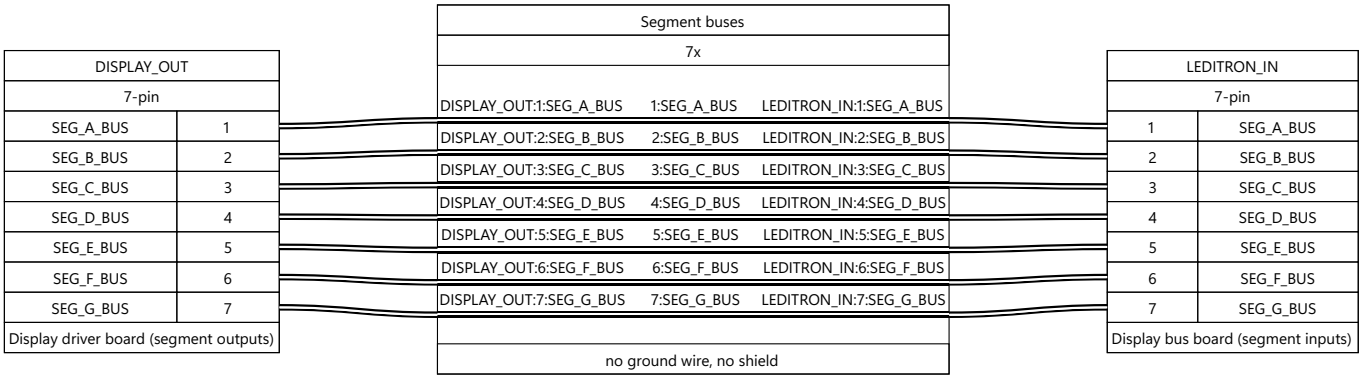
If it fails: a broken wire leaves one digit dark.

10.28 Segment buses — display driver board to the display bus board

Carries: the seven segment lines, A to G. Each is the cathode line of one segment on all four digits. The segment driver sinks it to ground through a 10 Ω and a 200 Ω resistor in series (chapter 7 (p. 38)).

Shape: one 7-wire cable, pin for pin: `DISPLAY_OUT` at the driver board, `LEDITRON_IN` at the display bus board.

Figure 62. The segment bus cable, by signal name, as on the cable sheet



Wire by wire.

Table 99. Segment buses — display driver board to the display bus board

Driver board end, pin	Signal	Display bus board end, pin
1	SEG_A_BUS	1
2	SEG_B_BUS	2
3	SEG_C_BUS	3
4	SEG_D_BUS	4
5	SEG_E_BUS	5
6	SEG_F_BUS	6
7	SEG_G_BUS	7

Shield: none. There is no ground wire.

If it fails: a broken wire leaves one segment dark on all four digits, like a broken segment drive wire (section 10.5 (p. 65)).

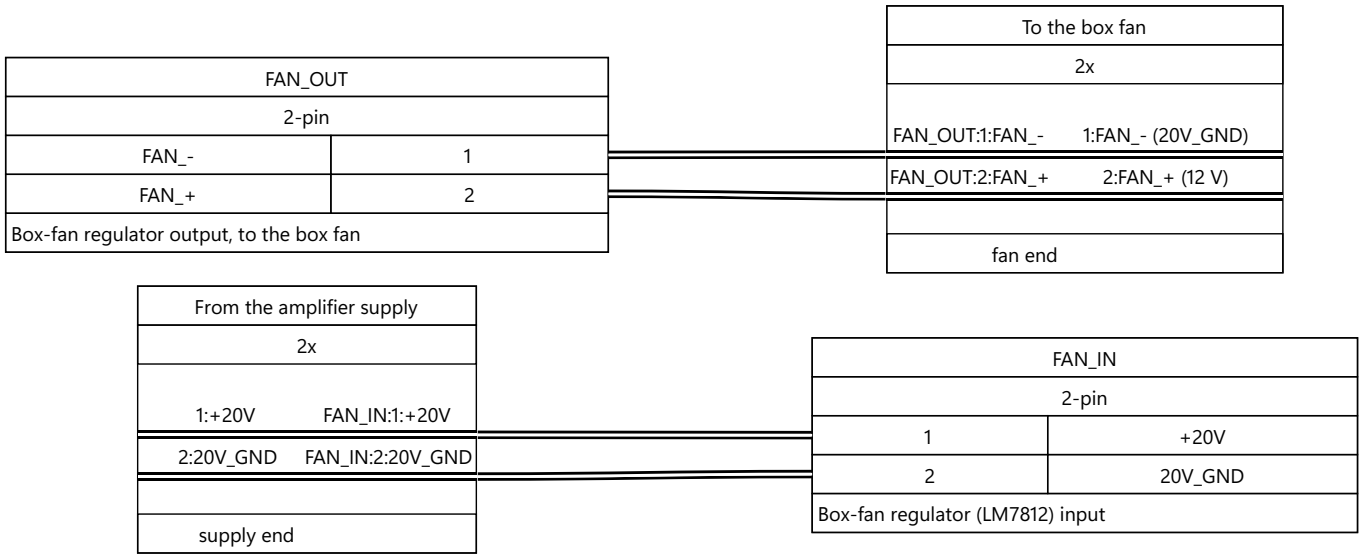
10.29 Box-fan harness — amplifier supply to the box fan

This page and the ones after it describe wiring that is not on the cable sheet (section 10.1 (p. 60)).

Carries: the amplifier supply's +20 V to the box fan's regulator, and the regulator's 12 V to the box fan. The box fan runs on 12 V made from the amplifier supply's +20 V by an LM7812 regulator (chapter 5 (p. 21), section 5.6 (p. 27)).

Shape: two 2-pin connectors. `FAN_IN` carries +20 V and its return to the regulator's input. `FAN_OUT` carries the regulator's 12 V output and its return to the fan.

Figure 63. The box-fan harness



Wire by wire.

Table 100. Box-fan harness — amplifier supply to the box fan

Connector, pin	Signal	What it is
`FAN_IN` 1	`+20V`	The regulator's input
`FAN_IN` 2	`20V_GND`	The amplifier supply's return
`FAN_OUT` 1	`FAN_−`	The fan's −, on `20V_GND`
`FAN_OUT` 2	`FAN_+`	The fan's +, the regulator's 12 V output

The fan runs only while the front power switch is on.

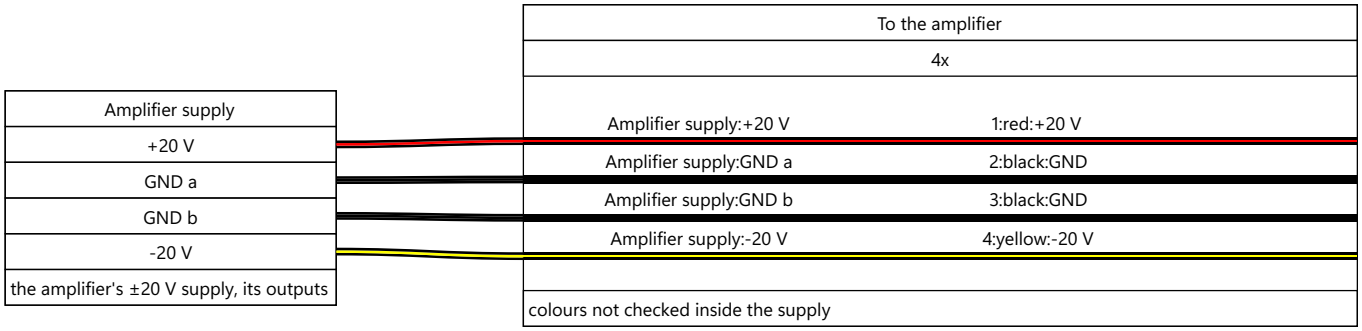
If it fails: the box fan stops while the radio plays (chapter 5 (p. 21), section 5.7 (p. 28)).

10.30 Amplifier supply wires — amplifier supply to the amplifier

Carries: the amplifier's ±20 V supply, from the supply to the amplifier.

Shape: four wires.

Figure 64. The amplifier supply's four wires



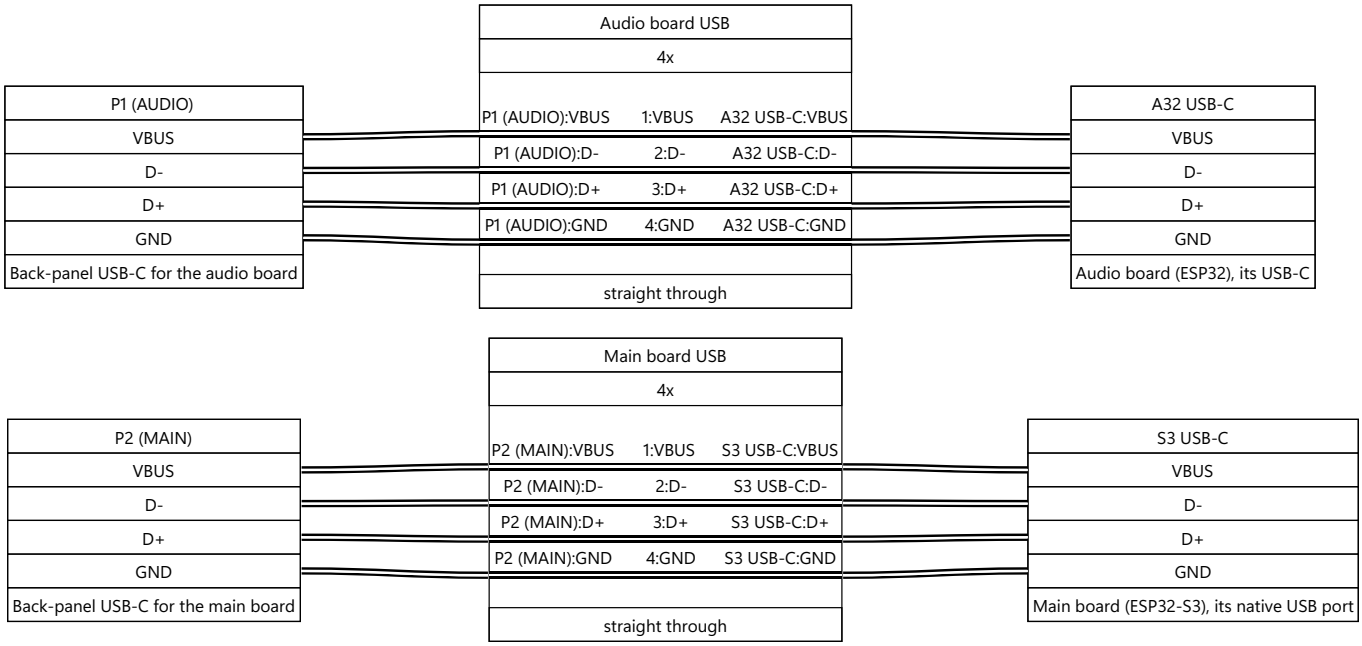
Wire by wire. The drawing gives each wire's colour and what it carries; chapter 5 (p. 21), section 5.6 (p. 27), has the same as a table. These are the only wires in the machine whose colours are recorded. The colours are not checked inside the supply.

10.31 USB — back panel to each board

Carries: USB, from the back panel to each board.

Shape: two USB-C connectors on the back panel lead to the two boards. `P2 (MAIN)` goes to the main board's native USB port. `P1 (AUDIO)` goes to the audio board's USB-C.

Figure 65. The two USB connections, straight through



Wire by wire. Each connection carries all four lines straight through: `VBUS`, `D−`, `D+` and `GND`. On each back-panel connector the other three ground pins (A12, B1 and B12) are joined together.

NOTICE VBUS is carried through, so a computer plugged into the back panel feeds a board's USB power input while the 5 V bus feeds its 5V pin. Chapter 2, section 2.5 (p. 9), quotes Espressif's guides on this: powering a board from both at once risks damaging it.

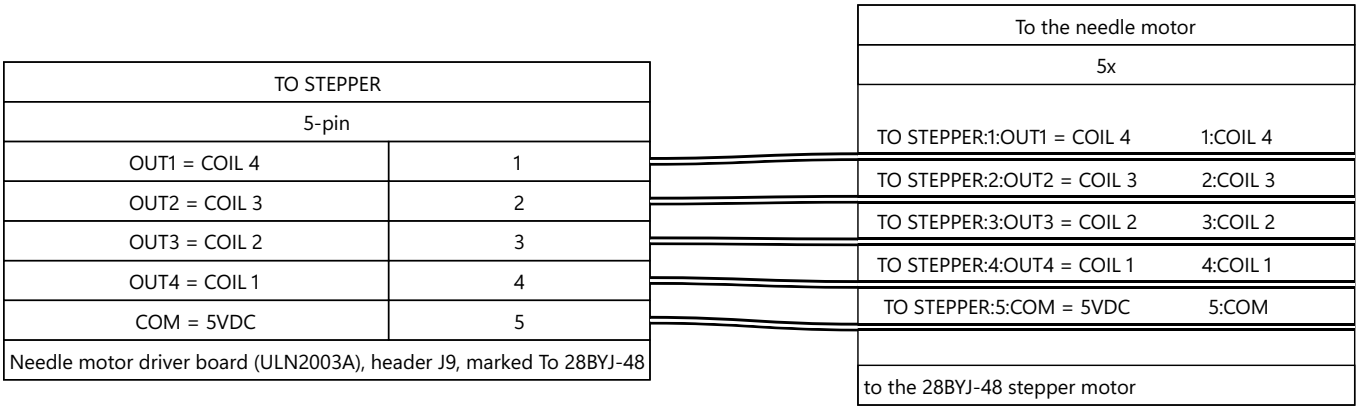
If it fails: that board can no longer be reached over USB from the back panel.

10.32 Needle motor lead — needle motor driver to the needle motor

Carries: the four coil drives of the needle motor, a 28BYJ-48 stepper, and the coils' common. The driver outputs sink current to ground; the coils' common sits on 5 V.

Shape: the motor plugs into a 5-pin header on the needle motor driver board, marked "To 28BYJ-48" (J9). It is the connector that comes with the common ULN2003A boards.

Figure 66. The needle motor header



Wire by wire.

Table 101. Needle motor lead — needle motor driver to the needle motor

Header pin	Signal	What it is
1	OUT1	Driver output 1, to coil 4
2	OUT2	Driver output 2, to coil 3
3	OUT3	Driver output 3, to coil 2
4	OUT4	Driver output 4, to coil 1
5	COM	The coils' common, on 5 V

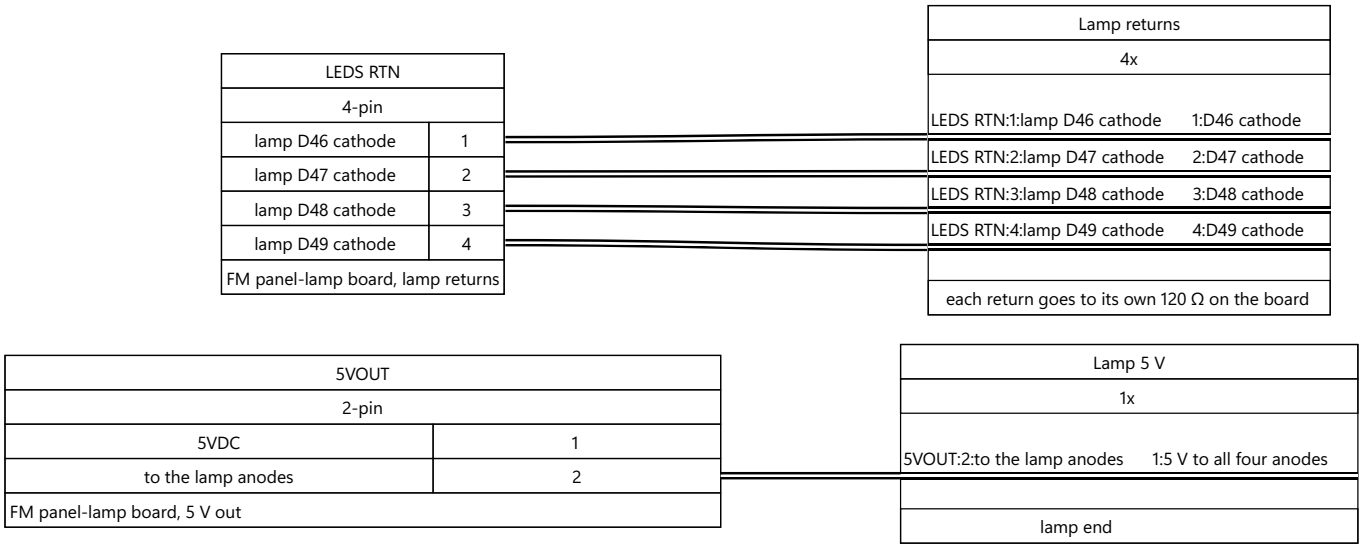
If it fails: the needle does not move, or moves erratically (chapter 8 (p. 46), section 8.7 (p. 52)).

10.33 Panel-lamp leads — panel-lamp board to the four FM lamps

Carries: 5 V to the four FM panel lamps (LEDs), and each lamp's return to the panel-lamp board.

Shape: two connectors on the panel-lamp board. 5VOUT (2 pins) carries 5 V to the four lamp anodes. LEDS RTN (4 pins) brings each lamp's cathode back to its own 120 Ω resistor on the board. The four resistors join on the MOSFET's drain (section 10.11 (p. 71)).

Figure 67. The panel-lamp leads



Wire by wire. On 5VOUT, as drawn, pin 1 is on the board's 5 V and pin 2 goes to the four anodes. On LEDS RTN, each pin takes one lamp:

Table 102. Panel-lamp leads — panel-lamp board to the four FM lamps

LEDS RTN pin	What it is
1	Return from lamp D46′s cathode
2	Return from lamp D47′s cathode
3	Return from lamp D48′s cathode
4	Return from lamp D49′s cathode

If it fails: a broken return leaves its one lamp dark; a broken 5 V lead leaves all four dark (chapter 9 (p. 52), section 9.6 (p. 60)).

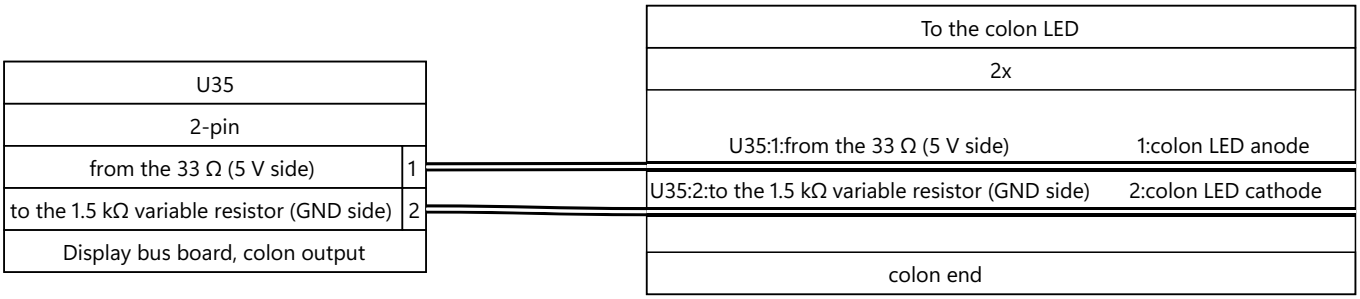
10.34 Inside the clock display — colon and digits

Carries: the colon LED's two leads, and each digit's seven segment lines and its common anode. The colon's chain is 5 V → 33 Ω → colon LED → variable resistor (glued at 1.5 kΩ) → ground; the variable resistor sits at the colon module. Each digit is built of seven segment LEDs whose anodes join on pin 8 of its 8-pin header, with each segment's cathode on pins 1 to 7.

Shape — the colon. On the display bus board, the colon output (U35) has two pins.

Shape — the digits. Each digit's header is drawn plugged into its own 8-pin socket, pin for pin; no cable is drawn. The display bus board has four 8-pin digit outputs, which carry the same signals on the same pin numbers.

Figure 68. The colon output



Wire by wire — the colon. Pin 1 of the colon output comes from the 33 Ω, and pin 2 goes to the variable resistor. The colon LED sits between them, anode on pin 1.

Wire by wire — the digits. Pins 1 to 7 carry the same signal on every socket and every output; pin 8 is each digit's own common:

Table 103. Inside the clock display — colon and digits

Pin	Signal on the socket and on the output	At the digit
1	SEG_A_BUS	Segment A cathode
2	SEG_B_BUS	Segment B cathode
3	SEG_C_BUS	Segment C cathode
4	SEG_D_BUS	Segment D cathode
5	SEG_E_BUS	Segment E cathode
6	SEG_F_BUS	Segment F cathode
7	SEG_G_BUS	Segment G cathode
8	D1, D2, D3 or D4	Common anode

Pin 8 is D1 on output 1 (U18), D2 on output 2 (U17), D3 on output 3 (U19) and D4 on output 4 (U32). Digit 1 is the rightmost.

If it fails: a broken colon lead leaves the colon dark. A bad contact on one pin of a digit's header leaves that segment dark in that digit only; on pin 8, the whole digit goes dark (chapter 7 (p. 38), section 7.8 (p. 46)).

10.35 FM receiver antenna and the antenna coax

Neither of these is drawn as a cable, so this page has no drawing and no wire table.

Carries: two antenna signals: the FM calibration receiver's antenna wire, and the FM antenna's coax into the machine.

Shape — the receiver's antenna. The FM calibration receiver's antenna is a 30 cm wire inside the tube radio's Faraday cage, bent in a U over the radio's FM oscillator section. It reaches the receiver board through a 2×3 pin grid (U71), so that you can unplug the receiver from its antenna during a repair.

Shape — the antenna coax. The antenna's coax arrives at a BNC male connector (J10) on the back panel: its centre carries the antenna signal ANT,

and its shell is on Earth. The tube radio's antenna input is a second coax connector (J53); chapter 13 (p. 98), section 13.11 (p. 138), gives the radio's side of it. No cable joining J10 to J53 is drawn. The antenna itself is in chapter 11 (p. 92) the earth path is in chapter 5 (p. 21), section 5.4 (p. 25).

If it fails: if the receiver's antenna wire or its pin grid fails, the receiver still answers the main board but finds no signal (chapter 8 (p. 46), section 8.7 (p. 52)).

10.36 Mains and earth wiring

The mains wiring (inlet, front power switch, 5 V supply, amplifier supply, line filter and tube radio) and the earth wiring (the inlet's earth, the amplifier's back plate, the three Faraday cages and the antenna's earth) are described in chapter 5 (p. 21), sections 5.2 (p. 23) to 5.4. They are not drawn here.

10.37 When something is wrong

Find what you see in the left column; the right column names the cable pages to check.

Table 104. When something is wrong

You see	Check
A cable needs replacing	Each wire is identified by its signal name, not by its position (section 10.1 (p. 60)). Each page says at which end the shield is joined to ground.
One segment dark on every digit	Segment drive (10.5), then segment buses (10.28)
One digit dark	Digit select (10.8), then digit rails (10.27)
Whole display dark, colon lit	Driver board 5 V (10.26)
Colon dark, digits lit	Colon 5 V (10.23), and its reversed polarity
Needle does not move	Needle motor control (10.6), motor driver 5 V (10.24), needle motor lead (10.32)
Needle never finds its reference point	Needle index sensor harness (10.9)
Main board reads "amplifier off" with the amplifier on	Amplifier power sensor cable (10.7)
Panel lamps dark	Panel-lamp control (10.11), then the lamp leads (10.33)
Radio silent, Bluetooth plays	Radio to ADC (10.22), ADC 5 V (10.25), the ADC branch of the audio harness (10.15)
Radio and Bluetooth silent, AUX plays	DAC to amplifier (10.21), the DAC branch of the audio harness and its power lead (10.15)
Source switch stuck on AUX in one or all positions	Source switch cables (10.20)
Volume knob does nothing	Front volume knob cables (10.13), audio board 3.3 V (10.19)
Bluetooth LED dark, or button dead	Bluetooth button and LED cable (10.14)
The two boards do not talk	Board link (10.12)

11. The FM antenna

The tube radio receives FM from an outdoor antenna built for this machine: a three-element Yagi on the roof. Any FM or VHF antenna will work on the radio's antenna input. This chapter describes this one, as built and installed,

so that you can check it, repair it or build your own. Its dimensions, mounting and aim suit this house and this location. Adapt them to yours.

Sections 11.1 and 11.2 give the figures of this antenna as built. Section 11.3 gives further details from the build notes; those details are not confirmed one by one, so they are weaker than 11.1 and 11.2.

11.1 The aerial

A Yagi is a directional antenna: one driven element, the only part wired to the cable, with unpowered parasitic elements beside it on a common support, the boom. The longer reflector sits behind the driven element and the shorter director in front, so the antenna listens best in the direction of the director.

This one is a three-element Yagi, mounted horizontally on the roof. The table gives each part's material and size.

Table 105. The aerial

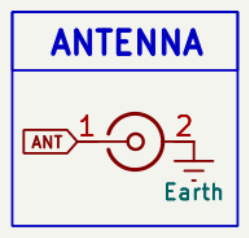
Part	Material	Size
Director	plain aluminium U-channel, 3.2 mm thick, 12.6 mm high, 25.5 mm wide	133.8 cm long
Driven element	½ in type L copper tube, folded (below)	139 cm per element; about 1390 mm tip to tip
Reflector	the same aluminium U-channel as the director	150.9 cm long
Boom	3030 aluminium extrusion	1170 mm long

The driven element is a folded dipole: two parallel copper tubes, joined at both ends by soldered 90-degree tubing, with the two conductors 50 mm apart, centre to centre. Only this element is wired to the coax.

Nothing touches the boom electrically. The director and the reflector touch neither the boom nor anything else. The driven element reaches the boom only through a non-conductive clamp.

11.2 The feedline and its earthing

Figure 69. Schematic: the antenna input, as drawn on the main sheet



The feedline is the coax cable that carries the signal from the antenna down to the radio. The table gives how the feedline, its earthing and the mast are installed.

Table 106. The feedline and its earthing

What	As installed
Feedline	Coax. At present it is at least 100 ft long. At its final place it is to be about 50 ft in total, from the bulkhead (the F-81 connector at the feedpoint, section 11.3 (p. 94)) to the radio.
Grounding block	Fitted outside, out of the weather. Its exact position does not matter.
Earth wire	8 AWG stranded copper, ferruled, run from the grounding block to the electricity meter box (Hydro-Québec) and landed on a lug on the meter base, beside one an electrician made for the external wiring.
Mast	The same 3030 aluminium extrusion as the boom, 4 ft high, fixed to the roof with continuous aluminium extrusion. The mast is centred on the boom.
Height and aim	About 4 ft over its base and 1 ft over the tip of the roof, clear of reflections from the roof. The roof's slope faces away from Mount Royal; the aerial points as near Mount Royal as it can.
Guys	Dacron lines, two from each end of the boom. As fitted, the antenna cannot move short of a very big accident.

The coax shield is earthed at the grounding block. The grounding block is outside. An 8 AWG stranded copper run goes from the grounding block to a lug on the meter base.

The boom is earthed through the structure. The boom is continuous metal through the mast to the roof's metal. The metal roof is electrically connected to the electrical mast, and so to earth. The boom and the coax shield's 8 AWG run therefore reach the same earth electrode system.

What follows from this. The elements sit on insulating bushings (section 11.3 (p. 94)), so they are isolated from that earthed structure. The coax shield is earthed by its own 8 AWG run from the grounding block to the meter-base lug, not by a separate wire from the boom.

At the radio end. The coax arrives at the machine's antenna connector, a BNC (J10) on the back panel. Its centre carries the antenna signal, ANT its shell, and so the coax shield, is on Earth. The tube radio's own antenna input is another coax connector (J53); no cable joining J10 to J53 is drawn. Chapter 5, section 5.4 (p. 25), gives the earth path inside the cabinet. Chapter 13, section 13.11 (p. 138), gives the radio's side, where the shield reaches the chassis only through one safety capacitor (C51).

> ⚠ WARNING the coax shield is on Earth, but the radio's chassis is on mains neutral and is never joined to Earth. The shield's only path to the radio's chassis is the safety capacitor (C51). Chapter 2 explains what the chassis is joined to and why it matters.

11.3 Build details from the notes

These details come from the build notes and are not confirmed one by one.

Element positions. The table gives where each element sits along the boom, with the reflector at 0 cm.

Table 107. Build details from the notes

Element	Position	Unit
Reflector	0	cm
Driven element	55	cm
Director	110	cm

Elements. Both U-channels are mounted opening downwards. The centres of the three elements lie in one plane, within 5 mm.

Feed gap. A 20 mm gap is cut in one tube of the folded dipole only. The feedpoint is there.

How each element is mounted. Each element sits in a stack of parts on the boom, listed here from the top down:

1. a nut;
2. a Belleville washer (a dished spring washer);
3. a neoprene washer, of the kind used with metal-roofing screws;
4. a G10 washer (glass-epoxy laminate);
5. the element, in an ASA bushing;
6. an ASA plate, continuous with that bushing;
7. the boom, through a 90-degree clamp.

The driven element is mounted the same way. No static-bleed resistors are fitted. These would be 1 MΩ resistors from the parasitic elements to the boom, to let static charge drain away.

Feedpoint. The feedpoint is where the coax joins the driven element. An F-81 bulkhead connector (an F-type coax connector made to pass through a panel) sits in the ASA clamp. Its shell is on one side of the 20 mm gap, its centre pin on the other. Six salvaged HDMI ferrite sleeves (17.4 mm outside, 9.7 mm inside, 28.5 mm long) sit on the feedpoint as a common-mode choke, to keep signal currents off the outside of the coax shield.

Feedline and grounding block. The feedline is RG-6 coax. The grounding block is a Perfect Vision PVGB1HFWS: UL listed, rated to 3 GHz, F-type, single, with a weather boot. It is a plain block. It has no gas discharge tube (no surge arrester).

11.5 When something is wrong

The table lists what you may notice and where to look first.

Table 108. When something is wrong

You see	Check
FM weak or noisy on every station, while AM and SW work	The FM antenna path, which only FM uses: the antenna connector (J10), the coax inside the cabinet, the radio's antenna input (J53) and its coupling capacitor (C3) (chapter 13 (p. 98), section 13.2 (p. 105)). Then the coax outside, the grounding block and the feedpoint.
The aerial has been moved or re-aimed	Its aim (as near Mount Royal as it can, here) and its height clear of the roof (section 11.2 (p. 93)).

12. The amplifier and speakers

Every sound the machine makes leaves through one amplifier and its two speakers. The DAC's output (the RADIO and BT sources) and the AUX input both arrive at the amplifier's input panel, which adds them together. This chapter says what the amplifier and the speakers are, what the input panel does, and what has been measured there. It describes this speaker set; adapt it to yours.

The amplifier's own circuit is not covered. No drawing shows its boards.

12.1 What the amplifier and speakers are

The amplifier and speakers came from a Kramer Tavor 5-O powered speaker pair, part number 60-00009010: active left, passive right, 2 × 30 W RMS, 4 Ω. The active speaker held the amplifier; the passive one did not. The table lists the parts.

Table 109. What the amplifier and speakers are

Part	What it is
Power stage	a TDA7265
Input board	three op-amp chips: a TL074, a TL072 and a C4558
Supply	the amplifier's own isolated mains switching supply, measured at ±19.33 V with no load. Two TO-220 regulators (a common three-pin power package) make ±12 V for the op-amps. It reaches the amplifier on four wires, the only wires in the machine whose colours are recorded (chapter 5 (p. 21), section 5.6 (p. 27)). It also feeds the box fan and the amplifier power sensor (chapter 5 (p. 21), section 5.6 (p. 27) chapter 9 (p. 52), section 9.5 (p. 59)).
Auto mode	a mode of the original amplifier; it has been removed.
Volume control	a B10k dual-gang potentiometer (section 12.2 (p. 96))
Speakers	on each side, a 5.25 in polypropylene woofer and a 0.5 in Mylar dome tweeter, with a first-order passive crossover at 8 kHz

Passive radiators, not ports. The original speaker cabinets were ported. In this machine the speakers work with passive radiators, added for this build, and are not ported. A passive radiator is a cone with no magnet or coil of its own. The air in the enclosure moves it, where a port would let that air out through a tube. Its printed parts (a holder, two protectors, a membrane weight and a TPU membrane) are listed in Appendix D (p. 152).

The earth plate. Of the original cabinets, only the active speaker's aluminium back plate remains. It is at the back of the cabinet. The machine's mains inlet came on this plate. The plate carries the earth star point, and the amplifier's ground is joined to it. Chapter 5, section 5.3 (p. 23), explains that joint and what it means for the DC ground.

12.2 The volume control

The amplifier's original dual-gang volume potentiometer has been replaced with a B10k dual-gang potentiometer: two potentiometers on one shaft, one per channel. Its two gangs are separate, as the original's were. It has a power switch, which is not used.

- It sits on the amplifier's circuit board and is reached only from the back.
- It does not reach silence at its minimum.
- It is not the front volume knob. The front knob is read digitally by the audio board and acts on RADIO and BT only (chapter 9 (p. 52), section 9.4 (p. 58)). Chapter 6, section 6.8 (p. 37), sets out all three volume controls in the chain.

12.3 The input panel

The input panel is a small circuit board carrying the amplifier's inputs. The cable sheet's AMP_ signals (chapter 10 (p. 60)) land on its pins.

It adds up its inputs; it does not choose between them. If two or three inputs carry sound at the same time, the amplifier plays their sum.

A balanced input takes each channel on two wires, a plus and a minus, and amplifies the difference between them. The table lists the panel's three inputs, their contacts, and whether each is grounded with nothing plugged in.

Table 110. The input panel

Input	Contacts	With nothing plugged in
Balanced	L+, L–, ground, R–, R+	not grounded
AUX	a TRS jack (tip, ring, sleeve)	grounded
RCA pair	two RCA jacks	grounded

The DAC reaches the panel through the balanced input. Its cable puts a 470 Ω resistor in each of its four signal legs and carries six ferrite cores (chapter 6 (p. 28), section 6.6 (p. 36) chapter 10 (p. 60), section 10.21 (p. 80)).

The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn, and it is not among the front-panel controls (chapter 3 (p. 9), section 3.2 (p. 10)). AUX never passes through the audio board (chapter 6 (p. 28), section 6.7 (p. 37)).

12.4 What was measured at the panel and on the cable

These readings show how the amplifier's balanced input behaves at DC, which matters if you chase a noise or a fault on the DAC-to-amplifier link. All the readings below are resistance, taken with a multimeter, one reading each.

At the amplifier's balanced input, with the DAC-to-amplifier cable unplugged at the amplifier:

Table 111. What was measured at the panel and on the cable

Between	Reading
L– and AMP_G (ground)	open: 3 MΩ and rising, as a capacitor charged
R– and AMP_G	open: 3 MΩ and rising, as a capacitor charged

On the DAC-to-amplifier cable, unplugged. These readings were taken before the 470 Ω resistors on the two plus legs (L+ and R+) were fitted:

Table 112. What was measured at the panel and on the cable

Between	Reading
L− and ground	470 Ω
R− and ground	470 Ω
L− and R−	a little under 1 kΩ
L+ and ground	4.2 MΩ
R+ and ground	4.2 MΩ

What this supports, and no more. At the amplifier's connector, neither minus leg has a DC path to the amplifier's ground. That is what a differential, AC-coupled input looks like on a meter. Such an input amplifies the difference between its two legs, through capacitors that pass audio but block DC. It is one meter reading, taken once; nothing inside the amplifier was examined.

The frying noise once heard at the amplifier, and the parts later added to the DAC-to-amplifier cable, are in chapter 6 (p. 28), section 6.6 (p. 36).

12.5 When something is wrong

The table lists what you may notice and where to look first.

Table 113. When something is wrong

You see	Check
No sound from any source, with the front power switch on	The amplifier supply: +20 V and −20 V on its four wires (chapter 5 (p. 21), section 5.6 (p. 27)). Then the amplifier's own volume control, at the back.
The amplifier is never fully silent	Normal: its volume control does not reach silence at its minimum.
Two sources heard at once	Normal: the input panel adds its inputs. Unplug whatever is playing into AUX (chapter 6 (p. 28), section 6.7 (p. 37)).
Radio and Bluetooth silent, AUX plays	Not the amplifier: the DAC side (chapter 6 (p. 28), section 6.9 (p. 38)).
A frying noise at the amplifier	The DAC-to-amplifier cable, its four 470 Ω resistors and its ferrites (chapter 6 (p. 28), section 6.9 (p. 38)).

13. The tube radio (LLOYDS TM-838N)

The tube radio is the part of Ambersong that receives. It is a LLOYDS TM-838N, an AM, short-wave (SW) and FM table set with seven tubes, kept on its own chassis inside the cabinet. When you listen to the radio, every station is found, amplified and detected by these tubes; the rest of the machine only digitises the result. This chapter describes the set stage by stage, as fitted in this machine, so that you can check or repair it without its schematic.

This is one TM-838N as fitted and repaired. Another set of the same model will differ in details.

⚠ DANGER the radio's chassis sits on mains neutral, and its high-voltage supply (B+) is made straight from the mains. Read chapter 2 (p. 7) before you work on it, and unplug the cord first.

13.1 Overview

13.1.1 What the set is

The set receives three bands, chosen with its four-position mode switch: SW, AM, FM and FM-AFC. FM-AFC is FM with automatic frequency control (AFC) switched on: a voltage from the FM detector keeps the FM oscillator on the station (section 13.3 (p. 109)).

It is a superheterodyne. A mixer combines the incoming station with the signal of a local oscillator inside the set, and passes on the difference, the intermediate frequency (IF). The IF is the same for every station, so the stages after the mixer stay tuned to it and never move.

FM has its own front end, the stages that take the station at its own radio frequency (RF) and mix it down to the IF. AM and SW share another front end, and use a different IF:

Table 114. What the set is

Band	Front end	IF	Unit
FM	the FM RF amplifier and mixer (V1), with the FM local oscillator (V2)	10.6	MHz
AM and SW	the converter (V3), a mixer tube with its own oscillator	455	kHz

The two paths meet at the first IF amplifier (V4), which amplifies both IFs. After it, the FM IF goes to the FM limiter (V5) and the ratio detector. The AM IF goes to the AM detector, the two diodes inside V6. The audio unit of the mode switch picks the audio of one of the two detectors. From there the sound passes through the radio's volume control, the first audio amplifier (V6′s triode), the output stage (V7) and the output transformer.

The FM IF is 10.6 MHz. It was set when a professional, Radio Hosvep on Park Ave., tuned the FM path. The readings of the FM calibration receiver, the single-chip FM receiver the main board uses to calibrate the dial (chapter 8 (p. 46)), agree: they find the FM local oscillator one IF below each station. 10.7 MHz, the figure the SAMS service data gives (section 13.1.3 (p. 100)), is the design value. Only the FM path was tuned: the three FM IF transformers (L10, L12 and L14) and what touches them. The AM and SW path was not.

The set is built the old "AC/DC" way. It has no mains transformer: its seven tube heaters are wired in series across the mains (section 13.9 (p. 130)), and a single rectifier makes its B+ straight from the mains (section 13.10 (p. 134)).

13.1.2 How it sits in Ambersong

The radio is a unit of its own inside the cabinet. The table lists every place where it meets the rest of the machine:

Table 115. How it sits in Ambersong

Where the radio meets the machine	What happens there	More in
Mains	The radio is fed from the front power switch through the mains line filter. It has its own power switch and a 0.75 A fuse.	chapter 5 (p. 21) 13.10
Chassis	The radio's circuit common, CHASSIS, is on mains neutral. It is never joined to Earth or to the Faraday cages.	chapter 2 (p. 7)
FM antenna	The antenna's coax arrives at the radio's coax input. Its shield is on Earth and reaches the chassis only through one safety capacitor.	13.2; chapter 11 (p. 92)
Sound out	The output transformer's secondary, loaded by a 7.5 Ω 5 W resistor, feeds the ADC (the analogue-to-digital converter board) through a shielded cable. The feed is mono.	13.11; chapter 6 (p. 28) chapter 10 (p. 60), section 10.22 (p. 81)
Speakers	SP1 and SP2 are the radio's original speakers as drawn by the factory. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.	13.8
Volume	The radio's own volume control sits at about one eighth of its travel in normal use. Above that the sound is too loud further down the chain; it is not clipped.	13.8; chapter 6 (p. 28)
Tuning	A printed front knob, on a printed shaft extension, turns the radio's original tuning knob, which drives the dial cord and the four-section tuning capacitor. A magnetic angle sensor reads its shaft; a motor, not the dial cord, moves the needle.	chapter 8 (p. 46)
FM calibration receiver	Its antenna is a 30 cm wire inside the radio's cage, bent in a U over the FM oscillator section.	13.3; chapter 8 (p. 46)
Cage and fan	The radio sits under its own earthed Faraday cage, with a 5 V fan blowing air into it.	chapters 3 (p. 9) and 5

13.1.3 Fitted values and original specifications

Each stage has a parts table with two value columns:

- Fitted is the part in this radio now.
- Original is what the two original documents give: the factory drawing and the SAMS service data. "Same" means both give the fitted value. Where they differ, the table gives both; the fitted value is what is in the radio. Where neither gives a value, the table says so.

Unless a table says otherwise, neither original gives a resistor's wattage or tolerance or a capacitor's voltage rating, and the fitted parts give none of these either.

Substitutes and selection notes. The original specifications come with notes on catalogue substitutes and on choosing a replacement, from a component list that is not published. That list is not in this book. A “listed substitute” in a parts table is one named in that list, and a note on what to match when you replace a part is that list’s selection note. They describe the original part, though a few notes mention the fitted one.

Three-lead capacitors. Four ceramic capacitors have three leads: the FM supply bypass (C5), the AFC shunt (C11), the AFC supply bypass (C14) and one heater-string bypass (C48). The middle lead is wired to the chassis, and the two outer leads have 0 Ω between them.

Terminal numbers are logical. The numbers and letters given here for coil terminals and mode-switch contacts identify connections on the drawing and in the SAMS data. They are not the numbers on the parts’ lugs.

IF transformers. Each IF transformer sits in a shielded metal can, together with its own tuning capacitors. Neither original gives the values of those capacitors.

Trimmers and padders. Two kinds of small capacitor recur in the tuned circuits. A trimmer is a small adjustable capacitor set once, during alignment. A padder is a fixed capacitor in series with an oscillator coil; it makes the oscillator track the station across the dial.

13.1.4 Tube line-up

The set has seven tubes. “New old stock” means a tube that was made in the period and never used. The three new-old-stock tubes (V1, V2 and V6) were fitted by Radio Hosvep on Park Ave.; their types and connections are the original ones.

Table 116. Tube line-up

Tube	Type	What it does	Section
V1	12DT8 twin triode, new old stock	first triode: FM RF amplifier; second triode: FM mixer	13.2
V2	12DT8 twin triode, new old stock	first triode: FM local oscillator; second triode: AFC triode	13.3
V3	12BE6 pentagrid converter	AM and SW mixer with its own oscillator	13.7
V4	12BA6 pentode	first IF amplifier, for FM and AM	13.4, 13.6
V5	12BA6 pentode	FM limiter	13.4
V6	12AV6 duplex-diode triode, new old stock	its two diodes: AM detector; its triode: first audio amplifier	13.6, 13.8
V7	50C5 beam power tube	audio output	13.8

The tube pin assignments and the order of the heater string given in this chapter were checked on the chassis. The tubes’ published ratings:

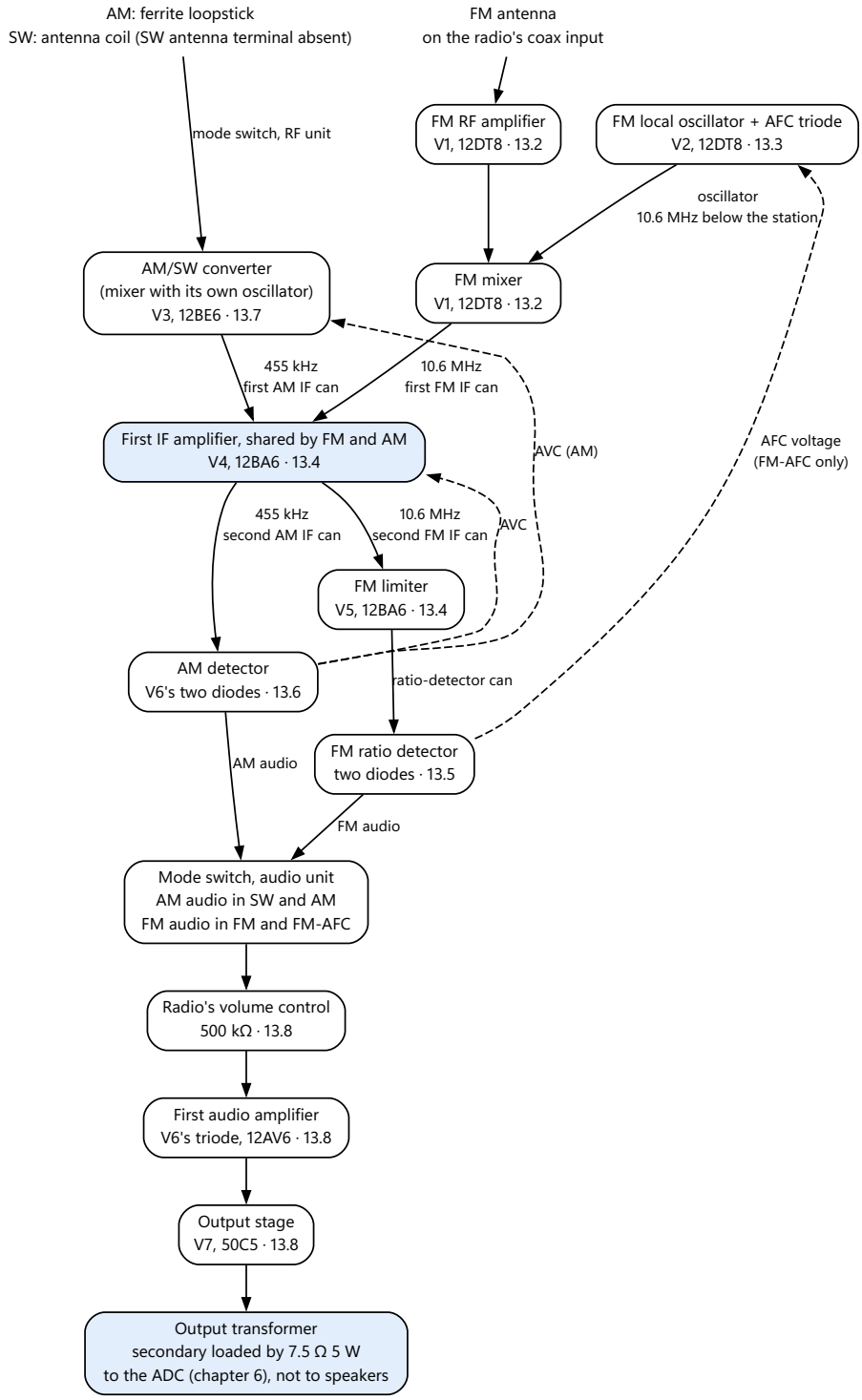
Table 117. Tube line-up

Type	Base	Heater
12DT8	9-pin miniature (noval, B9A), EIA basing 9DE	12.6 V, 0.15 A
12BE6	7-pin miniature (B7G), EIA basing 7CH	12.6 V, 0.15 A
12BA6	7-pin miniature (B7G), basing 7BK	12.6 V, 0.15 A
12AV6	7-pin miniature (B7G)	12.6 V, 0.15 A
50C5	7-pin miniature (B7G), EIA basing 7CV	50 V, 0.15 A, for series heater strings

NOTICE a 12DT8 is not a 12AT7. The two are alike except for their heater and heater-to-cathode ratings, their capacitances and their basing. Do not assume a 12AT7 pinout when you wire or test a 12DT8 socket. The 12DT8′s two triodes are separated by an internal shield on its own pin (pin 9).

13.1.5 Block diagram

Figure 70. The tube radio's stages, from the antennas to the output transformer



Dashed lines are control voltages, not signal. AVC (automatic volume control) is a voltage from the AM detector that sets the gain of the first IF amplifier, and in AM of the converter, so that strong and weak stations come out at similar levels (section 13.6 (p. 120)).

13.1.6 The mode switch

The mode switch is one rotary switch with four positions. The factory drawing calls it a set of mechanically linked band, supply, audio, AFC and neon contacts; SAMS (its part M5) gives four positions and three physical sections. The drawing splits it into seven units, S1A to S1G. They are seven functions of one switch, not seven switches, and they all turn together.

The table gives, for each unit, where its common goes and what the common is joined to in each position:

Table 118. The mode switch

Unit, and its job	Its common goes to	SW	AM	FM	FM-AFC	Contacts (common; SW, AM, FM, FM-AFC)
S1A, AFC off	chassis	not connected	not connected	the AFC control line: AFC off	not connected: AFC on	13; 1, 2, 3, 4
S1B, audio	top of the radio's volume control	AM audio	AM audio	FM audio	FM audio	14; 5, 6, 7, 8
S1C, RF selection	the converter's signal grid (V3 pin 7) and the AM/SW RF tuning section (VC3)	SW antenna coil and its trimmer	AM loopstick and its trimmer	not connected	not connected	15; 9, 10, 11, 12
S1D, B+	the last point of the B+ filter (`B115_A`)	AM supply, `AM_B`	`AM_B`	FM supply, `FM_B`	`FM_B`	14; 5, 6, 7, 8
S1E, mode lamps	the neon feed, from the fused live through a 50 kΩ resistor (R32)	SW lamp (NE1)	AM lamp (NE2)	FM lamp (NE3)	FM lamp (NE3)	15; 9, 10, 11, 12
S1F, oscillator tank	the AM/SW oscillator tuning section (VC4) and feedback capacitor (C25)	SW padder (C26, 0.003 µF)	AM padder (C27, 350 pF) and AM oscillator trimmer (C23B)	not connected	not connected	15; 9, 10, 11, 12
S1G, oscillator feedback	the converter's cathode (V3 pin 2)	tap of the SW oscillator coil (L9)	tap of the AM oscillator coil (L8)	not connected	not connected	13; 1, 2, 3, 4

The contact numbers are logical (section 13.1.3 (p. 100)), and the same number on two units is not the same contact. SAMS describes the switch as three physical sections, or wafers, and the numbers repeat from one to another. S1A's common 13 is on the chassis, for example, while S1G's common 13 is on the converter's cathode. S1C's numbers are SAMS's markings for the switch's second section; S1F's and S1G's are its markings for the third.

What the B+ unit decides. `AM_B` feeds the converter (V3), whose plate supply passes through the primary of the first AM IF transformer. `FM_B` feeds the FM RF amplifier and mixer (V1), the FM oscillator and AFC triode (V2), the limiter (V5) and the first FM IF transformer's primary. So V1, V2 and V5 run only in FM and FM-AFC, and V3 only in SW and AM. The first IF amplifier (V4), the first

audio amplifier (V6) and the output stage (V7) are fed from the B+ filter directly, in every position. Their supply is `B115_A`, and, through the output transformer, `B130`, the first filtered B+ point.

The mode lamps are not fitted. The neon bulbs NE1, NE2 and NE3 are absent from this radio. Their resistor (R32) and the S1E contacts are fitted.

> NOTICE no commercial substitute is listed for the mode switch. A replacement must match this whole truth table, the shaft and detents, and the insulation and current ratings.

13.1.7 How parts are named in this chapter

Every part is named by what it does, with its reference from the radio's drawing in brackets: "the FM RF trimmer (C8)". Always read the two together. Many of the radio's references are also used for different parts elsewhere in the machine: C13 to C30, J2 to J4 and R17 exist on the main drawing too, and M1 is both the radio's fuse and the box fan. Where a reference stands alone in this chapter, it is the radio's part.

The first letters of the radio's references mean:

Table 119. How parts are named in this chapter

Letter	In the radio
V	a tube
R, C, L	resistor, capacitor, coil or transformer
VC	a section of the four-section tuning capacitor (VC1 to VC4, one assembly, SAMS part M2)
S	a switch: S1 is the mode switch, S2 the power and tone switch (SAMS part M6)
M	a part numbered from the SAMS list: the fuse (M1), the FM detector diodes (M3, M4)
J	a terminal or connector
NE, SP, T, X	neon lamp, speaker, transformer, rectifier

The output transformer (T1) carries the same reference in the radio and in the rest of the machine: it is one part.

Signal names in `code font` are the drawing's names for a connection, such as `FM_B` for the FM B+ supply. They are used where a connection reaches beyond one stage.

13.2 FM RF amplifier and mixer (V1, 12DT8)

V1 holds two triodes. The first is the FM RF amplifier: it lifts the weak signal from the antenna. Its grid is held at the chassis and the signal enters at its cathode, a grounded-grid amplifier. The second triode is the FM mixer. It takes the amplified station and the FM local oscillator's signal from V2. It sends their difference, the 10.6 MHz IF, to the first FM IF transformer. Both triodes take their B+ from `FM_B`, so they work only in FM and FM-AFC.

13.2.1 How it is joined

Antenna input. The FM antenna's coax arrives at the radio's coax input (J53). Its centre conductor goes through the antenna coupling capacitor (C3) to the RF input point. From that point the FM antenna choke (L2) goes to the chassis. The RF cathode resistor (R2) and the RF cathode bypass capacitor (C4), side by side, go from the same point to the RF triode's cathode. The coax shield is on Earth and reaches the chassis only through the antenna shield capacitor (C51). The radio first had a simple wire antenna; the coax input replaced it.

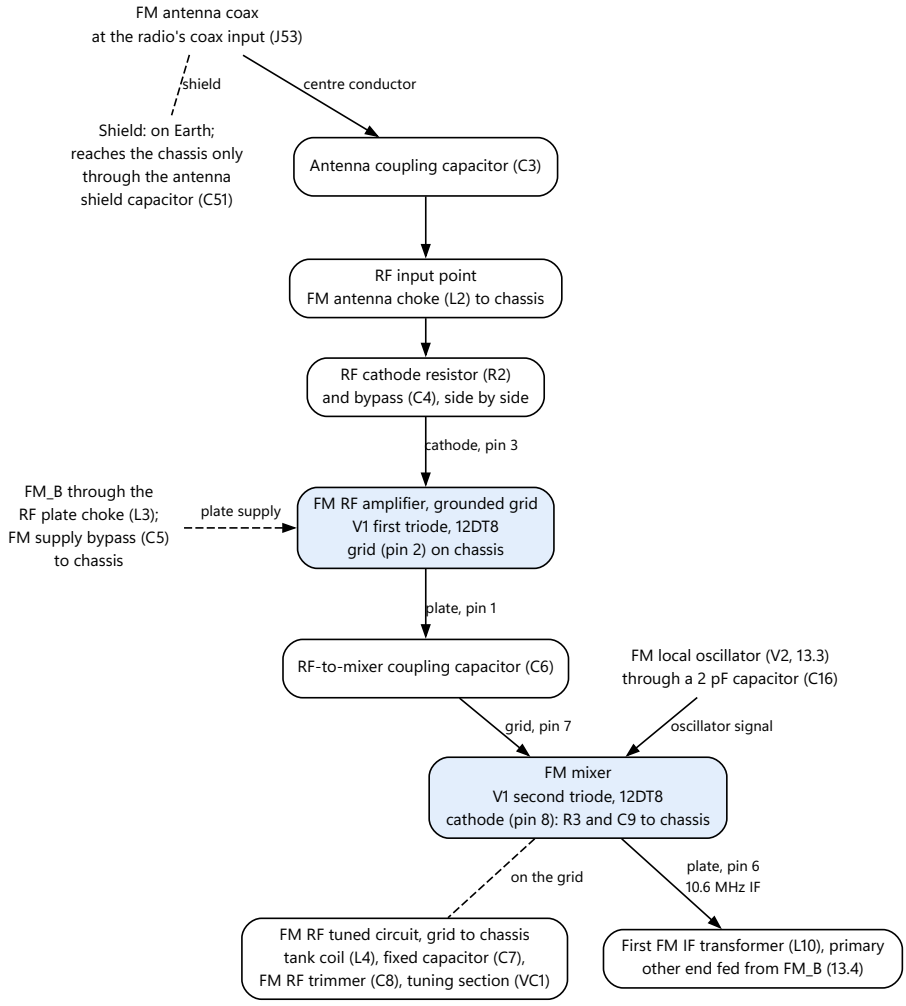
RF triode. The grid (pin 2) is on the chassis. The plate (pin 1) is fed from `FM_B` through the RF plate choke (L3), and `FM_B` is bypassed to the chassis there by the FM supply bypass capacitor (C5). The plate's signal goes to the mixer grid through the RF-to-mixer coupling capacitor (C6).

Mixer grid. The mixer grid (pin 7) sits on the FM RF tuned circuit, or tank. It is four parts, each joined from the grid to the chassis: the FM RF tank coil (L4), the tank's fixed capacitor (C7), the FM RF trimmer (C8) and the FM RF tuning section (VC1). The FM oscillator's signal arrives on the same grid through a 2 pF capacitor (C16, section 13.3 (p. 109)).

Mixer cathode and plate. The cathode (pin 8) goes to the chassis through the mixer cathode resistor (R3) and its bypass capacitor (C9), side by side. The plate (pin 6) goes to the primary of the first FM IF transformer (L10). The plate's B+ comes through that primary, whose other end is fed from `FM_B` (section 13.4 (p. 113)).

Heater and shield. The heater is on pins 4 and 5, in the series string (section 13.9 (p. 130)). The internal shield, pin 9, is on the chassis.

Figure 71. How the stage is joined (simplified)



13.2.2 V1's pins

The table gives each of V1's pins and what it is joined to.

Table 120. V1′s pins

Pin	What it is	Joined to
1	RF triode plate	RF plate choke (L3) to `FM_B` coupling capacitor (C6) to the mixer grid
2	RF triode grid	chassis
3	RF triode cathode	RF cathode resistor (R2) and bypass (C4), to the RF input point
4	heater	heater string, toward the FM oscillator tube's heater through a heater choke (L15)
5	heater	heater string, from the converter's heater through a heater choke (L16)
6	mixer plate	primary of the first FM IF transformer (L10)
7	mixer grid	FM RF tuned circuit (L4, C7, C8, VC1); oscillator injection (C16)
8	mixer cathode	mixer cathode resistor (R3) and bypass (C9), to chassis
9	internal shield	chassis

13.2.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 121. Parts

Part	Fitted	Original	Notes
FM RF amplifier and mixer tube (V1)	12DT8, new old stock	12DT8	
Radio's FM coax input (J53)	coax connector, RG179 coax	a simple wire antenna	centre to the antenna coupling capacitor (C3); shield to Earth
Antenna coupling capacitor (C3)	100 pF, safety class X1Y1	100 pF (factory and SAMS)	A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)). The substitutes listed for the original are DC-rated and do not establish a modern safety class or insulation design; choose a replacement with its own safety rating and an adequate voltage rating.
FM antenna choke (L2)	1 µH, 23 turns	factory 1 µH; SAMS adds 23 turns	
RF cathode resistor (R2)	70 Ω	factory 70 Ω; SAMS 68 Ω, with 70 Ω as an alternate	
RF cathode bypass capacitor (C4)	0.002 µF	same	
Antenna shield capacitor (C51)	0.001 µF, safety class X1Y2	—	The coax shield's only path to the chassis. A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)).
RF plate choke (L3)	3 µH	same	SAMS: DCR about 0.9 Ω.
FM supply bypass capacitor (C5)	0.002 µF, three-lead ceramic	factory 0.002 µF; SAMS schematic 0.01 µF (0.002	Middle lead to chassis (section 13.1.3 (p. 100)).

Part	Fitted	Original	Notes
		µF alternate); SAMS parts table 0.002 µF	
RF-to-mixer coupling capacitor (C6)	100 pF	same	
FM RF tank coil (L4)	tuned coil	no inductance in either original	
Tank's fixed capacitor (C7)	5 pF	same	
FM RF trimmer (C8)	trimmer	no value in either original	Range and dielectric not specified. Belongs with the tuning capacitor assembly.
FM RF tuning section (VC1)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Turns with the other three sections of the tuning capacitor. A replacement must match the minimum and maximum capacitance, tracking, shaft and gang geometry, and trimmers.
Mixer cathode resistor (R3)	1 kΩ	same	
Mixer cathode bypass capacitor (C9)	0.01 µF	same	

13.2.4 How to check it

- Coil resistance. SAMS gives a DCR (DC resistance) of about 0.9 Ω for the RF plate choke (L3). A DCR is a check for an open winding, not a measure of inductance.
- The three-lead bypass (C5). Its two outer leads read 0 Ω to each other, and the middle lead goes to the chassis.
- The antenna side. The coax shield reaches the chassis only through the antenna shield capacitor (C51). Nothing else joins them.
- Tube pins. Use the table in section 13.2.2 (p. 107), not a 12AT7 pinout.

13.3 FM local oscillator and AFC (V2, 12DT8)

V2 also holds two triodes. The first is the FM local oscillator. It runs one IF below the station: oscillator frequency = station − 10.6 MHz. For a station at 100.0 MHz it runs at 89.4 MHz. Its tuning section (VC2) turns with the FM RF tuning section (VC1), so the oscillator follows the dial. The second triode is the AFC triode. Its plate is coupled into the oscillator's tuned circuit, and its grid takes the AFC control voltage from the ratio detector. In FM-AFC that voltage nudges the oscillator so the station stays tuned. In FM the mode switch holds it at the chassis, and AFC is off. Both triodes work only in FM and FM-AFC.

13.3.1 How it is joined

Supply. `FM_B` reaches this stage only through the AFC plate feed resistor (R8). After it, a local supply point is bypassed to the chassis by the AFC supply bypass capacitor (C14). This point feeds both triodes: the AFC plate through an RF choke (L1), and the oscillator plate through the oscillator plate feed resistor (R10).

Oscillator tuned circuit. The oscillator's tank is four parts, each joined from the tank to the chassis: the tank winding of the FM oscillator coil (L5), the tank's fixed capacitor (C17), the FM oscillator trimmer (C18) and the FM oscillator tuning section (VC2). Three capacitors meet the tank: the oscillator plate's (C20, 20 pF), the AFC plate's (C15, 10 pF), and the one that carries the oscillator's signal out to the FM mixer grid, V1 pin 7 (C16, 2 pF).

Oscillator feedback. The FM oscillator coil (L5) has two coupled windings: the tank winding and a feedback winding. One end of each is on the chassis. The feedback winding's other end goes to the oscillator grid (pin 2) through the grid feedback capacitor (C19), with the grid leak resistor (R9) across it. A grid leak is the resistor that gives a grid its DC path to the chassis. The oscillator cathode (pin 3) goes to the chassis through the oscillator cathode resistor (R33) and its bypass capacitor (C50), side by side.

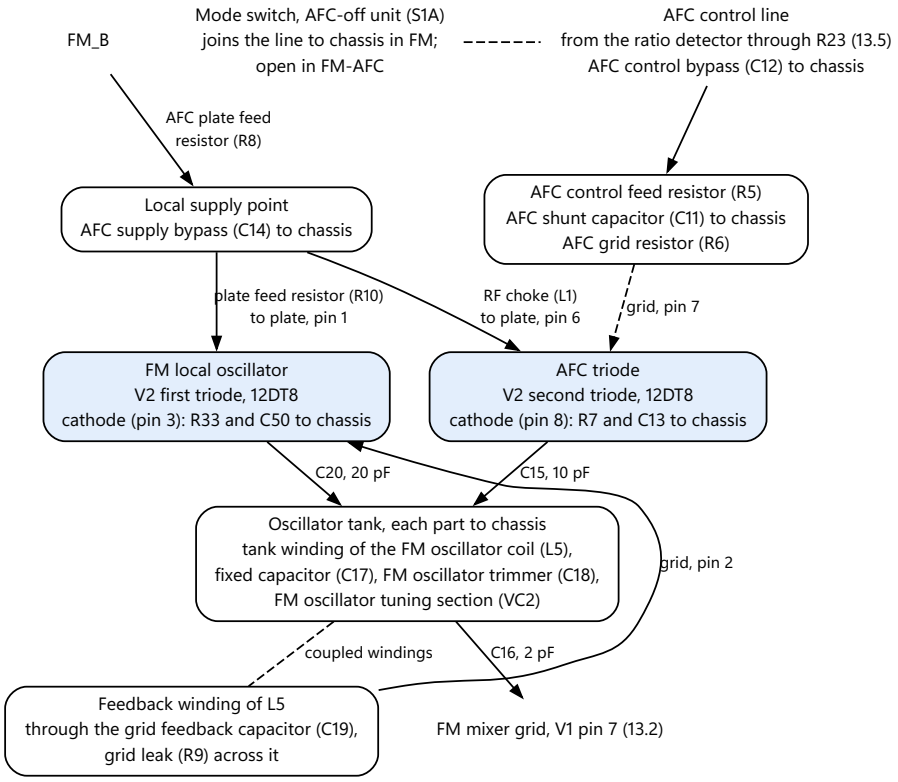
AFC triode. The AFC control line arrives from the ratio detector through a 3 MΩ resistor (R23, section 13.5 (p. 117)) and is bypassed to the chassis by the AFC control bypass capacitor (C12). From there the AFC control feed resistor (R5) leads to a point that the AFC shunt capacitor (C11) joins to the chassis. From that point the AFC grid resistor (R6) leads on to the AFC grid (pin 7). The grid has no other connection. The AFC cathode (pin 8) goes to the chassis through the AFC cathode resistor (R7) and its bypass capacitor (C13), side by side. The AFC plate (pin 6) is fed through the RF choke (L1) and coupled to the oscillator tank through its coupling capacitor (C15).

In FM, the AFC-off unit of the mode switch (S1A) joins the AFC control line to the chassis. In FM-AFC that contact is open (section 13.1.6 (p. 104)).

Heater and shield. The heater is on pins 5 and 4. Pin 5 comes from the FM RF tube's heater through a heater choke (L15). Pin 4 is the chassis end of the whole heater string: V2 is the last tube in it (section 13.9 (p. 130)). The internal shield, pin 9, is on the chassis.

Near this stage. The FM calibration receiver's antenna, a 30 cm wire, lies over this section, bent in a U, inside the radio's cage. It is not connected to the radio (chapter 8 (p. 46)).

Figure 72. How the stage is joined (simplified)



13.3.2 V2′s pins

The table gives each of V2′s pins and what it is joined to.

Table 122. V2′s pins

Pin	What it is	Joined to
1	oscillator plate	plate feed resistor (R10) to the local supply; plate coupling capacitor (C20) to the tank
2	oscillator grid	grid feedback capacitor (C19) with grid leak (R9) across it, to the feedback winding
3	oscillator cathode	oscillator cathode resistor (R33) and bypass (C50), to chassis
4	heater	chassis: the end of the heater string
5	heater	heater string, from the FM RF tube through a heater choke (L15)
6	AFC plate	RF choke (L1) to the local supply; coupling capacitor (C15) to the tank
7	AFC grid	AFC grid resistor (R6) only
8	AFC cathode	AFC cathode resistor (R7) and bypass (C13), to chassis
9	internal shield	chassis

13.3.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 123. Parts

Part	Fitted	Original	Notes
FM oscillator and AFC tube (V2)	12DT8, new old stock	12DT8	
AFC control feed resistor (R5)	250 kΩ	same	
AFC shunt capacitor (C11)	0.002 µF, three-lead ceramic, from the AFC grid network to chassis	factory 0.002 µF; SAMS 0.001 µF	The fitted C11 is a shunt from the AFC line (AFC_C, the point between the AFC control feed resistor R5 and the AFC grid resistor R6) to chassis. SAMS's substitute list is for its 0.001 µF part. Middle lead to chassis (section 13.1.3 (p. 100)).
AFC grid resistor (R6)	100 Ω	same	
AFC control bypass capacitor (C12)	0.01 µF	factory 0.01 µF; SAMS 0.001 µF	
Oscillator cathode resistor (R33)	600 Ω	—	
Oscillator cathode bypass capacitor (C50)	0.01 µF	—	
AFC cathode resistor (R7)	600 Ω	same	
AFC cathode bypass capacitor (C13)	0.01 µF	factory 0.01 µF; SAMS schematic 0.002 µF (0.01 µF alternate); SAMS parts table 0.01 µF	
AFC RF choke (L1)	3 µH	same	SAMS: DCR about 0.9 Ω.
AFC plate feed resistor (R8)	1 kΩ	same	
AFC supply bypass capacitor (C14)	0.002 µF, three-lead ceramic	same	Middle lead to chassis (section 13.1.3 (p. 100)).
AFC-to-oscillator coupling capacitor (C15)	10 pF	same	
Oscillator-to-mixer coupling capacitor (C16)	2 pF	same	Several listed substitutes, Sprague among them, are 2.2 pF. Prefer a low-loss NP0 part of exactly 2 pF.
Oscillator tank's fixed capacitor (C17)	5 pF	same	
FM oscillator trimmer (C18)	trimmer	no value in either original	Range and dielectric not specified. Belongs with the tuning capacitor assembly.
FM oscillator tuning section (VC2)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Turns with the other three sections of the tuning capacitor.
FM oscillator coil (L5)	two coupled windings, tank and feedback	factory: two coupled windings; SAMS: one tapped winding; no inductance in either	Do not replace it with a tapped oscillator coil on the strength of its name.
Grid feedback capacitor (C19)	50 pF	same	
Grid leak resistor (R9)	20 kΩ	same	
Oscillator plate feed resistor (R10)	30 kΩ	factory 30 kΩ, with a different supply connection; SAMS 10 kΩ	

Part	Fitted	Original	Notes
Oscillator plate coupling capacitor (C20)	20 pF, from the oscillator plate to the tank	factory 20 pF	SAMS's C20 is a different part, a 0.002 µF bypass; its substitutes do not apply here. The factory part's dielectric, tolerance and voltage rating are unknown.

13.3.4 How to check it

- The oscillator's frequency. The FM calibration receiver hears this oscillator: its readings put the oscillator one IF, 10.6 MHz, below each station (chapter 8 (p. 46)).
- AFC off in FM. With the cord out, the mode switch's AFC-off unit (S1A) joins the AFC control line to the chassis on FM, and that contact is open on FM-AFC.
- Coil resistance. SAMS gives a DCR of about 0.9 Ω for the AFC RF choke (L1).
- The three-lead capacitors (C11, C14). The two outer leads read 0 Ω to each other, and the middle lead goes to the chassis.

13.4 The FM IF path (V4, V5, 12BA6)

The IF amplifiers raise the 10.6 MHz IF from the mixer to a level the detector can use. The first, V4, is shared with AM. Its grid circuit holds the secondaries of the first FM and the first AM IF transformers in series. Its plate circuit holds the primaries of the second FM and the second AM IF transformers in series. So the one tube amplifies both IFs. The second, V5, is the FM limiter: it clips the IF to a steady level, so that the detector answers to changes of frequency rather than of strength. V5 feeds the ratio-detector transformer.

13.4.1 How it is joined

First FM IF transformer (L10). Its primary runs from terminal P, on the FM mixer plate (V1 pin 6), to terminal B, on a supply point. The first FM IF supply resistor (R4) feeds that point from `FM_B`, and the first FM IF supply bypass capacitor (C10) joins it to the chassis. Terminal G of L10′s secondary drives V4′s control grid (pin 1). Terminal E, the other end, is in series with the secondary of the first AM IF transformer (L11), joining L11′s terminal G; L11′s far end is on the AVC line. The letters are logical (section 13.1.3 (p. 100)). V4′s grid therefore returns to the AVC line through both secondaries.

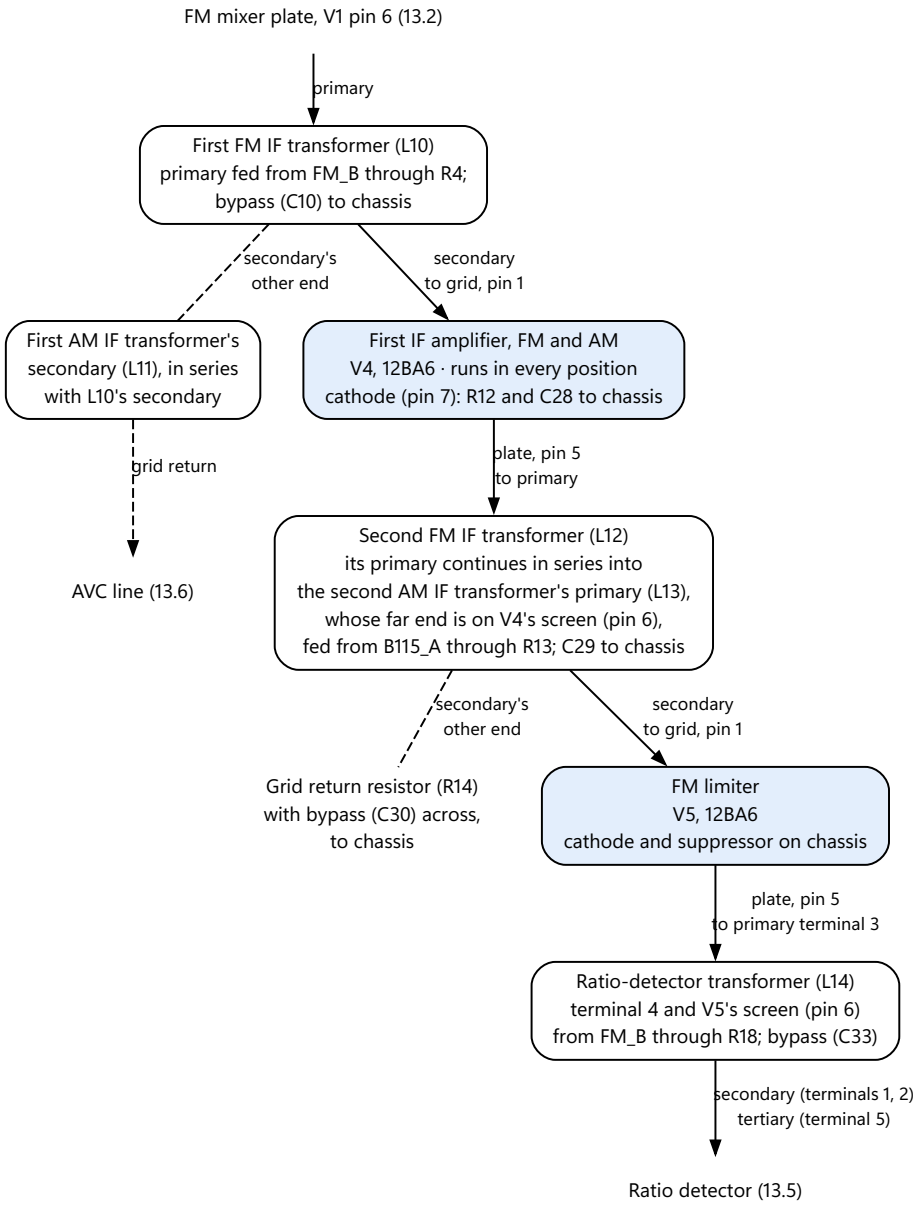
V4. The cathode (pin 7) goes to the chassis through the first IF cathode resistor (R12) and its bypass capacitor (C28), side by side. The suppressor grid (pin 2) is on the chassis. The plate (pin 5) feeds the primary of the second FM IF transformer (L12), and that primary continues in series into the primary of the second AM IF transformer (L13). The far end of that AM primary shares a point with V4′s screen grid (pin 6). The AM IF screen supply resistor (R13, 1 kΩ) feeds that point from `B115_A`, and its bypass capacitor (C29, 0.01 µF) joins it to the chassis (section 13.6 (p. 120)). None of V4′s supply comes from **`FM_B`**, so V4 runs in every position.

V5. One end of the second FM IF transformer's secondary (L12) drives V5′s control grid (pin 1). The other end goes to the chassis through the limiter grid return resistor (R14), with the limiter grid bypass capacitor (C30) across it. V5′s cathode (pin 7) and suppressor grid (pin 2) are both straight on the chassis. The plate (pin 5) feeds terminal 3 of the ratio-detector transformer's primary (L14). The primary's other end, terminal 4, and V5′s screen grid (pin 6) share a supply point. The ratio-detector supply resistor (R18) feeds it from `FM_B`, and the ratio-detector supply bypass capacitor (C33) joins it to the chassis.

Ratio-detector transformer (L14). Besides its primary it has a secondary (terminals 1 and 2) and a third winding, the tertiary (terminal 5). They feed the ratio detector (section 13.5 (p. 117)).

Cans and heaters. The cans of L10, L12 and L14 are on the chassis. V4′s and V5′s heaters are in the series string (section 13.9 (p. 130)).

Figure 73. How the stage is joined (simplified)



13.4.2 V4′s and V5′s pins

Both tubes are 12BA6 pentodes with the same basing. The table gives each pin for both tubes.

Table 124. V4′s and V5′s pins

Pin	What it is	V4, first IF	V5, limiter
1	control grid	first FM IF transformer's secondary (L10), then first AM IF transformer's secondary (L11), to the AVC line	second FM IF transformer's secondary (L12), then the grid return resistor (R14) with its bypass (C30), to chassis
2	suppressor grid and internal shield	chassis	chassis
3, 4	heater	heater string	heater string
5	plate	second FM IF transformer's primary (L12), then second AM IF transformer's primary (L13)	ratio-detector transformer's primary (L14), terminal 3
6	screen grid	far end of the second AM IF transformer's primary (L13); screen supply resistor (R13) from `B115_A` bypass (C29)	ratio-detector transformer's primary (L14), terminal 4; supply resistor (R18) from `FM_B` bypass (C33)
7	cathode	cathode resistor (R12) with its bypass (C28), to chassis	chassis

13.4.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 125. Parts

Part	Fitted	Original	Notes
First IF tube (V4)	12BA6	12BA6	
Limiter tube (V5)	12BA6	12BA6	
First FM IF transformer (L10)	IF can, tuned to 10.6 MHz	factory: FM IF (A); SAMS: 10.7 MHz	SAMS DCR: primary 0.8 Ω, secondary 0.8 Ω.
Second FM IF transformer (L12)	IF can, tuned to 10.6 MHz	factory: FM IF (B); SAMS: 10.7 MHz	SAMS DCR: primary 0.6 Ω, secondary 0.4 Ω.
Ratio-detector transformer (L14)	IF can with tertiary winding, tuned to 10.6 MHz	factory: ratio detector with tertiary winding; SAMS: 10.7 MHz	SAMS DCR: primary 1.2 Ω, secondary halves 0.4 Ω each, tertiary 0.5 Ω.
First FM IF supply resistor (R4)	1 kΩ	same	
First FM IF supply bypass capacitor (C10)	0.01 µF	same	
First IF cathode resistor (R12)	70 Ω	factory 70 Ω; SAMS 68 Ω, with 70 Ω as an alternate	
First IF cathode bypass capacitor (C28)	0.002 µF	factory 0.002 µF; SAMS 0.01 µF	
Limiter grid return resistor (R14)	250 kΩ	same	
Limiter grid bypass capacitor (C30)	50 pF	same	SAMS prints Centralab DD-501 as the replacement here, but DD-500 for the other 50 pF part (C19), so the DD-501 number is not approved.
Ratio-detector supply resistor (R18)	1 kΩ	same	
Ratio-detector supply bypass capacitor (C33)	0.01 µF	same	

For a replacement IF transformer, neither original gives what you need to choose one: the tuned frequency as fitted is 10.6 MHz, but neither gives the windings' coupling and impedance, the pinout, the internal capacitor values, the bandwidth or the mechanical fit.

13.4.4 How to check it

- Coil resistance. The parts table gives the SAMS DCR figures, which describe the original parts. A DCR is a check for an open winding, not a measure of inductance or tuning.
- Shared with AM. V4′s grid returns through the secondaries of both first IF transformers, FM and AM, and its plate is fed through the primaries of both second IF transformers. An open winding in any of these four stops V4 on both bands.
- Tuning. The FM IF transformers (L10, L12 and L14) were tuned to 10.6 MHz by a professional, Radio Hosvep on Park Ave. The AM IF transformers were not retuned.

13.5 The FM ratio detector and AFC control

The ratio detector turns the frequency swings of the FM IF into audio. It uses two diodes fed from the ratio-detector transformer (L14), a stabilising capacitor across their outputs, and a capacitor divider whose midpoint is the output. That output gives two things. One is the FM audio, through a de-emphasis filter that takes back the treble boost FM stations transmit. The other, through a 3 MΩ resistor, is the AFC control voltage for the FM oscillator (section 13.3 (p. 109)).

13.5.1 How it is joined

Diodes. One end of the ratio-detector transformer's secondary (L14, terminal 1) goes through the upper series resistor (R19) to the cathode of the upper detector diode (M3). M3′s anode is on the upper rail. The other end (terminal 2) goes through the lower series resistor (R21) to the anode of the lower detector diode (M4). M4′s cathode is on the lower rail.

Rails. The upper DC load resistor (R20) runs from the upper rail to the chassis, and the lower DC load resistor (R22) from the chassis to the lower rail, so the chassis is at their midpoint. The stabilising capacitor (C2), an electrolytic, sits across the two rails with its

1. on the lower rail

.

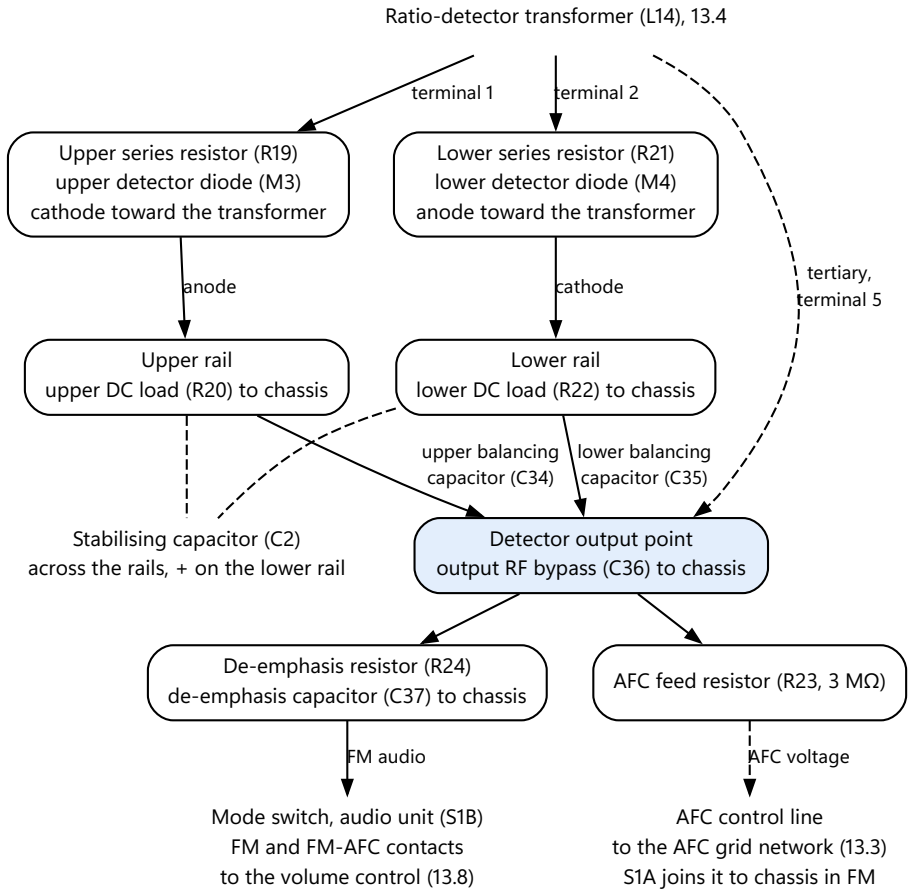
Output. The upper balancing capacitor (C34) runs from the upper rail to the detector output point, and the lower balancing capacitor (C35) from that point to the lower rail. The transformer's tertiary winding (terminal 5) is on the same point. From it:

- the output RF bypass capacitor (C36) goes to the chassis;

- the de-emphasis resistor (R24) leads to the FM audio, with the de-emphasis capacitor (C37) from there to the chassis. The FM audio goes to the FM and FM-AFC contacts of the mode switch's audio unit (S1B), whose common feeds the top of the radio's volume control (section 13.8 (p. 127)). In SW and AM, that unit takes the AM audio instead;
- the AFC feed resistor (R23) leads to the AFC control line, which goes to the AFC triode's grid network (section 13.3 (p. 109)).

AFC control. The AFC-off unit of the mode switch (S1A) has its common on the chassis and its FM contact on the AFC control line. In FM it joins the line to the chassis, and AFC is off. In FM-AFC, and in SW and AM, the contact is open.

Figure 74. How the stage is joined (simplified)



13.5.2 Which way round the diodes and the stabilising capacitor go

The diodes are fitted in the factory orientation, which is the opposite of the SAMS drawing's. The table gives which way each one faces:

Table 126. Which way round the diodes and the stabilising capacitor go

Part	Cathode	Anode
Upper detector diode (M3)	toward the transformer, through the upper series resistor (R19)	upper rail
Lower detector diode (M4)	lower rail	toward the transformer, through the lower series resistor (R21)

The stabilising capacitor (C2) has its

1. on the lower rail

and its − on the upper rail. SAMS shows its + on the upper rail; this radio does not follow SAMS here.

> NOTICE do not fit from the SAMS drawing. A detector diode or the stabilising capacitor fitted to that drawing is backwards in this radio. Note the orientation before you remove a part, and confirm it on the circuit before you fit the new one.

13.5.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 127. Parts

Part	Fitted	Original	Notes
Stabilising capacitor (C2)	10 µF electrolytic, + on the lower rail	10 µF (factory and SAMS); factory polarity not resolved; SAMS: at least 50 V, + on the upper rail	
Upper balancing capacitor (C34)	200 pF	same	
Lower balancing capacitor (C35)	200 pF	same	
Output RF bypass capacitor (C36)	200 pF	same	
De-emphasis capacitor (C37)	0.001 µF	same	1 nF, not 10 nF.
Upper detector diode (M3)	a diode	a diode, type unmarked (factory); no part number (SAMS)	No evidence for any particular replacement type, such as a 1N34 or 1N60.
Lower detector diode (M4)	a diode	a diode, type unmarked (factory); no part number (SAMS)	As the upper diode.
Upper series resistor (R19)	1 kΩ	same	
Upper DC load resistor (R20)	10 kΩ	same	
Lower series resistor (R21)	1 kΩ	same	
Lower DC load resistor (R22)	10 kΩ	same	
AFC feed resistor (R23)	3 MΩ	same	
De-emphasis resistor (R24)	50 kΩ	same	
Mode switch, AFC-off unit (S1A)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
Mode switch, audio unit (S1B)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

13.5.4 How to check it

- Orientation. Check both detector diodes and the stabilising capacitor against section 13.5.2 (p. 118).
- The rails. Each DC load resistor (R20, R22) runs from one rail to the chassis.
- AFC off in FM. See section 13.3.4 (p. 113).
- Transformer. The SAMS DCR figures for the ratio-detector transformer (L14) are in section 13.4.3 (p. 116).

13.6 The 455 kHz AM IF and AM detector

In SW and AM, the converter (V3, section 13.7 (p. 122)) sends its 455 kHz IF through the first AM IF transformer (L11) to the first IF amplifier (V4), the tube the FM path also uses (section 13.4 (p. 113)). From V4 the IF goes through the second AM IF transformer (L13) to the AM detector: the two diodes inside V6, a 12AV6. The detector recovers the audio from the IF and also makes the AVC voltage, which sets the gain of V4 and, in AM, of the converter (section 13.1.5 (p. 103)).

13.6.1 How it is joined

First AM IF transformer (L11). One end of its primary (terminal B) is on the converter's plate (V3 pin 5). The other end (terminal P) is on the AM supply, `AM_B`, together with the converter's screen grids (V3 pin 6). The mode switch's B+ unit (S1D) puts B+ on `AM_B` in SW and AM only (section 13.1.6 (p. 104)). No capacitor is on **`AM_B`**. SAMS shows a 0.002 µF bypass there; this radio has none.

Into V4. The secondary of L11 is in series with the secondary of the first FM IF transformer (L10): one end (terminal G) joins L10′s secondary, and the other end (terminal E) is on the AVC line. V4′s control grid (pin 1) is at the far end of L10′s secondary, so it returns to the AVC line through both secondaries (section 13.4.1 (p. 113)).

Out of V4. V4′s plate feeds the primary of the second FM IF transformer (L12), then, in series, the primary of the second AM IF transformer (L13). The far end of L13′s primary shares a point with V4′s screen grid (pin 6). The AM IF screen supply resistor (R13, 1 kΩ) feeds that point from the B+ filter (`B115_A`, section 13.10 (p. 134)), and its bypass capacitor (C29, 0.01 µF) joins it to the chassis. So L13 and V4 have B+ in every position of the mode switch.

Detector. One end of L13′s secondary (terminal G) goes to both diode plates of V6, pins 5 and 6. The diodes' cathode is pin 2, which V6′s triode shares, and it is on the chassis. The other end of the secondary (terminal E) is the detector load point. From there:

- the detector IF bypass capacitor (C31, 100 pF) goes to the chassis;
- the detector load resistor (R15, 50 kΩ) leads to the AM audio point.

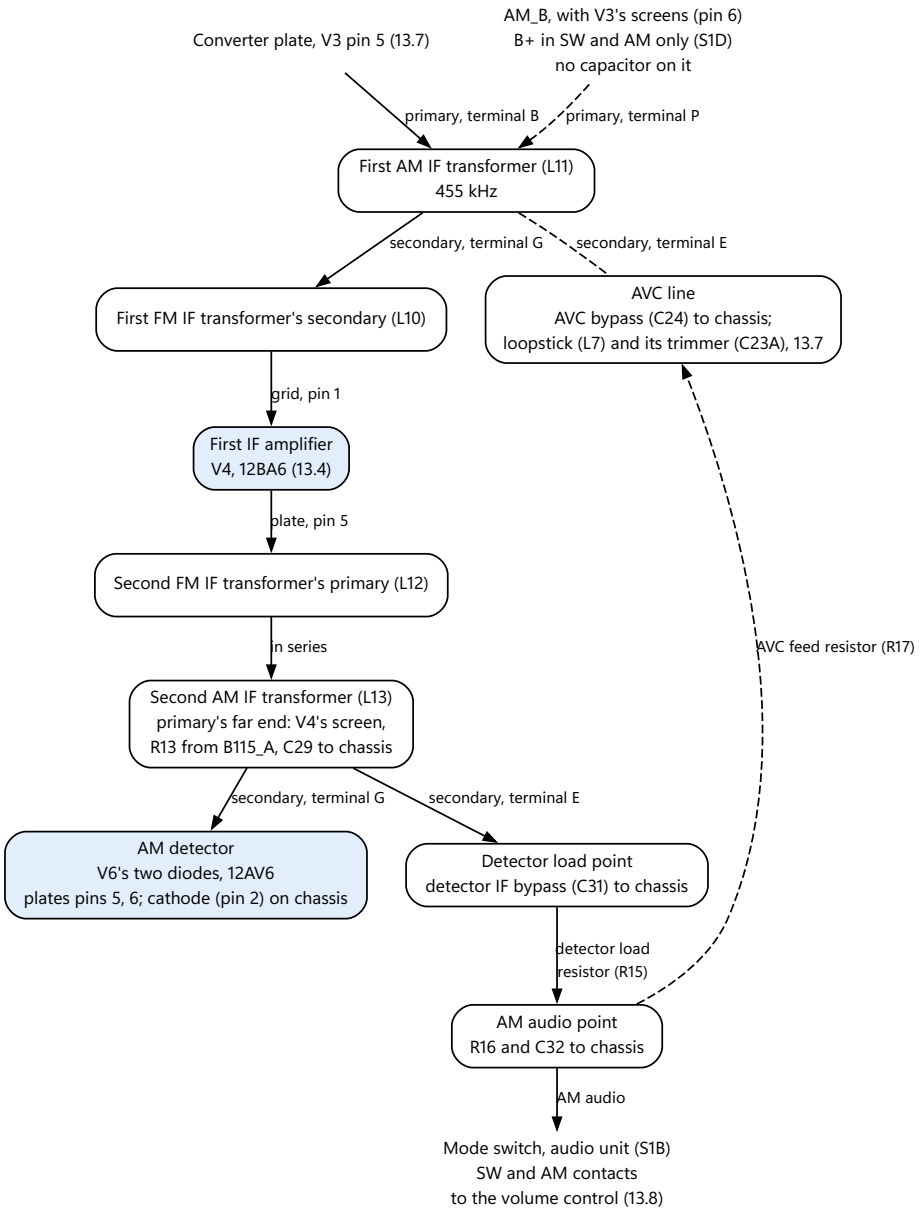
AM audio point. Three parts leave it: the detector DC return resistor (R16, 1 MΩ) and the audio IF bypass capacitor (C32, 100 pF), both to the chassis, and the AVC feed resistor (R17, 2 MΩ) to the AVC line. The AM audio goes to the

SW and AM contacts of the mode switch's audio unit (S1B), whose common feeds the top of the radio's volume control (section 13.8 (p. 127)).

AVC line. It joins five things: the far end of the AVC feed resistor (R17), terminal E of L11′s secondary, one end of the AM loopstick (L7), one side of the AM antenna trimmer (C23A), and the AVC bypass capacitor (C24, 0.05 µF), whose other side is on the chassis. In AM the converter's signal grid returns to this line through the loopstick (section 13.7 (p. 122)).

Cans. The cans of L11 and L13 are on the chassis.

Figure 75. How the stage is joined (simplified)



13.6.2 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 128. Parts

Part	Fitted	Original	Notes
First AM IF transformer (L11)	IF can, 455 kHz	factory: AM IF (A); SAMS: 455 kHz	SAMS DCR: primary 22 Ω, secondary 22 Ω. Not retuned.
Second AM IF transformer (L13)	IF can, 455 kHz	factory: AM IF (B); SAMS: 455 kHz	SAMS DCR: primary 22 Ω, secondary 20 Ω. Not retuned. Its secondary feeds V6′s diodes.
AM IF screen supply resistor (R13)	1 kΩ	same	Feeds the far end of L13′s primary and V4′s screen.
AM IF screen supply bypass capacitor (C29)	0.01 µF	same	
Detector IF bypass capacitor (C31)	100 pF	same	
Detector load resistor (R15)	50 kΩ	same	
Detector DC return resistor (R16)	1 MΩ	same	
Audio IF bypass capacitor (C32)	100 pF	same	
AVC feed resistor (R17)	2 MΩ	same	
AVC bypass capacitor (C24)	0.05 µF	0.05 µF (factory and SAMS); SAMS: 50 V	
AM detector (V6′s two diodes)	12AV6, new old stock	12AV6	The whole tube is in section 13.8 (p. 127).
Mode switch, audio unit (S1B)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

For a replacement AM IF transformer, neither original gives what you need: the windings' coupling and impedance, the pinout, the internal capacitor values, the bandwidth or the mechanical fit. The tuned frequency is 455 kHz.

13.6.3 How to check it

- Coil resistance. The parts table gives the SAMS DCR figures for L11 and L13, which describe the original parts. A DCR is a check for an open winding, not a measure of inductance or tuning.
- Shared with FM. An open secondary in L11, or an open primary in L13, also stops V4 on FM (section 13.4.4 (p. 117)).
- Tuning. The AM IF transformers were not retuned when the FM path was tuned (section 13.1.1 (p. 99)).
- Detector. V6′s diode plates are pins 5 and 6, both on the same end of L13′s secondary; the cathode, pin 2, is on the chassis.

13.7 AM and SW input, converter (V3, 12BE6) and the SW and AM local oscillator

V3 is a pentagrid converter: one tube with five grids that is both the mixer and the local oscillator for AM and SW. Its oscillator section runs on its first grid and cathode. The station enters on its third grid, the signal grid. The IF, 455 kHz, leaves from its plate into the first AM IF transformer (L11). The mode switch picks the input coil and the oscillator coil for each band. The input coil is the ferrite loopstick (a ferrite rod carrying an antenna winding) in AM,

and the SW antenna coil in SW. V3 takes its B+ from AM_B, so it works only in SW and AM.

The external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is. Every other input part is fitted too: the SW coupling capacitor (C21), the SW antenna coil (L6) and its trimmer (C22A), and the loopstick's trimmer (C23A).

SAMS gives the bands as 540 to 1620 kHz for AM (the loopstick) and 3.5 to 9 MHz for SW (the SW antenna coil).

The stage has its own trimmers and padders, the small capacitors that set its alignment and tracking (section 13.1.3 (p. 100)).

13.7.1 How it is joined

SW input. The SW antenna coupling capacitor (C21) has one lead where the SW antenna terminal (J2) would be; nothing else is on that lead. Its other lead goes to one end of the primary of the SW antenna coil (L6); the other end of that primary is on the chassis. One end of L6′s secondary is the SW RF point, which the SW antenna trimmer (C22A) joins to the chassis, and which goes to the SW contact of the mode switch's RF unit (S1C). The other end of the secondary is on the chassis.

AM input. One end of the loopstick (L7) is the AM RF point, which goes to the AM contact of S1C. The other end is on the AVC line (section 13.6 (p. 120)). The AM antenna trimmer (C23A) sits across the loopstick, from the AVC line to the AM RF point.

Signal grid. The common of S1C, the AM/SW RF tuning section (VC3, whose other side is on the chassis) and V3′s signal grid (pin 7) are one point. In SW, S1C puts the SW RF point on it; in AM, the AM RF point; in FM and FM-AFC, nothing. So in AM the grid returns to the AVC line through the loopstick, and in SW it returns to the chassis through the SW coil's secondary.

Plate and screens. The plate (pin 5) goes to the primary of the first AM IF transformer (L11). The screen grids (pin 6) are on AM_B with the other end of that primary. No capacitor is on AM_B (section 13.6.1 (p. 120)).

Oscillator grid. The oscillator grid (pin 1) goes to the chassis through the oscillator grid leak resistor (R11, 20 kΩ; grid leak: section 13.3.1 (p. 109)). The grid also goes through the oscillator feedback capacitor (C25, 100 pF) to the oscillator tuning point.

Oscillator tuning point. It joins C25, the AM/SW oscillator tuning section (VC4, whose other side is on the chassis) and the common of the mode switch's oscillator tank unit (S1F).

- In SW, S1F goes through the SW oscillator padder (C26, 0.003 µF) to the top of the SW oscillator coil (L9). The SW oscillator trimmer (C22B) joins that top to the chassis.

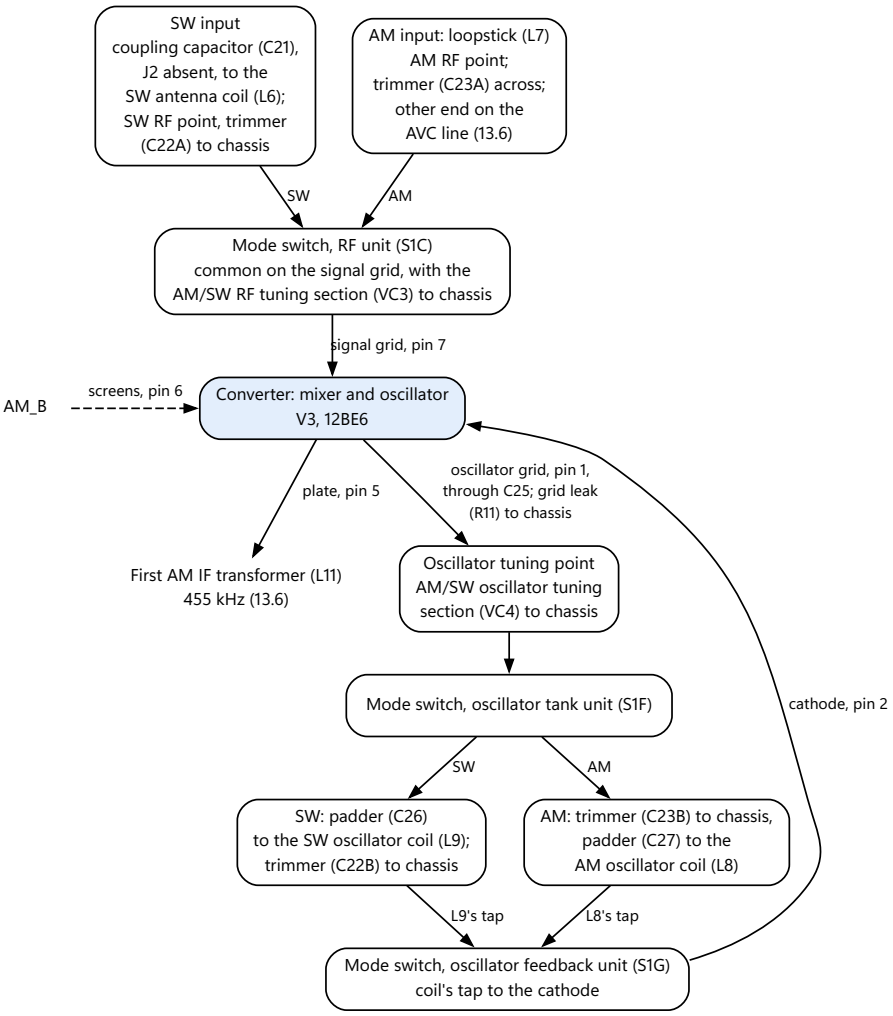
- In AM, S1F goes to a point that the AM oscillator trimmer (C23B) joins to the chassis. From there the AM oscillator padder (C27, 350 pF) leads to the top of the AM oscillator coil (L8).

The bottoms of both oscillator coils are on the chassis.

Cathode. The cathode (pin 2) goes only to the common of the mode switch's oscillator feedback unit (S1G). In SW, S1G puts it on the tap of the SW oscillator coil (L9); in AM, on the tap of the AM oscillator coil (L8). In FM and FM-AFC the cathode is connected to nothing, and the tube has no B+.

Heater. The heater is on pins 3 and 4, in the series string (section 13.9 (p. 130)).

Figure 76. How the stage is joined (simplified)



13.7.2 V3's pins

The table gives each of V3's pins and what it is joined to.

Table 129. V3′s pins

Pin	What it is	Joined to
1	oscillator grid (grid 1)	grid leak (R11) to chassis; feedback capacitor (C25) to the oscillator tuning point
2	cathode	oscillator feedback unit (S1G) only: SW oscillator coil's tap in SW, AM oscillator coil's tap in AM
3	heater	heater string, toward the FM RF tube's heater through a heater choke (L16)
4	heater	heater string, from the first IF tube's heater; bypass capacitor (C47) to chassis
5	plate	primary of the first AM IF transformer (L11)
6	screen grids (grids 2 and 4)	`AM_B`, with the other end of L11′s primary
7	signal grid (grid 3)	RF unit's common (S1C) and the AM/SW RF tuning section (VC3)

13.7.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 130. Parts

Part	Fitted	Original	Notes
Converter tube (V3)	12BE6	12BE6	
SW antenna terminal (J2)	absent	factory: antenna terminal with a series coupling capacitor; SAMS: a terminal, no rating	The external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is.
SW antenna coupling capacitor (C21)	450 pF	same	Several listed substitutes, Sprague among them, are 470 pF. The listed substitutes are DC-rated and give no safety class for an antenna input on a mains-connected chassis.
SW antenna coil (L6)	two-winding coil	factory: two-winding SW matching coil; SAMS: DCR 0.3 Ω each winding, band 3.5–9 MHz	Inductance not given.
SW antenna trimmer (C22A)	trimmer	no value in either original	Range and dielectric not specified. Part of the band alignment: do not choose one by appearance.
AM loopstick (L7)	loopstick	factory: ferrite bar with antenna winding; SAMS: DCR 0.6 Ω, band 540–1620 kHz	Inductance not given.
AM antenna trimmer (C23A)	trimmer	no value in either original	As the SW antenna trimmer.
AM/SW RF tuning section (VC3)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Serves both SW and AM. Turns with the other three sections of the tuning capacitor.
Mode switch, RF unit (S1C)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

Part	Fitted	Original	Notes
Oscillator grid leak resistor (R11)	20 kΩ	same	
Oscillator feedback capacitor (C25)	100 pF	same	
AM/SW oscillator tuning section (VC4)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Serves both SW and AM.
Mode switch, oscillator tank unit (S1F)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
SW oscillator padder (C26)	0.003 µF	0.003 µF (3000 pF), SAMS: ±5 %	A stable, low-loss RF capacitor. The listed substitute is silvered mica, 500 V.
SW oscillator trimmer (C22B)	trimmer	no value in either original	As the SW antenna trimmer.
SW oscillator coil (L9)	tapped coil	factory: tapped oscillator coil; SAMS: DCR 0.3 Ω and 0.1 Ω sections	
AM oscillator padder (C27)	350 pF	same	Do not round it to 330 pF: the padder sets the tracking. Sprague 5GA-T35 is 350 pF; Aerovox DI-330 and other listed substitutes are not.
AM oscillator trimmer (C23B)	trimmer	no value in either original	As the SW antenna trimmer.
AM oscillator coil (L8)	tapped coil	factory: tapped oscillator coil; SAMS: DCR 4.6 Ω and 0.9 Ω sections	SAMS: on the listed Merit, Miller and Stancor substitutes, ignore the extra tap.
Mode switch, oscillator feedback unit (S1G)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

The AVC bypass capacitor (C24) is in section 13.6 (p. 120).

13.7.4 How to check it

- Coil resistance. SAMS gives these DCRs for the original coils: the SW antenna coil (L6) 0.3 Ω each winding; the loopstick (L7) 0.6 Ω; the AM oscillator coil (L8) 4.6 Ω and 0.9 Ω for its two sections; the SW oscillator coil (L9) 0.3 Ω and 0.1 Ω. A DCR finds an open winding; it does not measure inductance.
- The mode switch. With the cord out, the mode switch's RF, oscillator tank and oscillator feedback units (S1C, S1F, S1G) each join their common to their SW contact in SW and their AM contact in AM. In FM and FM-AFC they join it to nothing (section 13.1.6 (p. 104)).
- Alignment. The AM and SW path was not retuned when the FM path was (section 13.1.1 (p. 99)).

13.8 The audio amplifier (V6, 12AV6, and V7, 50C5), tone switch and output transformer

The audio picked by the mode switch's audio unit (S1B) reaches the top of the radio's volume control (R1). Its wiper drives the triode inside V6, the first audio amplifier, which raises the audio's voltage. V7, a 50C5 beam power tube (a tube built to deliver power to a load), is the output stage: it drives the primary of the output transformer (T1). A tone switch can join one or two small capacitors from V6′s plate to the chassis, which takes some of the treble away. V6 and V7 have B+ in every position of the mode switch.

13.8.1 How it is joined

Volume control (R1). Its top (pin 1) comes from the common of the audio unit (S1B). Its bottom (pin 3) is on the chassis. Its wiper (pin 2) goes through the grid coupling capacitor (C38, 0.01 µF) to V6′s triode grid (pin 1). The audio grid leak resistor (R25, 2 MΩ) joins that grid to the chassis.

V6′s triode. The cathode (pin 2), shared with the AM detector diodes, is on the chassis. The plate (pin 7) is fed from the B+ filter (`B115_A`) through the audio plate load resistor (R26, 250 kΩ). Three more parts leave the plate: the audio plate shunt capacitor (C39, 100 pF) to the chassis, the output coupling capacitor (C42, 0.01 µF) to V7′s grid, and the first tone capacitor (C40).

Tone switch. The first tone capacitor (C40, 0.002 µF) runs from V6′s plate to a point the tone switch calls LOW. The second tone capacitor (C41, 0.002 µF) runs from LOW to a point called MID. The tone unit's common is on the chassis. The table shows what each tone position switches in:

Table 131. How it is joined

Tone position	What it switches in, from V6′s plate to the chassis
H (high)	nothing: the H contact is connected to nothing
M (medium)	C40 and C41 in series: 0.001 µF
L (low)	C40 alone: 0.002 µF

The factory drawing and SAMS agree on this. SAMS numbers the switch's lugs 13 for the common, 2 for OFF, 3 for H, 4 for M and 5 for L. The letters H, M and L above are the drawing's; no record maps them to the physical lugs.

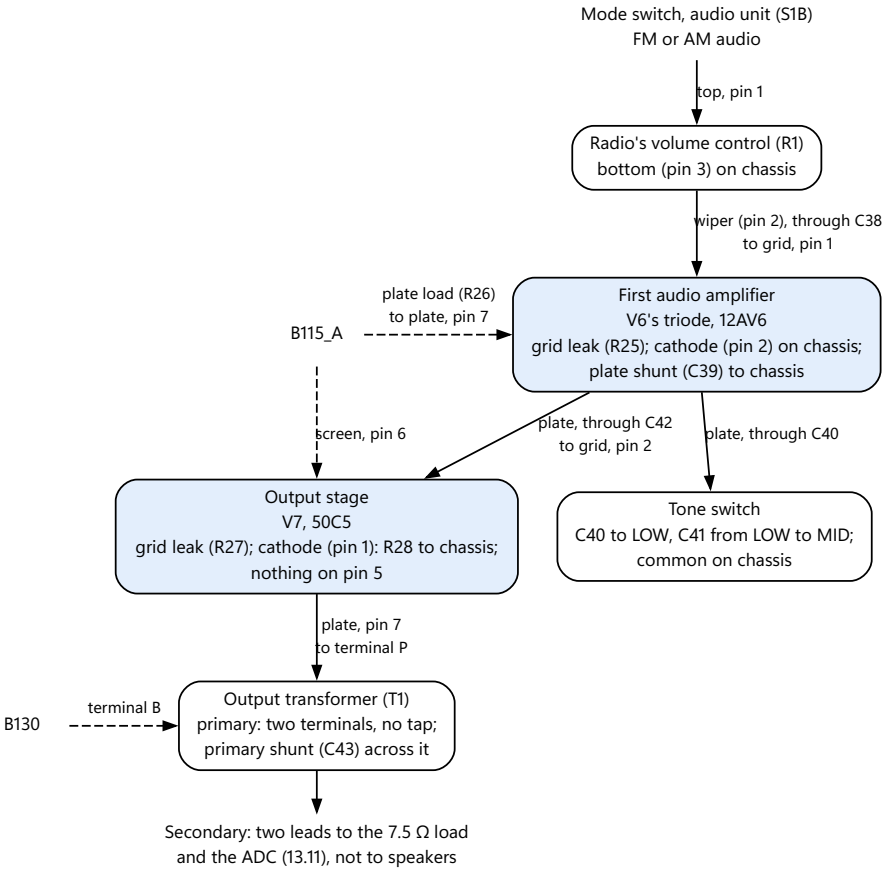
V7. The control grid (pin 2) takes the audio from C42 and returns to the chassis through the output grid leak resistor (R27, 500 kΩ). The cathode (pin 1) goes to the chassis through the output cathode resistor (R28, 150 Ω 1 W); nothing else is on the cathode. The screen grid (pin 6) is on the B+ filter (`B115_A`). The plate (pin 7) goes to one end of the output transformer's primary (terminal P) and to the primary shunt capacitor (C43, 0.0047 µF). Nothing is wired to pin 5.

Output transformer (T1). Its primary has two terminals and no tap. Terminal P is on V7′s plate. Terminal B is on `B130`, the first filtered point of the B+ supply (section 13.10 (p. 134)), and C43 sits across the two. The secondary's two leads leave the radio for the ADC (section 13.11 (p. 138)).

SP1 and SP2 are the radio's original speakers as drawn by the factory, in series across the secondary. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.

Heaters. V6′s heater is on pins 3 and 4, V7′s on pins 3 and 4, both in the series string (section 13.9 (p. 130)).

Figure 77. How the stage is joined (simplified)



13.8.2 V6′s and V7′s pins

The table gives each pin of both tubes and what it is joined to.

Table 132. V6′s and V7′s pins

Pin	V6, 12AV6	Joined to	V7, 50C5	Joined to
1	triode grid	grid coupling capacitor (C38) from the volume wiper; grid leak (R25) to chassis	cathode	output cathode resistor (R28) to chassis
2	cathode, shared by the triode and both diodes	chassis	control grid	output coupling capacitor (C42); grid leak (R27) to chassis
3	heater	heater string, toward the first IF tube's heater	heater	heater string, toward the limiter's heater
4	heater	heater string, from the limiter's heater	heater	the fused live: the start of the heater string
5	diode plate	second AM IF transformer's secondary (L13), terminal G	second control-grid pin	nothing
6	diode plate	the same point as pin 5	screen grid	B+ filter (`B115_A`)
7	triode plate	plate load (R26) from `B115_A` C39, C42, C40	plate	output transformer primary, terminal P; C43

The 50C5′s published ratings: plate at most 150 V, screen at most 130 V, plate dissipation 7 W, screen dissipation 1.4 W.

13.8.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 133. Parts

Part	Fitted	Original	Notes
AM detector and first audio tube (V6)	12AV6, new old stock	12AV6	
Output tube (V7)	50C5	50C5	
Radio's volume control (R1)	500 kΩ potentiometer	500 kΩ (factory and SAMS); SAMS: ½ W or less	Audio taper, as the original specification infers from its listed substitutes. No power switch is drawn on it. Sits at about one eighth of its travel in normal use (section 13.1.2 (p. 99)). For a replacement, measure the shaft, flat, bushing and rotation.
Grid coupling capacitor (C38)	0.01 µF	0.01 µF; SAMS: at least 450 V	A low-leakage coupling capacitor. The listed film substitute is 600 V, ±10 %.
Audio grid leak resistor (R25)	2 MΩ	same	
Audio plate load resistor (R26)	250 kΩ	same	
Audio plate shunt capacitor (C39)	100 pF	same	
Output coupling capacitor (C42)	0.01 µF	0.01 µF; SAMS: at least 450 V	As the grid coupling capacitor.
Output grid leak resistor (R27)	500 kΩ	same	

Part	Fitted	Original	Notes
Output cathode resistor (R28)	150 Ω, 1 W	same; SAMS: 1 W	Use at least 1 W, with room around it for the heat.
First tone capacitor (C40)	0.002 µF	same	
Second tone capacitor (C41)	0.002 µF	same	
Power and tone switch (S2)	one switch: a tone unit and a power unit	SAMS part M6; factory: OFF, H, M, L, with a linked power contact	No part number or contact rating is listed. A replacement needs mains-rated contacts and insulation, the same contact sequence, shaft and mounting. The power unit is in section 13.10 (p. 134).
Output transformer (T1)	output transformer, two-terminal primary, no tap	factory: single-ended, two-terminal primary, two speakers in series; SAMS: part 38.11.1, 7000 Ω primary tapped at 2500 Ω, 6–8 Ω secondary	The fitted transformer has the factory form. SAMS's DCRs (active primary 176 Ω, unused section 146 Ω, secondary 0.9 Ω) describe SAMS's tapped version, not the fitted part. Do not apply SAMS's full-primary 7000 Ω to its active tap.
Primary shunt capacitor (C43)	0.0047 µF	0.005 µF (factory and SAMS); SAMS: at least 450 V	For a replacement, also consider its pulse and transient rating.
Speakers (SP1, SP2)	not on the output: it feeds the 7.5 Ω load and the ADC	factory: two speakers in series; SAMS: 4 × 6 inch, permanent magnet, 3–4 Ω each, 6–8 Ω the pair	The radio's original speakers as drawn by the factory.

13.8.4 How to check it

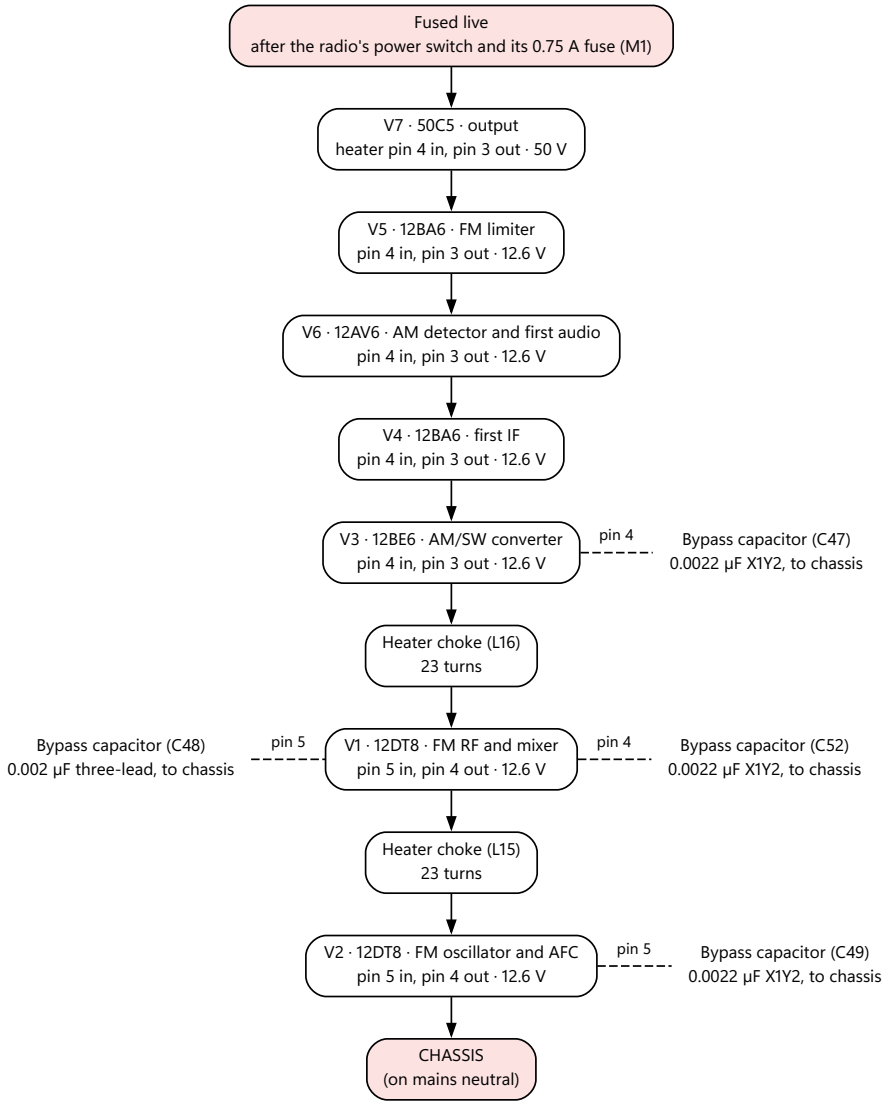
- Tone switch. With the cord out: in L, 0.002 µF (C40) is between V6′s plate and the chassis; in M, 0.001 µF (C40 and C41 in series); in H, neither.
- The output transformer. Its primary has two terminals only. SAMS's DCR figures do not describe it.
- V7, pin 5. Nothing is wired to it.
- Volume. The radio's own volume control belongs at about one eighth of its travel. Higher settings make the sound too loud further down the chain (chapter 6 (p. 28), section 6.8 (p. 37)).

13.9 The series heater string, the mode supply and the indicator lamps

The seven tube heaters are wired in series, one after the other, from the fused live to the chassis, so one current flows through all of them. Every tube here has a 0.15 A heater, the usual heater current of AC/DC sets. The 50C5 has a 50 V heater, the six others 12.6 V each (section 13.1.4 (p. 101)). Because the string is one chain, one open heater or choke stops all seven heaters. The same switch unit that sends B+ to the AM or the FM stages (S1D) and the unit that would light a mode lamp (S1E) are also described here.

13.9.1 The heater string

Figure 78. The series heater string, from the fused live to the chassis



The order was checked on the chassis:

Table 134. The heater string

Order	Tube or part	In	Out	At the junction after it
1	output tube V7, 50C5	pin 4, on the fused live	pin 3	
2	limiter V5, 12BA6	pin 4	pin 3	
3	detector and first audio tube V6, 12AV6	pin 4	pin 3	
4	first IF tube V4, 12BA6	pin 4	pin 3	the heater-string bypass capacitor (C47) to chassis
5	converter V3, 12BE6	pin 4	pin 3	
6	heater choke (L16)			the three-lead heater bypass capacitor (C48) to chassis
7	FM RF and mixer tube V1, 12DT8	pin 5	pin 4	a heater bypass capacitor (C52) to chassis
8	heater choke (L15)			a heater bypass capacitor (C49) to chassis
9	FM oscillator and AFC tube V2, 12DT8	pin 5	pin 4, on the chassis	

The two heater chokes (L15, L16) are RF chokes: coils that pass the heater current but block radio frequencies. The choke between V1 and V2 (L15) has a bypass capacitor to the chassis on each side (C52 and C49). The choke between V3 and V1 (L16) has one, on V1′s side (C48); its V3 side has none. With the heater-string bypass capacitor at V3′s pin 4 (C47), the string has four bypass capacitors. V1 sits between the two chokes, and V2 is the last tube: its pin 4 is the chassis end of the whole string. The internal shields of both 12DT8s (pin 9) are on the chassis.

13.9.2 Mode supply

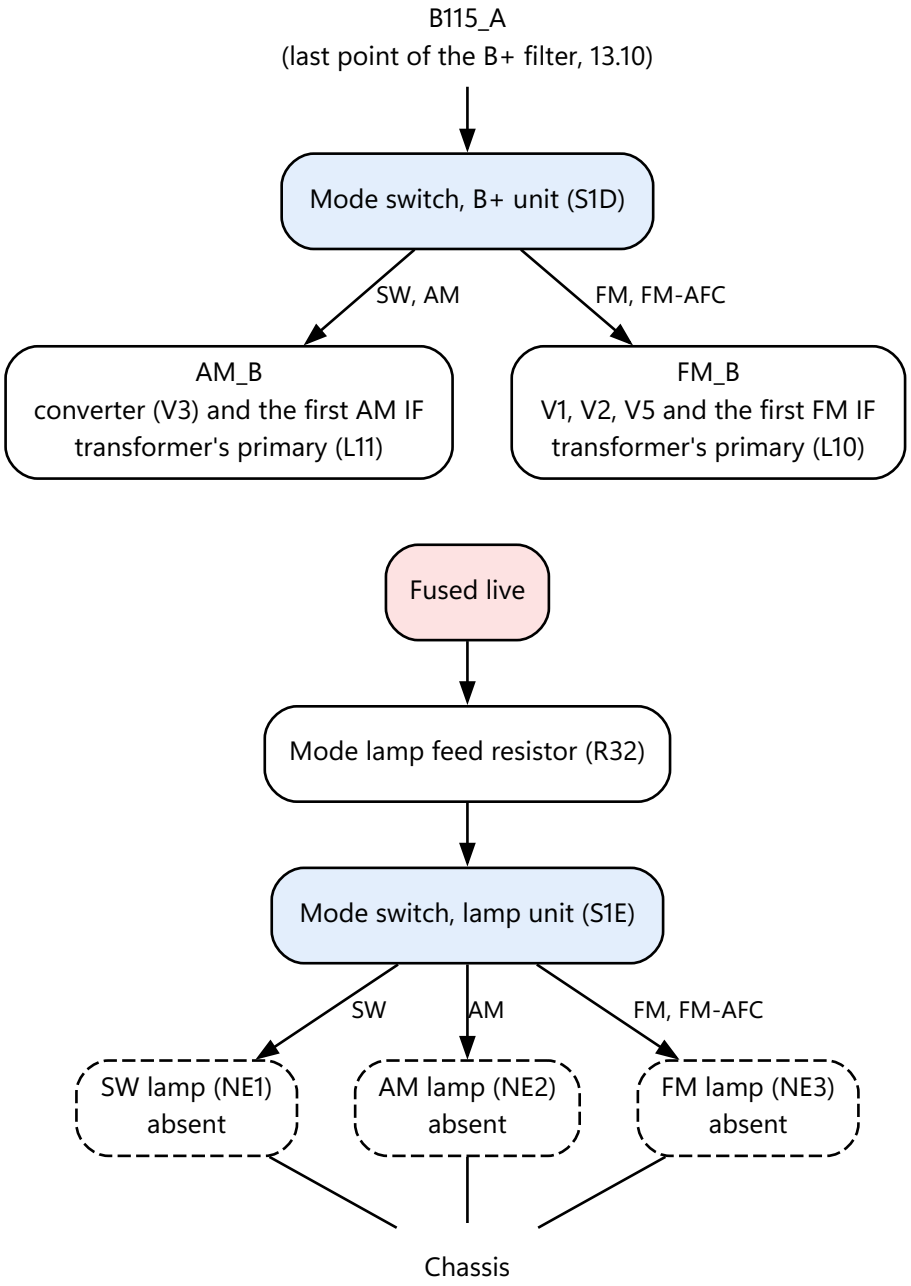
The mode switch's B+ unit (S1D) has its common on the B+ filter (`B115_A`). It puts B+ on `AM_B` in SW and AM, and on `FM_B` in FM and FM-AFC. Section 13.1.6 lists what each supply feeds.

13.9.3 Indicator lamps

The mode lamp feed resistor (R32, 50 kΩ) runs from the fused live to the common of the mode switch's lamp unit (S1E). S1E sends that feed to the SW lamp (NE1) in SW, the AM lamp (NE2) in AM, and the FM lamp (NE3) in FM and FM-AFC. Each lamp's other side goes to the chassis.

The neon bulbs NE1, NE2 and NE3 are absent from this radio. R32 and the S1E contacts are fitted.

Figure 79. How the stage is joined (simplified)



13.9.4 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 135. Parts

Part	Fitted	Original	Notes
Heater choke between V3 and V1 (L16)	RF heater choke	factory: series heater RF choke; SAMS: 23 turns	23 turns alone does not define a replacement: neither original gives the winding's diameter, length, wire gauge, core, current or insulation rating.
Heater choke between V1 and V2 (L15)	RF heater choke	factory: series heater RF choke; SAMS: 23 turns	As the choke between V3 and V1.
Heater-string bypass capacitor (C47), at V3′s pin 4	0.0022 µF, safety class X1Y2	0.002 µF (factory and SAMS)	Replaced the original. A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)).
Heater bypass capacitor (C48), at V1′s pin 5	0.002 µF, three-lead ceramic, the original part	0.002 µF (factory and SAMS)	Middle lead to chassis (section 13.1.3 (p. 100)).
Heater bypass capacitor (C52), at V1′s pin 4	0.0022 µF, safety class X1Y2	—	Replaced the original part. A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)).
Heater bypass capacitor (C49), at V2′s pin 5	0.0022 µF, safety class X1Y2	0.002 µF (factory and SAMS)	Replaced the original. A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)).
Mode switch, B+ unit (S1D)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
Mode switch, lamp unit (S1E)	one unit of the mode switch, contacts fitted	SAMS part M5	Section 13.1.6.
Mode lamp feed resistor (R32)	50 kΩ	same	Original wattage not printed. Work out the lamp's voltage and current, and the resistor's dissipation, before you buy one.
SW lamp (NE1)	absent	a neon lamp, sharing the 50 kΩ feed resistor	Neither original gives a lamp model, striking or running voltage, current, size or brightness.
AM lamp (NE2)	absent	as the SW lamp	
FM lamp (NE3)	absent	as the SW lamp	Lit in FM and FM-AFC.

A safety marking alone does not make a part equivalent at RF: match its construction and its place on the chassis.

13.9.5 How to check it

- All heaters cold. With the cord out, check the string end to end, in the order of the table in section 13.9.1 (p. 131): each heater, then each choke. One open part breaks the whole chain.
- The string's ends. V7′s pin 4 is on the fused live; V2′s pin 4 is on the chassis.
- The two 12DT8s. V1 is the tube between the two chokes; V2 is the tube whose heater returns to the chassis.
- The three-lead capacitor (C48). Its two outer leads read 0 Ω to each other, and the middle lead goes to the chassis.

13.10 The original transformerless AC supply and B+

"Original" here means the radio's own supply, not part of Ambersong; this section describes it as fitted now. The set has no mains transformer. Its live

terminal goes through the radio's power switch and fuse to the fused live, which feeds the heater string (section 13.9 (p. 130)) and the B+ supply. The neutral terminal is the chassis.

The B+ is made by one silicon diode used as a half-wave rectifier: it passes only one half of each mains cycle. Three electrolytic capacitors then smooth it, with two dropping resistors between them (series resistors that lower and filter the supply).

Everything in this section is connected to the mains: read chapter 2 (p. 7), section 2.1 (p. 7), first.

13.10.1 How it is joined

Mains in. The radio's live terminal (J3) goes to the input of the power unit of the power and tone switch (S2). That unit's output goes straight into the mains fuse (M1, 0.75 A), with no wire between them. The fuse's output is the fused live. The radio's neutral terminal (J4) is the chassis itself.

From the fused live, four things leave:

- the heater string, at V7′s pin 4 (section 13.9 (p. 130));
- the mode lamp feed resistor (R32, section 13.9 (p. 130));
- the mains bypass capacitor (C44, 0.01 µF, safety class X1Y2), to the chassis;
- the surge resistor (R29, 28 Ω 3 W), to the rectifier.

Rectifier. R29 goes to the anode of the silicon rectifier (X1). Its cathode is the raw B+ point, `B_RAW`. The rectifier shunt capacitor (C46, 0.0047 µF) sits across the rectifier.

Filter. Each B+ point has its own 47 µF capacitor, + to the point and – to the chassis:

Table 136. How it is joined

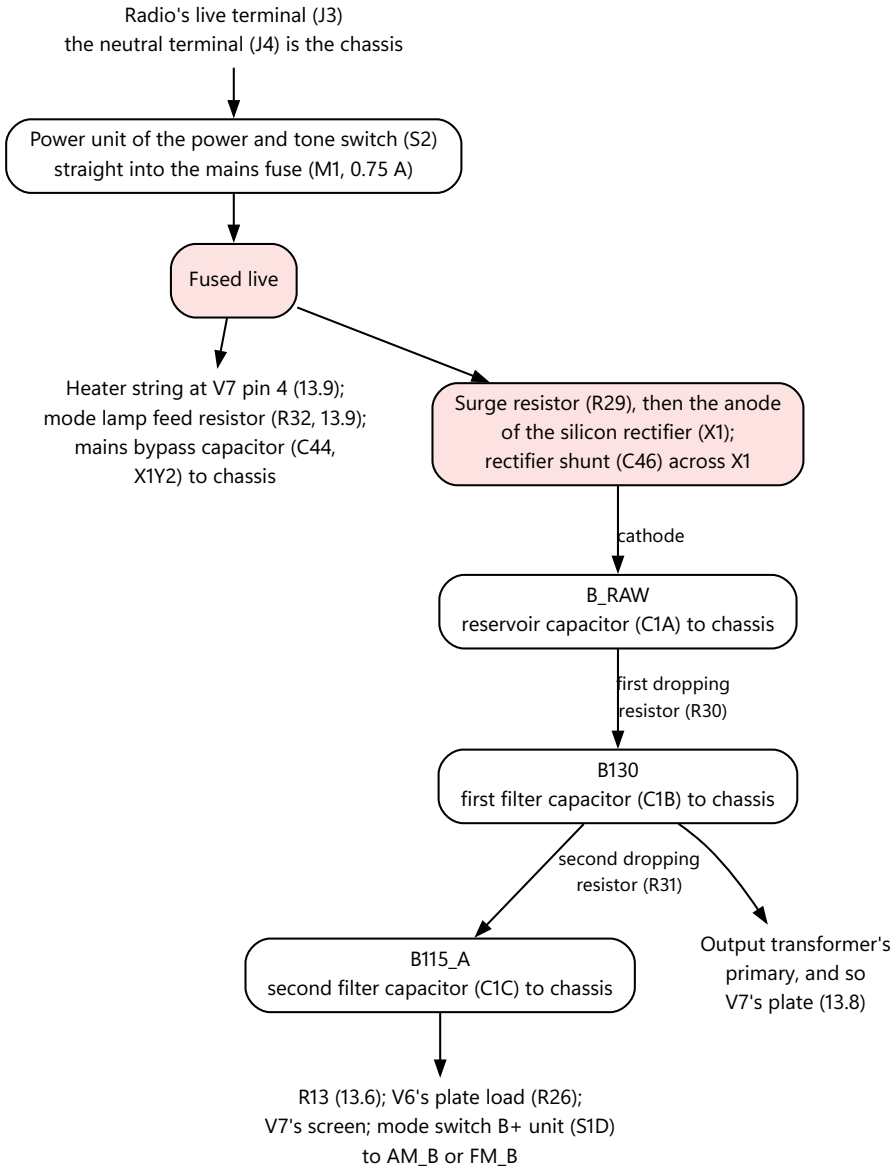
Point	Its capacitor	Reached through	Feeds
`B_RAW`	reservoir capacitor (C1A)	the rectifier (X1)	the first dropping resistor only
`B130`	first filter capacitor (C1B)	first dropping resistor (R30, 150 Ω 5 W)	the output transformer's primary, and so V7′s plate (section 13.8 (p. 127)); the second dropping resistor
`B115_A`	second filter capacitor (C1C)	second dropping resistor (R31, 300 Ω 1 W)	the AM IF screen supply resistor (R13, section 13.6 (p. 120)); V6′s plate load (R26); V7′s screen grid; the mode switch's B+ unit (S1D), which feeds `AM_B` or `FM_B` (section 13.1.6 (p. 104))

The reservoir capacitor is the first one after the rectifier: it holds the B+ up between the rectifier's pulses.

The names are not voltages. The numbers in `B130` and `B115_A` are the SAMS data's nominal names, not measured voltages of this radio.

Two coils are gone. Two of the radio's original coils, L17 and L18, were removed by accident. The external mains line filter took their place (chapter 5 (p. 21), section 5.2 (p. 23)).

Figure 80. How the stage is joined (simplified)



13.10.2 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 100)):

Table 137. Parts

Part	Fitted	Original	Notes
Radio's live terminal (J3)	fitted	—	Takes the switched live through the line filter (section 13.11 (p. 138)).
Radio's neutral terminal (J4)	fitted	—	It is the chassis. Takes neutral through the line filter.
Power and tone switch, power unit (S2)	one switch with the tone unit (section 13.8 (p. 127))	SAMS part M6: a separate AC power pole and tone contacts	
Mains fuse (M1)	0.75 A	factory: no fuse drawn; SAMS: 0.75 A, type GJV, in a different position, omitted in some versions	Here it sits between the power switch and the fused live. The original specification's selection note: use a matching speed, voltage, breaking capacity and package.
Mains bypass capacitor (C44)	0.01 µF, safety class X1Y2	factory: 0.01 µF, no voltage, tolerance or safety class given	SAMS numbers its version of this bypass C45; SAMS's 400 V for it does not set this part's rating. This radio has no C45. A safety-rated part (chapter 2 (p. 7), section 2.4 (p. 8)).
Surge resistor (R29)	28 Ω, 3 W	same; SAMS: 3 W	Use at least 3 W, with room for the heat. SAMS's listed substitute, IRC MR 2, is a configurable 10 W wirewound assembly, not a fixed 28 Ω part.
Rectifier (X1)	silicon rectifier	factory: silicon rectifier; SAMS: rating not printed, output current measured at 0.090 A	Choose at least the 400 V, 0.5 A class of the period, with suitable surge, heat and leakage ratings. A higher rating alone does not give the same B+. One listed substitute, the RCA 1N2861, carries a conflicting 105 V AC capacitor-input rating.
Rectifier shunt capacitor (C46)	0.0047 µF	0.005 µF (factory and SAMS); SAMS: at least 450 V	Consider its pulse and transient rating.
Reservoir capacitor (C1A)	47 µF, Rubycon, 315 V, a separate capacitor	40 µF, 150 V, one section of a common-negative can (factory and SAMS)	Match polarity, ripple current, temperature and start-up voltage.
First dropping resistor (R30)	150 Ω, 5 W	150 Ω; SAMS: 2 W	
First filter capacitor (C1B)	47 µF, Rubycon, 315 V, a separate capacitor	40 µF, 150 V, one section of the same can	As the reservoir capacitor.
Second dropping resistor (R31)	300 Ω, 1 W	same; SAMS: 1 W	
Second filter capacitor (C1C)	47 µF, Rubycon, 315 V, a separate capacitor	40 µF, 150 V, one section of the same can	As the reservoir capacitor.

13.10.3 How to check it

All with the cord out.

- Fuse. 0.75 A, between the power switch and the fused live.

- Rectifier direction. Anode toward the surge resistor (R29), cathode toward the reservoir capacitor (C1A).
- Capacitor polarity. The reservoir and the two filter capacitors (C1A, C1B, C1C) each have + on their B+ point and – on the chassis.
- Safety parts. The mains bypass capacitor (C44) is X1Y2 rated (chapter 2 (p. 7), section 2.4 (p. 8)).

13.11 Where the tube radio meets Ambersong

Section 13.1.2 lists the meeting points. This section gives the radio's side of each.

Mains. The radio's live terminal (J3) takes the switched live, after the front power switch, through the external mains line filter; its neutral terminal (J4) takes neutral through the same filter. The filter sits under its own earthed Faraday cage. There is no isolation transformer. So the radio needs both the front power switch and its own power switch (S2) on to run (chapter 5 (p. 21), section 5.2 (p. 23)). S2 is not on the front panel. The tube radio has its own 0.75 A fuse; there is no other fuse on the machine's mains side (M1, section 13.10 (p. 134)).

Chassis. Because J4 is the chassis, CHASSIS is on mains neutral. It is never joined to Earth, to the Faraday cages or to the DC ground (chapter 2 (p. 7), section 2.3 (p. 8) chapter 5 (p. 21), section 5.3 (p. 23)).

Antenna. The radio's FM input is a coax connector (J53, RG179 coax). It replaced the radio's original simple wire antenna. Its centre goes through the antenna coupling capacitor (C3, X1Y1) to the FM RF stage (section 13.2 (p. 105)). Its shield is on Earth and reaches the chassis only through the antenna shield capacitor (C51, 0.001 µF, X1Y2). The machine's antenna connector is J10, a BNC on the back panel. No cable joining J10 to J53 is drawn (chapter 10 (p. 60), section 10.35 (p. 91)). The antenna itself is in chapter 11 (p. 92) the earth path is in chapter 5 (p. 21), section 5.4 (p. 25). On the AM and SW side, the external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is (section 13.7 (p. 122)).

Sound out. The output transformer (T1) is one part: its primary is the radio's (section 13.8 (p. 127)), its secondary feeds Ambersong. The secondary's two leads carry RAD_L and RAD_R. A 7.5 Ω 5 W resistor (marked 5W7Ω5J) sits across them as the transformer's load. A shielded cable takes them to the radio input jack at the ADC (J51). Which lead goes to which contact does not matter. On the cable side of that jack the shield and foil are on the sleeve, RAD_L on the ring and RAD_R on the tip. On its output side the sleeve and ring are on ground, and the tip goes into the 4.7 kΩ resistor of the input attenuator. So the feed is mono: RAD_L goes to ground with the cable's shield and foil, and RAD_R goes through the 4.7 kΩ (chapter 6 (p. 28), section 6.4 (p. 33) chapter 10 (p. 60), section 10.22 (p. 81)).

Volume. Of the three volume controls in the chain, the radio's own (R1) is the first. It is not on the front panel, and sits at about one eighth of its travel in normal use (chapter 6 (p. 28), section 6.8 (p. 37)).

FM calibration receiver. Its antenna, a 30 cm wire bent in a U, lies over the FM oscillator section inside the radio's cage. It is not connected to the radio (section 13.3 (p. 109) chapter 8 (p. 46), section 8.5 (p. 51)).

13.12 When something is wrong

Find what you see in the left column; the right column says where to look first.

Table 138. When something is wrong

You see	Where to look
Radio silent on every band; no heater warms up	The radio's own power switch (S2) and its fuse (M1). Then the heater string, end to end (section 13.9.5 (p. 134)): one open heater or choke stops all seven.
Radio silent on every band; heaters warm	Parts every band shares: the B+ chain (section 13.10 (p. 134)), the first IF tube (V4), V6 and V7, the volume control and the output transformer.
FM silent, SW and AM work	What runs only in FM and FM-AFC: the mode switch's B+ unit on FM_B, V1, V2, V5, and the FM windings of the IF transformers (sections 13.2 (p. 105) to 13.5).
SW and AM silent, FM works	What runs only in SW and AM: AM_B, the converter (V3), the AM windings of the IF transformers and V6′s diodes (sections 13.6 (p. 120), 13.7).
FM works in FM but not in FM-AFC	The only difference between the two positions is the AFC-off unit (S1A). Look at the AFC path: the AFC feed from the ratio detector (section 13.5 (p. 117)), the AFC triode and its grid network (section 13.3 (p. 109)).
Radio too loud further down the chain	The radio's own volume control: it belongs at about one eighth of its travel (section 13.8 (p. 127)).
Treble missing	The tone switch: L and M shunt treble to the chassis; H switches nothing in (section 13.8 (p. 127)).

Appendix A. Datasheet facts used in this machine

This appendix gathers the makers' figures that matter for this machine: pinouts, limits, addresses and ratings, part by part. Only the facts the machine uses are here. You need it when you replace a part, check a reading against a limit, or adapt the design to other parts. The chapters tell you what each part does; abbreviations are explained in the glossary.

Every figure is the maker's, for the catalogue part. It is not a measurement of the part fitted here. Sometimes the maker's document describes a related part rather than the fitted one: a module the boards may not carry, or a reference development board the fitted board may not match. The entry then says so. There are no entries for the real-time clock module: its datasheet is not included in this book (Appendix C (p. 149)).

A.1 Main board: ESP32-S3

The fitted module is marked ESP32-S3-N16R8. The chip read itself as an ESP32-S3 (QFN56), revision v0.2, with 8 MB of PSRAM inside (chapter 4 (p. 12), section 4.1 (p. 12)). The module figures below are from Espressif's ESP32-S3-WROOM-1 module datasheet. That the fitted module is a WROOM-1 is not established.

The table gives each fact, the maker's figure, and why it matters on this machine.

Table 139. A.1 Main board: ESP32-S3

Fact	Maker's figure	Why it matters here
Memory, variant N16R8	16 MB flash (quad SPI) and 8 MB PSRAM (octal SPI); operating temperature −40 to 65 °C	the fitted module's marking
GPIO35, GPIO36, GPIO37	On modules with octal PSRAM, connected to the PSRAM and not available for other uses. Espressif's ESP32-S3-DevKitC-1 guide says the same for its boards.	not used here (chapter 4 (p. 12), section 4.2 (p. 13))
GPIO47, GPIO48 voltage	1.8 V only on modules whose chip is an ESP32-S3R16V. The chip datasheet gives the ESP32-S3R8 a 3.3 V flash supply (VDD_SPI).	They carry the FM calibration receiver's bus. That they work at the normal 3.3 V on this board is an inference from these two figures; the datasheet does not state it directly.
Supply (VDD33)	3.0 to 3.6 V, 3.3 V typical; absolute maximum 3.6 V	
Strapping pins	GPIO0 (weak pull-up) and GPIO46 (weak pull-down) set the boot mode; GPIO45 (weak pull-down) sets the flash supply voltage; GPIO3 (floating) sets the JTAG signal source. They are read at reset. GPIO1 is not a strapping pin.	none of the four is used; GPIO1 reads the amplifier power sensor
GPIO19, GPIO20	USB D− and D+, connected by default to the chip's USB Serial/JTAG controller. Reconfiguring them as ordinary GPIOs disables USB-JTAG.	the main board's USB port
GPIO39 to GPIO42	the JTAG interface: MTCK (GPIO39), MTDO (GPIO40), MTDI (GPIO41), MTMS (GPIO42)	used here for the angle sensor's clock and three digit selects
UART0	TXD0 on GPIO43, RXD0 on GPIO44	not used
RGB LED (DevKitC-1 guide)	on GPIO48 on the first version of the DevKitC-1 board, on GPIO38 on version 1.1	The fitted board's RGB LED is on GPIO48, which also carries the calibration receiver's clock.
Power inputs (DevKitC-1 guide)	the USB ports, the 5V pin and the 3V3 pin are "three mutually exclusive ways to provide power to the board"	see the caution in chapter 2 (p. 7), section 2.5 (p. 9)

A.2 Audio board: ESP32

The audio board is meant to carry an ESP32-WROOM-32E module; whether it does is not established. Its chip read itself as an ESP32-D0WD-V3. The module figures are from Espressif's ESP32-WROOM-32 datasheet. The board figures are from Espressif's ESP32-DevKitC V4 guide. Whether the fitted board matches that board is not established.

The table is laid out as in section A (p. 139).1.

Table 140. A.2 Audio board: ESP32

Fact	Maker's figure	Why it matters here
GPIO6 to GPIO11	connected to the module's SPI flash; not recommended for other uses	not connected here
Strapping pins	GPIO0 (pull-up) and GPIO2 (pull-down) set the boot mode; GPIO12 (MTDI, pull-down) selects the flash supply voltage at reset; GPIO15 (MTDO, pull-up); GPIO5 (pull-up)	GPIO0 carries the converters' master clock
GPIO34 to GPIO39	input only: no output driver and no internal pull-up or pull-down (ESP32 chip datasheet). On the module: GPIO36 is ADC1 channel 0, GPIO39 channel 3, GPIO34 channel 6, GPIO35 channel 7.	the volume knob (GPIO35) and the source switch (GPIO36)
Supply (VDD33)	3.0 to 3.6 V, 3.3 V typical; absolute maximum −0.3 to 3.6 V	
GPIO16, GPIO17 (DevKitC V4)	available only on boards with ESP32-WROOM or ESP32-SOLO-1 modules; boards with ESP32-WROVER modules reserve them	GPIO17 is the word clock here
Capacitor C15 (DevKitC V4)	On earlier DevKitC V4 boards, C15 may make the board boot into download mode, and may disturb a clock output on GPIO0.	GPIO0 is the master clock output here
Power inputs (DevKitC V4)	Micro USB, the 5V pin and the 3V3 pin are three mutually exclusive ways to power the board; power must come from one and only one of them, "otherwise the board and/or the power supply source can be damaged"	see chapter 2 (p. 7), section 2.5 (p. 9)

A.3 The ADC: PCM1802 (Texas Instruments)

The ADC turns the tube radio's audio into digital samples for the audio board (chapter 6 (p. 28)). The table gives the maker's figures for its supplies and setting pins.

Table 141. A.3 The ADC: PCM1802 (Texas Instruments)

Fact	Maker's figure
Supplies	VCC (pin 5): analogue, 5 V. VDD (pin 14): digital, 3.3 V.
Setting pins	PDWN (pin 7), BYPAS (pin 8), OSR (pin 16), FMT0, FMT1, MODE0 and MODE1 are Schmitt-trigger inputs with an internal pull-down of about 50 kΩ, and are 5 V tolerant.
PDWN	power-down, active low: the chip runs only while PDWN is high
Format	FMT1 low and FMT0 high: I2S, 24-bit
Interface mode	MODE1 and MODE0 both low: slave mode (system clock 256, 384, 512 or 768 times the sample rate)
Slave mode	BCK, LRCK and FSYNC are inputs. The chip sends its data only while FSYNC is high.
OSR	low: oversampling ×64; high: ×128
BYPAS	low: normal mode, with the high-pass filter cutting DC; high: filter bypassed
SCKI (pin 15)	the system clock input, 256, 384, 512 or 768 times the sample rate; Schmitt-trigger, 5 V tolerant

How each setting pin is tied on this machine's module is in chapter 6 (p. 28), section 6.3 (p. 32).

A.4 The DAC: PCM5102A (Texas Instruments)

The DAC turns the audio board's digital audio back into analogue audio for the amplifier (chapter 6 (p. 28)). The table gives the maker's figures for its pins, supplies and modes.

Table 142. A.4 The DAC: PCM5102A (Texas Instruments)

Fact	Maker's figure
Pins	OUTL 6, OUTR 7, AVDD 8, AGND 9, DEMP 10, FLT 11, SCK 12, BCK 13, DIN 14, LRCK 15, FMT 16, XSMT 17, DVDD 20
Digital inputs	failsafe LVCMOS Schmitt-trigger inputs
Supplies	AVDD: 3.3 V. DVDD: 1.8 V or 3.3 V. Absolute maximum 3.9 V on AVDD, CPVDD and DVDD.
XSMT	soft mute when low, un-mute when high. Where it is not needed, it can be tied straight to AVDD.
FMT	low: I2S; high: left-justified
FLT	low: normal-latency filter; high: low-latency filter
DEMP	de-emphasis for 44.1 kHz audio: off when low, on when high
Zero-data mute	After 1024 word-clock periods of zero data (21 ms at 48 kHz), on both channels by default, the chip puts its output into full analogue mute.
Master clock and PLL	The internal PLL is turned off as soon as an external system clock (SCK) is supplied. If BCK and LRCK start correctly while SCK stays at ground for 16 word-clock periods, the PLL starts and makes its own system clock from BCK.

A.5 Amplifier power sensor and box-fan regulator

PC817 optocoupler (Sharp). An optocoupler passes a signal across by light: an LED on one side lights a phototransistor on the other, with no wire between them.

Table 143. A.5 Amplifier power sensor and box-fan regulator

Fact	Maker's figure
Pins (DIP-4)	1 anode, 2 cathode, 3 emitter, 4 collector
Absolute maximum, input (LED)	forward current 50 mA; peak forward current 1 A; reverse voltage 6 V; power dissipation 70 mW
Absolute maximum, output	collector–emitter 80 V; emitter–collector 6 V; collector current 50 mA
Current transfer ratio (CTR)	50 to 400 % at 5 mA LED current and 5 V collector–emitter, 25 °C; 50 % minimum. The CTR is the output current as a share of the LED current.

LM7812 12 V regulator (Texas Instruments). It makes the box fan's 12 V from the amplifier supply's +20 V (chapter 5 (p. 21), section 5.6 (p. 27)).

Table 144. A.5 Amplifier power sensor and box-fan regulator

Fact	Maker's figure
Pins (TO-220)	1 input, 2 ground, 3 output
Output	11.5 to 12.5 V at 25 °C; 11.4 to 12.6 V over an input of 14.5 to 27 V, 5 mA to 1 A output, at most 15 W dissipated. The maker specifies it at 19 V in.

The amplifier supply that feeds it measured 19.33 V with no load (chapter 5 (p. 21), section 5.6 (p. 27)).

A.6 Display and motor drivers

ULN2003A (Texas Instruments). The chip holds seven Darlington transistor pairs, each able to pull a load to ground. It is used on the display driver board and on the needle motor driver.

Table 145. A.6 Display and motor drivers

Fact	Maker's figure
Inputs	1B to 7B on pins 1 to 7
Outputs	1C to 7C on pins 16 to 10 (open collector: each can only pull low)
E (pin 8)	common emitter of all channels, usually tied to ground
COM (pin 9)	common cathode of the built-in flyback diodes; required for inductive loads
Input resistor	2.7 kΩ in series with each base, so the inputs work straight from 5 V or 3.3 V logic
Absolute maximum	collector–emitter 50 V; input 30 V; peak collector current 500 mA; all referred to E

FQP27P06 P-channel MOSFET (onsemi). These are the digit switches of the clock display (chapter 7 (p. 38)).

Table 146. A.6 Display and motor drivers

Fact	Maker's figure
Drain–source voltage	−60 V
Drain current, continuous	−27 A (case at 25 °C)
Gate–source voltage	±25 V maximum
Gate threshold	−2.0 to −4.0 V
On-resistance	70 mΩ maximum at −10 V gate–source, −13.5 A
Package and pins	TO-220, pins G, D, S

A.7 Connectors

JST XH. These are the 2-pin sockets of the 5 V and 3.3 V buses. They are identified only by the drawing's text "JST XH".

Table 147. A.7 Connectors

Fact	Maker's figure
Pitch	2.5 mm
Current	3 A AC/DC, with 22 AWG wire
Voltage	250 V AC/DC
Temperature	−25 to +85 °C
Wire	30 to 22 AWG (0.05 to 0.33 mm²); insulation 0.9 to 1.9 mm outside diameter

IEC C14 mains inlet. No datasheet names the fitted inlet. One maker's C14 inlet, the Schurter 6100-4, is rated 10 A, protection class I.

A.8 Needle, tuning and lamp parts

AS5600 magnetic angle sensor (the tuning shaft).

Table 148. A.8 Needle, tuning and lamp parts

Fact	Maker's figure
I2C address	0x36 (7-bit)
DIR pin	on ground: the reading increases clockwise; on the supply: counter-clockwise
SDA, SCL	"consider external pull-up"
Supply	VDD3V3 absolute maximum 4.0 V. For 3.3 V operation, the VDD5V and VDD3V3 pins must be tied together.

RDA5807M FM receiver (the FM calibration receiver).

Table 149. A.8 Needle, tuning and lamp parts

Fact	Maker's figure
Pins	ground 1, 3 and 8; FM_IN 2; SCLK 4 (serial clock); SDIO 5; RCLK 6 (32.768 kHz crystal or reference clock input); VDD 7; ROUT and LOUT 9 and 10
Supply	1.8 to 3.3 V, 3.0 V typical, through an internal regulator
Control	I2C only
Address	0010000 binary (0x10). This revision of the datasheet documents no other address. The fitted chip answers at 0x11 (observed on the running machine, not a datasheet figure).
Bypass	the maker's reference design puts a 22 nF supply bypass capacitor close to pin 7

The decoupling fitted on this machine's module is in chapter 8 (p. 46), section 8.5 (p. 51).

IRL540N MOSFET (the panel-lamp switch).

Table 150. A.8 Needle, tuning and lamp parts

Fact	Maker's figure
Gate	logic-level gate drive
Gate threshold	1.0 to 2.0 V
On-resistance	0.044 Ω at 10 V gate, 18 A; 0.053 Ω at 5.0 V, 18 A; 0.063 Ω at 4.0 V, 15 A

A3144 Hall-effect switch (the needle index).

Table 151. A.8 Needle, tuning and lamp parts

Fact	Maker's figure
Supply	4.5 to 24 V, with a built-in regulator
Output	open collector, sinks up to 25 mA; needs a pull-up for logic circuits
Pins	viewed from the branded side: 1 supply, 2 ground, 3 output
Switching	the output (pin 3) goes low when the field at the sensor rises above the operate point, and high again when it falls below the release point
Availability	discontinued the maker points A3144 users to the A1104

A.9 Tubes

The tubes' bases, heater ratings and pins, and the 50C5′s maximum ratings, are in chapter 13 (p. 98) (sections 13.1.4 (p. 101) and 13.8.2). These figures are published tube data. Some of it comes from reference sites rather than from the tubes' makers.

50C5 typical class-A operation. The 50C5 is the radio's output tube (chapter 13 (p. 98), section 13.8 (p. 127)). The table gives two typical operating points: the voltages on its plate, control grid and screen grid, the plate current, the load and the output power.

Table 152. A.9 Tubes

Plate voltage	Unit	Grid voltage	Unit	Screen voltage	Unit	Plate current	Unit	Load	Unit	Output power	Unit
110	V	−7.5	V	110	V	50	mA	2.5	kΩ	1.9	W
120	V	−8	V	110	V	49	mA	2.5	kΩ	2.3	W

A.10 The documents behind these facts

The facts in this appendix were checked against these documents, in these versions.

Table 153. A.10 The documents behind these facts

Document	Link
ESP32-S3-WROOM-1 / -1U module datasheet (Espressif, version 1.8)	documentation.espressif.com
ESP32-S3 chip datasheet (Espressif, version 2.2)	documentation.espressif.com
ESP32-S3-WROOM-2 module datasheet (Espressif, version 1.7)	documentation.espressif.com
ESP32-S3-DevKitC-1 user guide v1.1 (Espressif), header pin tables	docs.espressif.com
ESP-IDF GPIO reference for the ESP32-S3 (USB-JTAG on GPIO19 and GPIO20)	docs.espressif.com
ESP32-WROOM-32 module datasheet (Espressif, version 3.7)	documentation.espressif.com
ESP32-DevKitC user guide (Espressif), power options	docs.espressif.com
PCM1802 ADC datasheet (Texas Instruments, SLES023D)	www.ti.com
PCM5102A DAC datasheet (Texas Instruments, SLAS859C)	www.ti.com
ULN2003A Darlington array datasheet (Texas Instruments, SLRS027T)	www.ti.com
FQP27P06 P-MOSFET datasheet (onsemi, rev. 4)	www.onsemi.com
JST XH connector series (JST)	www.jst-mfg.com
An IEC 60320 C14 inlet for reference, Schurter 6100-4	www.schurter.com
12DT8 tube data	www.radiomuseum.org
12DT8 tube data (base 9DE)	nj7p.org
12BA6 tube data (base 7BK)	www.nostalgiaair.org
12BA6 tube data	www.radiomuseum.org
12BE6 tube data	www.radiomuseum.org
12BE6 pin connections	www.r-type.org
12AV6 tube data	www.radiomuseum.org
50C5 tube data	www.radiomuseum.org
50C5 ratings and typical operation (RC-29 data)	www.qsl.net
Tube base pinout diagrams (7BK and others)	www.bustedgear.com

Appendix B. Where the schematics differ from this book

You do not need the KiCad drawings to use this book. This appendix is for readers who also open them: read it before you compare a drawing with the machine or with this book. The machine was drawn to show a person what was built, not to make a circuit board. So a drawing sometimes places a part where it reads best rather than where it sits. Where a drawing and this book differ, this book is right.

B.1 Drawing errors still in the published sheets

Two known errors are left in the drawings on purpose. Both are wrong names on library symbols, and neither changes any wiring. The table gives what the drawing shows and what is in the machine.

Table 154. B.1 Drawing errors still in the published sheets

On the drawing	In the machine
The main board's module is drawn with Espressif's ESP32-S3-WROOM-2 library symbol, with the value "ESP32-S3-WROOM-1_N16R8".	The fitted module is marked ESP32-S3-N16R8 (chapter 4 (p. 12), section 4.1 (p. 12)).
The tube radio's mode-switch units (S1A to S1G) use a library symbol named TM-838M_UNIT1.	The set is a TM-838N.

Do not take the WROOM-2's datasheet for this board. It describes the module named by the symbol, not the fitted one. That module:

- carries only the ESP32-S3R8V or ESP32-S3R16V chip, with octal flash on every variant;
- runs GPIO47 and GPIO48 at 1.8 V;
- has its pins 28 to 30 not connected, so it has no IO35 to IO37 pads, while the drawn symbol has pins 35, 36 and 37.

Appendix A (p. 139) gives the figures that do apply.

B.2 Parts drawn where they read best

On the main sheet, these parts are drawn for clarity, not where they sit. The sub-sheet or the cable sheet shows the real place. The table gives where each is drawn and where it really is.

Table 155. B.2 Parts drawn where they read best

On the main sheet	In the machine
The source switch's 22 kΩ and 68 kΩ resistors, drawn in the front switch's block	on the audio board (chapter 9 (p. 52), section 9.3 (p. 55))
The 0.1 µF filter capacitor on GPIO36, drawn at the ladder side of the 1 kΩ	on the pin side, at the audio board
The colon's 33 Ω and its 1.5 kΩ variable resistor, both drawn on the 5 V side of the colon LED	5 V → 33 Ω → colon → variable resistor → ground, as the display sub-sheet draws it. The variable resistor is glued at 1.5 kΩ (chapter 7 (p. 38), section 7.6 (p. 45)).
The four 470 Ω resistors of the DAC-to-amplifier link, drawn on the DAC's output pins	in the cable, as the cable sheet draws them: the two plus legs' at the DAC end, the two minus legs' in the cable (chapter 6 (p. 28), section 6.6 (p. 36) chapter 10 (p. 60), section 10.21 (p. 80))

B.3 Reading notes

Names and symbols

- Rails with two names are one conductor: `5VDC_S` and `5VDC PSU_GND` and `GND A_VDC` and `A_3V3` (About this book, Notation).
- **`A_IO0A`** and **`A_IO0D`** are both the audio board's GPIO0, named twice so each converter's wiring reads clearly. The two clock contacts of the audio harness both carry GPIO0; neither belongs to one converter.
- One label on several wires. On the Bluetooth and source-switch sheet, "GPIO" names both the H and the L leg. On the USB-C sheet, one "TO … USB-C" name sits on all four wires of each port (VBUS, D−, D+ and ground). These are visual aids, not shorts. Each is a separate wire, and this book describes each one separately.
- **`GND`** twice on one connector (the digit-rail cable's 2-pin connector) is the cable's shield and foil, ending on the driver board's common ground (chapter 10 (p. 60), section 10.27 (p. 85)).
- **`20V_GND`** is drawn on the main sheet for completeness only. The sheet's note beside it reads: "The AMP PSU '20V_GND' is shown for completeness only; the actual path goes to the AMP's −20V and +20V inputs." The amplifier supply's four wires, as recorded, are in chapter 5 (p. 21), section 5.6 (p. 27), and chapter 10 (p. 60), section 10.30 (p. 87).
- The real-time clock is drawn with a DS3231M symbol. The fitted clock is a DS3231 module of the ZS-042 style (chapter 4 (p. 12), section 4.7 (p. 20)). The symbol's eight ground pins are hidden on the drawing, and lie on the main sheet's `A_VDC` wire. They belong to the drawn symbol only and are disregarded: they are not a connection between 3.3 V and ground. The module's ground comes on its power cable (chapter 10 (p. 60), section 10.16 (p. 76)).
- The clock module's I2C connector is drawn with pins 1 and 2 not connected, pin 3 on SDA and pin 4 on SCL. That is right: a 4-pin socket with two pins used.

References

- The radio input jack at the ADC (`RADIO_TO_ADC`) is J51, as on the main sheet; the ADC's own sheet numbers it J42. This book uses J51.
- That jack's output side. The ADC's sheet labels both its ring and its tip ATTENUATOR. The ring's label is wrong: the ring is on ground with the sleeve, and only the tip feeds the 4.7 kΩ (chapter 6 (p. 28), section 6.4 (p. 33)).
- The DAC's output plug, named `DAC_OUT` on both, is J49 on the cable sheet and J40 on the DAC sheet.
- The same reference on two sheets can mean two different parts: C13 to C30, J2 to J4, R17 and M1, for example, are used both on the main sheet and in the tube radio. Chapter 13, section 13.1.7 (p. 105), explains how this book keeps them apart.

What to ignore

- The TL074 circuit off the page of the main sheet is a draft of the amplifier's input board, drawn to help visualise it. Disregard it.
- Unlabelled rectangles over modules are the printed shells that hold or cover them.

What a drawing does and does not tell you

- Parts drawn only on the main sheet are fitted.
- Capacitors drawn as polarised are fitted the right way round as drawn; the others are film or ceramic. In the tube radio's polarised capacitors (C1A to C1C and C2), pin 1 is the + terminal.
- Pin numbers on cable connectors count along the drawn symbol. They are not contact numbers, and a sub-sheet sometimes orders a connector's pins differently from the cable sheet. This book prints the cable sheet's order. Neither order says anything about the real connector (chapter 10 (p. 60), section 10.1 (p. 60)).

The tube radio sheet

- The FM IF is 10.6 MHz. The sheet's "10.7 MHz FM IF Path" section title and the fields of the first two FM IF transformers (L10, L12) read 10.7 MHz, the design value; the third, the ratio-detector transformer (L14), is drawn in its own section but its field reads 10.7 MHz too. The fitted IF is 10.6 MHz for all three (L10, L12 and L14), set when Radio Hosvep on Park Ave. professionally tuned the FM path (chapter 13 (p. 98), section 13.1.1 (p. 99)).
- Terminal numbers are logical. The identifiers on the coils L5, L6 and L8 to L14 and on the mode switch's units S1A to S1G name connections, not physical lugs.
- One part, several units. A tube is drawn as separate units: its signal sections in their circuits, its heater on the heater string. S1A to S1G are seven drawn units of one mode switch.
- Three-lead capacitors (C5, C11, C14, C48) are drawn as two-terminal capacitors. On the part, the middle lead goes to the chassis and the two

outer leads read 0 Ω to each other (chapter 13 (p. 98), section 13.1.3 (p. 100)).
- "50C5" on the output transformer's primary means the radio's output stage.
- The speakers SP1 and SP2 are the radio's original speakers as the factory drew them. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.
- Drawn but absent: the neon lamps NE1 to NE3 and the external SW antenna terminal (J2) are not in this radio.
- Values and specifications. A part's value on the sheet is its fitted value. Its hidden specification fields hold the original factory and SAMS values; chapter 13 (p. 98) prints both.
- Notes that point elsewhere. The sheet's notes that send you to "the audio sheet" or "the Main Sheet" are left as written, because the sheet is usually one of a larger set.

Appendix C. What this book does not cover, and open points

This book states only what is known about the fitted machine. This appendix gathers, in one place, what it leaves out on purpose and what is not known. Read it before you plan a repair or a change, or before you rely on a detail the book does not give. Each open point is stated where it matters, in the chapter named. It is listed here so that you can see the gaps at a glance.

C.1 Outside the book's scope

The table lists what the book leaves out on purpose.

Table 156. C.1 Outside the book's scope

Not covered	Notes
The firmware	What the two boards run is in the companion book, the Ambersong Firmware Gospel.
The amplifier's own circuit	Chapter 12 says what the amplifier is and how it connects. No drawing shows its boards.
Wire colours	Except the amplifier supply's four wires (chapter 5 (p. 21), section 5.6 (p. 27)).
Connector details	Which side of a connector is pin 1, the crimp and connector housings, and the order of the contacts inside a connector. Pin numbers in this book are the cable sheet's (chapter 10 (p. 60), section 10.1 (p. 60)).
Cable lengths and routing	The only length facts: every cable inside the cabinet is under 14 in, and the DAC-to-amplifier cable is under 12 in (chapter 10 (p. 60), section 10.21 (p. 80)).
Screws of the printed plates	Mentioned, not detailed (chapter 3 (p. 9)).
Where each module sits in the cabinet	Beyond "each board in its own printed case, or on a printed carrying plate, screwed to the cabinet" (chapter 3 (p. 9), section 3.1 (p. 10)). The amplifier's aluminium back plate, with the mains inlet, is at the back of the cabinet, and the antenna connector is on the back panel.
The tube radio's original documents	The factory schematic and the SAMS service data are scans, and the SAMS data is not included. Their values reach this book only through the radio drawing's specification fields and notes: the "Original" columns of chapter 13 (p. 98).
The real-time clock's datasheet	The fitted clock is a DS3231 module of the ZS-042 style. Its datasheet limits are not included in this book, so Appendix A (p. 139) has no entry for it (chapter 4 (p. 12), section 4.7 (p. 20)).

C.2 Open points

The open points are grouped by chapter. In each table, a row names a topic and gives the detail that remains open.

The boards (chapter 4)

Table 157. The boards (chapter 4)

Topic	Detail
Module makers	Neither module's maker is identified. The main board module's FCC grantee code (2AD6X: the part of its FCC radio-approval ID that names the maker) differs from Espressif's (2AC7Z). The audio board's code (2BB77) is not identified with a maker. Whether either module matches any Espressif module datasheet is not established.
Flash chips	The makers behind the flash manufacturer IDs 0x68 (main board) and 0x5e (audio board) are not identified.
Main board against the DevKitC-1	Compatible in firmware and pin-compatible for every pin the machine uses. The pins it does not use have not been checked.
Audio board against the DevKitC V4	Whether it matches, its header order included, is not established.

Power and grounds (chapter 5)

Table 158. Power and grounds (chapter 5)

Topic	Detail
DC ground to Earth	The DC ground reaches Earth through the amplifier's ground (a washer on its Bass pot, chapter 5 (p. 21), section 5.3 (p. 23)), measured at almost 0 Ω.
Bus positions	Which 5 V bus socket each power cable uses; which of the clock module, the volume knob and the source switch plug into 3.3 V bus sockets; which of the 3.3 V jacks are bus positions and which sit on the audio board (chapter 10 (p. 60)).
Amplifier supply wires	Their colours are not checked inside the supply.
Box-fan harness	Which board its two plugs sit on, and how the wires end at the supply and at the fan.
Mains wiring	The wires' gauges and colours.
Mains fuse	The tube radio has its own 0.75 A fuse (chapter 13 (p. 98), section 13.10 (p. 134)); there is no other fuse on the machine's mains side.

Audio (chapters 6, 9 and 12)

Table 159. Audio (chapters 6, 9 and 12)

Topic	Detail
ADC's PDWN and FSYNC	As drawn, they are joined on a `P_3V3` label that goes nowhere else. The converter runs, so PDWN is not held low (chapter 6 (p. 28), section 6.3 (p. 32)).
DAC data pull-down	Where the 100 kΩ to ground sits along the DAC data line (chapter 4 (p. 12), section 4.5 (p. 19)).
AUX and RCA	The AUX jack reaches the amplifier's input panel, but the run is not drawn (chapter 6 (p. 28), section 6.7 (p. 37)).
DAC-to-amplifier cable	Where along the cable the two minus legs' 470 Ω resistors sit (chapter 10 (p. 60), section 10.21 (p. 80)).
Radio-to-ADC cable	Whether it is soldered or plugged at the output transformer's secondary (chapter 10 (p. 60), section 10.22 (p. 81)).
Amplifier input	What is known rests on one meter reading, taken once; nothing inside the amplifier was examined (chapter 12 (p. 96), section 12.4 (p. 97)).

Clock display (chapter 7)

Table 160. Clock display (chapter 7)

Topic	Detail
Digit order	The order (GPIO42, GPIO2, GPIO40, GPIO41 for digits 1 to 4) is the one drawn in chapter 7 (p. 38), section 7.4 (p. 43), and chapter 10 (p. 60), section 10.8 (p. 68), and the one the display shows. Which physical board or cable puts the digits in this order is not established; it is believed to be the digit driver board.
Digit sockets	Whether the display sub-sheet's digit outputs are the sockets the digits plug into (chapter 10 (p. 60), section 10.34 (p. 90)).
Colon cable	Where its reversed polarity physically sits (chapter 10 (p. 60), section 10.23 (p. 82)).
Digit-rail cable	What its shield and foil do at the display end (chapter 10 (p. 60), section 10.27 (p. 85)).

Dial and sensors (chapter 8)

Table 161. Dial and sensors (chapter 8)

Topic	Detail
Index magnet	Which pole faces the Hall switch.
Angle sensor cable	The connector at the sensor end, and its pin order (chapter 10 (p. 60), section 10.10 (p. 70)).
Calibration receiver's antenna plug	Which pins of the 2 × 3 grid carry the antenna wire (chapter 10 (p. 60), section 10.35 (p. 91)).
Needle motor lead	Anything beyond the socket's five pins (chapter 10 (p. 60), section 10.32 (p. 89)).

FM antenna (chapter 11)

Table 162. FM antenna (chapter 11)

Topic	Detail
Performance	No SWR, no field strength and no reception comparison has been measured.
Antenna coax inside	The cable from the antenna connector, on the back panel, to the radio's antenna input (chapter 10 (p. 60), section 10.35 (p. 91)).
Build details	The details of section 11.3 (p. 94) come from the build notes and were not confirmed one by one.

The tube radio (chapter 13)

Table 163. The tube radio (chapter 13)

Topic	Detail
Lugs	Which physical lug carries each logical terminal number of the coils and the mode switch (section 13.1.3 (p. 100)).
The ratio detector's tertiary winding (L14)	Where its other end goes (section 13.4 (p. 113)).
Original specifications	The oscillator cathode resistor (R33) and its bypass capacitor (C50) have no original specification (section 13.3 (p. 109)).
The antenna shield capacitor (C51)	Where it came from (section 13.2 (p. 105)).
Fitted part types and ratings	Several parts have no documented type or rating: among them the FM detector diodes (M3, M4), the mains fuse's type and voltage (M1), the rectifier's part number and rating (X1), and some capacitors' dielectrics. Each stage's parts table says which (sections 13.2 (p. 105) to 13.10).
Power switch	Which lug of the power and tone switch (S2) carries the power contact (section 13.10 (p. 134)), which of its positions have power on, and where the switch sits (it is not on the front panel).

Appendix D. Printed parts

This appendix lists every printed part of the build. You need it when you reprint a broken or lost part, or when you build a cabinet of your own.

Every printed part is an STL file (the file a 3D printer works from) in the `hardware/STLs/` folder of the published Ambersong project files, named as in the tables below: 76 files. All are PLA, except the passive radiator's membrane, which is TPU (a flexible filament). They are the parts for this cabinet and these boards; adapt them to yours. In the file names, MCU means a microcontroller board, PSU a power supply, and LEDITRON the clock display.

Not printed: the source switch and the power switch (fixed in the wooden front panel), the mains inlet (on the amplifier's aluminium back plate), and the standoffs of the amplifier's ±20 V supply, which are plastic screws and standoffs.

D.1 Speakers

These are the passive radiators that replace the speakers' original ports, and the speaker holder (chapter 12 (p. 96), section 12.1 (p. 96)).

Table 164. D.1 Speakers

File	What it is
`AUDIO - Passive Radiator - Holder.stl`	passive radiator: holder
`AUDIO - Passive Radiator - Inside Protector.stl`	passive radiator: inside protector
`AUDIO - Passive Radiator - Membrane Weight.stl`	passive radiator: membrane weight
`AUDIO - Passive Radiator - Outside Protector.stl`	passive radiator: outside protector
`AUDIO - Passive Radiator - TPU Membrane.stl`	passive radiator: membrane, printed in TPU
`AUDIO - Speakers Holder and Seal.stl`	speaker holder and seal

D.2 Back

The files whose names begin with BACK are the back panel's anchors and vents, and the parts for the USB-C ports and the antenna connector (chapter 3 (p. 9), sections 3.1 (p. 10) and 3.4).

Table 165. D.2 Back

File	What it is
`BACK - Dual USB-C Ports.stl`	the two back-panel USB-C ports (chapter 4 (p. 12), section 4.6 (p. 20))
`BACK - Electronics - BNC Antenna Interface.stl`	the antenna connector's interface: the connector (J10) is a BNC on the back panel (chapter 11 (p. 92))
`BACK - Fan Exhaust.stl`	the box fan's exhaust
`BACK - Fan Intake.stl`	the louvred air intakes
`BACK - Panel Anchors Bottom.stl`	back panel anchors, bottom
`BACK - Panel Anchors Inner Corners.stl`	back panel anchors, inner corners
`BACK - Panel Anchors Outer Corners.stl`	back panel anchors, outer corners
`BACK - Panel Anchors.stl`	back panel anchors

D.3 Front panel

These are the parts of the front panel, the FM dial and the tuning knob (chapter 3 (p. 9), section 3.2 (p. 10) chapter 8 (p. 46), section 8.1 (p. 47)).

Table 166. D.3 Front panel

File	What it is
`FRONT - FM Dial - Back.stl`	FM dial back; it is also the stepper motor's mount
`FRONT - FM Dial - Cover.stl`	FM dial cover
`FRONT - FM Dial - Spacer.stl`	FM dial spacer
`FRONT - FM Dial - Stepper Needle and Magnet Holder.stl`	the mount on the stepper motor's shaft that carries the needle and the index magnet
`FRONT - Panel Anchors - Insidemost.stl`	front panel anchors, insidemost
`FRONT - Panel Anchors - Outsidemost.stl`	front panel anchors, outsidemost
`FRONT - Tuner Pot Knob.stl`	the printed front tuning knob. On the printed shaft extension, it turns the radio's original tuning knob, which drives the dial cord and the tuning capacitor (chapter 3 (p. 9), section 3.2 (p. 10)). "Pot" is the file's word: tuning is by the tuning capacitor.
`FRONT - Tuning Pot Extender PLA Shaft.stl`	the tuning shaft extension
`FRONT - Tuning Shaft Bearing Seat.stl`	the tuning shaft's bearing seat
`FRONT - Volume Pot Knob.stl`	the front volume knob (chapter 9 (p. 52), section 9.4 (p. 58))

D.4 Board cases and holders, inside the cabinet

Each board or module sits in its own printed case, or on a printed carrying plate, screwed to the cabinet (chapter 3 (p. 9), section 3.1 (p. 10)).

Table 167. D.4 Board cases and holders, inside the cabinet

File	What it is
`IN - Electronics - 5V BUS case Bottom.stl`	5 V bus case, bottom (chapter 5 (p. 21), section 5.5 (p. 26))
`IN - Electronics - 5V BUS case Top.stl`	5 V bus case, top

File	What it is
`IN - Electronics - 5V PSU Case Bottom.stl`	5 V power supply case, bottom
`IN - Electronics - 5V PSU Case Top.stl`	5 V power supply case, top
`IN - Electronics - 5V+3V3 BUS case Bottom.stl`	5 V and 3.3 V bus case, bottom
`IN - Electronics - 5V+3V3 BUS case Top.stl`	5 V and 3.3 V bus case, top
`IN - Electronics - AC Line Filter Case.stl`	mains line filter case (chapter 5 (p. 21), section 5.4 (p. 25))
`IN - Electronics - AS5600 Case - Bottom.stl`	the angle sensor's cage, bottom (chapter 5 (p. 21), section 5.4 (p. 25))
`IN - Electronics - AS5600 Case - Connecting Rod.stl`	the printed shaft that carries the angle sensor's magnet (chapter 8 (p. 46), section 8.1 (p. 47))
`IN - Electronics - AS5600 Case - Top.stl`	the angle sensor's cage, top
`IN - Electronics - Audio Listener Case BOTTOM.stl`	the source-switch board's case, bottom (chapter 9 (p. 52), section 9.3 (p. 55))
`IN - Electronics - Audio Listener Case TOP.stl`	the source-switch board's case, top
`IN - Electronics - Audio MCU Case BOTTOM.stl`	audio board case, bottom
`IN - Electronics - Audio MCU Case TOP.stl`	audio board case, top
`IN - Electronics - Backing Panel 1.stl`	a panel that holds several of the cases
`IN - Electronics - Bluetooth LED Holder.stl`	Bluetooth LED holder
`IN - Electronics - Bluetooth LED Pusher.stl`	Bluetooth LED pusher (the LED is also the button)
`IN - Electronics - Box Fan LM7812 Case Bottom.stl`	the box fan's regulator case, bottom (chapter 5 (p. 21), section 5.6 (p. 27))
`IN - Electronics - Box Fan LM7812 Case Top.stl`	the box fan's regulator case, top
`IN - Electronics - DS3231 RTC Case BOTTOM.stl`	real-time clock module case, bottom
`IN - Electronics - DS3231 RTC Case TOP.stl`	real-time clock module case, top
`IN - Electronics - Digit Drivers Case BOTTOM.stl`	digit driver case, bottom (chapter 7 (p. 38))
`IN - Electronics - Digit Drivers Case TOP.stl`	digit driver case, top
`IN - Electronics - Front Panel FM LEDs Case BOTTOM.stl`	FM panel lamp board case, bottom (chapter 9 (p. 52), section 9.1 (p. 53))
`IN - Electronics - Front Panel FM LEDs Case TOP.stl`	FM panel lamp board case, top
`IN - Electronics - Hall Sensor Stepper Limit Case.stl`	the needle index sensor's case (chapter 8 (p. 46), section 8.3 (p. 49))
`IN - Electronics - LEDITRON Back.stl`	clock display back
`IN - Electronics - LEDITRON Bus Case Bottom.stl`	clock display bus case, bottom
`IN - Electronics - LEDITRON Bus Case TOP.stl`	clock display bus case, top
`IN - Electronics - PCM1802 ADC Case BOTTOM.stl`	ADC case, bottom
`IN - Electronics - PCM1802 ADC Case TOP.stl`	ADC case, top
`IN - Electronics - PCM5102A DAC Case Bottom.stl`	DAC case, bottom
`IN - Electronics - PCM5102A DAC Case TOP.stl`	DAC case, top
`IN - Electronics - Power Detect Case BOTTOM.stl`	amplifier power sensor case, bottom (chapter 9 (p. 52), section 9.5 (p. 59))
`IN - Electronics - Power Detect Case TOP.stl`	amplifier power sensor case, top
`IN - Electronics - RDA5807M - Bottom.stl`	FM calibration receiver case, bottom (chapter 8 (p. 46), section 8.5 (p. 51))
`IN - Electronics - RDA5807M - Cage Holder.stl`	FM calibration receiver cage holder

File	What it is
`IN - Electronics - RDA5807M - Middle Bottom.stl`	FM calibration receiver case, middle bottom
`IN - Electronics - RDA5807M - Middle TOP.stl`	FM calibration receiver case, middle top
`IN - Electronics - RDA5807M - TOP.stl`	FM calibration receiver case, top
`IN - Electronics - S3 Main MCU Backing.stl`	main board backing
`IN - Electronics - S3 Main MCU Case BOTTOM.stl`	main board case, bottom
`IN - Electronics - S3 Main MCU Case TOP.stl`	main board case, top
`IN - Electronics - Stepper Driver ULN2003A Case - Bottom.stl`	needle motor driver case, bottom (chapter 8 (p. 46), section 8.2 (p. 48))
`IN - Electronics - Stepper Driver ULN2003A Case - Top.stl`	needle motor driver case, top

D.5 The tube radio's Faraday cage

These two parts make the cage over the tube radio (chapter 3 (p. 9), section 3.5 (p. 11) chapter 5 (p. 21), section 5.4 (p. 25)).

Table 168. D.5 The tube radio's Faraday cage

File	What it is
`IN - Tube Radio - Faraday Cage 1.stl`	the tube radio's Faraday cage, part 1; the cage fan's mount is part of it
`IN - Tube Radio - Faraday Cage 2.stl`	the tube radio's Faraday cage, part 2

D.6 Clock display

These are the clock display's housing and digits (chapter 7 (p. 38)).

Table 169. D.6 Clock display

File	What it is
`LEDITRON - Colon - Blanket.stl`	colon blanket
`LEDITRON - Colon.stl`	colon
`LEDITRON - Digit.stl`	digit
`LEDITRON - Front Case.stl`	front case
`LEDITRON - Glass Spacer.stl`	glass spacer

Appendix E. Schematic sheets

Each sheet of the KiCad schematic, as drawn, one per landscape page. The source files are in `hardware/schematics/`. Where a sheet and this book differ, this book is right (Appendix B).

Figure 81. Schematic sheet: Main sheet

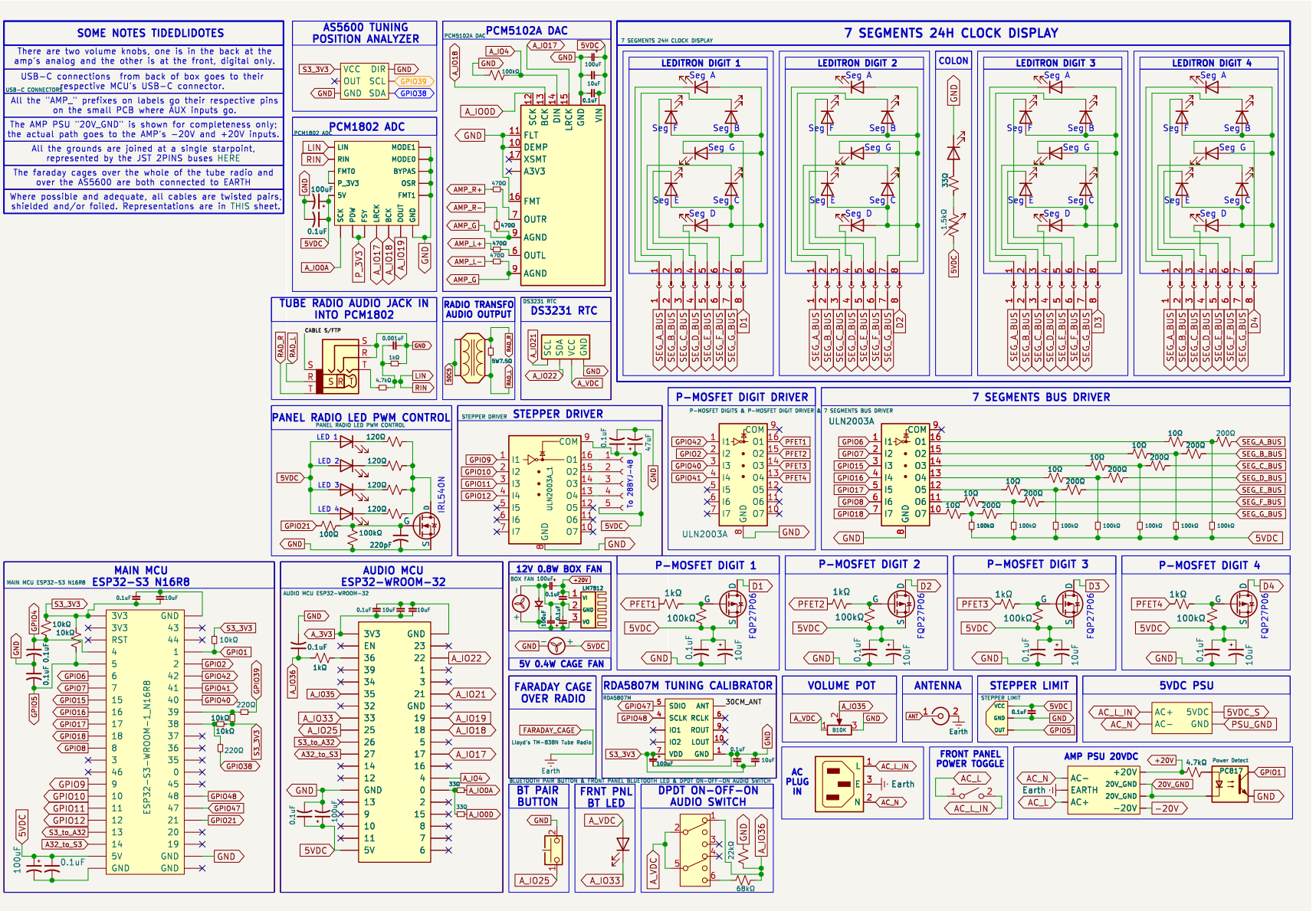


Figure 82. Schematic sheet: 7 SEGMENTS 24H CLOCK DISPLAY

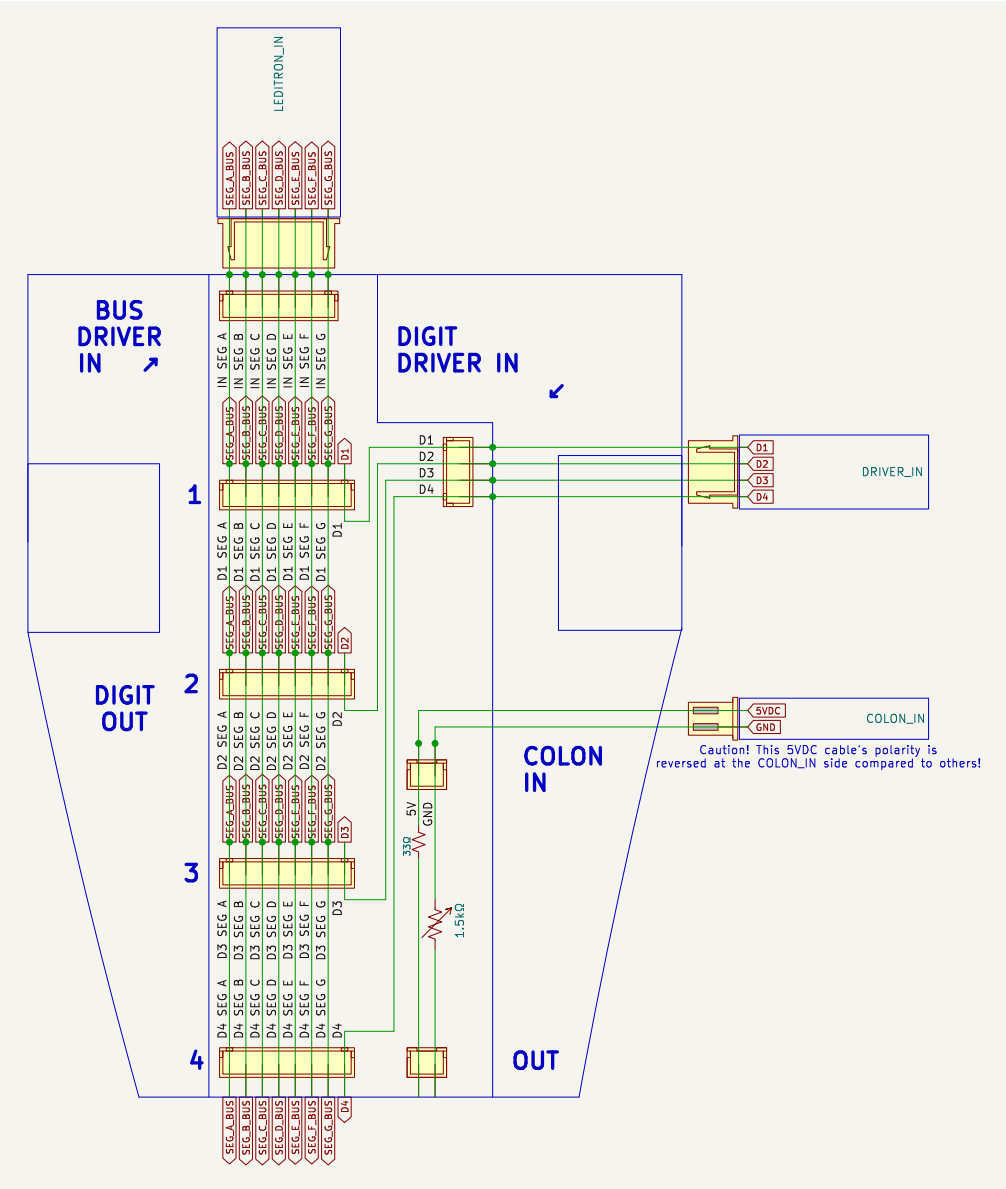


Figure 83. Schematic sheet: AUDIO MCU ESP32-WROOM-32

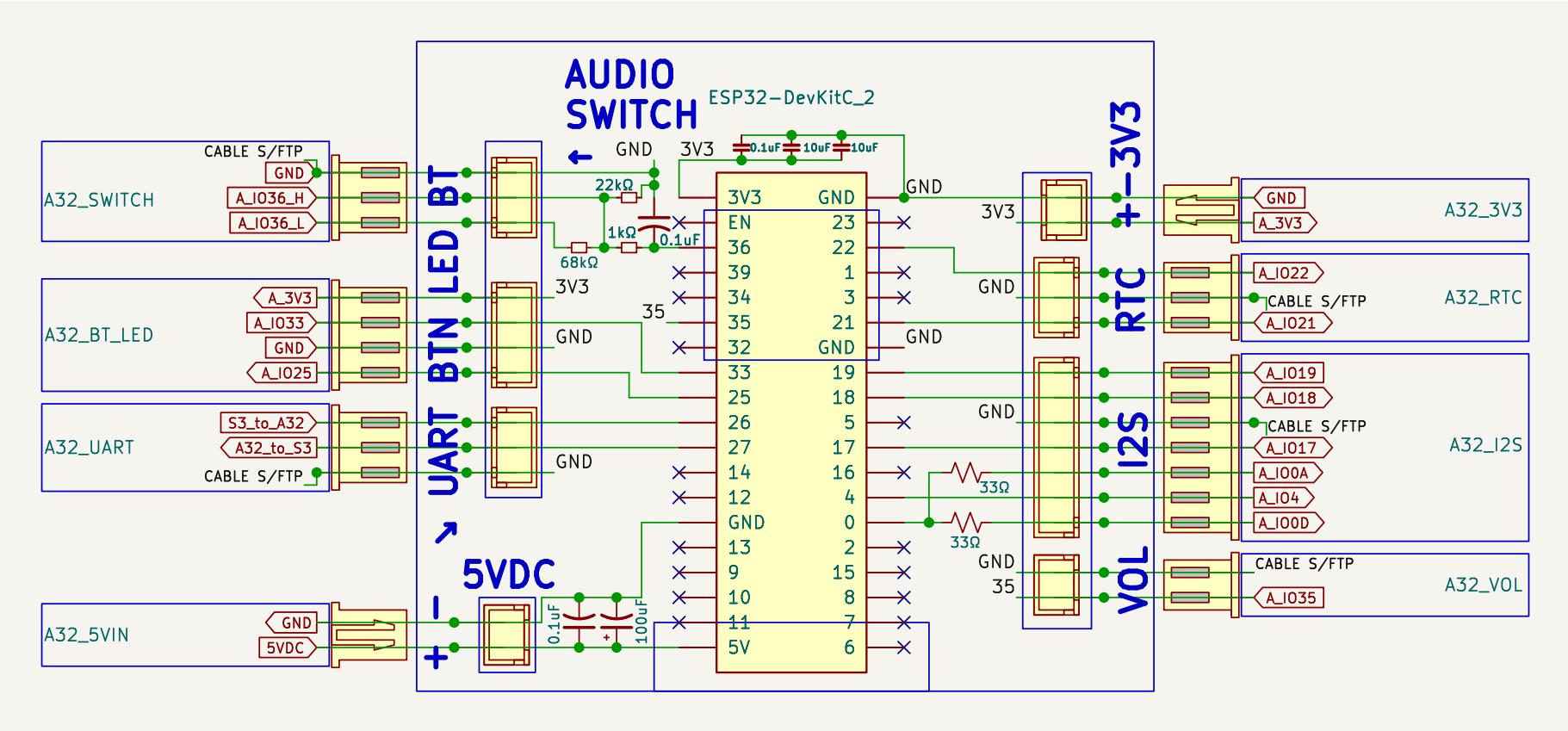


Figure 84. Schematic sheet: BLUETOOTH PAIR BUTTON & FRONT PANEL BLUETOOTH LED & DPDT ON-OFF-ON AUDIO SWITCH

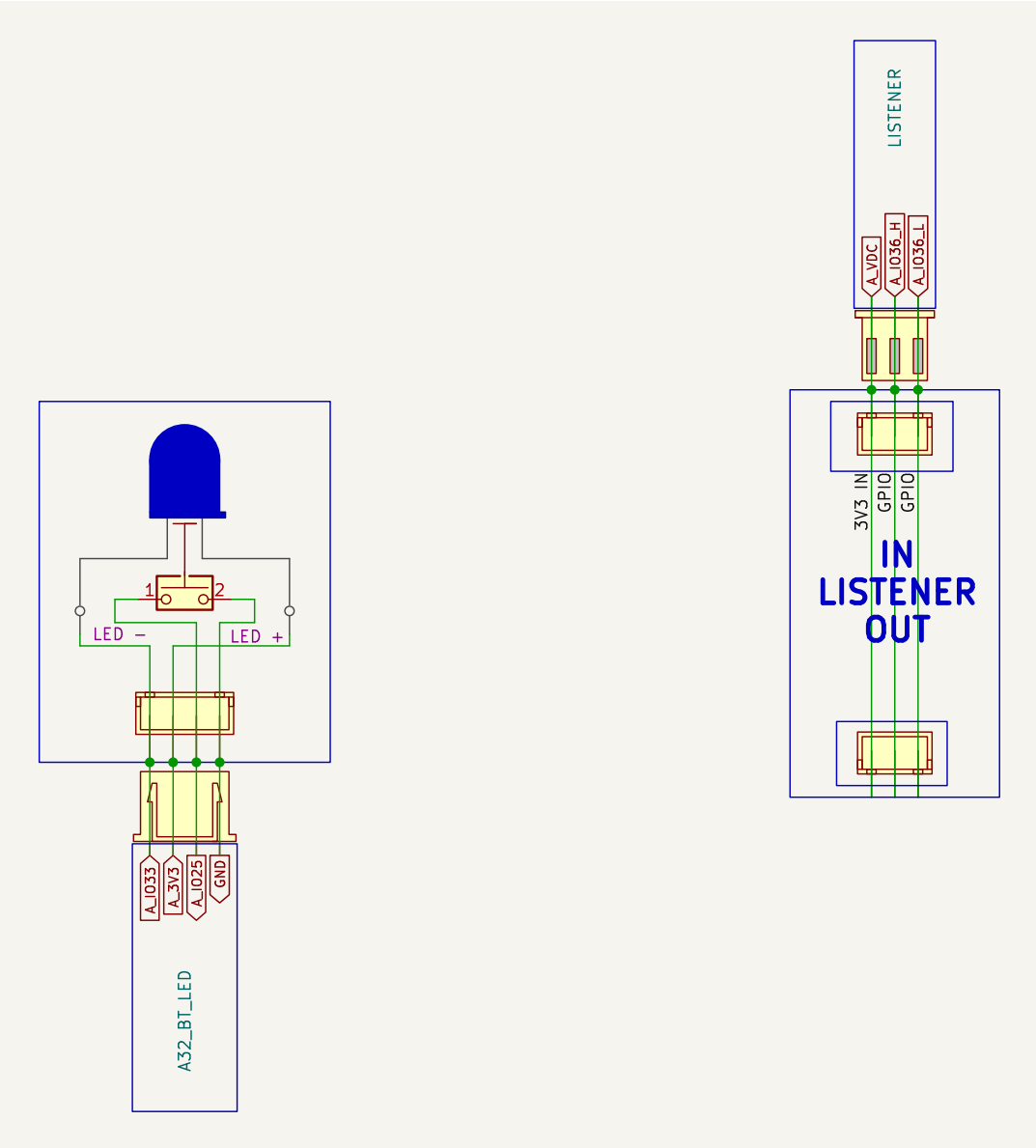


Figure 85. Schematic sheet: BOX FAN

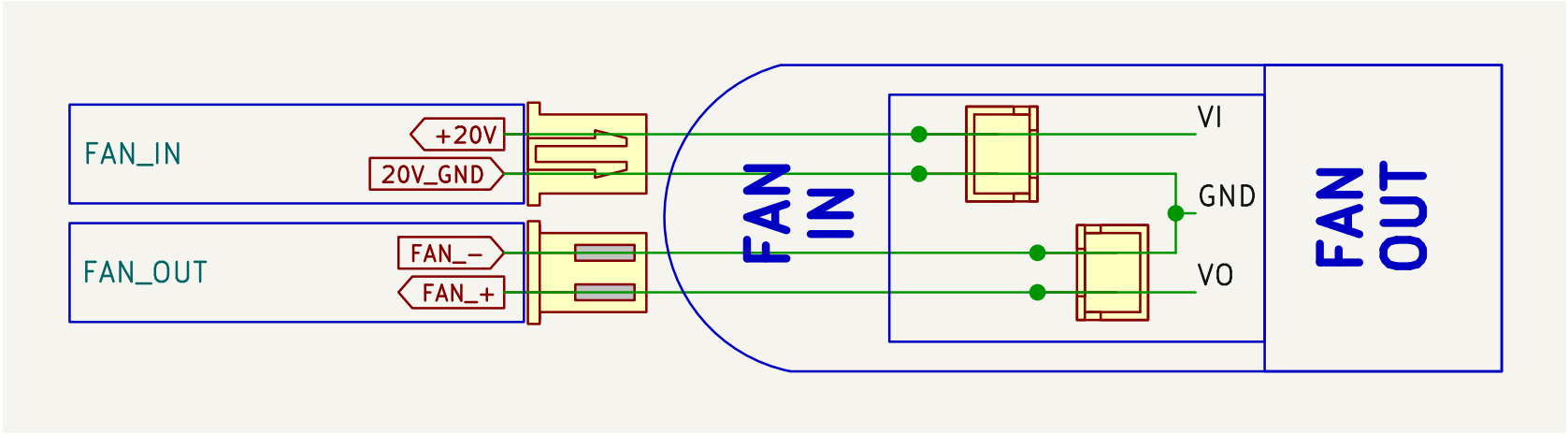


Figure 86. Schematic sheet: Cables (sheet “THIS”)

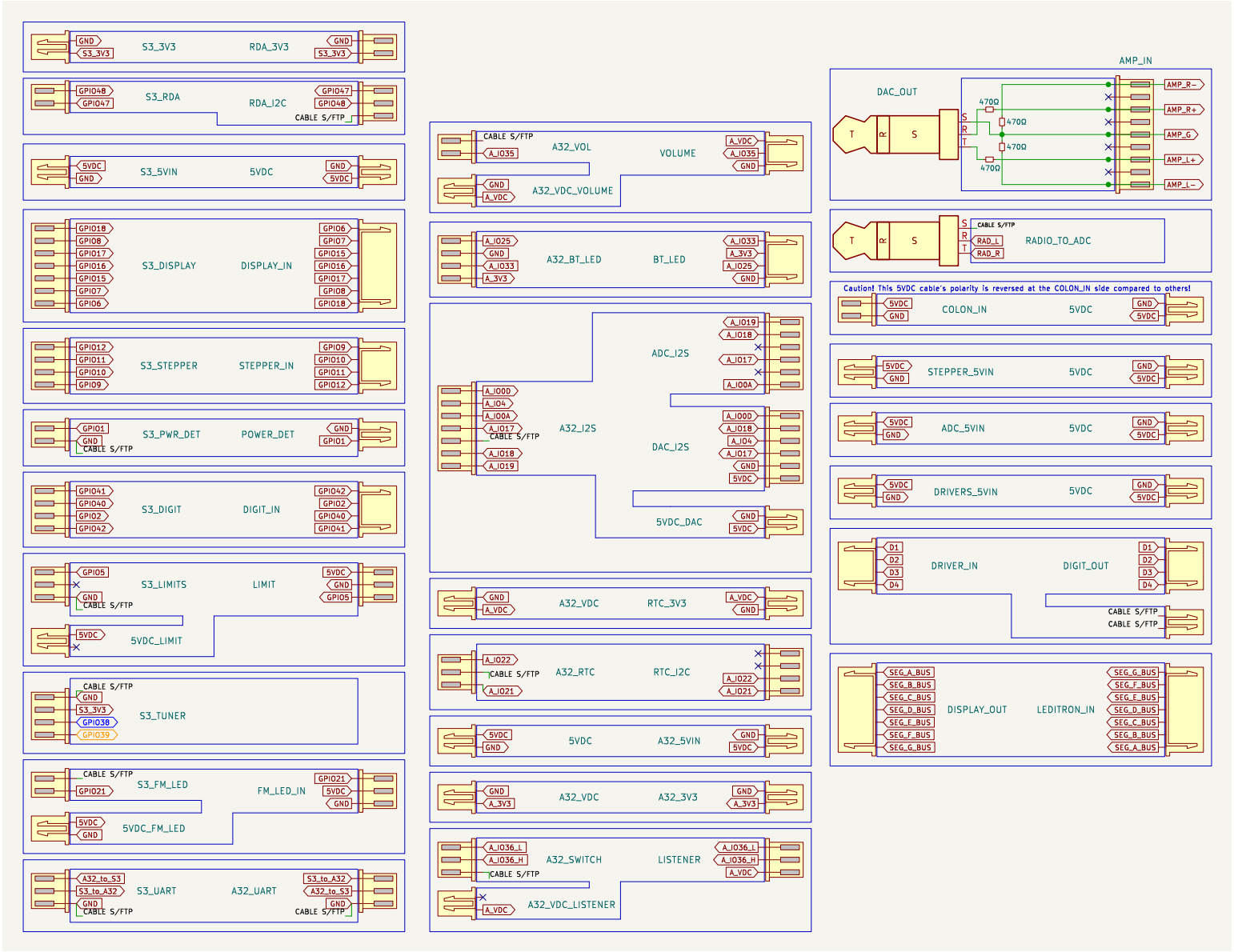


Figure 87. Schematic sheet: DS3231 RTC

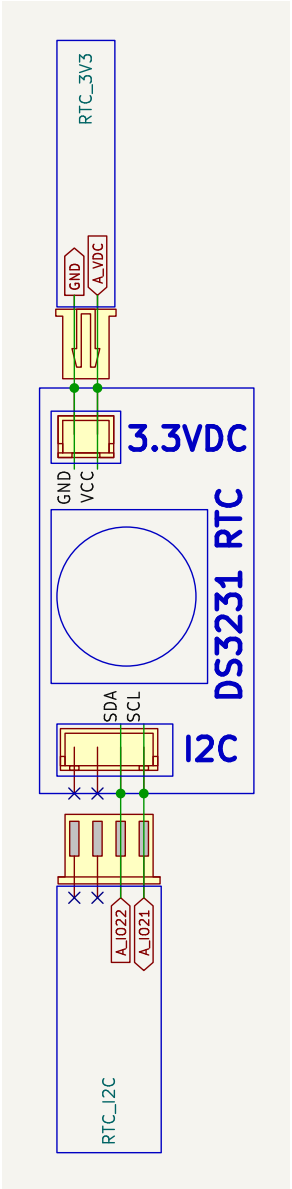


Figure 88. Schematic sheet: LLOYDS TM-838N tube radio

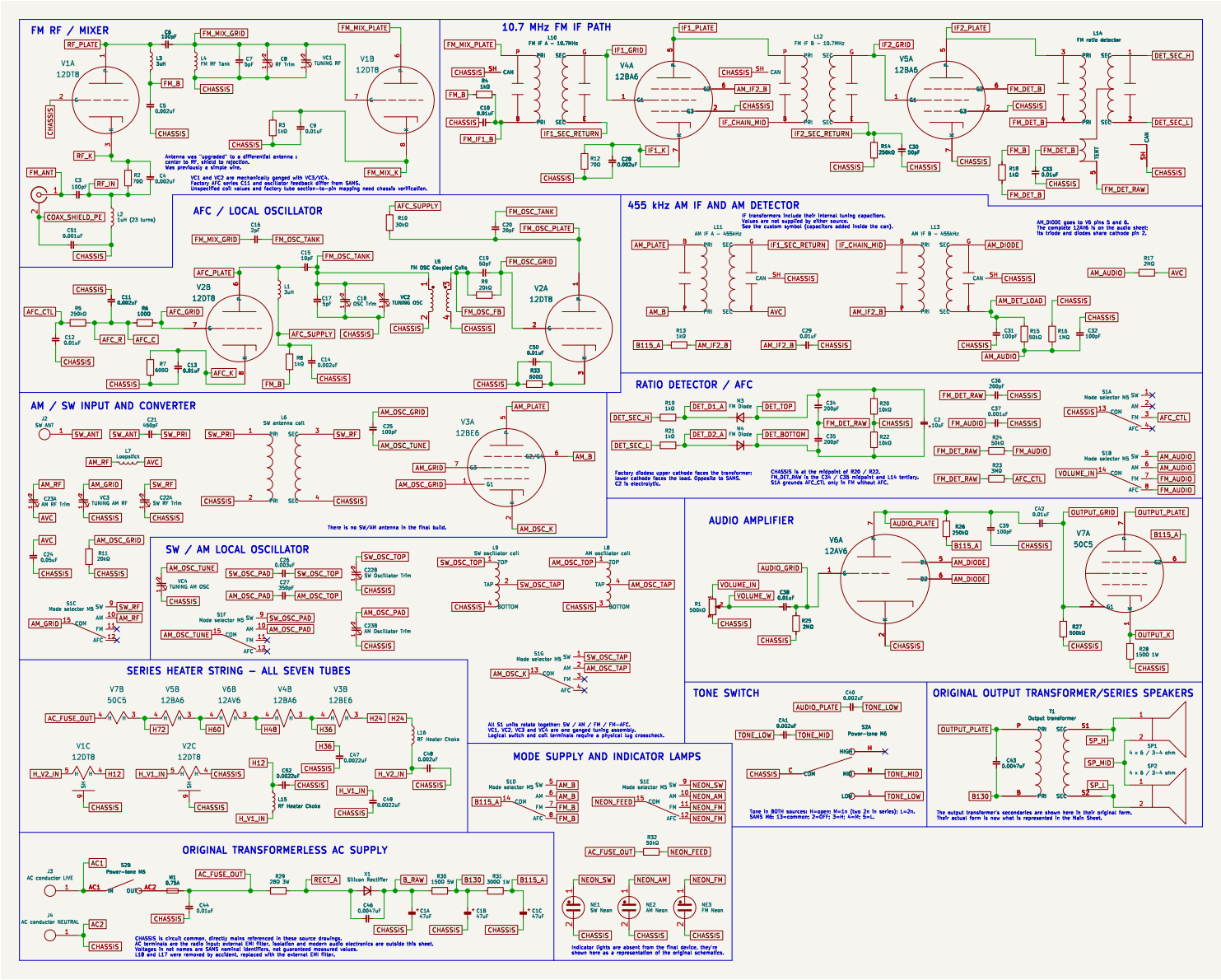


Figure 89. Schematic sheet: MAIN MCU ESP32-S3 N16R8

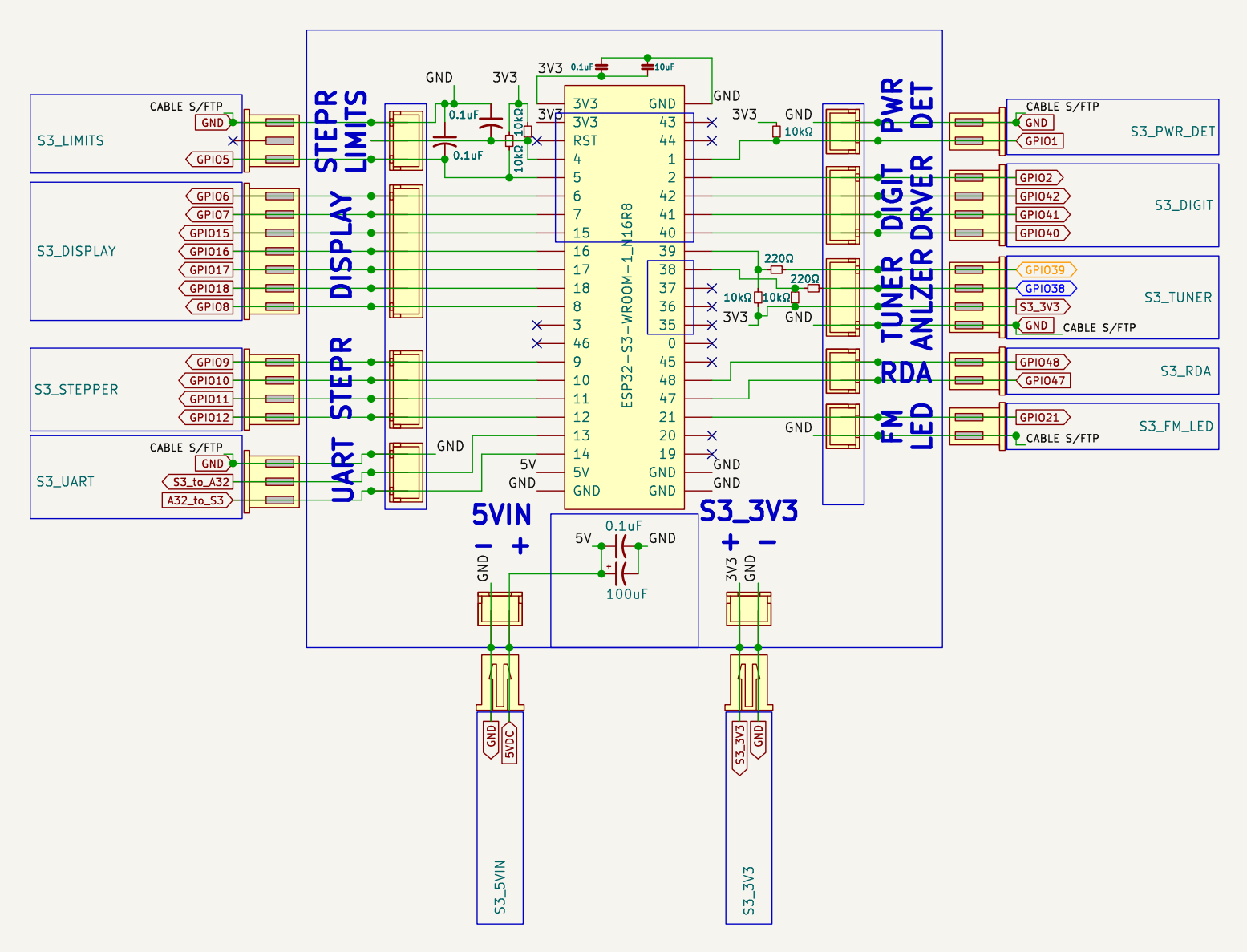


Figure 90. Schematic sheet: P-MOSFET DIGITS & P-MOSFET DIGIT DRIVER & 7 SEGMENTS BUS DRIVER

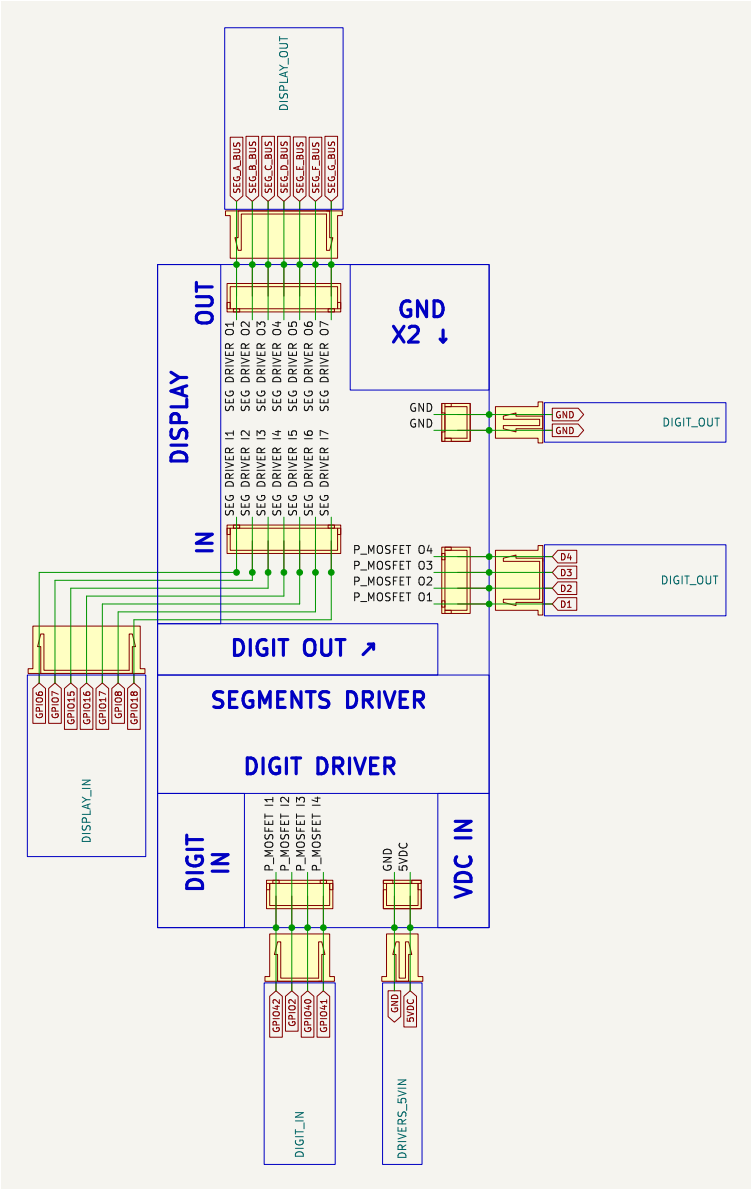


Figure 91. Schematic sheet: PANEL RADIO LED PWM CONTROL

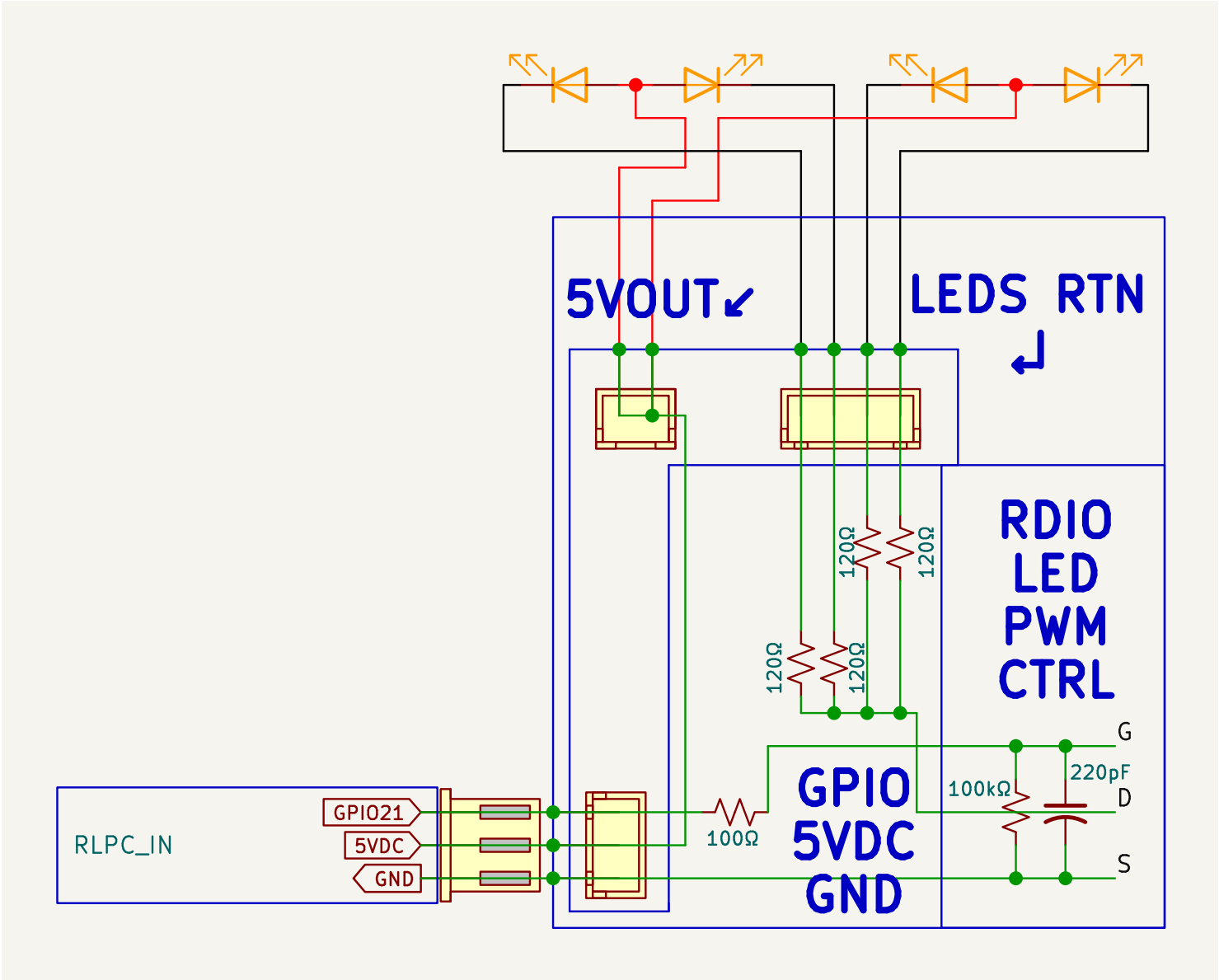


Figure 92. Schematic sheet: PCM1802 ADC

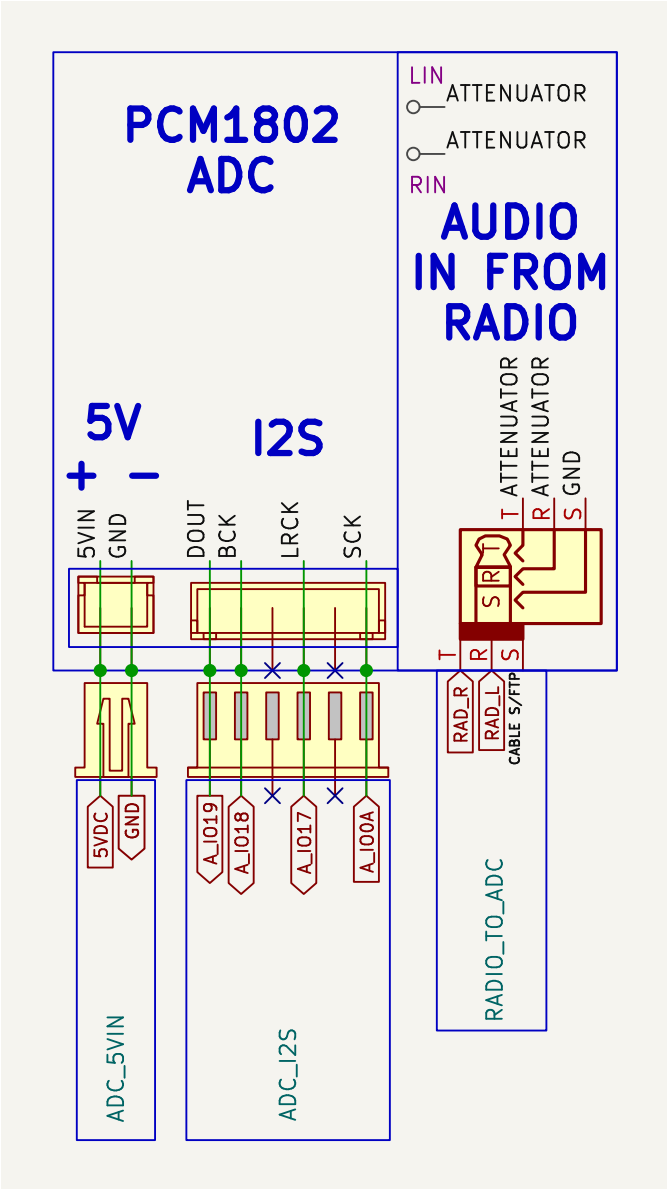


Figure 93. Schematic sheet: PCM5102A DAC

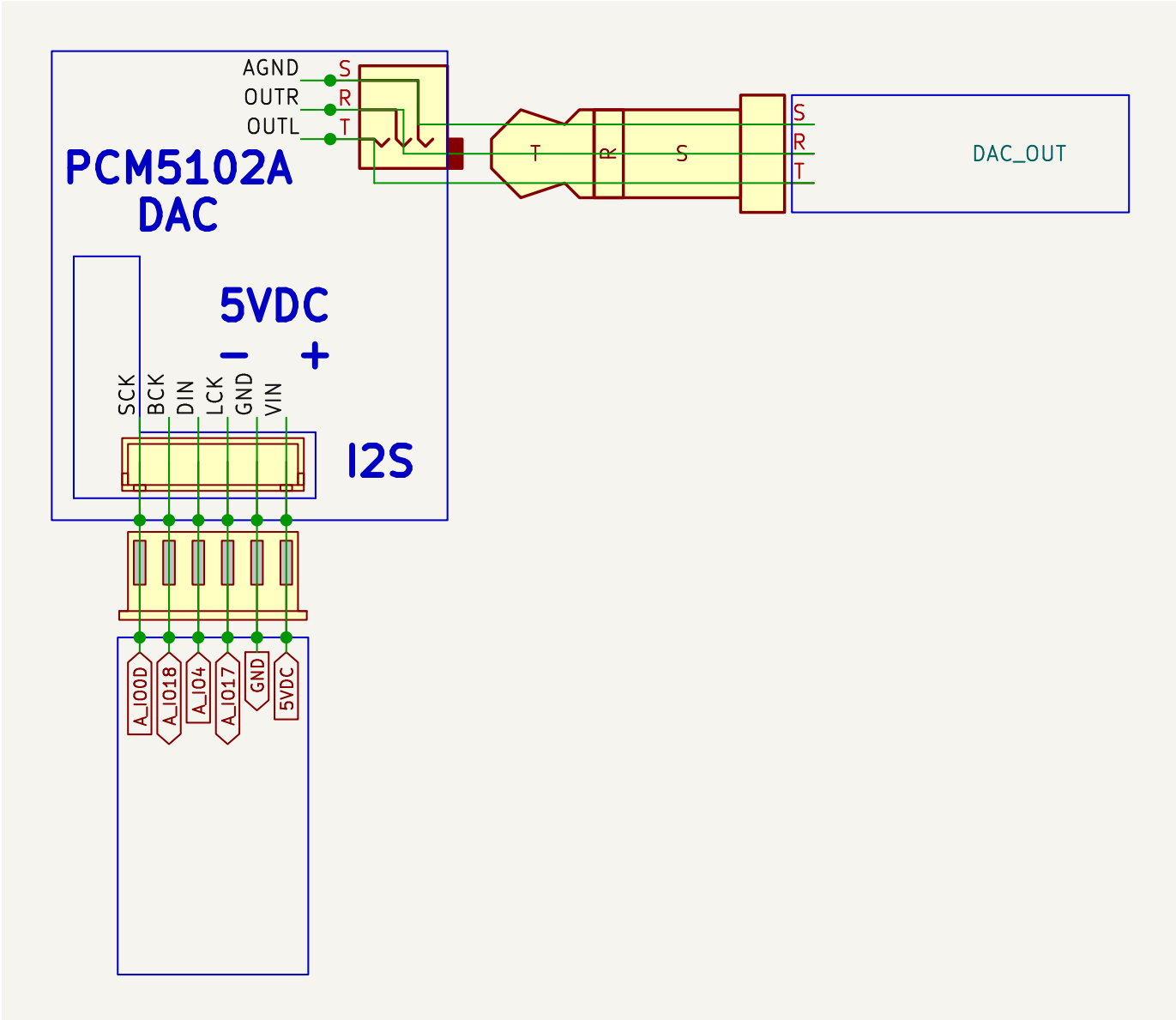


Figure 94. Schematic sheet: Power Detect

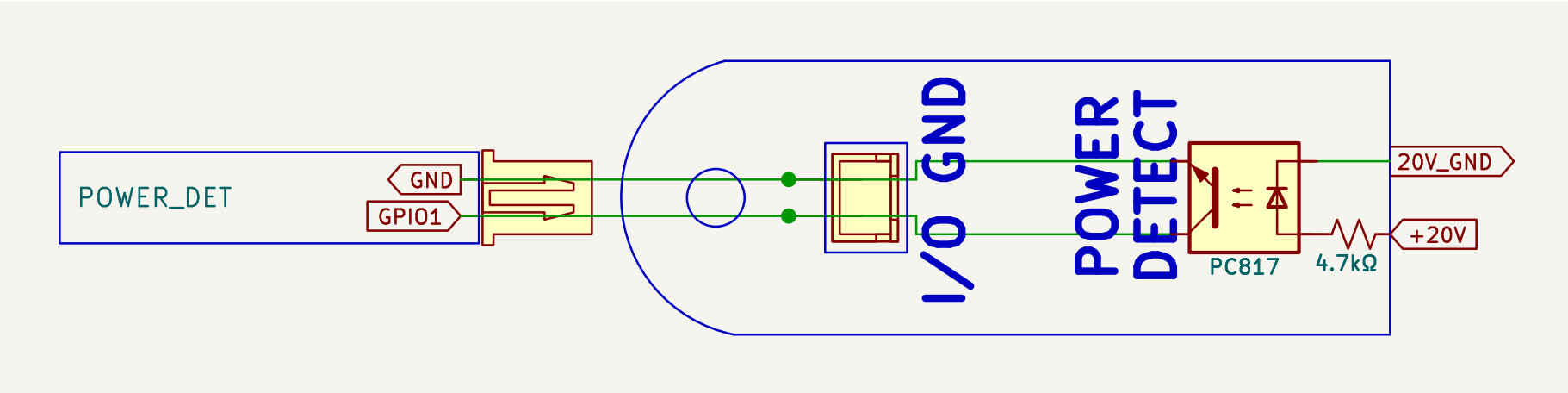


Figure 95. Schematic sheet: RDA5807M

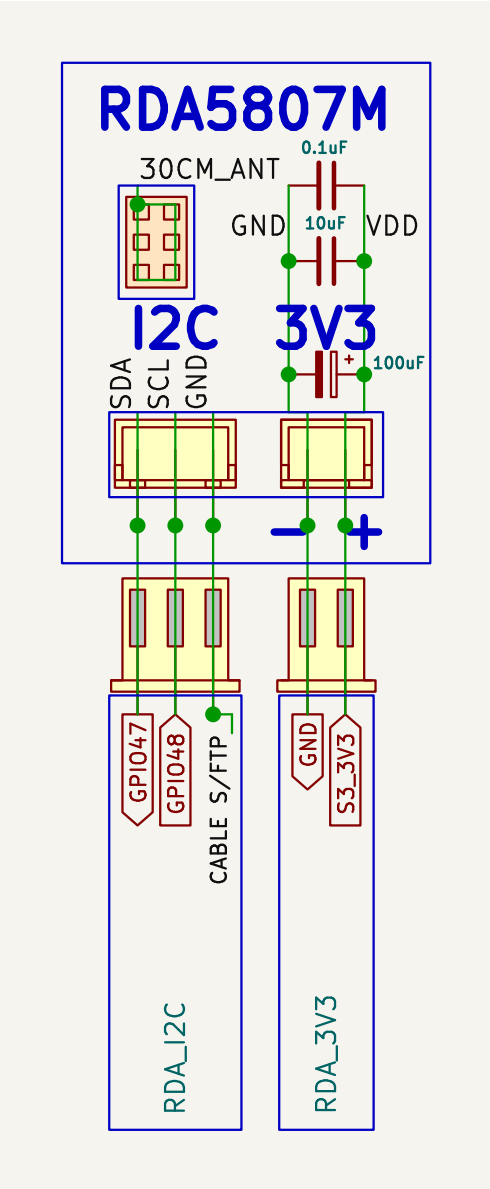


Figure 96. Schematic sheet: STEPPER DRIVER

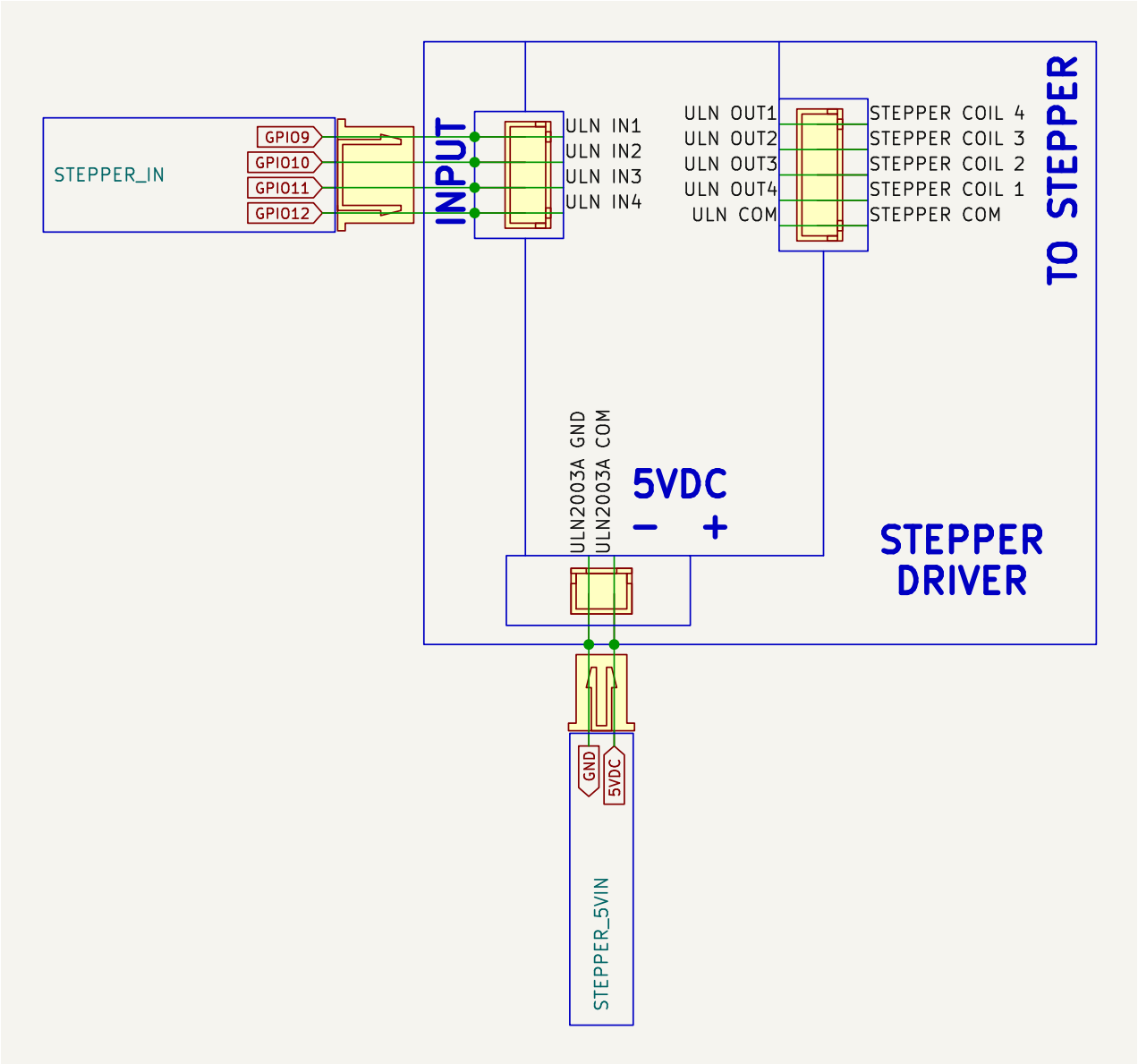


Figure 97. Schematic sheet: STEPPER LIMIT

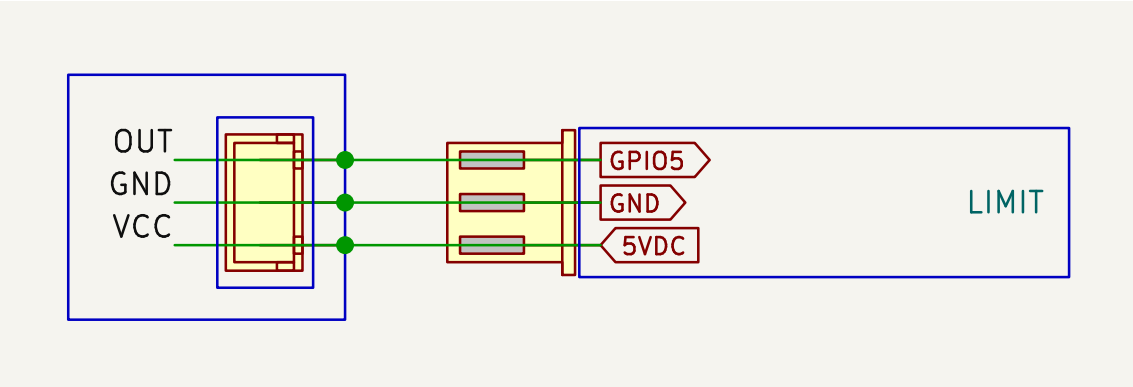


Figure 98. Schematic sheet: Star point, 5 V, 3.3 V and ground buses (sheet “HERE”)

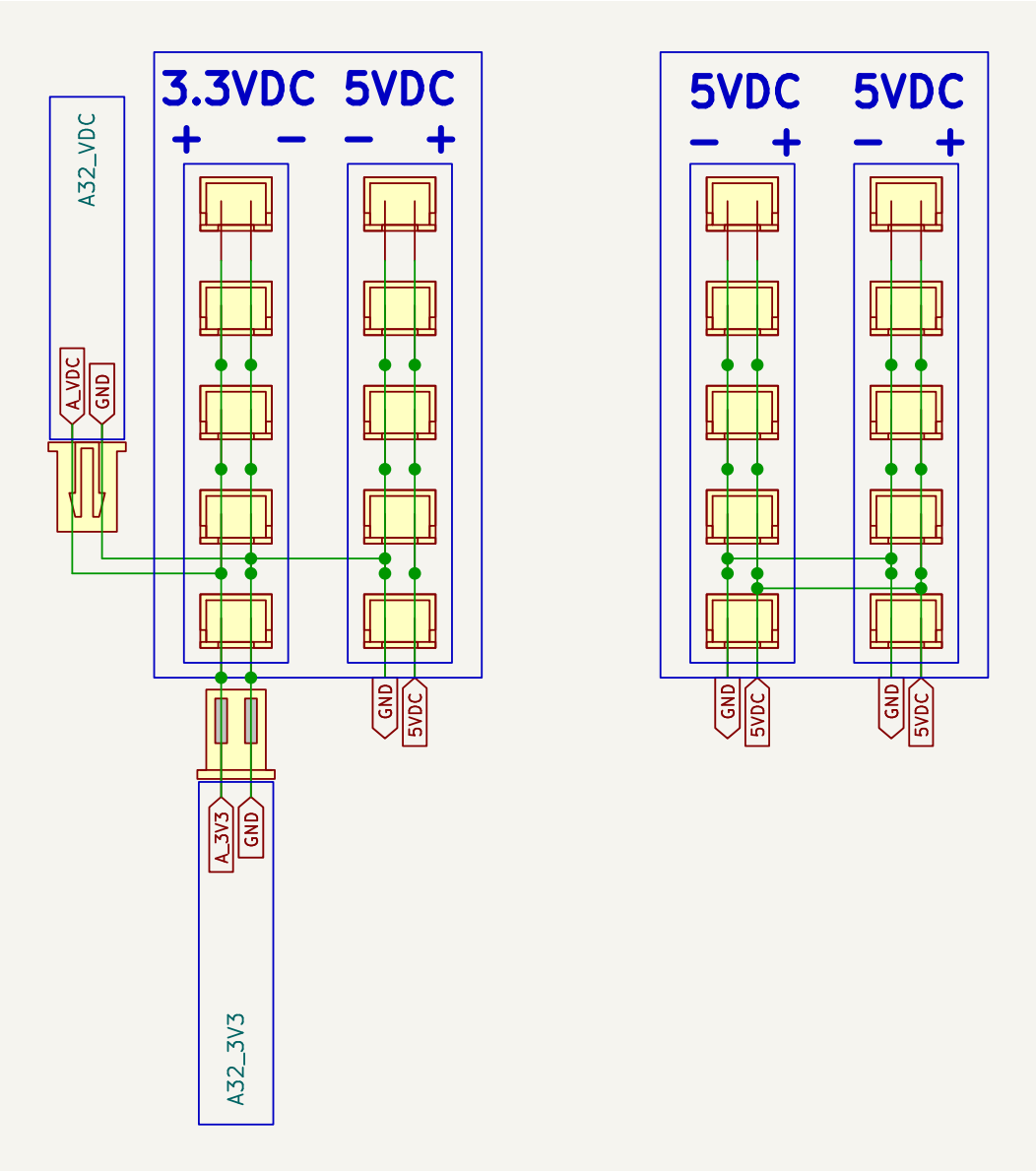
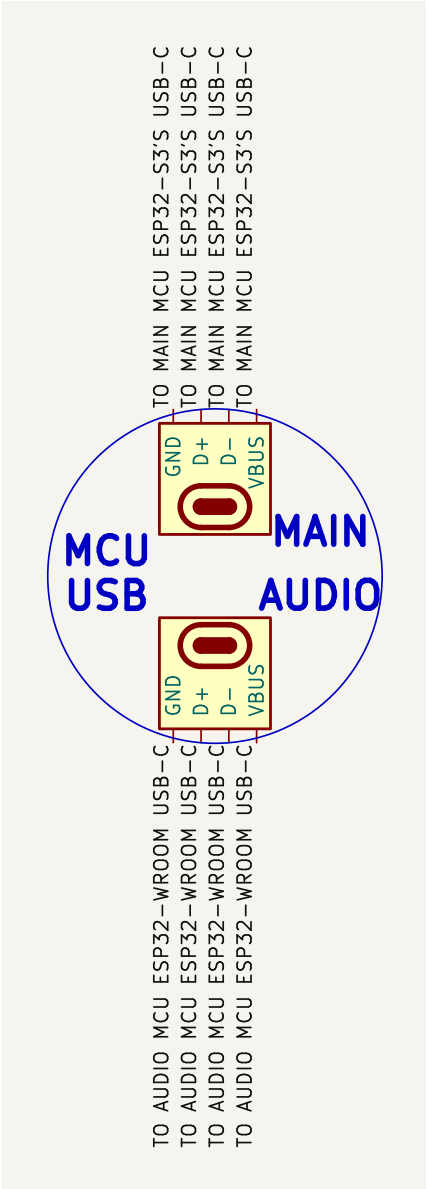


Figure 99. Schematic sheet: USB-C CONNECTORS



Glossary

The terms and names you meet in this book, in alphabetical order. Look a term up here whenever a chapter uses it without explaining it, or when you open an appendix. Signal names are in `code font`, as on the drawings. Where a term has its own section, the chapter is given.

`20V_GND` — The amplifier supply return: the middle (0 V) output of the amplifier's ±20 V supply. One of the machine's four grounds. Its two middle black wires go to the amplifier's ground, so it is joined to Earth and to the DC ground (chapter 5 (p. 21), section 5.3 (p. 23)).

`5VDC` — The 5 V bus, from the 5 V power supply. `5VDC_S` is the same conductor, named before the bus (chapter 5 (p. 21), section 5.5 (p. 26)).

A32 — The audio board: the ESP32 development board that handles sound, Bluetooth, the front controls and the real-time clock (chapter 4 (p. 12)).

`A_IOn` — A signal on pin GPIOn of the audio board: `A_IO22` is its GPIO22.

`A_VDC`, **`A_3V3`** — Two names for one rail: the audio side's 3.3 V, from the audio board's 3.3 V output, shared out by the 3.3 V bus.

Absolute maximum — A maker's limit that must not be exceeded, even briefly. It is not a working value (Appendix A (p. 139)).

AC/DC set — A tube radio with no mains transformer. Its tube heaters are wired in series across the mains, and its B+ is rectified straight from the mains, so its chassis is tied to the mains. The tube radio here is one (chapters 2 (p. 7) and 13).

ADC — Analogue-to-digital converter: here a PCM1802 board that turns the tube radio's audio into digital samples for the audio board (chapter 6 (p. 28)).

AFC — Automatic frequency control: a voltage from the FM detector that keeps the FM oscillator on the station. Switched on in the radio's FM-AFC position (chapter 13 (p. 98)).

Alignment — Adjusting a radio's trimmers, padders and IF transformers so that its tuned circuits track together across the band. Only the tube radio's FM path was professionally tuned; its AM and SW path was not retuned (chapter 13 (p. 98), sections 13.1.1 (p. 99) and 13.7).

AM — Amplitude modulation; on this radio, the medium-wave band. See also SW and FM.

`AMP_` signals — The cable sheet's names for the pins of the amplifier's input panel: `AMP_L+`, `AMP_L-`, `AMP_R+`, `AMP_R-` and `AMP_G`, the ground (chapters 6 (p. 28) and 12).

ASA — A weather-resistant plastic. The antenna's element bushings, their plates and the feedpoint clamp are ASA (chapter 11 (p. 92), section 11.3 (p. 94)).

Attenuator — A resistor network that lowers a signal's level. The radio's output passes one on its way into the ADC, starting with a 4.7 kΩ resistor (chapter 6 (p. 28), section 6.4 (p. 33)).

Audio board — The ESP32 development board that handles the sound, Bluetooth, the front controls and the real-time clock. Also written A32 (chapter 4 (p. 12)).

AUX — The auxiliary analogue input. The AUX jack reaches the amp's input panel, which sums it with the other inputs; the run is not drawn. It never passes through the audio board (chapter 6 (p. 28), section 6.7 (p. 37)).

AVC — Automatic volume control: a voltage from the AM detector that lowers the gain of earlier stages on strong stations, so that strong and weak stations come out at similar levels (chapter 13 (p. 98)).

AWG — American Wire Gauge: a wire-size scale in which a larger number means a thinner wire.

B+ — The tube radio's high-voltage DC supply for its tubes, made by rectifying the mains. The numbers in its point names (`B130`, `B115_A`) are the original service data's nominal figures, not measured voltages (chapter 2 (p. 7) chapter 13 (p. 98), section 13.10 (p. 134)).

Balanced input — An input that takes each channel on two wires, a plus and a minus, and amplifies the difference between them. The DAC reaches the amplifier through one (chapter 12 (p. 96), section 12.3 (p. 97)).

BCK, LRCK, SCK — The three clocks of the I2S audio bus: the bit clock, the word clock (left and right samples) and the master or system clock (chapter 6 (p. 28), section 6.2 (p. 31)).

BNC — A coaxial connector with a bayonet lock. The machine's antenna connector, on the back panel, is one (chapter 11 (p. 92)).

Boom — The metal bar that carries an antenna's elements (chapter 11 (p. 92)).

Bus — Here, a star-point bus: a row of 2-pin power sockets in which every power pin is one wire and every ground pin is one wire. The machine has a 5 V bus and a 3.3 V bus (chapter 5 (p. 21), section 5.5 (p. 26)).

Bypass capacitor — A capacitor that gives AC a short path to ground or chassis around a part, while the part keeps its DC level. Many of the tube radio's stages use them (chapter 13 (p. 98)).

CABLE S/FTP — On a connector pin of the drawings: the cable's shield and foil end there, on ground, and are left unconnected at the other end unless the cable's page says otherwise (chapter 10 (p. 60), section 10.1 (p. 60)).

CBB22 — A common type of metallised polypropylene film capacitor. One sits at the ADC's input (chapter 6 (p. 28), section 6.4 (p. 33)).

CHASSIS — The tube radio's circuit common: its metal chassis and everything joined to it. It is on mains neutral and is joined to none of the other three grounds: Earth, the DC ground and the amplifier supply return (chapter 2 (p. 7), section 2.3 (p. 8) chapter 5 (p. 21), section 5.3 (p. 23)).

Class A — An amplifier stage whose tube conducts through the whole signal cycle. Appendix A (p. 139) gives the 50C5′s typical class-A operating points.

Coax — Coaxial cable: a centre conductor inside a tubular shield, made for radio signals. The FM antenna's feedline is RG-6 coax; the radio's FM input takes RG179 coax (chapters 11 (p. 92) and 13).

Common anode — A display digit whose segment LEDs share one anode. The digit lights when its anode is switched to the supply and a segment's cathode is pulled low (chapter 7 (p. 38)).

Common-mode choke — Here, ferrite cores on a cable or a feedpoint. They resist currents that flow the same way on all the conductors, such as noise picked up along a shield, while passing the signal.

Coupling capacitor — A capacitor that passes a signal from one point to the next while blocking DC. The radio's FM antenna input reaches its first stage through one (C3) (chapter 13 (p. 98), section 13.2 (p. 105)).

CP210x — A Silicon Labs USB-to-serial bridge chip. The audio board's USB port uses one (chapter 4 (p. 12)).

CTR — Current transfer ratio: an optocoupler's output current as a share of its LED current (Appendix A (p. 139)).

DAC — Digital-to-analogue converter: here a PCM5102A board that turns the audio board's digital audio back into analogue audio for the amplifier (chapter 6 (p. 28)).

Darlington — Two transistors in tandem that act as one with a very high gain. The ULN2003A holds seven (Appendix A (p. 139)).

DC ground (**GND**) — The common return of everything on 5 V and 3.3 V. It reaches the amplifier's ground and so is on Earth at almost 0 Ω (chapter 5 (p. 21), section 5.3 (p. 23)).

DCR — The DC resistance of a coil, measured with an ohmmeter (chapter 13 (p. 98)).

De-emphasis — A treble cut that undoes the treble boost added to FM broadcasts before they are sent. The DAC's de-emphasis is switched off; the tube radio has its own (chapters 6 (p. 28) and 13).

Decoupling capacitor — A capacitor from a supply pin to ground, close to a part, that keeps the supply steady when the part draws current in bursts.

Detector — The stage that recovers the audio from the IF signal. The tube radio has an AM detector and, for FM, a ratio detector (chapter 13 (p. 98), sections 13.5 (p. 117) and 13.6).

Development board — A module on a small board with pin headers and a USB port. Both microcontroller boards are development boards (chapter 4 (p. 12)).

Dielectric — The insulating material inside a capacitor, which gives the capacitor its type: film, ceramic, mica or electrolytic. Chapter 13 gives it where it is recorded.

Director, reflector, driven element — The elements of a Yagi antenna. Only the driven element is wired to the cable; the reflector sits behind it and the director in front (chapter 11 (p. 92)).

Download mode — The boot mode in which an ESP32 waits to receive new firmware instead of running its program (Appendix A (p. 139)).

Dual-gang, gang — A gang is one section of a potentiometer or capacitor; a dual-gang part has two sections turned by one shaft. The amplifier's volume control is a dual-gang potentiometer (chapter 12 (p. 96), section 12.2 (p. 96)).

Earth — The mains protective earth, from the mains inlet's earth pin. The Faraday cages, the antenna connector's shell, the amplifier's ground, the DC ground and the amplifier supply return are on it (chapter 5 (p. 21), section 5.3 (p. 23)).

Earth star point — The earth point on the amplifier's aluminium back plate. The amplifier's ground sits on it, through a washer on the Bass pot (chapter 5 (p. 21), section 5.3 (p. 23)). Not the same as the DC ground's star point.

eFuse — A one-time setting burned into a chip. The main board's flash voltage is set by eFuse (chapter 4 (p. 12), section 4.1 (p. 12)).

EMI filter — See mains line filter.

ESP32, ESP32-S3 — Espressif's microcontroller families. The main board is built on an ESP32-S3, the audio board on an ESP32 (chapter 4 (p. 12)).

F-type — The screw-on coax connector of TV and satellite cables. The antenna's feedpoint connector (an F-81 bulkhead) and its grounding block are F-type (chapter 11 (p. 92)).

Faraday cage — An earthed metal enclosure that shields what is inside from electrical noise, and the outside from it. The machine has three, made of aluminium tape (chapter 5 (p. 21), section 5.4 (p. 25)).

FCC ID — The US radio-approval code printed on a module. Its first part, the grantee code, names the company that obtained the approval (chapter 4 (p. 12) Appendix C (p. 149)).

Feedline, feedpoint — The feedline is the coax from the antenna to the radio. The feedpoint is where the feedline joins the driven element, at a 20 mm gap in the folded dipole (chapter 11 (p. 92)).

Ferrite core — A ring or sleeve of ferrite slipped over a cable to damp high-frequency noise on it (chapter 10 (p. 60)).

Fitted, original — In chapter 13 (p. 98), fitted is the part in this radio now; original is what the factory drawing and the SAMS service data give.

Flash — Memory that keeps the program with the power off. The main board has 16 MB, the audio board 4 MB (chapter 4 (p. 12), section 4.1 (p. 12)).

Flyback diode — A diode across a coil that absorbs the voltage spike when the coil is switched off. The ULN2003A has them built in (Appendix A (p. 139)).

FM, FM-AFC — Frequency modulation, the VHF broadcast band. FM-AFC is the radio's FM position with automatic frequency control switched on (chapter 13 (p. 98), section 13.1 (p. 99)).

Folded dipole — A driven element made of two parallel conductors joined at both ends, fed at a gap in one of them. The antenna's driven element is one (chapter 11 (p. 92)).

Gas discharge tube (GDT) — A surge arrester often fitted in coax grounding blocks. The antenna's grounding block has none (chapter 11 (p. 92)).

GPIO — General-purpose input/output: a pin of a microcontroller that its program can read or drive.

Grid leak — A high-value resistor that sets a tube grid's DC level by letting charge leak away (chapter 13 (p. 98)).

Grounding block — A block that joins two lengths of coax. The antenna's is fitted outside, out of the weather. An 8 AWG stranded copper run goes from the grounding block to a lug on the meter base, earthing the coax shield (chapter 11 (p. 92), section 11.2 (p. 93)).

Hall-effect switch — A sensor whose output switches when a magnet comes close enough. The needle's index sensor, an A3144, is one (chapter 8 (p. 46), section 8.3 (p. 49)).

Header — A row of pins on a board, onto which a connector plugs.

Heater string — The tube radio's seven tube heaters wired in series, one after the other, so one current flows through all of them. One open heater stops them all (chapter 13 (p. 98), section 13.9 (p. 130)).

I2C — A two-wire bus, data and clock, that a board uses to talk to several chips by address. The angle sensor, the FM calibration receiver and the real-time clock use it.

I2S — A bus that carries digital audio between chips, with a bit clock, a word clock, a master clock and a data line per direction (chapter 6 (p. 28), section 6.2 (p. 31)).

IEC C14 — The standard three-pin mains inlet, the same kind as on a computer. The machine's mains inlet is one (chapter 5 (p. 21)).

IF — Intermediate frequency: the fixed frequency to which a superheterodyne turns every station. The tube radio's FM IF is 10.6 MHz, its AM IF 455 kHz (chapter 13 (p. 98)).

IF transformer — A tuned transformer that passes the IF from one stage to the next, in a shielded metal can with its own tuning capacitors (chapter 13 (p. 98)).

IN/LISTENER/OUT board — The drawings' name for the source-switch board: the small board, in its own printed case, that passes the audio board's 3.3 V and the two source-switch legs on to the front switch. Its connector is named `LISTENER` (chapter 10 (p. 60), section 10.20 (p. 79)).

Index — The needle's one fixed reference point: a magnet on the needle mount passing over a Hall-effect switch (chapter 8 (p. 46), section 8.3 (p. 49)).

Isolation transformer — A transformer that separates a circuit from the mains. The tube radio has none, which is why its chassis is on mains neutral.

JST XH — A family of small connectors with a 2.5 mm pitch, used for the 5 V and 3.3 V buses (Appendix A (p. 139)).

JTAG — A debug interface for chips. On the main board, GPIO39 to GPIO42 are the chip's JTAG pins; the machine uses them for other jobs (chapter 4 (p. 12) Appendix A (p. 139)).

Kapton — A heat-resistant insulating tape, used to cover the aluminium tape of the Faraday cages.

Ladder — Here, the resistor divider that turns the three positions of the source switch into three voltages for the audio board (chapter 9 (p. 52), section 9.3 (p. 55)).

LED — Light-emitting diode.

Leditron — The name of the clock display: four seven-segment digits and a colon, made from single LEDs (chapter 7 (p. 38)).

Limiter — An FM stage that clips the IF signal to a steady level, so that changes in strength do not reach the detector (chapter 13 (p. 98)).

Listed substitute — A catalogue replacement part named for one of the tube radio's original parts in an unpublished component list. The list is not in this book; chapter 13 (p. 98) names a substitute only where it tells something about the part (chapter 13 (p. 98), section 13.1.3 (p. 100)).

Local oscillator — The oscillator inside a superheterodyne whose signal the mixer combines with the station. The tube radio's FM local oscillator runs one IF below the station (chapter 13 (p. 98)).

Logic-level MOSFET — A MOSFET that turns fully on with a gate voltage a logic pin can give. The panel-lamp switch, an IRL540N, is one (chapter 9 (p. 52), section 9.1 (p. 53)).

Loopstick — A coil wound on a ferrite rod that serves as an AM antenna. The tube radio's AM input uses its internal loopstick (chapter 13 (p. 98)).

LVCMOS — Low-voltage CMOS: a family of logic input and output levels (Appendix A (p. 139)).

Main board — The ESP32-S3 development board that runs the clock display, the dial needle and its sensors, the FM calibration receiver and the panel lamps. Also written S3 (chapter 4 (p. 12)).

Mains line filter — The EMI (electromagnetic interference) filter between the switched mains and the tube radio, under its own Faraday cage (chapter 5 (p. 21), section 5.2 (p. 23)).

Microcontroller, MCU — A small computer on one chip, with its own inputs and outputs. The machine has two, on the main board and the audio board (chapter 4 (p. 12)).

Mixer — The stage of a superheterodyne that combines the station with the local oscillator's signal to make the IF (chapter 13 (p. 98)).

Mode switch — The tube radio's four-position switch: SW, AM, FM and FM-AFC. The drawing splits it into seven units, S1A to S1G, that turn together (chapter 13 (p. 98), section 13.1.6 (p. 104)).

Module — The small shielded board that carries a microcontroller chip with its flash memory. Each development board carries one (chapter 4 (p. 12)).

MOSFET, P-channel MOSFET — A transistor switched by the voltage on its gate. A P-channel MOSFET switches a load to the supply; the clock digits are switched by four of them, FQP27P06 (chapter 7 (p. 38)).

Neon lamp — A small glow lamp that runs from mains voltage. The tube radio's mode lamps (NE1 to NE3) are drawn but not fitted (chapter 13 (p. 98), section 13.1.6 (p. 104)).

New old stock — A part made long ago and never used. Three of the radio's tubes are new old stock (chapter 13 (p. 98)).

NP0 — A ceramic capacitor dielectric whose value hardly changes with temperature, used where a value must stay exact (chapter 13 (p. 98)).

Op-amp — Operational amplifier: a small amplifier chip used for signal stages. The amplifier's input board holds three (chapter 12 (p. 96), section 12.1 (p. 96)).

Open collector — An output that can only pull its line low; a pull-up resistor sets the high level. The index sensor and the optocoupler have one.

Optocoupler — A part that passes a signal across by light, an LED lighting a phototransistor, with no wire between the two sides. The amplifier power sensor uses a PC817 (chapter 9 (p. 52), section 9.5 (p. 59)).

Output transformer — The transformer between the radio's output tube and its load. In this machine its secondary feeds a 7.5 Ω resistor and the ADC, not speakers (chapter 6 (p. 28), section 6.4 (p. 33)).

Package (TO-220, DIP-4, QFN56) — The body and pin layout of a chip or transistor. TO-220 is a power package with three legs and a metal tab; DIP-4 a

four-pin through-hole package; QFN56 a flat chip package with 56 pads (Appendix A (p. 139)).

Padder, trimmer — Small capacitors in a tuned circuit. A trimmer is adjustable and sets the end of a tuning range; a padder, in series with the oscillator's tuning, keeps the oscillator in step with the signal circuits across the band (chapter 13 (p. 98)).

Passive radiator — A speaker cone with no magnet or coil of its own, moved by the air in the enclosure, used in place of a port (chapter 12 (p. 96), section 12.1 (p. 96)).

Pentagrid converter, pentode, triode, beam power tube — Tube types. A triode has one grid, a pentode three; a pentagrid converter has five grids and is mixer and oscillator in one tube; a beam power tube is built to deliver power to a load (chapter 13 (p. 98)).

Perfboard — A board with a grid of holes for hand-wired circuits. Several of the machine's small circuits are built on perfboard.

PLA, TPU, STL — PLA is the plastic of the printed parts; TPU is a flexible printing plastic, used only for the passive radiators' membranes; an STL is the file a printed part is made from (Appendix D (p. 152)).

Plate, cathode, control grid — Electrodes of a tube. The heated cathode gives off electrons, the plate (the anode) collects them, and the control grid between them sets how many get through (chapter 13 (p. 98)).

PLL — Phase-locked loop: a circuit that makes one clock from another. The DAC can make its own master clock this way from the bit clock (Appendix A (p. 139)).

Polarised capacitor — A capacitor with a + and a − terminal, such as an electrolytic, that must be fitted the right way round. In the tube radio's polarised capacitors (C1A to C1C and C2), pin 1 is the + terminal (Appendix B (p. 146)).

PSRAM — Extra memory beside a microcontroller's own. The main board's chip carries 8 MB (chapter 4 (p. 12)).

PSU — Power supply unit. Used in drawing names and printed-part file names.

Pull-up, pull-down — A resistor that holds a line high (to the supply) or low (to ground) when nothing else drives it.

PWM — Pulse-width modulation: switching a load on and off fast, with the on-time setting the average power. The FM panel lamps are dimmed by PWM (chapter 9 (p. 52), section 9.1 (p. 53)).

Ratio detector — The tube radio's FM detector: two diodes and a transformer that turn frequency changes into audio and largely ignore changes in strength (chapter 13 (p. 98), section 13.5 (p. 117)).

RCA — The common phono jack for line-level audio. The amplifier's input panel has an RCA pair (chapter 12 (p. 96)).

Rectifier — A part that turns AC into DC. The tube radio's rectifier (X1) makes its B+ from the mains (chapter 13 (p. 98)).

RF — Radio frequency: the station's signal as it arrives, before the mixer turns it into the IF. On FM, the RF amplifier (V1) raises it first (chapter 13 (p. 98), section 13.2 (p. 105)).

RMS — Root mean square. A power rating in W RMS is a continuous power, not a peak.

RTC — Real-time clock: a chip that keeps the time; a backup cell on its module keeps it running while the power is off. Here a DS3231 module (chapter 4 (p. 12), section 4.7 (p. 20)).

S3 — The main board, an ESP32-S3 development board (chapter 4 (p. 12)).

`S3_3V3` — The main board's own 3.3 V output. It is a separate rail, not the 3.3 V bus.

S/FTP — See shield and foil.

Sample rate — The number of audio samples per second on each channel.

SAMS — SAMS Photofact, the published service data for the tube radio. With the factory drawing, it is one of the radio's two original documents.

Schmitt trigger — An input with two switching thresholds, one for a rising and one for a falling signal, so that a slow or noisy signal switches cleanly (Appendix A (p. 139)).

Screen grid — The grid of a pentode or beam tube between the control grid and the plate, held at a steady positive voltage (chapter 13 (p. 98)).

SDA, SCL — The data and clock lines of an I2C bus.

Shield and foil — The braided shield and the foil wrap of a shielded cable. The machine's shielded cables are S/FTP: a braided shield and a foil around twisted pairs (chapter 10 (p. 60), section 10.1 (p. 60)).

Slave mode — A converter mode in which the chip takes its clocks from outside instead of making them. The ADC runs in slave mode, clocked by the audio board (chapter 6 (p. 28)).

Source switch — The front three-position switch: RADIO, BT and AUX. It only tells the audio board which source was chosen; it switches no audio (chapter 9 (p. 52), section 9.3 (p. 55)).

SPI, quad SPI, octal SPI — A serial link between a chip and its memory, with four (quad) or eight (octal) data lines. On the main board's module variant (N16R8), the maker gives quad SPI flash and octal SPI PSRAM (Appendix A (p. 139)).

Standoff — A spacer post that holds a board or a supply off a surface. The amplifier's ±20 V supply is held off the back plate by plastic screws and standoffs (chapter 5 (p. 21), section 5.3 (p. 23)).

Star point — The one place where all the DC ground wires meet: the ground pins of the 5 V and 3.3 V buses (chapter 5 (p. 21), section 5.5 (p. 26)). See also earth star point.

Stepper motor — A motor that turns in fixed steps, one for each change of its coil currents. The needle's motor is a 28BYJ-48 (chapter 8 (p. 46), section 8.2 (p. 48)).

Strapping pin — A microcontroller pin whose level is read at power-up to set how the chip starts (Appendix A (p. 139)).

Superheterodyne — A receiver that turns every station into one fixed intermediate frequency (IF) before amplifying and detecting it (chapter 13 (p. 98), section 13.1.1 (p. 99)).

Suppressor grid — The grid of a pentode nearest the plate. It stops electrons knocked off the plate from reaching the screen grid. In the radio's 12BA6 IF tubes (V4, V5) it is on the chassis (chapter 13 (p. 98), section 13.4 (p. 113)).

SW — Short wave, one of the tube radio's bands.

SWR — Standing-wave ratio: a measure of how well an antenna matches its cable. Not measured on this antenna (chapter 11 (p. 92)).

Taper — How a potentiometer's resistance changes along its travel. An audio taper changes slowly at first, to suit the ear. The radio's volume control is audio taper, as its original specification infers from its listed substitutes (chapter 13 (p. 98)).

Tertiary winding — A third winding on a transformer, besides the primary and the secondary. The ratio-detector transformer (L14) has one (chapter 13 (p. 98), section 13.4 (p. 113)).

TRS — Tip, ring, sleeve: a three-contact jack or plug, as on stereo headphones (chapter 6 (p. 28)).

Tube base — The pin layout of a tube. B7G is the 7-pin miniature base; B9A, or noval, the 9-pin miniature. An EIA basing code (such as 7CH or 9DE) says which electrode is on which pin (chapter 13 (p. 98), section 13.1.4 (p. 101)).

UART — A two-wire serial link, one wire each way, plus ground. The two boards talk over one (chapter 10, section 10.12 (p. 72)).

USB-Serial/JTAG — The ESP32-S3 chip's own USB port, which serves as both a serial link and a debug interface. The main board's USB port uses it (chapter 4 (p. 12)).

VBUS — The 5 V power wire of a USB cable (chapter 4 (p. 12), section 4.6 (p. 20)).

VHF — Very high frequency: the radio band that holds the FM broadcast band (chapters 11 (p. 92) and 13).

Wiper — The sliding contact of a potentiometer. Its voltage follows the knob's position (chapter 9 (p. 52), section 9.4 (p. 58)).

X1Y1, X1Y2 — Safety ratings for capacitors connected to the mains. An X rating is for a capacitor across the mains, a Y rating for one between the mains and a part a person may touch; X1Y1 and X1Y2 parts carry both. Replace such a part only with one of the same class (chapter 2 (p. 7), section 2.4 (p. 8)).

Yagi — A directional antenna: a driven element with unpowered elements beside it on a boom. The FM antenna is a three-element Yagi (chapter 11 (p. 92)).

Zero-data mute — The DAC's own mute: after a run of all-zero audio on both channels, it silences its output (Appendix A (p. 139)).

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